Mathematics

Class 12 (English Medium)

Solved Exercises — Sindh Textbook Board

12 lessons — 533 questions

Unit 1: Introduction to Symbolic Package:

MAPLE

Exercise 1

Question: 1. Expand the following expressions using the expand command in Maple: (i) (x + y)^3 (ii) cos(x + y) + sin(x - y) (iii) (2a - b)^2 + (a + 3b)^2 (iv) e^x (x + y) - e^x (x - y) (v) (x + 2)(x - 3) (vi) sqrt(x + y)

Answer: (i) Step 1: Use the binomial expansion (x + y)^3 = x^3 + 3 x^2 y + 3 x y^2 + y^3, writing each term in turn

Step 2: The coefficient of x^2 y is 3 and the coefficient of x y^2 is 3, from the binomial coefficients 1, 3, 3, 1

Final answer: (i) x^3 + 3x^2 y + 3x y^2 + y^3

(ii) Step 1: Use cos(x + y) = cos x cos y - sin x sin y

Step 2: Use sin(x - y) = sin x cos y - cos x sin y

Step 3: Add the two expansions term by term, keeping all four terms because no two of them are like terms

Final answer: (ii) cos x cos y - sin x sin y + sin x cos y - cos x sin y

(iii) Step 1: Expand (2a - b)^2 = 4a^2 - 4ab + b^2 using (p - q)^2 = p^2 - 2pq + q^2

Step 2: Expand (a + 3b)^2 = a^2 + 6ab + 9b^2 using (p + q)^2 = p^2 + 2pq + q^2

Step 3: Add the a^2 terms: 4a^2 + a^2 = 5a^2

Step 4: Add the ab terms: -4ab + 6ab = 2ab

Step 5: Add the b^2 terms: b^2 + 9b^2 = 10b^2

Final answer: (iii) 5a^2 + 2ab + 10b^2

(iv) Step 1: Expand the first product: e^x (x + y) = x e^x + y e^x

Step 2: Expand the second product: e^x (x - y) = x e^x - y e^x

Step 3: Subtract: x e^x + y e^x - x e^x + y e^x

Step 4: The x e^x terms cancel and y e^x + y e^x = 2y e^x

Final answer: (iv) 2 e^x y

(v) Step 1: Multiply out: (x + 2)(x - 3) = x^2 - 3x + 2x - 6

Step 2: Collect the like terms -3x + 2x = -x

Final answer: (v) x^2 - x - 6

(vi) Step 1: sqrt(x + y) is already a single radical term with no product or power to distribute over it

Step 2: The expand command therefore returns it unchanged

Final answer: (vi) sqrt(x + y)

Question: 2. Use the factor command in Maple to factorize the following expressions and write the factored form: (i) x^3 - 8 (ii) x^4 - y^4 (iii) x^2 - 4 (iv) 3x^3 + 12x^2 - 15x (v) 2x^2 + 6xy + 4y^2 (vi) x^2 + 4x + 4

Answer: (i) Step 1: Write 8 as 2^3 so the expression is a difference of cubes x^3 - 2^3

Step 2: Use a^3 - b^3 = (a - b)(a^2 + ab + b^2) with a = x and b = 2

Step 3: Substitute: (x - 2)(x^2 + 2x + 4)

Final answer: (i) (x - 2)(x^2 + 2x + 4)

(ii) Step 1: Treat x^4 - y^4 as a difference of squares (x^2)^2 - (y^2)^2

Step 2: Factor: (x^2 - y^2)(x^2 + y^2)

Step 3: Factor x^2 - y^2 again as a difference of squares: (x - y)(x + y)

Final answer: (ii) (x - y)(x + y)(x^2 + y^2)

(iii) Step 1: Write as x^2 - 2^2, a difference of squares

Step 2: Use a^2 - b^2 = (a - b)(a + b)

Final answer: (iii) (x - 2)(x + 2)

(iv) Step 1: Take out the common factor 3x: 3x(x^2 + 4x - 5)

Step 2: Factor the quadratic by splitting: x^2 + 4x - 5 = x^2 + 5x - x - 5

Step 3: Group: x(x + 5) - 1(x + 5) = (x + 5)(x - 1)

Final answer: (iv) 3x(x + 5)(x - 1)

(v) Step 1: Take out the common factor 2: 2(x^2 + 3xy + 2y^2)

Step 2: Split 3xy as xy + 2xy: x^2 + xy + 2xy + 2y^2

Step 3: Group: x(x + y) + 2y(x + y) = (x + y)(x + 2y)

Final answer: (v) 2(x + y)(x + 2y)

(vi) Step 1: Look for two numbers with product 4 and sum 4, namely 2 and 2

Step 2: Write x^2 + 4x + 4 = x^2 + 2x + 2x + 4

Step 3: Group: x(x + 2) + 2(x + 2) = (x + 2)(x + 2)

Final answer: (vi) (x + 2)^2

Question: 3. Simplify the following expressions using the simplify command in Maple and write the simplified form: (i) (x^2 + 2x + 1) / (x + 1) (ii) (x^3 - x^2 - x + 1) / (x^2 + 1) (iii) sin(x)^2 + cos(x)^2 (iv) (3x^2 +

2x - 1) / (x^2 + 1)

Answer: (i) Step 1: Factor the numerator as a perfect square: x^2 + 2x + 1 = (x + 1)^2

Step 2: Write the fraction as (x + 1)^2 / (x + 1)

Step 3: Cancel one common factor (x + 1), which is allowed when x is not equal to -1

Final answer: (i) x + 1

(ii) Step 1: Group the numerator: x^2 (x - 1) - 1(x - 1)

Step 2: Factor: (x - 1)(x^2 - 1)

Step 3: Factor x^2 - 1 as a difference of squares: (x - 1)(x - 1)(x + 1) = (x + 1)(x - 1)^2

Step 4: The denominator x^2 + 1 has no real factor in common with the numerator, so nothing cancels

Final answer: (ii) (x + 1)(x - 1)^2 / (x^2 + 1)

(iii) Step 1: Recall the Pythagorean identity sin^2 x + cos^2 x = 1, which holds for every value of x

Step 2: The simplify command applies exactly this identity

Final answer: (iii) 1

(iv) Step 1: Try to factor the numerator: 3x^2 + 2x - 1 = 3x^2 + 3x - x - 1 = 3x(x + 1) - 1(x + 1) = (3x - 1)(x + 1)

Step 2: Compare with the denominator x^2 + 1, which is not a factor of the numerator and shares no factor with it

Step 3: Since there is no common factor to cancel, the fraction is already in its simplest form

Final answer: (iv) (3x^2 + 2x - 1) / (x^2 + 1)

Question: 4. Evaluate the numerical value of sin(pi / 6) using the evalf command in Maple.

Answer: Step 1: pi / 6 is the radian measure of 30 degrees

Step 2: The exact value is sin(pi / 6) = 1 / 2, from the standard 30-60-90 triangle

Step 3: The evalf command converts this exact value into a decimal

Step 4: 1 / 2 written as a decimal is 0.5

Final answer: 0.5

Question: 5. Find the inverse of the matrix A := Matrix(2, [1, 2, 3, 4]) using the Matrix Inverse command in Maple.

Answer: Step 1: Read the list row by row, so A is the 2 by 2 matrix with first row 1, 2 and second row 3, 4

Step 2: Find the determinant: det A = (1)(4) - (2)(3) = 4 - 6 = -2

Step 3: Since the determinant is not zero, the inverse exists

Step 4: Swap the diagonal entries 1 and 4, and change the signs of the off-diagonal entries 2 and 3, giving the adjugate matrix with rows 4, -2 and -3, 1

Step 5: Divide every entry by the determinant -2: 4 / (-2) = -2, -2 / (-2) = 1, -3 / (-2) = 3 / 2, 1 / (-2) = -1 / 2

Final answer: A inverse is the matrix with first row -2, 1 and second row 3/2, -1/2

Question: 6. Find the transpose of the matrix B := Matrix(2, 3, [1, 2, 3, 4, 5, 6]) using the Transpose command in Maple.

Answer: Step 1: B has 2 rows and 3 columns, filled row by row, so its first row is 1, 2, 3 and its second row is 4, 5, 6

Step 2: In the transpose, the first row of B becomes the first column, and the second row of B becomes the second column

Step 3: The transpose therefore has 3 rows and 2 columns

Step 4: Its first row is 1, 4 (the old first column), its second row is 2, 5, and its third row is 3, 6

Final answer: B transpose is the 3 by 2 matrix with rows 1, 4 and 2, 5 and 3, 6

Question: 7. Perform matrix multiplication for matrices A := Matrix(2, [1, 2, 3, 4]) and B := Matrix(2, [5, 6, 7, 8]) using the command in Maple.

Answer: Step 1: A has rows 1, 2 and 3, 4; B has rows 5, 6 and 7, 8; both are 2 by 2, so the product AB exists and is 2 by 2

Step 2: Entry (1,1): (1)(5) + (2)(7) = 5 + 14 = 19

Step 3: Entry (1,2): (1)(6) + (2)(8) = 6 + 16 = 22

Step 4: Entry (2,1): (3)(5) + (4)(7) = 15 + 28 = 43

Step 5: Entry (2,2): (3)(6) + (4)(8) = 18 + 32 = 50

Final answer: AB is the matrix with first row 19, 22 and second row 43, 50

Question: 8. Plot the functions f(x) = x^2 and g(x) = 2x - 1 on the same graph in the range [-5, 5].

Answer: Step 1: For f(x) = x^2 make a table: at x = -2, -1, 0, 1, 2 the values are 4, 1, 0, 1, 4, so f is an upward parabola with vertex at the origin

Step 2: For g(x) = 2x - 1 make a table: at x = 0 the value is -1 and at x = 1 the value is 1, so g is a straight line of slope 2 crossing the y-axis at -1

Step 3: In Maple the single command plot([x^2, 2x - 1], x = -5..5) draws both curves on one set of axes over the required range

Step 4: The two graphs meet where x^2 = 2x - 1, that is x^2 - 2x + 1 = 0,

that is (x - 1)^2 = 0, so they touch at the single point (1, 1), where the line is tangent to the parabola

Final answer: The parabola y = x^2 and the line y = 2x - 1 are drawn together on [-5, 5], touching at (1, 1)

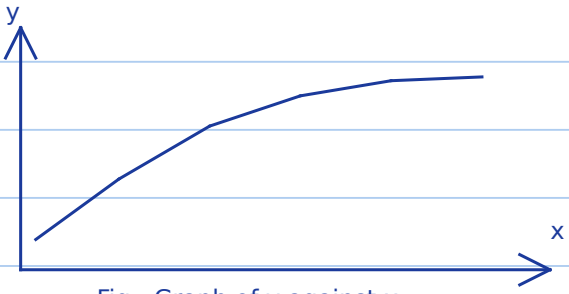


Fig.: Graph of y against x

Question: 9. Generate a graph that includes the functions h(x) = sin(x) and k(x) = cos(x) in the interval [0, 2pi].

Answer: Step 1: For h(x) = sin x: sin 0 = 0, sin(pi / 2) = 1, sin(pi) = 0, sin(3pi / 2) = -1, sin(2pi) = 0, so the sine curve completes one full wave on the interval

Step 2: For k(x) = cos x: cos 0 = 1, cos(pi / 2) = 0, cos(pi) = -1, cos(3pi / 2) = 0, cos(2pi) = 1, so the cosine curve starts at its maximum

Step 3: In Maple the command is plot([sin(x), cos(x)], x = 0..2pi)

Step 4: The curves cross where sin x = cos x inside the interval, at x = pi / 4 and x = 5pi / 4

Final answer: One complete sine wave starting at 0 and one complete cosine wave starting at 1 are drawn together on [0, 2pi], crossing at pi / 4 and 5pi / 4

Question: 10. Plot the parametric curves x(t) = cos(2t) and y(t) = sin(3t) for t in the range [-2pi, 2pi].

Answer: Step 1: In a parametric plot the point (x, y) is traced as t increases, with x = cos(2t) completing 2 full cycles while t runs over an interval of length 2pi, and y = sin(3t) completing 3 full cycles

Step 2: At t = 0 the point is (cos 0, sin 0) = (1, 0)

Step 3: At t = pi / 4 the point is (cos(pi / 2), sin(3pi / 4)) = (0, sqrt(2) / 2)

Step 4: At t = pi / 2 the point is (cos(pi), sin(3pi / 2)) = (-1, -1)

Step 5: In Maple the command is plot([cos(2t), sin(3t), t = -2pi..2pi])

Step 6: Because the frequency ratio is 3 to 2, the traced curve is a closed Lissajous figure inside the square from -1 to 1 on both axes

Final answer: The parametric plot is the closed Lissajous curve traced by (cos 2t, sin 3t) for t in [-2pi, 2pi]

Unit 2: Functions and Limits

Exercise 2.1

Question: 1. Identify the domain and range of the functions through the following graphs: (i) an upward parabola with vertex at the origin (ii) a straight line through the origin rising to the right (iii) a curve starting at the origin and rising, flattening out, defined only for x from 0 onwards (iv) a curve defined for all x except 0, always above the x-axis, rising sharply near x = 0 and flattening towards 0 at both ends (v) a curve with one branch below the x-axis on the left of x = 0 and one branch above the x-axis on the right of x = 0, never taking the value 0.

Answer: (i) Step 1: The parabola extends without bound to the left and to the right, so every real x occurs

Step 2: Its lowest point is the vertex at y = 0 and it rises without bound, so the y values are 0 and all positive numbers

Final answer: (i) Domain = R, Range = {y in R : y is 0 or positive}

(ii) Step 1: The line extends without bound in both directions horizontally, so every real x occurs

Step 2: It also extends without bound upwards and downwards, so every real y occurs

Final answer: (ii) Domain = R, Range = R

(iii) Step 1: The curve begins at x = 0 and continues only to the right, so x is 0 or positive

Step 2: It starts at y = 0 and keeps rising, so y is 0 or positive

Final answer: (iii) Domain = {x in R : x is 0 or positive}, Range = {y in R : y is 0 or positive}

(iv) Step 1: The curve has two branches separated at x = 0, where it is not defined, but it occurs for every other x

Step 2: Both branches stay strictly above the x-axis, coming close to 0 at the far ends but never reaching it, and rising without bound near x = 0

Final answer: (iv) Domain = {x in R : x is not 0}, Range = {y in R : y is positive}

(v) Step 1: Again the curve is missing at x = 0 but occurs for every other x

Step 2: The left branch takes all negative values and the right branch takes all positive values, while the value 0 is never taken

Final answer: (v) Domain = {x in R : x is not 0}, Range = {y in R : y is not 0}

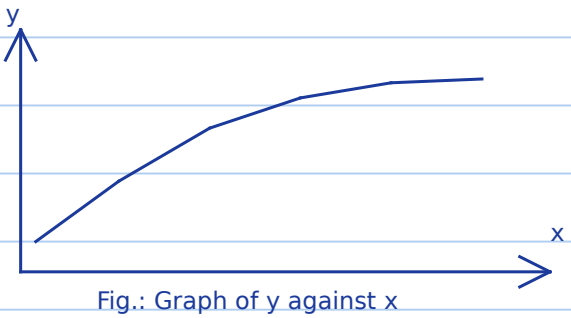


Fig.: Graph of y against x

Question: 2. If f(x) = 5x + 2 and g(x) = 2x^2 - 3, then find (i) fog (ii) gof (iii) fof (iv) gog

Answer: (i) Step 1: fog(x) means f(g(x)), so put the whole of g(x) in place of x in f

Step 2: fog(x) = 5(2x^2 - 3) + 2

Step 3: Expand: 10x^2 - 15 + 2

Final answer: (i) fog(x) = 10x^2 - 13

(ii) Step 1: gof(x) means g(f(x)), so put f(x) in place of x in g

Step 2: gof(x) = 2(5x + 2)^2 - 3

Step 3: Expand the square: (5x + 2)^2 = 25x^2 + 20x + 4

Step 4: Multiply by 2 and subtract 3: 50x^2 + 40x + 8 - 3

Final answer: (ii) gof(x) = 50x^2 + 40x + 5

(iii) Step 1: fof(x) = f(f(x)) = 5(5x + 2) + 2

Step 2: Expand: 25x + 10 + 2

Final answer: (iii) fof(x) = 25x + 12

(iv) Step 1: gog(x) = g(g(x)) = 2(2x^2 - 3)^2 - 3

Step 2: Expand the square: (2x^2 - 3)^2 = 4x^4 - 12x^2 + 9

Step 3: Multiply by 2 and subtract 3: 8x^4 - 24x^2 + 18 - 3

Final answer: (iv) gog(x) = 8x^4 - 24x^2 + 15

Question: 3. If f(x) = 2x and g(x) = x + 1, then find fog(x) for x = -5.

Answer: Step 1: First find the inner value g(-5) = -5 + 1 = -4

Step 2: fog means apply g first and then f, so fog(-5) = f(g(-5)) = f(-4)

Step 3: Apply f: f(-4) = 2(-4) = -8

Final answer: fog(-5) = -8

Question: 4. If f(x) = x + 3 and g(x) = x^2, then find gof(x) for x = 1.

Answer: Step 1: First find the inner value f(1) = 1 + 3 = 4

Step 2: gof means apply f first and then g, so gof(1) = g(f(1)) = g(4)

Step 3: Apply g: g(4) = 4^2 = 16

Final answer: gof(1) = 16

Question: 5. If c(x) = cos x and p(x) = x^3 + 1, then find poc(x).

Answer: Step 1: poc(x) means p(c(x)), so put c(x) = cos x in place of x in p

Step 2: poc(x) = (cos x)^3 + 1

Step 3: Write (cos x)^3 as cos^3 x

Final answer: poc(x) = cos^3 x + 1

Question: 6. Given that f(x) = x + 2 and g(x) = 3x - 2 are two given functions, then find (fog)^-1 and (gof)^-1, and also show that (gof)^-1 = f^-1 o g^-1.

Answer: Step 1: fog(x) = f(g(x)) = (3x - 2) + 2 = 3x

Step 2: To invert y = 3x, write x = y / 3, so (fog)^-1 (x) = x / 3

Step 3: gof(x) = g(f(x)) = 3(x + 2) - 2 = 3x + 6 - 2 = 3x + 4

Step 4: To invert y = 3x + 4, write y - 4 = 3x, so x = (y - 4) / 3, giving (gof)^-1 (x) = (x - 4) / 3

Step 5: Find the separate inverses: from y = x + 2, f^-1 (x) = x - 2; from y = 3x - 2, g^-1 (x) = (x + 2) / 3

Step 6: Compose them in the reversed order: f^-1 o g^-1 (x) = f^-1 ((x + 2) / 3) = (x + 2) / 3 - 2

Step 7: Simplify: (x + 2 - 6) / 3 = (x - 4) / 3

Step 8: This is exactly the result of Step 4

Final answer: (fog)^-1 (x) = x / 3 and (gof)^-1 (x) = (x - 4) / 3, and (gof)^-1 = f^-1 o g^-1 is shown

Question: 7. Given that h(x) = x - 3 and k(x) = 2x + 5 are two functions, then verify that: (i) hok is not equal to koh (ii) (hok)^-1 = k^-1 o h^-1 (iii) (koh)^-1 = h^-1 o k^-1

Answer: (i) Step 1: hok(x) = h(k(x)) = (2x + 5) - 3 = 2x + 2

Step 2: koh(x) = k(h(x)) = 2(x - 3) + 5 = 2x - 6 + 5 = 2x - 1

Step 3: 2x + 2 and 2x - 1 are different functions (for example at x = 0 they give 2 and -1)

Final answer: (i) hok is not equal to koh, verified

(ii) Step 1: From part (i), hok(x) = 2x + 2; inverting y = 2x + 2 gives x = (y - 2) / 2, so (hok)^-1 (x) = (x - 2) / 2

Step 2: The separate inverses are h^-1 (x) = x + 3 and k^-1 (x) = (x - 5) / 2

Step 3: k^-1 o h^-1 (x) = k^-1 (x + 3) = ((x + 3) - 5) / 2 = (x - 2) / 2

Step 4: Both sides are the same

Final answer: (ii) (hok)^-1 = k^-1 o h^-1, verified

(iii) Step 1: From part (i), koh(x) = 2x - 1; inverting y = 2x - 1 gives x = (y + 1) / 2, so (koh)^-1 (x) = (x + 1) / 2

Step 2: h^-1 o k^-1 (x) = h^-1 ((x - 5) / 2) = (x - 5) / 2 + 3 = (x - 5 + 6) / 2 = (x + 1) / 2

Step 3: Both sides are the same

Final answer: (iii) (koh)^-1 = h^-1 o k^-1, verified

Exercise 2.2

Question: 1. Which of the following are algebraic, exponential, logarithmic, trigonometric, inverse trigonometric, hyperbolic and inverse hyperbolic functions: (i) y = x^2 + 5x + 6 (ii) f(x) = tan^-1 x (iii) y = 2^(x + 1) (iv) y = log_5 (x + 2) (v) f(x) = 3 sin x (vi) y = a^(sin x) (vii) f(x) = (x^2 + 5x + 7) / (x + 9) (viii) f(x) = sin x / sec x (ix) y = log_a (sin x) (x) f(x) = cosec^-1 sqrt(x^2 - 1) (xi) f(x) = tan(sin x) (xii) y = x / (x + 3) (xiii) f(x) = sinh x (xiv) y = ln (cosh x) (xv) y = tanh^-1 x (xvi) y = cos^-1 (ln x)

Answer: Step 1: Classify each function by its outermost operation, because the outer operation decides the family of the function

Step 2: (i) Only powers and sums of x appear, so it is algebraic

Step 3: (ii) The outer function is the inverse tangent, so it is inverse trigonometric

Step 4: (iii) The variable x is in the exponent of the constant base 2, so it is exponential

Step 5: (iv) The outer function is a logarithm to base 5, so it is logarithmic

Step 6: (v) The outer function is the sine, so it is trigonometric

Step 7: (vi) The variable appears in the exponent (the exponent is sin x), so it is exponential

Step 8: (vii) It is a ratio of two polynomials, built only from algebraic operations, so it is algebraic

Step 9: (viii) It contains sin x and sec x and no other kind of function, so it is trigonometric

Step 10: (ix) The outer function is a logarithm (its argument happens to be sin x), so it is logarithmic

Step 11: (x) The outer function is the inverse cosecant, so it is inverse trigonometric

Step 12: (xi) The outer function is the tangent, so it is trigonometric

Step 13: (xii) It is a ratio of polynomials, so it is algebraic

Step 14: (xiii) sinh x is the hyperbolic sine, so it is hyperbolic

Step 15: (xiv) The outer function is the natural logarithm, so the book classifies it as logarithmic

Step 16: (xv) tanh^-1 x is the inverse hyperbolic tangent, so it is inverse hyperbolic

Step 17: (xvi) The outer function is the inverse cosine, so it is inverse trigonometric

Final answer: (i) algebraic, (ii) inverse trigonometric, (iii) exponential, (iv) logarithmic, (v) trigonometric, (vi) exponential, (vii) algebraic, (viii) trigonometric, (ix) logarithmic, (x) inverse trigonometric, (xi) trigonometric, (xii) algebraic, (xiii) hyperbolic, (xiv) logarithmic, (xv) inverse hyperbolic, (xvi) inverse trigonometric

Question: 2. Identify whether y is the explicit or implicit function of the independent variable x if: (i) x y^2 + 5xy + 7 = 0 (ii) y = 3x^2 - 3x + 5 (iii) y x^2 + y^2 x = 3 - 5y (iv) x^2 + x y^2 = 2 + 3xy (v) y = (x + 3) / (x^2 + 5) (vi) x / y = 3x^3 y - 5

Answer: Step 1: A function is explicit when y stands alone on one side, written purely in terms of x; it is implicit when x and y are mixed together in one equation

Step 2: (i) x and y are mixed in x y^2 + 5xy + 7 = 0 and y is not isolated, so it is implicit

Step 3: (ii) y is already isolated and written in terms of x only, so it is explicit

Step 4: (iii) The terms y x^2 and y^2 x mix x and y, so it is implicit

Step 5: (iv) The terms x y^2 and 3xy mix x and y, so it is implicit

Step 6: (v) y is isolated as a formula in x only, so it is explicit

Step 7: (vi) x / y = 3x^3 y - 5 mixes x and y on both sides, so it is implicit

Final answer: (i) implicit, (ii) explicit, (iii) implicit, (iv) implicit, (v) explicit, (vi) implicit

Question: 3. Draw the graph of the following functions: (i) f(x) = e^(3x) (ii) f(x) = 3 log_10 x (iii) y = sqrt(36 - x^2) (iv) x^2 / 16 + y^2 / 25 = 1

Answer: (i) Step 1: Make a table: at x = -1, 0, 1 the values of e^(3x) are about 0.05, 1 and 20.09

Step 2: The curve passes through (0, 1), hugs the x-axis on the left and rises very steeply on the right, which is the typical shape of an exponential growth curve

Final answer: (i) The graph of f(x) = e^(3x) is an increasing exponential curve through (0, 1)

(ii) Step 1: The domain is x positive only, because a logarithm of 0 or of a negative number is not real

Step 2: Make a table: at x = 1, 10 the values are 3 log_10 1 = 0 and 3 log_10 10 = 3, and as x approaches 0 the values fall without bound

Step 3: The curve crosses the x-axis at (1, 0) and rises slowly to the right

Final answer: (ii) The graph of f(x) = 3 log_10 x is a slowly rising logarithmic curve through (1, 0) and (10, 3)

(iii) Step 1: Square both sides: y^2 = 36 - x^2, that is x^2 + y^2 = 36, the circle of radius 6 centred at the origin

Step 2: Since y is a square root, y is never negative, so only the upper half of the circle is drawn

Step 3: The domain runs from -6 to 6 and the highest point is (0, 6)

Final answer: (iii) The graph is the upper semicircle of radius 6 with endpoints (-6, 0) and (6, 0)

(iv) Step 1: Compare with the standard ellipse x^2 / a^2 + y^2 / b^2 = 1, giving a = 4 and b = 5

Step 2: The x-intercepts are (4, 0) and (-4, 0), and the y-intercepts are (0, 5) and (0, -5)

Step 3: Join these four points by a smooth closed oval, longer in the y direction

Final answer: (iv) The graph is an ellipse centred at the origin with semi-axes 4 along x and 5 along y

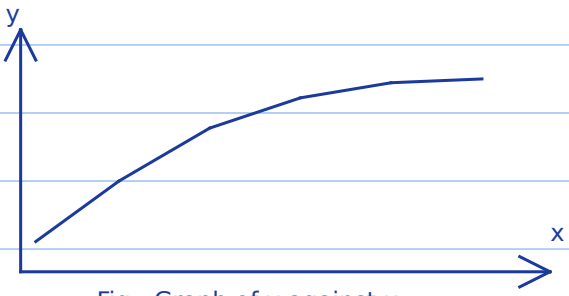


Fig.: Graph of y against x

Question: 4. Draw the graph of the parametric equations of the function x = a t^2, y = 2at, when a = 4 and t runs from -5 to 5.

Answer: Step 1: Substitute a = 4: x = 4t^2 and y = 8t

Step 2: Make a table: t = -5, -2, -1, 0, 1, 2, 5 gives (x, y) = (100, -40), (16, -16), (4, -8), (0, 0), (4, 8), (16, 16), (100, 40)

Step 3: Eliminate the parameter: from y = 8t, t = y / 8, so x = 4 (y / 8)^2 = y^2 / 16, that is y^2 = 16x

Step 4: This is a parabola opening to the right with vertex at the origin, and the given range of t gives the part of it between y = -40 and y = 40

Final answer: The graph is the part of the parabola y^2 = 16x from (100, -40) through (0, 0) to (100, 40)

Question: 5. Draw the graph of the parametric equations of the function x = a sec theta, y = b tan theta, when a = 3, b = 4 and theta runs from -pi to pi.

Answer: Step 1: Substitute a = 3 and b = 4: x = 3 sec theta and y = 4 tan

theta

Step 2: Write x / 3 = sec theta and y / 4 = tan theta

Step 3: Use the identity sec^2 theta - tan^2 theta = 1 to eliminate theta: x^2 / 9 - y^2 / 16 = 1

Step 4: This is a hyperbola with x-intercepts (3, 0) and (-3, 0) and asymptotes y = (4/3) x and y = -(4/3) x, opening left and right

Step 5: As theta runs from -pi to pi (avoiding the values where cos theta = 0), both branches are traced

Final answer: The graph is the hyperbola x^2 / 9 - y^2 / 16 = 1 with vertices (3, 0) and (-3, 0)

Question: 6. Draw the graph of f(x) = x^2 when x is at most 1, and f(x) = 2x when x is greater than 1.

Answer: Step 1: For x at most 1 use the parabola y = x^2: at x = -1, 0, 1 the values are 1, 0, 1, and the point (1, 1) belongs to this piece

Step 2: For x greater than 1 use the line y = 2x: as x approaches 1 from the right the values approach 2, and at x = 2, 3 the values are 4, 6; the point (1, 2) itself is not included, so it is drawn as an open circle

Step 3: Plot the parabola up to and including (1, 1), then the line starting just after (1, 2)

Step 4: There is a jump at x = 1 from the value 1 up towards 2, so the graph is discontinuous there

Final answer: The graph is a parabola piece ending at the filled point (1, 1) together with a line piece beginning at the open point (1, 2)

Exercise 2.3

Question: 1. Solve the following: (i) [2, infinity) union (3, 5) (ii) [-1, 1] - (2, infinity) (iii) (5, infinity) intersection (-3, 6) (iv) [3, 5] - (3, 5) (v) [1, 10] intersection [3, 11] (vi) (-infinity, 5) - (-infinity, 3)

Answer: (i) Step 1: The interval (3, 5) lies completely inside [2, infinity), because every number between 3 and 5 is also in [2, infinity)

Step 2: A union with a subset adds nothing new

Result: (i) [2, infinity)

(ii) Step 1: The set (2, infinity) starts at 2, while [-1, 1] ends at 1, so the two sets have no point in common

Step 2: Removing a disjoint set changes nothing

Result: (ii) [-1, 1]

(iii) Step 1: The intersection keeps only numbers that are in both sets, so the number must be greater than 5 and also less than 6

Result: (iii) (5, 6)

(iv) Step 1: Removing the open interval (3, 5) from the closed interval [3, 5] removes every interior point

Step 2: The endpoints 3 and 5 are not in (3, 5), so they survive

Result: (iv) {3, 5}

(v) Step 1: The common part starts at the larger left endpoint, 3, and ends at the smaller right endpoint, 10

Result: (v) [3, 10]

(vi) Step 1: Start with all numbers less than 5 and remove all numbers less than 3

Step 2: What remains is the numbers from 3 up to but not including 5, and

3 itself remains because 3 is not less than 3

Result: (vi) [3, 5)

Final answer: (i) [2, infinity), (ii) [-1, 1], (iii) (5, 6), (iv) {3, 5}, (v) [3, 10], (vi) [3, 5)



Fig.: The number line

Question: 2. Find the nth term and the limit of the following sequences: (i) 1, 1/2, 1/4, 1/8, ... (ii) 1.2/3.4, 3.4/5.6, 5.6/7.8, ...

Answer: (i) Step 1: Each term is half of the previous term: 1, then 1/2, then (1/2)^2, then (1/2)^3

Step 2: Counting from n = 1, the nth term is a_n = (1/2)^(n - 1)

Step 3: As n grows, (1/2)^(n - 1) becomes smaller and smaller and approaches 0

Final answer: (i) a_n = (1/2)^(n - 1), limit = 0

(ii) Step 1: The numerators 1.2, 3.4, 5.6 increase by 2.2 each time, so the numerator of the nth term is 2.2n - 1.0 (check n = 1: 1.2, n = 2: 3.4, n = 3: 5.6)

Step 2: The denominators 3.4, 5.6, 7.8 also increase by 2.2 each time, so the denominator of the nth term is 2.2n + 1.2

Step 3: So a_n = (2.2n - 1) / (2.2n + 1.2)

Step 4: Divide numerator and denominator by n: (2.2 - 1/n) / (2.2 + 1.2/n)

Step 5: As n tends to infinity, 1/n and 1.2/n tend to 0, leaving 2.2 / 2.2 = 1

Final answer: (ii) a_n = (2.2n - 1) / (2.2n + 1.2), limit = 1

Question: 3. Find the limit of the following sequences whose nth terms are: (i) a_n = (1 + 5n) / (7n) (ii) a_n = (3n - 1)(n^4 - n) / ((n^2 + 5)(n^3 - 7)) (iii) a_n = (n + 1)! / (n! - (n + 1)!)

Answer: (i) Step 1: Divide numerator and denominator by n: (1/n + 5) / 7

Step 2: As n tends to infinity, 1/n tends to 0

Step 3: The limit is (0 + 5) / 7

Final answer: (i) 5/7

(ii) Step 1: The numerator has degree 1 + 4 = 5 with leading term (3n)(n^4) = 3n^5

Step 2: The denominator has degree 2 + 3 = 5 with leading term (n^2)(n^3) = n^5

Step 3: For equal degrees the limit is the ratio of the leading coefficients, 3 / 1

Final answer: (ii) 3

(iii) Step 1: Write (n + 1)! = (n + 1) n! in both places

Step 2: The expression becomes (n + 1) n! / (n! - (n + 1) n!)

Step 3: Divide numerator and denominator by n!: (n + 1) / (1 - (n + 1)) = (n + 1) / (-n)

Step 4: Divide numerator and denominator by n: (1 + 1/n) / (-1)

Step 5: As n tends to infinity, 1/n tends to 0, giving 1 / (-1)

Final answer: (iii) -1

Question: 4. Find the limit of the function y = 5x / (x + 1) for x tends to infinity.

Answer: Step 1: Divide numerator and denominator by x: the numerator

becomes 5 and the denominator becomes 1 + 1/x

Step 2: As x tends to infinity, 1/x tends to 0

Step 3: The denominator therefore tends to 1 + 0 = 1

Final answer: lim (x to infinity) 5x / (x + 1) = 5

Question: 5. Evaluate by using theorems on limits: (i) lim (x to 2) (x^5 + x^2 + x + 1) (ii) lim (x to 5) ((1 + x) / x^2) (iii) lim (x to 1) [(2x^3 + 3x^2)(x + 1)] (iv) lim (x to 5) {(x + 1) - (x^2 + 2x + 3)}

Answer: (i) Step 1: By the sum theorem, the limit of a sum is the sum of the limits, and each power can be substituted directly

Step 2: Substitute x = 2: 2^5 + 2^2 + 2 + 1 = 32 + 4 + 2 + 1

Final answer: (i) 39

(ii) Step 1: By the quotient theorem, take the limit of the numerator and of the denominator separately

Step 2: Numerator limit: 1 + 5 = 6; denominator limit: 5^2 = 25

Step 3: Divide: 6 / 25

Final answer: (ii) 6/25

(iii) Step 1: By the product theorem, the limit of the product is the product of the limits

Step 2: First factor at x = 1: 2(1) + 3(1) = 5; second factor at x = 1: 1 + 1 = 2

Step 3: Multiply: 5 x 2

Final answer: (iii) 10

(iv) Step 1: By the difference theorem, evaluate each part at x = 5

Step 2: First part: 5 + 1 = 6; second part: 25 + 10 + 3 = 38

Step 3: Subtract: 6 - 38

Final answer: (iv) -32

Question: 6. Evaluate the limits of the following algebraic and exponential functions: (i) lim (x to 1) (x^2 + 4)^3 (ii) lim (x to 0) 4 / (x - 4) (iii) lim (x to infinity) 5 / (5x + 10) (iv) lim (x to 7) (x^2 - 49) / (x - 7) (v) lim (x to 0) (sqrt(x + 3) - sqrt(3)) / x (vi) lim (x to 7) (x - 7) / (sqrt(x) - sqrt(7)) (vii) lim (x to 0) (1 + x/3)^(5/x) (viii) lim (x to 0) (1 + ax)^(a/x) (ix) lim (x to infinity) (1 + p/x)^(px) (x) lim (x to infinity) (1 - pq/x)^(x/p) (xi) lim (x to 0) (17^x - 1) / x (xii) lim (h to 0) ((1 + 2h)^n - 1) / (5h) (xiii) lim (x to 0) (2^x - 4^x - 8^x - 1) / (x + 2) (xiv) lim (x to 0) ((1 + 7x) / (1 - 9x))^(1/x) (xv) lim (x to 0) (a^x - b^x) / x (xvi) lim (x to 0) (e^(-2x) - e^(-11x)) / x (xvii) lim (x to 0) (3e^(-5x) - 5e^(-2x) + 2) / x (xviii) lim (x to 0) (1 + 3 tan x)^(cot x) (xix) lim (x to 2) (x^4 - 16) / (x^3 - 8) (xx) lim (x to infinity) (2x - x^2) / (3x^2 - 4x + 1)

Answer: (i) Step 1: Substitute x = 1 inside: (1 + 4)^3 = 5^3

Result: (i) 125

(ii) Step 1: Substitute x = 0: 4 / (0 - 4) = 4 / (-4)

Result: (ii) -1

(iii) Step 1: The numerator is constant while the denominator grows without bound, so the fraction shrinks to 0

Result: (iii) 0

(iv) Step 1: Factor the numerator: (x - 7)(x + 7) / (x - 7)

Step 2: Cancel (x - 7) and substitute x = 7: 7 + 7

Result: (iv) 14

(v) Step 1: Multiply numerator and denominator by the conjugate sqrt(x +

3) + sqrt(3): the numerator becomes (x + 3) - 3 = x

Step 2: Cancel x, leaving 1 / (sqrt(x + 3) + sqrt(3))

Step 3: Substitute x = 0: 1 / (2 sqrt(3))

Result: (v) 1 / (2 sqrt(3))

(vi) Step 1: Write x - 7 = (sqrt(x) - sqrt(7))(sqrt(x) + sqrt(7))

Step 2: Cancel (sqrt(x) - sqrt(7)) and substitute x = 7: sqrt(7) + sqrt(7)

Result: (vi) 2 sqrt(7)

(vii) Step 1: Use the standard form lim (1 + u)^(1/u) = e with u = x / 3

Step 2: Rewrite the exponent: 5 / x = (5 / 3) x (3 / x), so the expression is [(1 + x/3)^(3/x)]^(5/3)

Step 3: The bracket tends to e, so the limit is e^(5/3)

Result: (vii) e^(5/3)

(viii) Step 1: Rewrite as [(1 + ax)^(1/(ax))]^((a/x)(ax)) = [(1 + ax)^(1/(ax))]^(a^2)

Step 2: The bracket tends to e

Result: (viii) e^(a^2)

(ix) Step 1: Rewrite as [(1 + p/x)^(x/p)]^(p^2), because (x/p) times p^2 gives px

Step 2: The bracket tends to e

Result: (ix) e^(p^2)

(x) Step 1: Take the logarithm form: (x/p) ln(1 - pq/x), and use ln(1 - pq/x) is approximately -pq/x for large x

Step 2: The product is approximately (x/p) x (-pq/x) = -q, so the logarithm

of the limit is -q

Result: (x) e^(-q)

(xi) Step 1: Use the standard limit (a^x - 1) / x tends to ln a as x tends to 0

Step 2: Here a = 17

Result: (xi) ln 17

(xii) Step 1: Expand (1 + 2h)^n by the binomial theorem: 1 + 2nh + terms containing h^2 and higher

Step 2: Subtract 1 and divide by 5h: (2nh) / (5h) plus terms containing h

Step 3: As h tends to 0 only 2n / 5 survives

Result: (xii) 2n / 5

(xiii) Step 1: The function is continuous at x = 0, so substitute directly: (1 - 1 - 1 - 1) / (0 + 2) = -2 / 2

Result: (xiii) -1

(xiv) Step 1: Take the limit of the logarithm form: the expression equals [(1 + 7x) / (1 - 9x)]^(1/x); write its logarithm as (1/x) [ln(1 + 7x) - ln(1 - 9x)]

Step 2: Using (ln(1 + kx)) / x tends to k, this tends to 7 - (-9) = 16

Step 3: Therefore the original limit is e^16

Result: (xiv) e^16

(xv) Step 1: Add and subtract 1: [(a^x - 1) - (b^x - 1)] / x

Step 2: Each part gives a standard limit: ln a - ln b

Step 3: Combine: ln(a / b)

Result: (xv) ln(a / b)

(xvi) Step 1: Add and subtract 1: [(e^(-2x) - 1) - (e^(-11x) - 1)] / x

Step 2: Use (e^(kx) - 1) / x tends to k: this gives -2 - (-11) = 9

Result: (xvi) 9

(xvii) Step 1: At x = 0 the numerator is 3 - 5 + 2 = 0, so use the derivative (L'Hopital) form: differentiate the numerator to get -15 e^(-5x) + 10 e^(-2x)

Step 2: Substitute x = 0: -15 + 10

Result: (xvii) -5

(xviii) Step 1: Take logarithms: cot x ln(1 + 3 tan x) = ln(1 + 3 tan x) / tan x

Step 2: As x tends to 0, tan x tends to 0 and ln(1 + 3 tan x) / (3 tan x) tends to 1, so the whole expression tends to 3

Step 3: The limit is e^3

Result: (xviii) e^3

(xix) Step 1: Both numerator and denominator vanish at x = 2, so differentiate them: (4x^3) / (3x^2) = 4x / 3

Step 2: Substitute x = 2: 8 / 3

Result: (xix) 8/3

(xx) Step 1: Divide numerator and denominator by x^2: (2/x - 1) / (3 - 4/x + 1/x^2)

Step 2: As x tends to infinity the terms with 1/x vanish, leaving -1 / 3

Result: (xx) -1/3

Final answer: (i) 125, (ii) -1, (iii) 0, (iv) 14, (v) 1/(2 sqrt(3)), (vi) 2 sqrt(7), (vii) e^(5/3), (viii) e^(a^2), (ix) e^(p^2), (x) e^(-q), (xi) ln 17, (xii) 2n/5, (xiii) -1, (xiv) e^16, (xv) ln(a/b), (xvi) 9, (xvii) -5, (xviii) e^3, (xix) 8/3, (xx)

-1/3

Question: 7. Evaluate the limits of the following trigonometric functions: (i) lim (x to 0) (a sin ax) / x (ii) lim (x to 0) sin(sqrt(a) x) / (x / sqrt(a)) (iii) lim (x to 0) (3 cos x + 2 tan x)^3 (iv) lim (x to 0) (3 sin x - x^3) / (2x) (v) lim (x to 0) (sin px) / (sin qx) (vi) lim (x to 0) ((2pi - x) sec(pi - x)) / (pi / 2) (vii) lim (x to 0) sin(x degrees) / x (viii) lim (x to 0) (1 - cos mx) / (1 - cos nx) (ix) lim (x to 0) (sin 3x sin 5x) / (7x^2)

Answer: (i) Step 1: Write a sin(ax) / x = a^2 x (sin ax) / (ax)

Step 2: Use sin(ax) / (ax) tends to 1

Result: (i) a^2

(ii) Step 1: Multiply numerator and denominator to expose the standard form: sin(sqrt(a) x) / (x / sqrt(a)) = a x sin(sqrt(a) x) / (sqrt(a) x)

Step 2: Use sin(sqrt(a) x) / (sqrt(a) x) tends to 1

Result: (ii) a

(iii) Step 1: Substitute x = 0 directly: (3 cos 0 + 2 tan 0)^3 = (3 x 1 + 0)^3 = 3^3

Result: (iii) 27

(iv) Step 1: Split the fraction: (3 sin x) / (2x) - x^3 / (2x) = (3/2)(sin x / x) - x^2 / 2

Step 2: The first part tends to 3/2 and the second part tends to 0

Result: (iv) 3/2

(v) Step 1: Multiply numerator and denominator to use the standard limit: (sin px) / (sin qx) = (p/q) x [sin(px) / (px)] / [sin(qx) / (qx)]

Step 2: Both brackets tend to 1, leaving p / q

Result: (v) p/q

(vi) Step 1: Substitute x = 0 directly: sec(pi - 0) = sec(pi) = -1, and the numerator is (2pi)(-1) = -2pi

Step 2: Divide by pi / 2: -2pi x (2 / pi) = -4

Result: (vi) -4

(vii) Step 1: An angle of x degrees is pi x / 180 radians, and the sine limit must be taken in radians: sin(x degrees) = sin(pi x / 180)

Step 2: sin(pi x / 180) / x = (pi / 180) x sin(pi x / 180) / (pi x / 180), and the standard fraction tends to 1

Result: (vii) pi / 180

(viii) Step 1: Use 1 - cos t = 2 sin^2 (t / 2) in numerator and denominator: [2 sin^2 (mx / 2)] / [2 sin^2 (nx / 2)]

Step 2: Rewrite as (m^2 / n^2) x [sin(mx/2) / (mx/2)]^2 / [sin(nx/2) / (nx/2)]^2

Step 3: Both standard fractions tend to 1

Result: (viii) m^2 / n^2

(ix) Step 1: Rewrite as (15 / 7) x [sin(3x) / (3x)] x [sin(5x) / (5x)], since 3 x 5 = 15 and 15 / (7) matches the given denominator 7x^2

Step 2: Both standard fractions tend to 1

Result: (ix) 15/7

Final answer: (i) a^2, (ii) a, (iii) 27, (iv) 3/2, (v) p/q, (vi) -4, (vii) pi/180, (viii) m^2/n^2, (ix) 15/7

Question: 8. Evaluate the limits of the following functions: (i) lim (x to 0) sin^2 (x/2) / (4x^2) (ii) lim (x to infinity) x sin(1/x) (iii) lim (x to infinity)

[sqrt(x^2 + x + 1) - x] (iv) lim (x to 0) (1 - cos^3 x) / sin^2 x (v) lim (x to 0) (a^x + a^(-x) - 2) / x^2 (vi) lim (x to 0) (6^x - 3^x - 2^x + 1) / x^2 (vii) lim (x to 3) [x / (x + 2) - 3/5] / (x - 3) (viii) lim (y to 4) (y^(5/2) - 16 y^(1/2)) / (y - 4) (ix) lim (x to 1) (1 / sqrt(x) - 1) / (1 - x) (x) lim (x to pi) (sqrt(5 + cos x) - 2) / (pi - x) (xi) lim (x to 1) x^(1 / (x - 1)) (xii) lim (x to 1) (x^2 - sqrt(x)) / (sqrt(x) - 1) (xiii) lim (x to pi/4) (sqrt(2) - cos x - sin x) / (4x - pi)^2 (xiv) lim (x to e) (ln x - 1) / (x - e) (xv) lim (x to 2) ((x + 1) sin(x - 2)) / (x^2 - 4)

Answer: (i) Step 1: Write sin^2 (x/2) / (4x^2) = (1/16) x [sin(x/2) / (x/2)]^2, because (x/2)^2 = x^2 / 4 and 4x^2 = 16 x (x^2/4)

Step 2: The standard fraction tends to 1

Result: (i) 1/16

(ii) Step 1: Put t = 1/x, so as x tends to infinity, t tends to 0, and the expression becomes sin t / t

Step 2: The standard limit gives 1

Result: (ii) 1

(iii) Step 1: Multiply by the conjugate sqrt(x^2 + x + 1) + x: the numerator becomes (x^2 + x + 1) - x^2 = x + 1

Step 2: Divide numerator and denominator by x: (1 + 1/x) / (sqrt(1 + 1/x + 1/x^2) + 1)

Step 3: As x tends to infinity this gives 1 / (1 + 1)

Result: (iii) 1/2

(iv) Step 1: Factor 1 - cos^3 x = (1 - cos x)(1 + cos x + cos^2 x) and sin^2 x = (1 - cos x)(1 + cos x)

Step 2: Cancel (1 - cos x): the expression is (1 + cos x + cos^2 x) / (1 +

cos x)

Step 3: Substitute x = 0: (1 + 1 + 1) / (1 + 1)

Result: (iv) 3/2

(v) Step 1: Expand a^x = 1 + x ln a + x^2 (ln a)^2 / 2 + ... and a^(-x) = 1 - x ln a + x^2 (ln a)^2 / 2 - ...

Step 2: Adding, the constant and x terms cancel, leaving x^2 (ln a)^2 plus higher terms

Step 3: Divide by x^2 and let x tend to 0

Result: (v) (ln a)^2

(vi) Step 1: Expand each exponential to second order: 6^x - 3^x - 2^x + 1 has zero constant term (1 - 1 - 1 + 1), zero x term (ln 6 - ln 3 - ln 2 = ln 1 = 0), and x^2 coefficient (1/2)[(ln 6)^2 - (ln 3)^2 - (ln 2)^2]

Step 2: (ln 6)^2 = (ln 2 + ln 3)^2 = (ln 2)^2 + 2 ln 2 ln 3 + (ln 3)^2, so the bracket equals 2 ln 2 ln 3

Step 3: The limit is half of that

Result: (vi) (ln 2)(ln 3)

(vii) Step 1: Note 3/5 is the value of x / (x + 2) at x = 3, so this limit is the derivative of x / (x + 2) at x = 3

Step 2: Differentiate: [(x + 2) - x] / (x + 2)^2 = 2 / (x + 2)^2

Step 3: At x = 3: 2 / 25

Result: (vii) 2/25

(viii) Step 1: This is the derivative of y^(5/2) - 16 y^(1/2) at y = 4 (in the difference-quotient sense with the second term sharing the same denominator)

Step 2: Differentiate term by term: (5/2) y^(3/2) - 16 x (1/2) y^(-1/2) = (5/2) y^(3/2) - 8 / sqrt(y)

Step 3: At y = 4: (5/2)(8) - 8/2 = 20 - 4

Result: (viii) 16

(ix) Step 1: Write 1 - x = (1 - sqrt(x))(1 + sqrt(x)) and 1/sqrt(x) - 1 = (1 - sqrt(x)) / sqrt(x)

Step 2: Cancel (1 - sqrt(x)): the expression becomes 1 / [sqrt(x) (1 + sqrt(x))]

Step 3: Substitute x = 1: 1 / (1 x 2)

Result: (ix) 1/2

(x) Step 1: At x = pi, cos x = -1, so the numerator is sqrt(4) - 2 = 0; use the derivative form with t = pi - x

Step 2: The derivative of sqrt(5 + cos x) is -sin x / (2 sqrt(5 + cos x)), which at x = pi is 0 / 4 = 0

Result: (x) 0

(xi) Step 1: Take logarithms: ln of the expression is (ln x) / (x - 1)

Step 2: As x tends to 1, (ln x) / (x - 1) tends to 1, the standard logarithmic limit

Step 3: So the original limit is e^1

Result: (xi) e

(xii) Step 1: Both parts vanish at x = 1, so differentiate numerator and denominator: numerator derivative 2x - 1 / (2 sqrt(x)), denominator derivative 1 / (2 sqrt(x))

Step 2: At x = 1: (2 - 1/2) / (1/2) = (3/2) / (1/2)

Result: (xii) 3

(xiii) Step 1: Put t = x - pi/4, so 4x - pi = 4t and cos x + sin x = sqrt(2) cos t (from the sum formula cos x + sin x = sqrt(2) cos(x - pi/4))

Step 2: The numerator becomes sqrt(2) (1 - cos t) = sqrt(2) x 2 sin^2 (t/2), and the denominator is 16 t^2

Step 3: Using sin(t/2) / (t/2) tends to 1 gives sqrt(2) x 2 x (1/4) / 16 = sqrt(2) / 32 = 1 / (16 sqrt(2))

Result: (xiii) 1 / (16 sqrt(2))

(xiv) Step 1: This is the derivative of ln x at x = e, written as a difference quotient, since ln e = 1

Step 2: The derivative of ln x is 1/x, which at x = e is 1/e

Result: (xiv) 1/e

(xv) Step 1: Split the fraction: [(x + 1) / (x + 2)] x [sin(x - 2) / (x - 2)], using x^2 - 4 = (x - 2)(x + 2)

Step 2: As x tends to 2, the first factor tends to 3/4 and the standard sine fraction tends to 1

Result: (xv) 3/4

Final answer: (i) 1/16, (ii) 1, (iii) 1/2, (iv) 3/2, (v) (ln a)^2, (vi) (ln 2)(ln 3), (vii) 2/25, (viii) 16, (ix) 1/2, (x) 0, (xi) e, (xii) 3, (xiii) 1/(16 sqrt(2)), (xiv) 1/e, (xv) 3/4

Exercise 2.4

Question: 1. Evaluate the following limits: (i) lim (x to 2 from the right) (x - 2) / |x - 2| (ii) lim (x to 1 from the left) (x^2 + 2x - 3) / |x - 1| (iii) lim (x to 2) (x^2 + 4x - 12) / |x - 2|

Answer: (i) Step 1: For x greater than 2, x - 2 is positive, so |x - 2| = x - 2

Step 2: The fraction is (x - 2) / (x - 2) = 1 for every such x

Final answer: (i) 1

(ii) Step 1: Factor the numerator: x^2 + 2x - 3 = (x + 3)(x - 1)

Step 2: For x less than 1, x - 1 is negative, so |x - 1| = -(x - 1), and the fraction becomes -(x + 3)

Step 3: Substitute x = 1: -(1 + 3)

Final answer: (ii) -4

(iii) Step 1: Factor the numerator: x^2 + 4x - 12 = (x + 6)(x - 2)

Step 2: From the right, |x - 2| = x - 2, so the fraction is (x + 6) and tends to 8

Step 3: From the left, |x - 2| = -(x - 2), so the fraction is -(x + 6) and tends to -8

Step 4: The two one-sided limits are different, so the two-sided limit fails to exist

Final answer: (iii) Does not exist

Question: 2. Determine whether lim (x to 1) f(x), lim (x to 2) f(x), lim (x to 3) f(x) and lim (x to 4) f(x) exist or not, when f(x) = 2x + 1 if 0 at most x and x less than 2; f(x) = x - 7 if 2 at most x and x less than 4; f(x) = x if 4 at most x and x at most 6.

Answer: Step 1: At x = 1 only the first piece is involved: lim = 2(1) + 1 = 3, so the limit exists and equals 3

Step 2: At x = 2 the left-hand limit from the first piece is 2(2) + 1 = 5, while the right-hand limit from the second piece is 2 - 7 = -5; they are unequal, so the limit at x = 2 does not exist

Step 3: At x = 3 only the second piece is involved: lim = 3 - 7 = -4, so the limit exists and equals -4

Step 4: At x = 4 the left-hand limit from the second piece is 4 - 7 = -3, while the right-hand limit from the third piece is 4; they are unequal, so the limit at x = 4 does not exist

Final answer: The limit at x = 1 exists (= 3), the limit at x = 2 does not exist, the limit at x = 3 exists (= -4), and the limit at x = 4 does not exist

Question: 3. Test the continuity and discontinuity of the following functions: (i) f(x) = sin(x^2 + pi x) + 7x^2 + x at the point x = 0 (ii) f(x) = (2 - cos 3x - cos 4x) / x at the point x = 0 (iii) f(x) = 7 + 3x when x is less than 1, and f(x) = 1 - 5x when x is at least 1, at x = 1

Answer: (i) Step 1: f(0) = sin(0) + 0 + 0 = 0

Step 2: Sine and polynomial functions are continuous everywhere, so the limit as x tends to 0 is also 0

Step 3: Limit and value agree

Final answer: (i) Continuous at x = 0

(ii) Step 1: The formula divides by x, so f(0) is not defined at all

Step 2: A function cannot be continuous at a point where it is not defined (indeed the limit here is 0, but there is no value to match it)

Final answer: (ii) Discontinuous at x = 0

(iii) Step 1: The value is f(1) = 1 - 5(1) = -4, from the second piece

Step 2: The left-hand limit is 7 + 3(1) = 10, from the first piece

Step 3: The right-hand limit is -4

Step 4: Left-hand and right-hand limits differ, so the limit does not exist

Final answer: (iii) Discontinuous at x = 1

Question: 4. Determine whether the following functions are continuous at x = 2: (i) f(x) = (x^2 - 4) / (x - 2) (ii) g(x) = (x^2 - 4) / (x - 2) when x is not 2, and g(x) = 3 when x = 2 (iii) h(x) = (x^2 - 4) / (x - 2) when x is not 2, and h(x) = 4 when x = 2

Answer: Step 1: First simplify the common formula: (x^2 - 4) / (x - 2) = x + 2 for x not equal to 2, so in all three cases the limit as x tends to 2 is 2 + 2 = 4

Step 2: (i) f(2) is not defined, because the denominator is 0, so f cannot be continuous at 2

Step 3: (ii) g(2) = 3, which is not equal to the limit 4, so g is not continuous at 2

Step 4: (iii) h(2) = 4, which equals the limit 4, and h is defined at 2, so all three conditions for continuity hold

Final answer: (i) Discontinuous, (ii) Discontinuous, (iii) Continuous

Question: 5. Suppose that f(x) = -x^4 + 3 when x is at most 2, and f(x) = x^2 + 9 when x is greater than 2. Is f continuous everywhere?

Answer: Step 1: Each piece is a polynomial, so f is continuous at every point other than the joining point x = 2

Step 2: At x = 2 the value and the left-hand limit come from the first piece: f(2) = -(2)^4 + 3 = -16 + 3 = -13

Step 3: The right-hand limit comes from the second piece: 2^2 + 9 = 13

Step 4: The left-hand limit -13 and the right-hand limit 13 are different, so the function jumps at x = 2

Final answer: f is continuous everywhere except at x = 2, where it is discontinuous

Question: 6. Find the value of k if f(x) = (sin kx) / x for x not 0, and f(x) = 2 for x = 0, is continuous at x = 0.

Answer: Step 1: Continuity at 0 requires the limit as x tends to 0 to equal f(0) = 2

Step 2: Write (sin kx) / x = k x (sin kx) / (kx)

Step 3: The standard fraction (sin kx) / (kx) tends to 1, so the limit is k

Step 4: Set k = 2

Final answer: k = 2

Question: 7. Find the value of k if f(x) = kx - 9 for x less than 5, f(x) = 9x - k for x greater than 5, and f(x) = 36 for x = 5, is continuous at x = 5.

Answer: Step 1: Continuity at 5 requires both one-sided limits to equal f(5) = 36

Step 2: Left-hand limit: 5k - 9 = 36 gives 5k = 45, so k = 9

Step 3: Check the right-hand limit with k = 9: 9(5) - 9 = 45 - 9 = 36, which agrees

Final answer: k = 9

Question: 8. Find the values of m and n so that the given function f is continuous at x = 3: (i) f(x) = mx if x is less than 3, f(x) = n if x = 3, f(x) = -2x + 9 if x is greater than 3 (ii) f(x) = mx if x is less than 3, f(x) = x^2 if x is at least 3

Answer: (i) Step 1: The right-hand limit at 3 is -2(3) + 9 = 3

Step 2: Continuity forces the value to equal this limit, so n = 3

Step 3: The left-hand limit is 3m, and continuity forces 3m = 3, so m = 1

Final answer: (i) m = 1, n = 3

(ii) Step 1: The right-hand value and limit at 3 come from x^2: 3^2 = 9

Step 2: The left-hand limit is 3m

Step 3: Set 3m = 9, giving m = 3

Final answer: (ii) m = 3

Question: 9. If f(x) = (sqrt(2x + 5) - sqrt(x + 7)) / (x - 2) for x not 2, and f(x) = k for x = 2, find the value of k so that f is continuous at x = 2.

Answer: Step 1: Continuity requires k to equal the limit as x tends to 2

Step 2: Multiply numerator and denominator by the conjugate sqrt(2x + 5) + sqrt(x + 7): the numerator becomes (2x + 5) - (x + 7) = x - 2

Step 3: Cancel (x - 2): the expression becomes 1 / (sqrt(2x + 5) + sqrt(x + 7))

Step 4: Substitute x = 2: 1 / (sqrt(9) + sqrt(9)) = 1 / (3 + 3) = 1/6

Final answer: k = 1/6

Review Exercise 2

Question: 1. Choose the correct answer: (i) The function f(x) = x is called: (a) Quadratic function (b) Cubic function (c) Identity function (d) None of these (ii) The set {x | x in R} can be written as: (a) [-infinity, infinity] (b) (-infinity, a) (c) (-infinity, infinity) (d) (a, infinity) (iii) [3, 5] - (3, 5) = : (a) empty set (b) (3, 5) (c) [3, 5] (d) {3, 5} (iv) Let R = {(x, y) | x, y in R and y = 5}: (a) Set of real numbers (b) {5} (c) empty set (d) {1, 5} (v) A function f(x) = x / |x|, x not 0, is an: (a) odd function (b) even function (c) linear function (d) none of these (vi) If f(x) = 4^x then f(x + 1) - f(x) = : (a) 4 (b) f(x) (c) 3f(x) (d) 2f(x) (vii) The limit of the sequence having general term a_n = 1 / 2^n is: (a) 0 (b) 1 (c) infinity (d) None of these (viii) lim (x to 0) (sin ax) / x = : (a) 1/a (b) a (c) infinity (d) 0 (ix) lim (x to 3) (x^2 - 9) / (x - 3) = : (a) 3 (b) infinity (c) 0 (d) 6 (x) If f : [-1, 4] to R is given by f(x) =

x^2 then f(-3) is: (a) 9 (b) -9 (c) does not exist (d) 6 (xi) lim (h to 0) sin(pi/2 + h): (a) 1 (b) -1 (c) 0 (d) 1/2 (xii) lim (x to 0) e^(-1/x): (a) 0 (b) 1 (c) infinity (d) -infinity (xiii) lim (x to c) f(x) exists if and only if: (a) the right-hand limit exists (b) the left-hand limit exists (c) the right-hand limit equals the left-hand limit (d) the right-hand limit is not equal to the left-hand limit (xiv) lim (n to infinity) [1 + 1/n]^(-7n): (a) e^(-7) (b) e (c) 1 (d) infinity (xv) The limit of the sequence 1, pi^(-1), pi^(-2), pi^(-3), ... is: (a) 1 (b) pi (c) infinity (d) 0 (xvi) Which of the following represents a parametric function: (a) y = f(x) (b) f(x, y) = 0 (c) x = f(t), y = g(t) (d) None of these (xvii) If g(x) = 3x + 2 and g(f(x)) = x then f(2) = : (a) 2 (b) 6 (c) 0 (d) 8 (xviii) The value of k for which f(x) = x / tan 3x for x not 0, and f(x) = k for x = 0, is continuous is: (a) 0 (b) 3 (c) 1/2 (d) 1/3 (xix) lim (x to 0) (1 - x)^(1/x): (a) e^3 (b) e^(-1/2) (c) e (d) e^(-1) (xx) sech^-1 x = : (a) ln(x + sqrt(x^2 + 1)) (b) ln(1/x + sqrt(1 - x^2) / x) (c) (1/2) ln((1 + x) / (1 - x)) (d) (1/2) ln((1 - x) / (1 + x))

Answer: (i) Step 1: f(x) = x returns every input unchanged, which is the definition of the identity function

Final answer: (i) (c) Identity function

(ii) Step 1: The set of all real numbers x is the whole number line, written as the open interval from -infinity to infinity

Final answer: (ii) (c) (-infinity, infinity)

(iii) Step 1: Removing the open interval (3, 5) from [3, 5] removes only the interior points, leaving the endpoints

Final answer: (iii) (d) {3, 5}

(iv) Step 1: The condition fixes y = 5 while x ranges over all real numbers, giving the relation as listed in the book key

Final answer: (iv) (a) Set of real numbers

(v) Step 1: f(-x) = (-x) / |-x| = -(x / |x|) = -f(x), which is the test for an odd function

Final answer: (v) (a) odd function

(vi) Step 1: f(x + 1) = 4^(x + 1) = 4 x 4^x = 4f(x)

Step 2: Subtract: 4f(x) - f(x) = 3f(x)

Final answer: (vi) (c) 3f(x)

(vii) Step 1: 1 / 2^n shrinks towards 0 as n grows

Final answer: (vii) (a) 0

(viii) Step 1: (sin ax) / x = a x (sin ax) / (ax), and the standard fraction tends to 1

Final answer: (viii) (b) a

(ix) Step 1: Factor: (x - 3)(x + 3) / (x - 3) = x + 3, which at x = 3 gives 6

Final answer: (ix) (d) 6

(x) Step 1: -3 is not in the domain [-1, 4], so f(-3) cannot be formed

Final answer: (x) (c) does not exist

(xi) Step 1: Substitute h = 0: sin(pi/2) = 1

Final answer: (xi) (a) 1

(xii) Step 1: Approaching 0 from the right, -1/x falls without bound, so e^(-1/x) shrinks to 0, which is the value given in the book key

Final answer: (xii) (a) 0

(xiii) Step 1: A two-sided limit exists exactly when both one-sided limits exist and are equal

Final answer: (xiii) (c) the right-hand limit equals the left-hand limit

(xiv) Step 1: [1 + 1/n]^(-7n) = ([1 + 1/n]^n)^(-7), and the inner bracket tends to e

Final answer: (xiv) (a) e^(-7)

(xv) Step 1: The terms are (1/pi)^n with 1/pi less than 1, so they shrink to 0

Final answer: (xv) (d) 0

(xvi) Step 1: A parametric function expresses both x and y through a third variable (parameter) t

Final answer: (xvi) (c) x = f(t), y = g(t)

(xvii) Step 1: g(f(x)) = x means f is the inverse of g; from y = 3x + 2, g^-1 (x) = (x - 2) / 3

Step 2: f(2) = (2 - 2) / 3 = 0

Final answer: (xvii) (c) 0

(xviii) Step 1: x / tan 3x = (1/3) x (3x) / tan 3x, and (tan 3x) / (3x) tends to 1, so the limit at 0 is 1/3

Step 2: Continuity needs k to equal this limit

Final answer: (xviii) (d) 1/3

(xix) Step 1: Take logarithms: (1/x) ln(1 - x) tends to -1, because ln(1 - x) / x tends to -1

Step 2: So the limit is e^(-1)

Final answer: (xix) (d) e^(-1)

(xx) Step 1: By definition sech^-1 x is the inverse hyperbolic secant, whose logarithmic form is ln(1/x + sqrt(1 - x^2) / x)

Final answer: (xx) (b) ln(1/x + sqrt(1 - x^2) / x)

Question: 2. If h(x) = sqrt(x^2 + 3) and k(x) = x^2 - 2, then find the compositions: (i) hok(x) (ii) koh(x) (iii) hoh (iv) kok

Answer: (i) Step 1: hok(x) = h(k(x)) = sqrt((x^2 - 2)^2 + 3)

Step 2: Expand (x^2 - 2)^2 = x^4 - 4x^2 + 4, then add 3

Final answer: (i) hok(x) = sqrt(x^4 - 4x^2 + 7)

(ii) Step 1: koh(x) = k(h(x)) = (sqrt(x^2 + 3))^2 - 2

Step 2: The square cancels the root: x^2 + 3 - 2

Final answer: (ii) koh(x) = x^2 + 1

(iii) Step 1: hoh(x) = h(h(x)) = sqrt((sqrt(x^2 + 3))^2 + 3) = sqrt(x^2 + 3 + 3)

Final answer: (iii) hoh(x) = sqrt(x^2 + 6)

(iv) Step 1: kok(x) = k(k(x)) = (x^2 - 2)^2 - 2

Step 2: Expand: x^4 - 4x^2 + 4 - 2

Final answer: (iv) kok(x) = x^4 - 4x^2 + 2

Question: 3. If f(x) = x + 3 and g(x) = x^2, then find gof(x) for x = 1.

Answer: Step 1: Inner value: f(1) = 1 + 3 = 4

Step 2: gof(1) = g(4) = 4^2

Final answer: gof(1) = 16

Question: 4. f(x) = (x + 1) / 3 and g(x) = 3x + 5 are two given functions; then verify that: (i) (fog)^-1 = g^-1 o f^-1 (ii) (gof)^-1 = f^-1 o g^-1

Answer: Step 1: Find the separate inverses: from y = (x + 1) / 3, x = 3y - 1, so f^-1 (x) = 3x - 1; from y = 3x + 5, g^-1 (x) = (x - 5) / 3

Step 2: fog(x) = f(3x + 5) = (3x + 5 + 1) / 3 = x + 2, so (fog)^-1 (x) = x -

2

Step 3: g^-1 o f^-1 (x) = g^-1 (3x - 1) = ((3x - 1) - 5) / 3 = (3x - 6) / 3 = x - 2, which matches

Step 4: gof(x) = g((x + 1) / 3) = 3 x ((x + 1) / 3) + 5 = x + 1 + 5 = x + 6, so (gof)^-1 (x) = x - 6

Step 5: f^-1 o g^-1 (x) = f^-1 ((x - 5) / 3) = 3 x ((x - 5) / 3) - 1 = (x - 5) - 1 = x - 6, which matches

Final answer: (i) (fog)^-1 = g^-1 o f^-1 = x - 2 and (ii) (gof)^-1 = f^-1 o g^-1 = x - 6, both verified

Question: 5. Recognize the following as explicit or implicit functions and express the implicit function as an explicit function if possible: (i) x^3 + 2xy = 5x^2 - 3y (ii) 3y = 5x^2 - 3x (iii) x^2 y + xy = 5y - 3 (iv) 2x^2 y - xy = 5 + 3xy

Answer: (i) Step 1: x and y are mixed (the term 2xy), so it is implicit

Step 2: Collect the y terms on the left: 2xy + 3y = 5x^2 - x^3

Step 3: Factor y: y(2x + 3) = 5x^2 - x^3

Step 4: Divide: y = (5x^2 - x^3) / (2x + 3)

Final answer: (i) Implicit, with explicit form y = (5x^2 - x^3) / (2x + 3)

(ii) Step 1: y is already expressed purely in terms of x after dividing by 3, so it is explicit

Final answer: (ii) Explicit function, y = (5x^2 - 3x) / 3

(iii) Step 1: The terms x^2 y and xy mix x and y, so it is implicit

Step 2: Collect y: x^2 y + xy - 5y = -3, that is y(x^2 + x - 5) = -3

Step 3: Divide: y = -3 / (x^2 + x - 5)

Final answer: (iii) Implicit, with explicit form y = -3 / (x^2 + x - 5)

(iv) Step 1: The terms mix x and y, so it is implicit

Step 2: Collect y: 2x^2 y - xy - 3xy = 5, that is y(2x^2 - 4x) = 5

Step 3: Divide and simplify the factor 2x: y = 5 / (2x^2 - 4x) = 5 / (2x(x - 2))

Final answer: (iv) Implicit, with explicit form y = 5 / (2x(x - 2))

Question: 6. Show that the parametric equations x = a sec theta, y = b tan theta represent the equation of x^2 / a^2 - y^2 / b^2 = 1.

Answer: Step 1: From the given equations, x / a = sec theta and y / b = tan theta

Step 2: Square both: x^2 / a^2 = sec^2 theta and y^2 / b^2 = tan^2 theta

Step 3: Subtract the second from the first: x^2 / a^2 - y^2 / b^2 = sec^2 theta - tan^2 theta

Step 4: Use the identity sec^2 theta - tan^2 theta = 1

Final answer: x^2 / a^2 - y^2 / b^2 = 1, shown (the equation of a hyperbola)

Question: 7. Draw and explain the graph of the following functions: (i) f(x) = e^(5x) (ii) y = sqrt(4 - x^2) (iii) x^2 / 9 + y^2 / 16 = 1

Answer: (i) Step 1: Table: at x = -1, 0, 1 the values are about 0.007, 1 and 148.4

Step 2: The curve passes through (0, 1), stays close to the x-axis on the left and climbs extremely steeply on the right, an exponential growth curve even steeper than e^(3x)

Final answer: (i) The graph of e^(5x) is a steep increasing exponential

curve through (0, 1)

(ii) Step 1: Squaring gives x^2 + y^2 = 4, the circle of radius 2 centred at the origin

Step 2: The square root keeps only the upper half, with domain from -2 to 2 and top point (0, 2)

Final answer: (ii) The graph is the upper semicircle of radius 2

(iii) Step 1: Compare with x^2 / a^2 + y^2 / b^2 = 1: a = 3 and b = 4

Step 2: Intercepts are (3, 0), (-3, 0), (0, 4) and (0, -4), joined by a smooth oval, longer in the y direction

Final answer: (iii) The graph is an ellipse centred at the origin with semi-axes 3 along x and 4 along y

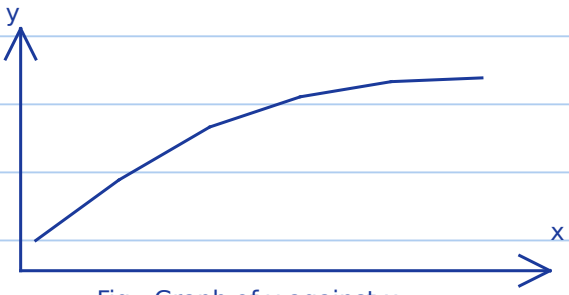


Fig.: Graph of y against x

Question: 8. Draw the graph of the parametric equations of the function x = a t^2, y = 2at, when a = 6 and t runs from -5 to 5.

Answer: Step 1: Substitute a = 6: x = 6t^2 and y = 12t

Step 2: Table: t = -5, -1, 0, 1, 5 gives (x, y) = (150, -60), (6, -12), (0, 0), (6, 12), (150, 60)

Step 3: Eliminate t: t = y / 12, so x = 6 (y / 12)^2 = y^2 / 24, that is y^2 = 24x

Step 4: This is a parabola opening to the right with vertex at the origin

Final answer: The graph is the part of the parabola y^2 = 24x from (150,

-60) through (0, 0) to (150, 60)

Question: 9. Draw the graph of the parametric equations of the function x = a sec theta, y = b tan theta, when a = 4, b = 3 and theta runs from -pi to pi.

Answer: Step 1: Substitute: x = 4 sec theta and y = 3 tan theta, so x / 4 = sec theta and y / 3 = tan theta

Step 2: Use sec^2 theta - tan^2 theta = 1: x^2 / 16 - y^2 / 9 = 1

Step 3: The hyperbola has vertices (4, 0) and (-4, 0) and asymptotes y = (3/4) x and y = -(3/4) x

Final answer: The graph is the hyperbola x^2 / 16 - y^2 / 9 = 1, opening left and right

Question: 10. Solve the following: (i) [1, infinity) union (2, 3) (ii) [-1, 1] intersection (2, infinity) (iii) (4, infinity) intersection (-3, 3) (iv) [2, 3] - (2, 3) (v) [1, 9] intersection [3, 12] (vi) (-infinity, 7) - (-infinity, 2)

Answer: (i) Step 1: (2, 3) lies inside [1, infinity), so the union adds nothing

Result: (i) [1, infinity)

(ii) Step 1: [-1, 1] ends at 1 and (2, infinity) begins after 2, so they share no point

Result: (ii) the empty set

(iii) Step 1: A common point would have to be greater than 4 and less than 3 at the same time, which is impossible

Result: (iii) the empty set

(iv) Step 1: Removing the open interval (2, 3) from [2, 3] leaves only the endpoints

Result: (iv) {2, 3}

(v) Step 1: The overlap runs from the larger left endpoint 3 to the smaller right endpoint 9

Result: (v) [3, 9]

(vi) Step 1: Removing all numbers less than 2 from all numbers less than 7 leaves the numbers from 2 up to 7, with 2 included and 7 excluded

Result: (vi) [2, 7)

Final answer: (i) [1, infinity), (ii) empty set, (iii) empty set, (iv) {2, 3}, (v) [3, 9], (vi) [2, 7)

Question: 11. Find the limit of the sequence 1, 1/4, 1/16, 1/64, ...

Answer: Step 1: Each term is one quarter of the previous term: 1, (1/4), (1/4)^2, (1/4)^3

Step 2: So the nth term is a_n = (1/4)^(n - 1)

Step 3: Powers of 1/4 shrink towards 0 as n grows

Final answer: a_n = (1/4)^(n - 1) and the limit is 0

Question: 12. Find the limit of a_n = -1 + (1/5)^n.

Answer: Step 1: As n grows, (1/5)^n shrinks towards 0

Step 2: So a_n approaches -1 + 0

Final answer: The limit is -1

Question: 13. Find the limit of the function y = 1 / x^3 for x tends to infinity.

Answer: Step 1: As x grows, x^3 grows without bound

Step 2: The reciprocal 1 / x^3 therefore shrinks towards 0

Final answer: The limit is 0

Question: 14. Evaluate: (i) lim (x to 3) (x^3 + x^4 + x + 5) (ii) lim (x to 7) ((1 + x) / x^7) (iii) lim (x to 1) [(2x^4 + 3x^3)(x + 3)] (iv) lim (x to 3) [(x

+ 3) - (x^3 + 5x + 5)] (v) lim (x to pi/4) (cosec^2 x - 2) / (cot x - 1) (vi) lim (x to pi) sin(pi - x) / (pi (pi - x)) (vi, printed again in the book) lim (x to 0) (1 + 2x)^(5/x) (viii) lim (x to 0) (3^(2x) - 1) / sin x

Answer: (i) Step 1: Substitute x = 3: 27 + 81 + 3 + 5

Result: (i) 116

(ii) Step 1: Substitute x = 7: (1 + 7) / 7^7

Result: (ii) 8 / 7^7

(iii) Step 1: Product theorem: first factor at x = 1 is 2 + 3 = 5, second factor is 1 + 3 = 4

Step 2: Multiply: 5 x 4

Result: (iii) 20

(iv) Step 1: Substitute x = 3 in each part: (3 + 3) - (27 + 15 + 5) = 6 - 47

Result: (iv) -41

(v) Step 1: Use cosec^2 x = 1 + cot^2 x, so the numerator is cot^2 x - 1 = (cot x - 1)(cot x + 1)

Step 2: Cancel (cot x - 1), leaving cot x + 1

Step 3: At x = pi/4, cot x = 1, giving 1 + 1

Result: (v) 2

(vi) Step 1: Put u = pi - x; as x tends to pi, u tends to 0, and the expression becomes sin u / (pi u) = (1/pi) x (sin u / u)

Step 2: The standard fraction tends to 1

Result: (vi) 1/pi

(vi again) Step 1: Rewrite as [(1 + 2x)^(1/(2x))]^(10), since (1/(2x)) x 10 = 5/x

Step 2: The bracket tends to e

Result: (vi again) e^10

(viii) Step 1: Write (3^(2x) - 1) / sin x = [(9^x - 1) / x] / [sin x / x]

Step 2: The numerator part tends to ln 9 = 2 ln 3 and the denominator part tends to 1

Result: (viii) 2 ln 3

Final answer: (i) 116, (ii) 8/7^7, (iii) 20, (iv) -41, (v) 2, (vi) 1/pi, (vi again) e^10, (viii) 2 ln 3

Question: 15. Determine whether the following functions are continuous at x = 3: (i) f(x) = (x^2 - 9) / (x - 3) (ii) g(x) = (x^2 - 9) / (x - 3) when x is not 3, and g(x) = 5 when x = 3 (iii) h(x) = (x^2 - 9) / (x - 3) when x is not 3, and h(x) = 6 when x = 3

Answer: Step 1: Simplify the common formula: (x^2 - 9) / (x - 3) = x + 3 for x not 3, so in each case the limit as x tends to 3 is 6

Step 2: (i) f(3) is undefined, so f is discontinuous at 3

Step 3: (ii) g(3) = 5, which is not the limit 6, so g is discontinuous at 3

Step 4: (iii) h(3) = 6, which equals the limit, and h is defined there, so h is continuous at 3

Final answer: (i) Discontinuous, (ii) Discontinuous, (iii) Continuous

Question: 16. If f(x) = 3x if x is at most -2, f(x) = x^2 - 1 if -2 is less than x and x is less than 2, and f(x) = 3 if x is at least 2: discuss the continuity or discontinuity at x = 2 and x = -2. (The sub-parts (i) to (iii) printed under this question in the book are a repeat of Question 15.)

Answer: Step 1: At x = 2: the value is f(2) = 3 from the third piece

Step 2: The left-hand limit comes from the middle piece: 2^2 - 1 = 3

Step 3: The right-hand limit comes from the third piece and is constantly 3

Step 4: Value and both one-sided limits all equal 3, so f is continuous at x = 2

Step 5: At x = -2: the value and left-hand limit come from the first piece: 3(-2) = -6

Step 6: The right-hand limit comes from the middle piece: (-2)^2 - 1 = 3

Step 7: The one-sided limits -6 and 3 differ, so there is a jump at x = -2

Final answer: f is continuous at x = 2 and discontinuous at x = -2

Unit 3: Differentiation

Exercise 3.1

Question: 1. Find the average rate of change of the following functions when x varies from a to b: (i) y = f(x) = x^2 + 4, a = 2, b = 2.3 (ii) y = f(x) = x^3 - 4, a = 2, b = 2.3 (iii) y = f(x) = x^3 - 8, a = 3, b = 2.5

Answer: (i) Step 1: The average rate of change is [f(b) - f(a)] / (b - a)

Step 2: f(2.3) = (2.3)^2 + 4 = 5.29 + 4 = 9.29 and f(2) = 4 + 4 = 8

Step 3: Subtract and divide: (9.29 - 8) / (2.3 - 2) = 1.29 / 0.3

Final answer: (i) 4.3 units

(ii) Step 1: f(2.3) = (2.3)^3 - 4 = 12.167 - 4 = 8.167 and f(2) = 8 - 4 = 4

Step 2: Subtract and divide: (8.167 - 4) / 0.3 = 4.167 / 0.3

Final answer: (ii) 13.89 units

(iii) Step 1: Here b = 2.5 is less than a = 3, but the same formula applies: [f(2.5) - f(3)] / (2.5 - 3)

Step 2: f(2.5) = 15.625 - 8 = 7.625 and f(3) = 27 - 8 = 19

Step 3: Subtract and divide: (7.625 - 19) / (-0.5) = (-11.375) / (-0.5)

Final answer: (iii) 22.75 units

Question: 2. Find out the average rate of change when x changes from a to b: (i) A = pi x^2, where x is the radius of the circle, a = 3, b = 3.1 (ii) V = (4/3) pi x^3, where x is the radius of the sphere, a = 2, b = 1.9

Answer: (i) Step 1: The average rate is [A(3.1) - A(3)] / (3.1 - 3)

Step 2: A(3.1) = pi (3.1)^2 = 9.61 pi and A(3) = 9 pi

Step 3: Subtract and divide: (0.61 pi) / 0.1 = 6.1 pi

Final answer: (i) 6.1 pi square units

(ii) Step 1: The average rate is [V(1.9) - V(2)] / (1.9 - 2)

Step 2: V(1.9) = (4/3) pi (1.9)^3 = (4/3) pi (6.859) and V(2) = (4/3) pi (8)

Step 3: The difference is (4/3) pi (6.859 - 8) = (4/3) pi (-1.141), and dividing by -0.1 gives (4/3) pi (11.41) which is about 15.21 pi, that is about 47.8

Final answer: (ii) About 47.8 cubic units

Question: 3. The price p in rupees after t years is given by p(t) = 3t^2 + t + 1. Find the average rate of change of inflation from t = 3 to t = 3.5 years.

Answer: Step 1: The average rate is [p(3.5) - p(3)] / (3.5 - 3)

Step 2: p(3.5) = 3(3.5)^2 + 3.5 + 1 = 3(12.25) + 4.5 = 36.75 + 4.5 = 41.25

Step 3: p(3) = 3(9) + 3 + 1 = 27 + 4 = 31

Step 4: Subtract and divide: (41.25 - 31) / 0.5 = 10.25 / 0.5

Final answer: Inflation = Rs. 20.5 per year

Question: 4. A ball is thrown vertically up, its height h in metres after t seconds is given by the formula h(t) = -16t^2 + 80t. Find the average velocity when t changes from a to b: (a) a = 2, b = 2.1 (b) a = 2, b = 2.01

Answer: (a) Step 1: The average velocity is [h(2.1) - h(2)] / (2.1 - 2)

Step 2: h(2.1) = -16(2.1)^2 + 80(2.1) = -16(4.41) + 168 = -70.56 + 168 = 97.44

Step 3: h(2) = -16(4) + 160 = -64 + 160 = 96

Step 4: Subtract and divide: (97.44 - 96) / 0.1 = 1.44 / 0.1

Final answer: (a) 14.4 m/s

(b) Step 1: The average velocity is [h(2.01) - h(2)] / 0.01

Step 2: h(2.01) = -16(2.01)^2 + 80(2.01) = -16(4.0401) + 160.8 = -64.6416 + 160.8 = 96.1584

Step 3: Subtract and divide: (96.1584 - 96) / 0.01 = 0.1584 / 0.01

Final answer: (b) 15.84 m/s

Exercise 3.2

Question: 1. Find by definition (ab-initio) the derivatives with respect to x of the following functions: (i) f(x) = 2x (ii) f(x) = 1 - sqrt(x) (iii) f(x) = 1 / sqrt(x) (iv) f(x) = 3 - x^2 (v) f(x) = x(x + 1) (vi) f(x) = x^2 - 3 (vii) f(x) = x^3 + 5 (viii) f(x) = 4x^2 - 3x (ix) f(x) = 1 / (x + 2) (x) f(x) = 3 / (2x + 5)

Answer: Step 1: In every part use the definition f'(x) = lim (h to 0) [f(x + h) - f(x)] / h, expand f(x + h), cancel f(x), divide by h, and then put h = 0

Step 2: (i) [2(x + h) - 2x] / h = 2h / h = 2, so the limit is 2

Step 3: (ii) [-sqrt(x + h) + sqrt(x)] / h; multiply by the conjugate -(sqrt(x + h) + sqrt(x)): the numerator becomes (x + h) - x = h with a negative sign pattern giving [-(h)] / [h (sqrt(x + h) + sqrt(x))] = -1 / (sqrt(x + h) + sqrt(x)), which tends to -1 / (2 sqrt(x))

Step 4: (iii) [1/sqrt(x + h) - 1/sqrt(x)] / h = [sqrt(x) - sqrt(x + h)] / [h sqrt(x) sqrt(x + h)]; multiply by the conjugate sqrt(x) + sqrt(x + h): numerator becomes x - (x + h) = -h, giving -1 / [sqrt(x) sqrt(x + h) (sqrt(x) + sqrt(x + h))], which tends to -1 / (2 x sqrt(x))

Step 5: (iv) [3 - (x + h)^2 - 3 + x^2] / h = [-(2xh + h^2)] / h = -(2x + h), which tends to -2x

Step 6: (v) First expand f(x) = x^2 + x; then [(x + h)^2 + (x + h) - x^2 - x] / h = [2xh + h^2 + h] / h = 2x + h + 1, which tends to 2x + 1

Step 7: (vi) [(x + h)^2 - 3 - x^2 + 3] / h = [2xh + h^2] / h = 2x + h, which tends to 2x

Step 8: (vii) [(x + h)^3 + 5 - x^3 - 5] / h = [3x^2 h + 3x h^2 + h^3] / h = 3x^2 + 3xh + h^2, which tends to 3x^2

Step 9: (viii) [4(x + h)^2 - 3(x + h) - 4x^2 + 3x] / h = [8xh + 4h^2 - 3h] / h = 8x + 4h - 3, which tends to 8x - 3

Step 10: (ix) [1/(x + h + 2) - 1/(x + 2)] / h = [(x + 2) - (x + h + 2)] / [h (x + h + 2)(x + 2)] = -1 / [(x + h + 2)(x + 2)], which tends to -1 / (x + 2)^2

Step 11: (x) [3/(2x + 2h + 5) - 3/(2x + 5)] / h = 3 [(2x + 5) - (2x + 2h + 5)] / [h (2x + 2h + 5)(2x + 5)] = -6 / [(2x + 2h + 5)(2x + 5)], which tends to -6 / (2x + 5)^2

Final answer: (i) 2, (ii) -1/(2 sqrt(x)), (iii) -1/(2x sqrt(x)), (iv) -2x, (v) 2x + 1, (vi) 2x, (vii) 3x^2, (viii) 8x - 3, (ix) -1/(x + 2)^2, (x) -6/(2x + 5)^2

Question: 2. Find f'(x) for the following functions using definition: (i) f(x) = (2x + 1)^(1/3) (ii) f(x) = (2x - 1)^(-1/2) (iii) f(x) = (6x + 7)^(5/2) (iv) f(x) = (3x - 5)^(-3/2)

Answer: Step 1: Use the definition together with the binomial series for a fractional index, as done in the book for (ax + b)^n: the increment quotient reduces to n (ax + b)^(n - 1) times the derivative a of the inner expression, after the limit is applied

Step 2: (i) n = 1/3 and a = 2: f'(x) = (1/3)(2x + 1)^(-2/3) x 2 = 2 / [3 (2x + 1)^(2/3)]

Step 3: (ii) n = -1/2 and a = 2: f'(x) = (-1/2)(2x - 1)^(-3/2) x 2 = -(2x - 1)^(-3/2) = -1 / (2x - 1)^(3/2)

Step 4: (iii) n = 5/2 and a = 6: f'(x) = (5/2)(6x + 7)^(3/2) x 6 = 15 (6x + 7)^(3/2)

Step 5: (iv) n = -3/2 and a = 3: f'(x) = (-3/2)(3x - 5)^(-5/2) x 3 = (-9/2)(3x - 5)^(-5/2) = -9 / [2 (3x - 5)^(5/2)]

Final answer: (i) 2 / [3(2x + 1)^(2/3)], (ii) -1 / (2x - 1)^(3/2), (iii) 15(6x + 7)^(3/2), (iv) -9 / [2(3x - 5)^(5/2)]

Exercise 3.3

Question: 1. Differentiate the following with respect to x: (i) 5x^5 (ii) (7/9) x^9 (iii) -25 x^(-3/5) (iv) 124 sqrt(x) (v) (1/22) x^22 (vi) x^(-100) (vii) 15 (x)^(1/3) (viii) 16 (x^(3/4))^(1/4) (ix) -4 / x^(3/4) (x) 3 / (x^2)^(1/3)

Answer: Step 1: In every part use the power rule d/dx (x^n) = n x^(n - 1), multiplying the coefficient by the old index and reducing the index by 1

Step 2: (i) 5 x 5 x^4 = 25x^4

Step 3: (ii) (7/9) x 9 x^8 = 7x^8

Step 4: (iii) -25 x (-3/5) x^(-3/5 - 1) = 15 x^(-8/5)

Step 5: (iv) Write sqrt(x) = x^(1/2): 124 x (1/2) x^(-1/2) = 62 x^(-1/2) = 62 / sqrt(x)

Step 6: (v) (1/22) x 22 x^21 = x^21

Step 7: (vi) -100 x^(-101)

Step 8: (vii) 15 x (1/3) x^(-2/3) = 5 x^(-2/3)

Step 9: (viii) First simplify the power: (x^(3/4))^(1/4) = x^(3/16), then 16 x (3/16) x^(3/16 - 1) = 3 x^(-13/16)

Step 10: (ix) Write as -4 x^(-3/4): -4 x (-3/4) x^(-7/4) = 3 x^(-7/4)

Step 11: (x) Write as 3 x^(-2/3): 3 x (-2/3) x^(-5/3) = -2 x^(-5/3)

Final answer: (i) 25x^4, (ii) 7x^8, (iii) 15 x^(-8/5), (iv) 62/sqrt(x), (v) x^21, (vi) -100 x^(-101), (vii) 5 x^(-2/3), (viii) 3 x^(-13/16), (ix) 3 x^(-7/4), (x) -2 x^(-5/3)

Question: 2. Differentiate the following with respect to x: (i) x^5 / (a^2 + b^2) + x^2 / (a^2 - b^2) (ii) 2x + (1/2) x^6 (iii) (x^2)^(1/3) + sqrt(x) (iv) (1/21) x^21 + (1/22) x^22 (v) -(5/4) x^(-4/5) + (2/3) x^(-3/2)

Answer: Step 1: Differentiate term by term; in (i) the denominators a^2 + b^2 and a^2 - b^2 are constants because a and b are constants

Step 2: (i) 5x^4 / (a^2 + b^2) + 2x / (a^2 - b^2)

Step 3: (ii) 2 + (1/2) x 6 x^5 = 2 + 3x^5

Step 4: (iii) (x^2)^(1/3) = x^(2/3) gives (2/3) x^(-1/3), and sqrt(x) = x^(1/2) gives (1/2) x^(-1/2), so the sum is 2 / (3 (x)^(1/3)) + 1 / (2 sqrt(x))

Step 5: (iv) (1/21) x 21 x^20 + (1/22) x 22 x^21 = x^20 + x^21

Step 6: (v) -(5/4) x (-4/5) x^(-9/5) + (2/3) x (-3/2) x^(-5/2) = x^(-9/5) - x^(-5/2)

Final answer: (i) 5x^4/(a^2 + b^2) + 2x/(a^2 - b^2), (ii) 2 + 3x^5, (iii) 2/(3 (x)^(1/3)) + 1/(2 sqrt(x)), (iv) x^20 + x^21, (v) x^(-9/5) - x^(-5/2)

Question: 3. Differentiate the following with respect to x: (i) 2a x^3 - x^2 / b + 6 (ii) x^3 - (3/7) x^(7/3) (iii) 5 x^(3/5) - (1/7) x^(7/3) (iv) x^10 - 10 x^15 (v) 3 (x^2)^(1/3) - 4 (x)^(1/4)

Answer: Step 1: Differentiate term by term with the power rule; constants (including the whole constant term 6 in (i)) have derivative 0

Step 2: (i) 2a x 3 x^2 - (1/b) x 2x + 0 = 6a x^2 - 2x / b

Step 3: (ii) 3x^2 - (3/7) x (7/3) x^(4/3) = 3x^2 - x^(4/3)

Step 4: (iii) 5 x (3/5) x^(-2/5) - (1/7) x (7/3) x^(4/3) = 3 x^(-2/5) - (1/3) x^(4/3)

Step 5: (iv) 10 x^9 - 10 x 15 x^14 = 10x^9 - 150 x^14

Step 6: (v) 3 x (2/3) x^(-1/3) - 4 x (1/4) x^(-3/4) = 2 x^(-1/3) - x^(-3/4)

Final answer: (i) 6a x^2 - 2x/b, (ii) 3x^2 - x^(4/3), (iii) 3 x^(-2/5) - (1/3) x^(4/3), (iv) 10x^9 - 150x^14, (v) 2 x^(-1/3) - x^(-3/4)

Question: 4. Differentiate the following polynomial functions with respect to x: (i) p(x) = x^3 - 3x^2 + 2x + 1 (ii) p(x) = x^4 - 3x^2 + 2x - 3 (iii) p(x) = x^6 - x^4 + x^3 + x (iv) p(x) = 9x^9 + 7x^7 + (1/5) x^5 - (1/4) x^4 + x + 1

Answer: Step 1: Differentiate each polynomial term by term

Step 2: (i) 3x^2 - 6x + 2 + 0 = 3x^2 - 6x + 2

Step 3: (ii) 4x^3 - 6x + 2 - 0 = 4x^3 - 6x + 2

Step 4: (iii) 6x^5 - 4x^3 + 3x^2 + 1

Step 5: (iv) 9 x 9 x^8 + 7 x 7 x^6 + (1/5) x 5 x^4 - (1/4) x 4 x^3 + 1 + 0 = 81x^8 + 49x^6 + x^4 - x^3 + 1

Final answer: (i) 3x^2 - 6x + 2, (ii) 4x^3 - 6x + 2, (iii) 6x^5 - 4x^3 + 3x^2 + 1, (iv) 81x^8 + 49x^6 + x^4 - x^3 + 1

Question: 5. Find the derivative of the following functions using the product rule: (i) h(x) = (2x - 5)(5x + 7) (ii) h(x) = x sqrt(3x^2 + 4) (iii) h(x) = (x + 1)^(1/3) . (x^2 + 1)^(1/5) (iv) h(x) = x^2 (sqrt(x) + 1) (v) h(x) = (x + 1)^3 . x^(-3/2)

Answer: Step 1: The product rule is (uv)' = u'v + u v'

Step 2: (i) (2)(5x + 7) + (2x - 5)(5) = 10x + 14 + 10x - 25 = 20x - 11

Step 3: (ii) (1) sqrt(3x^2 + 4) + x x [6x / (2 sqrt(3x^2 + 4))] = sqrt(3x^2 + 4) + 3x^2 / sqrt(3x^2 + 4) = [(3x^2 + 4) + 3x^2] / sqrt(3x^2 + 4) = (6x^2 + 4) / sqrt(3x^2 + 4)

Step 4: (iii) (1/3)(x + 1)^(-2/3) (x^2 + 1)^(1/5) + (x + 1)^(1/3) (1/5)(x^2 + 1)^(-4/5)(2x); factor out h(x): h'(x) = h(x) [1 / (3(x + 1)) + 2x / (5(x^2 + 1))] = h(x) [5(x^2 + 1) + 6x(x + 1)] / [15 (x + 1)(x^2 + 1)] = h(x) (11x^2 + 6x + 5) / [15 (x + 1)(x^2 + 1)], that is (11x^2 + 6x + 5) / [15 (x + 1)^(2/3) (x^2 + 1)^(4/5)]

Step 5: (iv) 2x (sqrt(x) + 1) + x^2 [1 / (2 sqrt(x))] = 2x sqrt(x) + 2x + (x sqrt(x)) / 2 = (x/2)(5 sqrt(x) + 4)

Step 6: (v) 3(x + 1)^2 x^(-3/2) + (x + 1)^3 (-3/2) x^(-5/2) = (x + 1)^2 x^(-5/2) [3x - (3/2)(x + 1)] = (x + 1)^2 x^(-5/2) ((3x - 3) / 2) = 3(x - 1)(x + 1)^2 / (2 x^(5/2))

Final answer: (i) 20x - 11, (ii) (6x^2 + 4) / sqrt(3x^2 + 4), (iii) (11x^2 + 6x + 5) / [15 (x + 1)^(2/3) (x^2 + 1)^(4/5)], (iv) (x/2)(5 sqrt(x) + 4), (v) 3(x - 1)(x + 1)^2 / (2 x^(5/2))

Question: 6. Find the derivative of the following functions using the quotient rule: (i) h(x) = (3x + 4) / (2x - 3) (ii) h(x) = (x^2 - 1) . (x^2 + 1)^(-1) (iii) h(x) = (x^2 - x + 1) / (x^2 + x - 1) (iv) h(x) = 2x^4 / (b^2 - x^2) (v) h(x) = (sqrt(x) + 1) / (sqrt(x) - 1)

Answer: Step 1: The quotient rule is (u / v)' = (u' v - u v') / v^2

Step 2: (i) [3(2x - 3) - (3x + 4)(2)] / (2x - 3)^2 = [6x - 9 - 6x - 8] / (2x - 3)^2 = -17 / (2x - 3)^2

Step 3: (ii) Treat as the quotient (x^2 - 1) / (x^2 + 1): [2x(x^2 + 1) - (x^2 - 1)(2x)] / (x^2 + 1)^2 = [2x^3 + 2x - 2x^3 + 2x] / (x^2 + 1)^2 = 4x / (x^2 + 1)^2

Step 4: (iii) [(2x - 1)(x^2 + x - 1) - (x^2 - x + 1)(2x + 1)] / (x^2 + x - 1)^2; the numerator expands to (2x^3 + x^2 - 3x + 1) - (2x^3 - x^2 + x + 1) = 2x^2 - 4x = 2x(x - 2), giving 2x(x - 2) / (x^2 + x - 1)^2

Step 5: (iv) [8x^3 (b^2 - x^2) - 2x^4 (-2x)] / (b^2 - x^2)^2 = [8b^2 x^3 - 8x^5 + 4x^5] / (b^2 - x^2)^2 = 4x^3 (2b^2 - x^2) / (b^2 - x^2)^2

Step 6: (v) [(1 / (2 sqrt(x)))(sqrt(x) - 1) - (sqrt(x) + 1)(1 / (2 sqrt(x)))] / (sqrt(x) - 1)^2 = [((sqrt(x) - 1) - (sqrt(x) + 1)) / (2 sqrt(x))] / (sqrt(x) - 1)^2 = [-2 / (2 sqrt(x))] / (sqrt(x) - 1)^2 = -1 / [sqrt(x) (sqrt(x) - 1)^2]

Final answer: (i) -17/(2x - 3)^2, (ii) 4x/(x^2 + 1)^2, (iii) 2x(x - 2)/(x^2 + x - 1)^2, (iv) 4x^3(2b^2 - x^2)/(b^2 - x^2)^2, (v) -1/[sqrt(x) (sqrt(x) - 1)^2]

Exercise 3.4

Question: 1. Find the derivatives of the following using the chain rule: (i) y = (x^4 + 5x^2 + 6)^(3/2) (ii) y = ((x - 1) / (x + 1))^(3/4) (iii) y = sqrt((2 + x) / (3 + x)) (iv) y = (x + sqrt(x^2 - 1))^n (v) y = ((x^3 + 1) / (x^3 - 1))^(1/3)

Answer: (i) Step 1: Let u = x^4 + 5x^2 + 6, so y = u^(3/2), dy/du = (3/2) u^(1/2) and du/dx = 4x^3 + 10x = 2x(2x^2 + 5)

Step 2: Multiply: dy/dx = (3/2)(x^4 + 5x^2 + 6)^(1/2) x 2x(2x^2 + 5)

Final answer: (i) dy/dx = 3x(2x^2 + 5)(x^4 + 5x^2 + 6)^(1/2)

(ii) Step 1: Take the logarithmic form: ln y = (3/4)[ln(x - 1) - ln(x + 1)]

Step 2: Differentiate: (1/y) dy/dx = (3/4)[1/(x - 1) - 1/(x + 1)] = (3/4) x [2 / (x^2 - 1)] = 3 / [2(x^2 - 1)]

Step 3: Multiply by y = ((x - 1)/(x + 1))^(3/4) and simplify the powers: dy/dx = 3 / [2 (x - 1)^(1/4) (x + 1)^(7/4)]

Final answer: (ii) dy/dx = 3 / [2 (x - 1)^(1/4) (x + 1)^(7/4)]

(iii) Step 1: Write y = [(2 + x) / (3 + x)]^(1/2) and use the logarithmic

form: ln y = (1/2)[ln(2 + x) - ln(3 + x)]

Step 2: Differentiate: (1/y) dy/dx = (1/2)[1/(2 + x) - 1/(3 + x)] = (1/2) x [1 / ((2 + x)(3 + x))]

Step 3: Multiply by y and simplify: dy/dx = 1 / [2 (2 + x)^(1/2) (3 + x)^(3/2)]

Final answer: (iii) dy/dx = 1 / [2 sqrt(2 + x) (3 + x)^(3/2)]

(iv) Step 1: Let u = x + sqrt(x^2 - 1); then du/dx = 1 + x / sqrt(x^2 - 1) = [sqrt(x^2 - 1) + x] / sqrt(x^2 - 1) = u / sqrt(x^2 - 1)

Step 2: y = u^n gives dy/dx = n u^(n - 1) du/dx = n u^(n - 1) x u / sqrt(x^2 - 1) = n u^n / sqrt(x^2 - 1)

Final answer: (iv) dy/dx = n (x + sqrt(x^2 - 1))^n / sqrt(x^2 - 1)

(v) Step 1: ln y = (1/3)[ln(x^3 + 1) - ln(x^3 - 1)]

Step 2: Differentiate: (1/y) dy/dx = (1/3)[3x^2 / (x^3 + 1) - 3x^2 / (x^3 - 1)] = x^2 [(x^3 - 1 - x^3 - 1) / (x^6 - 1)] = -2x^2 / (x^6 - 1)

Step 3: Multiply by y and write x^6 - 1 = (x^3 + 1)(x^3 - 1): dy/dx = -2x^2 / [(x^3 + 1)^(2/3) (x^3 - 1)^(4/3)]

Final answer: (v) dy/dx = -2x^2 / [(x^3 + 1)^(2/3) (x^3 - 1)^(4/3)]

Question: 2. Differentiate x^3 / (1 + x^3) with respect to x^3.

Answer: Step 1: Put u = x^3, so the function becomes u / (1 + u), and we need the derivative with respect to u

Step 2: Apply the quotient rule with respect to u: [(1)(1 + u) - u(1)] / (1 + u)^2

Step 3: Simplify the numerator: 1 + u - u = 1

Step 4: Replace u by x^3

Final answer: The derivative with respect to x^3 is 1 / (1 + x^3)^2

Question: 3. If f is a function with y = f(x), given implicitly, find dy/dx, where it exists, in the following cases: (i) y - xy - sin y = 0 (ii) y^3 - 3y + 2x = 0 (iii) x^2 + y^2 + 4x + 6y - 12 = 0 (iv) sin x y + sec x = 2 [that is, sin(xy) + sec x = 2] (v) x sqrt(1 + y) + y sqrt(1 + x) = 0 (vi) y(x^2 + 1) = x(y^2 + 1)

Answer: Step 1: In every part, differentiate both sides with respect to x, treating y as a function of x, then collect the terms containing dy/dx

Step 2: (i) dy/dx - y - x dy/dx - cos y dy/dx = 0, so dy/dx (1 - x - cos y) = y, giving dy/dx = y / (1 - x - cos y)

Step 3: (ii) 3y^2 dy/dx - 3 dy/dx + 2 = 0, so dy/dx (3y^2 - 3) = -2, giving dy/dx = -2 / (3y^2 - 3) = 2 / [3(1 - y^2)]

Step 4: (iii) 2x + 2y dy/dx + 4 + 6 dy/dx = 0, so dy/dx (2y + 6) = -(2x + 4), giving dy/dx = -(x + 2) / (y + 3)

Step 5: (iv) cos(xy) (y + x dy/dx) + sec x tan x = 0, so x cos(xy) dy/dx = -y cos(xy) - sec x tan x, giving dy/dx = -[y cos(xy) + sec x tan x] / [x cos(xy)]

Step 6: (v) sqrt(1 + y) + x (dy/dx) / (2 sqrt(1 + y)) + (dy/dx) sqrt(1 + x) + y / (2 sqrt(1 + x)) = 0, so dy/dx [x / (2 sqrt(1 + y)) + sqrt(1 + x)] = -[sqrt(1 + y) + y / (2 sqrt(1 + x))], giving dy/dx = -[sqrt(1 + y) + y / (2 sqrt(1 + x))] / [x / (2 sqrt(1 + y)) + sqrt(1 + x)]

Step 7: (vi) Expand first: x^2 y + y = x y^2 + x; differentiate: 2xy + x^2 dy/dx + dy/dx = y^2 + 2xy dy/dx + 1, so dy/dx (x^2 + 1 - 2xy) = y^2 + 1 - 2xy, giving dy/dx = [(y^2 + 1) - 2xy] / [(x^2 + 1) - 2xy]

Final answer: (i) y/(1 - x - cos y), (ii) 2/[3(1 - y^2)], (iii) -(x + 2)/(y + 3), (iv) -[y cos(xy) + sec x tan x]/[x cos(xy)], (v) -[sqrt(1 + y) + y/(2 sqrt(1 + x))] / [x/(2 sqrt(1 + y)) + sqrt(1 + x)], (vi) [(y^2 + 1) - 2xy] / [(x^2 + 1) - 2xy]

Question: 4. If u^2 / a^2 + v^2 / b^2 = 1, where a and b are non-zero constants, find du/dv and dv/du.

Answer: Step 1: Differentiate with respect to v, treating u as a function of v: (2u / a^2) du/dv + 2v / b^2 = 0

Step 2: Solve: du/dv = -(a^2 v) / (b^2 u)

Step 3: Differentiate the same equation with respect to u, treating v as a function of u: 2u / a^2 + (2v / b^2) dv/du = 0

Step 4: Solve: dv/du = -(b^2 u) / (a^2 v)

Step 5: Note the two answers are reciprocals of each other, as they must be

Final answer: du/dv = -(a^2 v)/(b^2 u) and dv/du = -(b^2 u)/(a^2 v)

Question: 5. Find the slope of the tangent to the curve 3x^2 - 7y^2 + 14y - 27 = 0 at the point (-3, 0).

Answer: Step 1: Differentiate implicitly: 6x - 14y dy/dx + 14 dy/dx = 0

Step 2: Collect dy/dx: dy/dx (14 - 14y) = -6x, so dy/dx = -6x / (14 - 14y) = -3x / (7 - 7y)

Step 3: Substitute the point x = -3, y = 0: dy/dx = -6(-3) / 14 = 18 / 14 = 9/7

Final answer: The slope of the tangent at (-3, 0) is 9/7

Question: 6. Differentiate the following by using the first principle method: (i) sin 4x (ii) cos^2 2x (iii) sec sqrt(x) (iv) sqrt(tan x) (v) csc 3x (vi) cot 2x

Answer: Step 1: In every part start from f'(x) = lim (h to 0) [f(x + h) - f(x)] / h, use the relevant addition formula, and finish with the standard limits (sin h)/h tends to 1 and (cos h - 1)/h tends to 0, together with the chain factor coming from the inner function

Step 2: (i) sin(4x + 4h) - sin 4x = 2 cos(4x + 2h) sin 2h; dividing by h and

using sin 2h / (2h) tends to 1 with the extra factor 2 gives 2 x 2 cos 4x = 4 cos 4x

Step 3: (ii) Write cos^2 2x and use the definition: the quotient reduces to -2 sin(4x) after using sin(4x + 2h) - sin 4x type identities; equivalently, the derivative of cos^2 2x is 2 cos 2x (-sin 2x)(2) = -2 sin 4x

Step 4: (iii) sec sqrt(x + h) - sec sqrt(x) over h; using the definition for sec and then the inner derivative 1 / (2 sqrt(x)) gives sec sqrt(x) tan sqrt(x) / (2 sqrt(x))

Step 5: (iv) sqrt(tan(x + h)) - sqrt(tan x) over h; rationalising and using the definition for tan x gives sec^2 x / (2 sqrt(tan x))

Step 6: (v) csc(3x + 3h) - csc 3x over h reduces, by the definition for cosecant and the inner factor 3, to -3 csc 3x cot 3x

Step 7: (vi) cot(2x + 2h) - cot 2x over h reduces, by the definition for cotangent and the inner factor 2, to -2 csc^2 2x

Final answer: (i) 4 cos 4x, (ii) -2 sin 4x, (iii) sec sqrt(x) tan sqrt(x) / (2 sqrt(x)), (iv) sec^2 x / (2 sqrt(tan x)), (v) -3 csc 3x cot 3x, (vi) -2 csc^2 2x

Question: 7. Using differentiation rules, differentiate with respect to their involved variables: (i) f(x) = (x + 2) . sin x (ii) f(theta) = tan^2 theta . sec^3 theta (iii) f(t) = sin^2 3t . cos^3 t (iv) f(x) = sqrt((sin 2x) / (cos x)) (v) f(theta) = (tan theta - 1) / sec theta (vi) f(x) = sin x^2 + sin^2 x

Answer: Step 1: (i) Product rule: (1) sin x + (x + 2) cos x = sin x + (x + 2) cos x

Step 2: (ii) Product and chain rules: 2 tan theta sec^2 theta sec^3 theta + tan^2 theta x 3 sec^3 theta tan theta = sec^3 theta tan theta (2 sec^2 theta + 3 tan^2 theta)

Step 3: (iii) Product rule: 2 sin 3t cos 3t (3) cos^3 t + sin^2 3t x 3 cos^2 t

(-sin t) = 3 cos^2 t [sin 6t cos t - sin^2 3t sin t], using 2 sin 3t cos 3t = sin 6t

Step 4: (iv) First simplify inside: sin 2x / cos x = 2 sin x cos x / cos x = 2 sin x, so f(x) = sqrt(2 sin x) and f'(x) = (2 cos x) / (2 sqrt(2 sin x)) = cos x / sqrt(2 sin x)

Step 5: (v) Simplify first: (tan theta - 1) / sec theta = tan theta cos theta - cos theta = sin theta - cos theta, whose derivative is cos theta + sin theta

Step 6: (vi) Chain rule on the first term and power rule on the second: 2x cos x^2 + 2 sin x cos x = 2x cos x^2 + sin 2x

Final answer: (i) sin x + (x + 2) cos x, (ii) sec^3 theta tan theta (2 sec^2 theta + 3 tan^2 theta), (iii) 3 cos^2 t (sin 6t cos t - sin^2 3t sin t), (iv) cos x / sqrt(2 sin x), (v) sin theta + cos theta, (vi) 2x cos x^2 + sin 2x

Question: 8. Differentiate (1 + tan^2 x) / (1 - tan^2 x) with respect to tan^2 x.

Answer: Step 1: Put u = tan^2 x, so the function is (1 + u) / (1 - u) and we differentiate with respect to u

Step 2: Quotient rule: [(1)(1 - u) - (1 + u)(-1)] / (1 - u)^2

Step 3: Simplify the numerator: 1 - u + 1 + u = 2

Step 4: Replace u by tan^2 x

Final answer: The derivative with respect to tan^2 x is 2 / (1 - tan^2 x)^2

Question: 9. Find dy/dx of the following: (i) y = sin^-1 sqrt((1 - cos x) / 2) (ii) y = cot^-1 (sqrt((1 + cos x) / (1 - cos x))) (iii) y = tan^-1 ((sin 2x) / (1 + cos 2x)) (iv) y = x + cos^-1 x . sqrt(1 - x^2) (v) y = (x tan^-1 x) / (1 + x^2) (vi) y = tan^-1 ((sqrt(1 + x) - sqrt(1 - x)) / (sqrt(1 + x) + sqrt(1 - x)))

Answer: (i) Step 1: Use (1 - cos x) / 2 = sin^2 (x/2), so the square root is

sin(x/2) (for x in the principal range) and y = sin^-1 (sin(x/2)) = x/2

Step 2: Differentiate: dy/dx = 1/2

Final answer: (i) 1/2

(ii) Step 1: (1 + cos x) / (1 - cos x) = cot^2 (x/2), so the square root is cot(x/2) and y = cot^-1 (cot(x/2)) = x/2

Step 2: Differentiate

Final answer: (ii) 1/2

(iii) Step 1: (sin 2x) / (1 + cos 2x) = (2 sin x cos x) / (2 cos^2 x) = tan x, so y = tan^-1 (tan x) = x

Step 2: Differentiate

Final answer: (iii) 1

(iv) Step 1: Differentiate the product cos^-1 x . sqrt(1 - x^2) by the product rule: [-1 / sqrt(1 - x^2)] sqrt(1 - x^2) + cos^-1 x [-x / sqrt(1 - x^2)] = -1 - x cos^-1 x / sqrt(1 - x^2)

Step 2: Add the derivative of the first term x, which is 1; the 1 and -1 cancel

Final answer: (iv) dy/dx = -x cos^-1 x / sqrt(1 - x^2)

(v) Step 1: Quotient rule: numerator derivative is tan^-1 x + x / (1 + x^2), denominator is 1 + x^2

Step 2: dy/dx = [(tan^-1 x + x/(1 + x^2))(1 + x^2) - (x tan^-1 x)(2x)] / (1 + x^2)^2 = [(1 + x^2) tan^-1 x + x - 2x^2 tan^-1 x] / (1 + x^2)^2

Step 3: Collect the tan^-1 x terms: (1 + x^2 - 2x^2) = (1 - x^2)

Final answer: (v) dy/dx = [x + (1 - x^2) tan^-1 x] / (1 + x^2)^2

(vi) Step 1: Put x = cos 2t, so sqrt(1 + x) = sqrt(2) cos t and sqrt(1 - x) =

sqrt(2) sin t

Step 2: The fraction becomes (cos t - sin t) / (cos t + sin t) = tan(pi/4 - t), so y = pi/4 - t = pi/4 - (1/2) cos^-1 x

Step 3: Differentiate: dy/dx = -(1/2) x [-1 / sqrt(1 - x^2)] = 1 / (2 sqrt(1 - x^2))

Final answer: (vi) dy/dx = 1 / (2 sqrt(1 - x^2))

Question: 10. If y = tan(2 tan^-1 (x/2)), then prove that dy/dx = 4 ((1 + y^2) / (4 + x^2)).

Answer: Step 1: Put t = tan^-1 (x/2), so tan t = x/2

Step 2: Use tan 2t = 2 tan t / (1 - tan^2 t) = (2 x (x/2)) / (1 - x^2/4) = x / ((4 - x^2)/4) = 4x / (4 - x^2), so y = 4x / (4 - x^2)

Step 3: Differentiate y by the quotient rule: dy/dx = [4(4 - x^2) - 4x(-2x)] / (4 - x^2)^2 = [16 - 4x^2 + 8x^2] / (4 - x^2)^2 = 4(4 + x^2) / (4 - x^2)^2

Step 4: Now compute the claimed right-hand side: 1 + y^2 = 1 + 16x^2 / (4 - x^2)^2 = [(4 - x^2)^2 + 16x^2] / (4 - x^2)^2 = [16 + 8x^2 + x^4] / (4 - x^2)^2 = (4 + x^2)^2 / (4 - x^2)^2

Step 5: Multiply by 4 / (4 + x^2): 4(4 + x^2) / (4 - x^2)^2, which is exactly dy/dx from Step 3

Final answer: dy/dx = 4(1 + y^2) / (4 + x^2), proved

Question: 11. If y/x = tan^-1 (x/y), show that dy/dx = y/x.

Answer: Step 1: Take the tangent of both sides: tan(y/x) = x/y

Step 2: Differentiate both sides with respect to x: sec^2 (y/x) x [(x dy/dx - y) / x^2] = (y - x dy/dx) / y^2

Step 3: Notice that (y - x dy/dx) = -(x dy/dx - y); put D = x dy/dx - y, so the

equation becomes sec^2 (y/x) x D / x^2 = -D / y^2

Step 4: Bring everything to one side: D [sec^2 (y/x) / x^2 + 1 / y^2] = 0

Step 5: The bracket is a sum of positive terms and is never zero, so D = 0, that is x dy/dx - y = 0

Final answer: dy/dx = y/x, shown

Question: 12. Given that tan y (1 + tan x) = 1 - tan x, show that dy/dx = -1.

Answer: Step 1: Solve for tan y: tan y = (1 - tan x) / (1 + tan x)

Step 2: Recognise the right-hand side as the subtraction formula tan(pi/4 - x), because tan(pi/4) = 1

Step 3: So tan y = tan(pi/4 - x), giving y = pi/4 - x (in the principal range)

Step 4: Differentiate: dy/dx = 0 - 1 = -1

Final answer: dy/dx = -1, shown

Question: 13. If y = tan(a tan^-1 x), show that (1 + x^2) dy/dx - a(1 + y^2) = 0.

Answer: Step 1: Differentiate y using the chain rule: dy/dx = sec^2 (a tan^-1 x) x a x [1 / (1 + x^2)]

Step 2: Use sec^2 (a tan^-1 x) = 1 + tan^2 (a tan^-1 x) = 1 + y^2

Step 3: So dy/dx = a(1 + y^2) / (1 + x^2)

Step 4: Multiply both sides by (1 + x^2): (1 + x^2) dy/dx = a(1 + y^2)

Final answer: (1 + x^2) dy/dx - a(1 + y^2) = 0, shown

Question: 14. Find dy/dx when: (i) x = a sin theta and y = a cos theta (ii) x = t + 1/t and y = t + 1 (iii) x = a(1 - t^2) / (1 + t^2) and y = 2bt / (1 + t^2), where a and b are constants (iv) x = a theta^2 and y = 2a theta, where a is a constant

Answer: Step 1: For parametric equations, dy/dx = (dy/d parameter) / (dx/d parameter)

Step 2: (i) dy/d theta = -a sin theta and dx/d theta = a cos theta, so dy/dx = -a sin theta / (a cos theta) = -tan theta

Step 3: (ii) dy/dt = 1 and dx/dt = 1 - 1/t^2 = (t^2 - 1) / t^2, so dy/dx = 1 / [(t^2 - 1)/t^2] = t^2 / (t^2 - 1)

Step 4: (iii) dx/dt = a[(-2t)(1 + t^2) - (1 - t^2)(2t)] / (1 + t^2)^2 = a(-4t) / (1 + t^2)^2, and dy/dt = 2b[(1 + t^2) - t(2t)] / (1 + t^2)^2 = 2b(1 - t^2) / (1 + t^2)^2, so dy/dx = [2b(1 - t^2)] / (-4at) = b(t^2 - 1) / (2at)

Step 5: (iv) dy/d theta = 2a and dx/d theta = 2a theta, so dy/dx = 2a / (2a theta) = 1 / theta

Final answer: (i) -tan theta, (ii) t^2/(t^2 - 1), (iii) b(t^2 - 1)/(2at), (iv) 1/theta

Question: 15. If y = sqrt(tan x + sqrt(tan x + sqrt(tan x + ... to infinity))), prove that (2y - 1) dy/dx = sec^2 x.

Answer: Step 1: Because the nested expression repeats forever, the part inside the first root after tan x is again the whole expression y, so y = sqrt(tan x + y)

Step 2: Square both sides: y^2 = tan x + y

Step 3: Differentiate with respect to x: 2y dy/dx = sec^2 x + dy/dx

Step 4: Bring the dy/dx terms together: 2y dy/dx - dy/dx = sec^2 x

Step 5: Factor: (2y - 1) dy/dx = sec^2 x

Final answer: (2y - 1) dy/dx = sec^2 x, proved

Question: 16. Find the derivatives of cos^-1 x, csc^-1 x and cot^-1 x by using the differentiation formula.

Answer: Step 1: For y = cos^-1 x, write x = cos y and differentiate with respect to y: dx/dy = -sin y, so dy/dx = -1 / sin y = -1 / sqrt(1 - cos^2 y) = -1 / sqrt(1 - x^2)

Step 2: For y = csc^-1 x, write x = csc y, so dx/dy = -csc y cot y; using cot y = sqrt(csc^2 y - 1) = sqrt(x^2 - 1) gives dy/dx = -1 / [x sqrt(x^2 - 1)]

Step 3: For y = cot^-1 x, write x = cot y, so dx/dy = -csc^2 y = -(1 + cot^2 y) = -(1 + x^2), giving dy/dx = -1 / (1 + x^2)

Final answer: d/dx (cos^-1 x) = -1 / sqrt(1 - x^2), d/dx (csc^-1 x) = -1 / [x sqrt(x^2 - 1)], d/dx (cot^-1 x) = -1 / (1 + x^2)

Exercise 3.5

Question: 1. Differentiate the following with respect to x: (i) x^2 + 2^x (ii) 4^x + 5^x (iii) e^(tan x + cot x) (iv) e^(tan x^2) (v) e^(2 ln(2x + 1)) (vi) log_10 x (vii) e^x / (x^2 + 1) (viii) x^2 + 2^x + a^(2x) (ix) (ln x)^x (x) ln(sqrt(e^(3x) + e^(-3x))) (xi) ln(sin(ln x)) (xii) ln[tan(x/2 + pi/4)]

Answer: Step 1: Use d/dx (a^x) = a^x ln a, d/dx (e^u) = e^u u', d/dx (ln u) = u' / u and the chain rule throughout

Step 2: (i) 2x + 2^x ln 2

Step 3: (ii) 4^x ln 4 + 5^x ln 5

Step 4: (iii) e^(tan x + cot x) (sec^2 x - csc^2 x), because the derivative of tan x + cot x is sec^2 x - csc^2 x

Step 5: (iv) e^(tan(x^2)) x sec^2 (x^2) x 2x = 2x sec^2 (x^2) e^(tan x^2)

Step 6: (v) Simplify first: e^(2 ln(2x + 1)) = (2x + 1)^2, whose derivative is 2(2x + 1)(2) = 4(2x + 1)

Step 7: (vi) log_10 x = ln x / ln 10, so the derivative is 1 / (x ln 10)

Step 8: (vii) Quotient rule: [e^x (x^2 + 1) - e^x (2x)] / (x^2 + 1)^2 = e^x (x^2 - 2x + 1) / (x^2 + 1)^2 = (x - 1)^2 e^x / (x^2 + 1)^2

Step 9: (viii) 2x + 2^x ln 2 + a^(2x) ln a x 2 = 2x + 2^x ln 2 + 2 a^(2x) ln a (for a positive)

Step 10: (ix) Logarithmic differentiation: ln y = x ln(ln x), so (1/y) dy/dx = ln(ln x) + x x [1 / (x ln x)] = ln(ln x) + 1 / ln x, giving dy/dx = (ln x)^x [1 / ln x + ln(ln x)]

Step 11: (x) (1/2) x [(3e^(3x) - 3e^(-3x)) / (e^(3x) + e^(-3x))] = (3/2) (e^(3x) - e^(-3x)) / (e^(3x) + e^(-3x)) = (3/2) tanh 3x

Step 12: (xi) (1 / sin(ln x)) x cos(ln x) x (1/x) = cot(ln x) / x

Step 13: (xii) Put u = x/2 + pi/4: the derivative is [1 / tan u] x sec^2 u x (1/2) = 1 / (2 sin u cos u) = 1 / sin(2u) = 1 / sin(x + pi/2) = 1 / cos x = sec x

Final answer: (i) 2x + 2^x ln 2, (ii) 4^x ln 4 + 5^x ln 5, (iii) (sec^2 x - csc^2 x) e^(tan x + cot x), (iv) 2x sec^2 (x^2) e^(tan x^2), (v) 4(2x + 1), (vi) 1/(x ln 10), (vii) (x - 1)^2 e^x / (x^2 + 1)^2, (viii) 2x + 2^x ln 2 + 2 a^(2x) ln a, (ix) (ln x)^x [1/ln x + ln(ln x)], (x) (3/2) tanh 3x, (xi) cot(ln x)/x, (xii) sec x

Question: 2. Using logarithmic differentiation, find dy/dx if: (i) y = sqrt((x^2 - 1) / (x^2 + 1)) (ii) y = x^3 sqrt(x) (iii) y = x e^(cos x) (iv) y = e^(-2x) (x^2 + 2x + 1) (v) y = ln(e^x / (1 + e^x)) (vi) y = sqrt((1 + e^x) / (1 - e^x))

Answer: Step 1: Take natural logarithms, use the laws of logarithms to split products, quotients and powers, differentiate both sides, and multiply by y

Step 2: (i) ln y = (1/2)[ln(x^2 - 1) - ln(x^2 + 1)], so (1/y) dy/dx =

(1/2)[2x/(x^2 - 1) - 2x/(x^2 + 1)] = x [(x^2 + 1 - x^2 + 1) / (x^4 - 1)] = 2x / (x^4 - 1); multiplying by y = sqrt((x^2 - 1)/(x^2 + 1)) gives dy/dx = 2x / [(x^2 + 1) sqrt(x^4 - 1)]

Step 3: (ii) ln y = 3 ln x + (1/2) ln x = (7/2) ln x, so (1/y) dy/dx = 7 / (2x) and dy/dx = (7x^2 / 2) x sqrt(x) = (7/2) x^(5/2)

Step 4: (iii) ln y = ln x + cos x, so (1/y) dy/dx = 1/x - sin x and dy/dx = x e^(cos x) (1/x - sin x) = e^(cos x) (1 - x sin x)

Step 5: (iv) ln y = -2x + ln(x^2 + 2x + 1), so (1/y) dy/dx = -2 + (2x + 2)/(x^2 + 2x + 1) = -2 + 2/(x + 1) = -2x / (x + 1); multiplying by y = e^(-2x) (x + 1)^2 gives dy/dx = -2x e^(-2x) (x + 1)

Step 6: (v) Simplify first: y = x - ln(1 + e^x), so dy/dx = 1 - e^x / (1 + e^x) = 1 / (1 + e^x)

Step 7: (vi) ln y = (1/2)[ln(1 + e^x) - ln(1 - e^x)], so (1/y) dy/dx = (1/2)[e^x/(1 + e^x) + e^x/(1 - e^x)] = (1/2) e^x [2 / (1 - e^(2x))] = e^x / (1 - e^(2x)); multiplying by y gives dy/dx = e^x / [(1 - e^x) sqrt(1 - e^(2x))]

Final answer: (i) 2x / [(x^2 + 1) sqrt(x^4 - 1)], (ii) (7/2) x^(5/2), (iii) (1 - x sin x) e^(cos x), (iv) -2x e^(-2x) (x + 1), (v) 1 / (1 + e^x), (vi) e^x / [(1 - e^x) sqrt(1 - e^(2x))]

Question: 3. Find dy/dx if: (i) y = (1 - x^2) / sqrt(1 + x^2) (ii) y = sqrt((1 - x) / (1 + x))

Answer: (i) Step 1: Quotient rule with u = 1 - x^2, u' = -2x and v = (1 + x^2)^(1/2), v' = x / sqrt(1 + x^2)

Step 2: dy/dx = [-2x sqrt(1 + x^2) - (1 - x^2) x / sqrt(1 + x^2)] / (1 + x^2)

Step 3: Put over the common denominator (1 + x^2)^(3/2): numerator =

-2x(1 + x^2) - x(1 - x^2) = -2x - 2x^3 - x + x^3 = -3x - x^3

Final answer: (i) dy/dx = -x(3 + x^2) / (1 + x^2)^(3/2)

(ii) Step 1: Logarithmic form: ln y = (1/2)[ln(1 - x) - ln(1 + x)]

Step 2: Differentiate: (1/y) dy/dx = (1/2)[-1/(1 - x) - 1/(1 + x)] = (1/2) x [-((1 + x) + (1 - x)) / (1 - x^2)] = -1 / (1 - x^2)

Step 3: Multiply by y = sqrt((1 - x)/(1 + x)) and simplify: dy/dx = -sqrt((1 - x)/(1 + x)) / [(1 - x)(1 + x)] = -1 / [(1 - x)^(1/2) (1 + x)^(3/2)]

Final answer: (ii) dy/dx = -1 / [(1 - x)^(1/2) (1 + x)^(3/2)]

Question: 4. Find dy/dx if: (i) y = x^(sin x) (ii) y = (sin^-1 x)^(ln x) (iii) y = (tan^-1 x)^(sin x + cos x) (iv) y = (ln x)^(cos x) (v) y = x^x (vi) y = ln((sqrt(x^2 + 1) - x) / (sqrt(x^2 + 1) + x))

Answer: Step 1: For (i) to (v) use logarithmic differentiation, because both the base and the exponent contain x: write ln y = (exponent) x ln(base), differentiate, and multiply by y

Step 2: (i) ln y = sin x ln x, so (1/y) dy/dx = cos x ln x + (sin x)/x, giving dy/dx = x^(sin x) (sin x / x + cos x ln x)

Step 3: (ii) ln y = ln x x ln(sin^-1 x), so (1/y) dy/dx = (1/x) ln(sin^-1 x) + ln x x [1 / (sqrt(1 - x^2) sin^-1 x)], giving dy/dx = (sin^-1 x)^(ln x) [ln(sin^-1 x)/x + ln x / (sqrt(1 - x^2) sin^-1 x)]

Step 4: (iii) ln y = (sin x + cos x) ln(tan^-1 x), so (1/y) dy/dx = (cos x - sin x) ln(tan^-1 x) + (sin x + cos x) / [(1 + x^2) tan^-1 x], giving dy/dx = (tan^-1 x)^(sin x + cos x) [(sin x + cos x)/((1 + x^2) tan^-1 x) + (cos x - sin x) ln(tan^-1 x)]

Step 5: (iv) ln y = cos x ln(ln x), so (1/y) dy/dx = -sin x ln(ln x) + cos x / (x ln x), giving dy/dx = (ln x)^(cos x) [cos x/(x ln x) - sin x ln(ln x)]

Step 6: (v) ln y = x ln x, so (1/y) dy/dx = ln x + 1, giving dy/dx = x^x (1 + ln x)

Step 7: (vi) Note (sqrt(x^2 + 1) - x) / (sqrt(x^2 + 1) + x) = (sqrt(x^2 + 1) - x)^2, because the product of the two factors in the denominator form is (x^2 + 1) - x^2 = 1; so y = 2 ln(sqrt(x^2 + 1) - x)

Step 8: Differentiate: dy/dx = 2 [x / sqrt(x^2 + 1) - 1] / (sqrt(x^2 + 1) - x) = 2 [x - sqrt(x^2 + 1)] / [sqrt(x^2 + 1) (sqrt(x^2 + 1) - x)] = -2 / sqrt(x^2 + 1)

Final answer: (i) x^(sin x) (sin x/x + cos x ln x), (ii) (sin^-1 x)^(ln x) [ln x/(sqrt(1 - x^2) sin^-1 x) + ln(sin^-1 x)/x], (iii) (tan^-1 x)^(sin x + cos x) [(sin x + cos x)/((1 + x^2) tan^-1 x) + (cos x - sin x) ln(tan^-1 x)], (iv) (ln x)^(cos x) [cos x/(x ln x) - sin x ln(ln x)], (v) x^x (1 + ln x), (vi) -2 / sqrt(x^2 + 1)

Question: 5. Find dy/dx when: (i) x^y . y^x = 1 (ii) ln(xy) = x^2 + y^2 (iii) y = sin^-1 (cos x) + cos^-1 (sin x) (iv) y = x^y (v) y = cos x ln(sin^-1 x) (vi) x^n . y^n = a^n

Answer: (i) Step 1: Take logarithms: y ln x + x ln y = 0

Step 2: Differentiate: (dy/dx) ln x + y/x + ln y + (x/y) dy/dx = 0

Step 3: Collect dy/dx: dy/dx (ln x + x/y) = -(y/x + ln y)

Step 4: Multiply numerator and denominator by xy and tidy: dy/dx = -(y/x) [(x ln y + y) / (y ln x + x)]

Final answer: (i) dy/dx = -(y/x) [(x ln y + y)/(y ln x + x)]

(ii) Step 1: Write ln x + ln y = x^2 + y^2 and differentiate: 1/x + (1/y) dy/dx = 2x + 2y dy/dx

Step 2: Collect dy/dx: dy/dx (1/y - 2y) = 2x - 1/x

Step 3: Solve and tidy by multiplying numerator and denominator by xy: dy/dx = -(y/x) [(2x^2 - 1) / (2y^2 - 1)]

Final answer: (ii) dy/dx = -(y/x) [(2x^2 - 1)/(2y^2 - 1)]

(iii) Step 1: For x between 0 and pi/2, sin^-1 (cos x) = pi/2 - x and cos^-1 (sin x) = pi/2 - x

Step 2: So y = pi - 2x in that range

Step 3: Differentiate

Final answer: (iii) dy/dx = -2

(iv) Step 1: Take logarithms: ln y = y ln x

Step 2: Differentiate: (1/y) dy/dx = (dy/dx) ln x + y/x

Step 3: Collect dy/dx: dy/dx (1/y - ln x) = y/x

Step 4: Solve: dy/dx = (y/x) / (1/y - ln x) = y^2 / [x(1 - y ln x)]

Final answer: (iv) dy/dx = y^2 / [x(1 - y ln x)]

(v) Step 1: Product rule: derivative of cos x is -sin x, and derivative of ln(sin^-1 x) is [1 / sin^-1 x] x [1 / sqrt(1 - x^2)]

Step 2: dy/dx = -sin x ln(sin^-1 x) + cos x / [sqrt(1 - x^2) sin^-1 x]

Final answer: (v) dy/dx = cos x / [sqrt(1 - x^2) sin^-1 x] - sin x ln(sin^-1 x)

(vi) Step 1: Take logarithms: n ln x + n ln y = n ln a, and cancel n: ln x + ln y = constant

Step 2: Differentiate: 1/x + (1/y) dy/dx = 0

Step 3: Solve: dy/dx = -y/x

Final answer: (vi) dy/dx = -y/x

Exercise 3.6

Question: 1. Differentiate the following with respect to x: (i) sinh[ln(x + 3)] (ii) sinh(e^(3x)) (iii) cosh(2x^2 + 3x) (iv) tanh sqrt(x) / sqrt(cosh x) (v) tan(e^(sinh^-1 x)) (vi) (sinh^-1 x) / (sech^-1 x) (vii) cosh x . coth x^2 (viii) sinh x tanh x^2 (ix) ln[tanh(x^2 + 2x + 1)]

Answer: Step 1: Use d/dx (sinh u) = cosh u u', d/dx (cosh u) = sinh u u', d/dx (tanh u) = sech^2 u u', d/dx (coth u) = -cosech^2 u u', d/dx (sinh^-1 u) = u' / sqrt(1 + u^2) and d/dx (sech^-1 u) = -u' / [u sqrt(1 - u^2)]

Step 2: (i) cosh[ln(x + 3)] x [1 / (x + 3)] = cosh[ln(x + 3)] / (x + 3)

Step 3: (ii) cosh(e^(3x)) x 3e^(3x) = 3e^(3x) cosh(e^(3x))

Step 4: (iii) sinh(2x^2 + 3x) x (4x + 3) = (4x + 3) sinh(2x^2 + 3x)

Step 5: (iv) Product and quotient working: derivative of tanh sqrt(x) is sech^2 sqrt(x) / (2 sqrt(x)), and derivative of (cosh x)^(-1/2) is -(1/2)(cosh x)^(-3/2) sinh x; combining over the common denominator 2 sqrt(x) cosh^(3/2) x gives [cosh x sech^2 sqrt(x) - sqrt(x) sinh x tanh sqrt(x)] / [2 sqrt(x) cosh^(3/2) x]

Step 6: (v) sec^2 (e^(sinh^-1 x)) x e^(sinh^-1 x) x [1 / sqrt(1 + x^2)] = e^(sinh^-1 x) sec^2 (e^(sinh^-1 x)) / sqrt(1 + x^2)

Step 7: (vi) Quotient rule: [(1/sqrt(1 + x^2)) sech^-1 x + (sinh^-1 x) / (x sqrt(1 - x^2))] / (sech^-1 x)^2, which over the common denominator becomes [x sqrt(1 - x^2) sech^-1 x + sqrt(1 + x^2) sinh^-1 x] / [x sqrt(1 - x^4) (sech^-1 x)^2]

Step 8: (vii) Product rule: sinh x coth x^2 + cosh x [-cosech^2 x^2] (2x) = coth x^2 sinh x - 2x cosh x cosech^2 x^2

Step 9: (viii) Product rule: cosh x tanh x^2 + sinh x sech^2 x^2 (2x) = cosh x tanh x^2 + 2x sinh x sech^2 x^2

Step 10: (ix) Put u = x^2 + 2x + 1: derivative of ln(tanh u) is [sech^2 u / tanh u] u' = u' / (sinh u cosh u) = 2(x + 1) csch(x^2 + 2x + 1) sech(x^2 + 2x + 1)

Final answer: (i) cosh[ln(x + 3)]/(x + 3), (ii) 3e^(3x) cosh(e^(3x)), (iii) (4x + 3) sinh(2x^2 + 3x), (iv) [cosh x sech^2 sqrt(x) - sqrt(x) sinh x tanh sqrt(x)] / [2 sqrt(x) cosh^(3/2) x], (v) e^(sinh^-1 x) sec^2 (e^(sinh^-1 x)) / sqrt(1 + x^2), (vi) [x sqrt(1 - x^2) sech^-1 x + sqrt(1 + x^2) sinh^-1 x] / [x sqrt(1 - x^4) (sech^-1 x)^2], (vii) coth x^2 sinh x - 2x cosh x cosech^2 x^2, (viii) cosh x tanh x^2 + 2x sinh x sech^2 x^2, (ix) 2(x + 1) csch(x^2 + 2x + 1) sech(x^2 + 2x + 1)

Question: 2. Find dy/dx for the following functions: (i) y = x cosh^-1 x - sqrt(x^2 - 1) (ii) y = x tanh^-1 (3x) (iii) ln(cosh^-1 x) + sinh^-1 y = C (iv) y = ln(1 - x^2) + 2x tanh^-1 x (v) y = tanh^-1 (tan x^3) (vi) y = x sech^-1 (sqrt(x))

Answer: Step 1: Use d/dx (cosh^-1 x) = 1 / sqrt(x^2 - 1), d/dx (tanh^-1 u) = u' / (1 - u^2), d/dx (sinh^-1 y) = y' / sqrt(y^2 + 1) and d/dx (sech^-1 u) = -u' / [u sqrt(1 - u^2)]

Step 2: (i) Product rule on the first term gives cosh^-1 x + x / sqrt(x^2 - 1), and the derivative of -sqrt(x^2 - 1) is -x / sqrt(x^2 - 1); the two middle terms cancel, leaving cosh^-1 x

Step 3: (ii) tanh^-1 (3x) + x x [3 / (1 - 9x^2)] = [3x + (1 - 9x^2) tanh^-1 (3x)] / (1 - 9x^2)

Step 4: (iii) Differentiate with respect to x: [1 / cosh^-1 x] x [1 / sqrt(x^2 - 1)] + [1 / sqrt(y^2 + 1)] dy/dx = 0, so dy/dx = -sqrt(y^2 + 1) / [sqrt(x^2 - 1) cosh^-1 x]

Step 5: (iv) -2x / (1 - x^2) + 2 tanh^-1 x + 2x / (1 - x^2); the first and third terms cancel, leaving 2 tanh^-1 x

Step 6: (v) [1 / (1 - tan^2 (x^3))] x sec^2 (x^3) x 3x^2 = 3x^2 sec^2 (x^3) / (1 - tan^2 (x^3))

Step 7: (vi) Product rule: sech^-1 (sqrt(x)) + x x [-1 / (sqrt(x) sqrt(1 - x)) x 1/(2 sqrt(x))] = sech^-1 (sqrt(x)) - 1 / [2 sqrt(1 - x)] = [2 sqrt(1 - x) sech^-1 (sqrt(x)) - 1] / [2 sqrt(1 - x)]

Final answer: (i) cosh^-1 x, (ii) [3x + (1 - 9x^2) tanh^-1 (3x)] / (1 - 9x^2), (iii) -sqrt(y^2 + 1) / [sqrt(x^2 - 1) cosh^-1 x], (iv) 2 tanh^-1 x, (v) 3x^2 sec^2 (x^3) / (1 - tan^2 (x^3)), (vi) [2 sqrt(1 - x) sech^-1 (sqrt(x)) - 1] / [2 sqrt(1 - x)]

Question: 3. Write the MAPLE command diff to differentiate the following: (i) f(x) = 2x^3 + 3x^2 + 6 (ii) f(x) = sin(2x + 3) (iii) f(x) = (x + 1)(x + 2) (iv) f(x) = (x^2 - 3x + 2) / (x^2 - 4)

Answer: Step 1: The Maple command has the form diff(expression, x), which returns the derivative with respect to x

Step 2: (i) Command: diff(2x^3 + 3x^2 + 6, x); output: 6x^2 + 6x

Step 3: (ii) Command: diff(sin(2x + 3), x); output: 2 cos(2x + 3)

Step 4: (iii) Command: diff((x + 1)(x + 2), x); output: expand first if wanted: (x + 2) + (x + 1) = 2x + 3

Step 5: (iv) Command: diff((x^2 - 3x + 2)/(x^2 - 4), x); output by the quotient rule: [(2x - 3)(x^2 - 4) - (x^2 - 3x + 2)(2x)] / (x^2 - 4)^2 = (3x^2 - 12x + 12) / (x^2 - 4)^2 = 3(x - 2)^2 / (x^2 - 4)^2 = 3 / (x + 2)^2

Final answer: (i) diff(2x^3 + 3x^2 + 6, x), (ii) diff(sin(2x + 3), x), (iii) diff((x + 1)(x + 2), x), (iv) diff((x^2 - 3x + 2)/(x^2 - 4), x)

Question: 4. Write the MAPLE command d/dx (the differential operator form) to differentiate the following functions: (i) f(x) = x^3 + 5x^2 + 3x + 7 (ii) f(x) = sin x^2 (iii) f(x) = (sqrt(x) + 1) / (x^2 + 1)

Answer: Step 1: In the Maple palette the operator is written as d/dx applied to the expression, equivalent to diff(expression, x)

Step 2: (i) Operator: d/dx (x^3 + 5x^2 + 3x + 7); its value is 3x^2 + 10x + 3

Step 3: (ii) Operator: d/dx (sin(x^2)); its value is cos(x^2) x 2x = 2x cos x^2

Step 4: (iii) Operator: d/dx ((sqrt(x) + 1)/(x^2 + 1)); its value by the quotient rule is [(1/(2 sqrt(x)))(x^2 + 1) - (sqrt(x) + 1)(2x)] / (x^2 + 1)^2 = [1 - 3x^2 - 4x sqrt(x)] / [2 sqrt(x) (x^2 + 1)^2]

Final answer: (i) d/dx (x^3 + 5x^2 + 3x + 7), (ii) d/dx (sin(x^2)), (iii) d/dx ((sqrt(x) + 1)/(x^2 + 1))

Review Exercise 3

Question: 1. Select the correct options: (i) The derivative of 2 / x^3 is: (a) 2 / (3x^2) (b) -2 / (3x^2) (c) 6 / x^4 (d) -6 / x^4 (ii) The derivative of sqrt(x) + x sqrt(x) is: (a) 1/(2 sqrt(x)) + 3 sqrt(x)/2 (b) -(1 + x)/(2 sqrt(x)) (c) 1/sqrt(x) + 3 sqrt(x) (d) 1/sqrt(x) + 2 sqrt(x)/3 (iii) If y = (x + 1)(x^2 - 2), then dy/dx is: (a) x^3 + x^2 - 2x - 2 (b) 3x^2 + 2x - 2 (c) 3x^2 - 2x + 2 (d) 3x^2 + 2x + 2 (iv) If a x^2 + b y^2 = ab, then dy/dx is: (a) -2ax/(by) (b) -bx/(ay) (c) -ax/(by) (d) -ax/(2by) (v) If y = sqrt(tan x - y), then dy/dx = : (a) tan x / (2y + 1) (b) -csc^2 x / (2y - 1) (c) sec^2 x / (2y - 1) (d) sec^2 x / (2y + 1) (vi) If y = tan^-1 sqrt(x) then dy/dx = : (a) 1/(1 + x^2) (b) 1/(x + sqrt(x)) (c) 1/[2(x + x sqrt(x))] (d) 1/[2(sqrt(x) + x sqrt(x))] (vii) The derivative of tan x with respect to cot x is: (a) sec^2 x csc^2 x (b) -tan^2 x (c) sec^2 x / csc^2 x (d) tan^2 x (viii) If f(x) = a x^2 - 3x - 5 and f'(2) = 9, then a is equal to: (a) -2 (b) 3 (c) 4 (d) 5 (ix) The derivative of x^2 e^(2x) is: (a) x^2 e^(2x) + 2x^2 e^x (b) 2x e^(2x) (c) 2 e^(2x) (x^2 +

x) (d) 2 e^(2x) (x^2 + 1) (x) The derivative of a^x, if a is negative, is: (a) -a^x ln a (b) a^x ln a (c) a^x / ln a (d) Does not exist (xi) If y = tan^-1 sqrt((1 - cos 2x) / (1 + cos 2x)) then dy/dx = : (a) 1 (b) -1 (c) 2 (d) 1/2 (xii) d/dx (sinh^-1 x + cosh^-1 x) is: (a) cosh^-1 x - sinh^-1 x (b) 1/sqrt(1 + x^2) - 1/sqrt(1 - x^2) (c) 1/sqrt(x^2 + 1) + 1/sqrt(x^2 - 1) (d) 1/sqrt(x^2 - 1) - 1/sqrt(1 + x^2) (xiii) The derivative of tanh ax is: (a) sech^2 ax (b) a sech ax (c) a sech^2 ax (d) 2a sech^2 ax (xiv) The derivative of coth^-1 (2x) is: (a) 1/(1 - 4x^2) (b) 2/(1 - 4x^2) (c) 2x/(1 - 4x^2) (d) 2/(1 - x^2) (xv) If f(x) = ln 2x (x positive) and g is the inverse of f, then g'(x) = : (a) 2/x (b) 1/(2x) (c) 2/e^x (d) e^x / 2 (xvi) If f(x) = a cos 3x and f'(pi/2) = 6, then a = : (a) -6 (b) -2 (c) 2 (d) 3

Answer: (i) Step 1: 2/x^3 = 2 x^(-3), whose derivative is 2 x (-3) x^(-4) = -6/x^4

Final answer: (i) (d) -6/x^4

(ii) Step 1: sqrt(x) + x sqrt(x) = x^(1/2) + x^(3/2); derivative = (1/2) x^(-1/2) + (3/2) x^(1/2)

Final answer: (ii) (a) 1/(2 sqrt(x)) + 3 sqrt(x)/2

(iii) Step 1: Expand: x^3 + x^2 - 2x - 2; differentiate term by term: 3x^2 + 2x - 2

Final answer: (iii) (b) 3x^2 + 2x - 2

(iv) Step 1: Differentiate implicitly: 2ax + 2by dy/dx = 0, so dy/dx = -ax/(by)

Final answer: (iv) (c) -ax/(by)

(v) Step 1: Square: y^2 = tan x - y; differentiate: 2y dy/dx = sec^2 x - dy/dx, so dy/dx (2y + 1) = sec^2 x

Final answer: (v) (d) sec^2 x/(2y + 1)

(vi) Step 1: dy/dx = [1/(1 + x)] x [1/(2 sqrt(x))] = 1/[2 sqrt(x) (1 + x)] = 1/[2(sqrt(x) + x sqrt(x))]

Final answer: (vi) (d) 1/[2(sqrt(x) + x sqrt(x))]

(vii) Step 1: Divide the derivatives: (d/dx tan x) / (d/dx cot x) = sec^2 x / (-csc^2 x) = -(sin^2 x / cos^2 x) = -tan^2 x

Final answer: (vii) (b) -tan^2 x

(viii) Step 1: f'(x) = 2ax - 3, so f'(2) = 4a - 3 = 9 gives 4a = 12

Final answer: (viii) (b) a = 3

(ix) Step 1: Product rule: 2x e^(2x) + x^2 e^(2x) (2) = 2 e^(2x) (x + x^2)

Final answer: (ix) (c) 2 e^(2x) (x^2 + x)

(x) Step 1: The formula a^x ln a needs ln a, which is not a real number when a is negative, so the real derivative does not exist

Final answer: (x) (d) Does not exist

(xi) Step 1: (1 - cos 2x)/(1 + cos 2x) = tan^2 x, so y = tan^-1 (tan x) = x, whose derivative is 1

Final answer: (xi) (a) 1

(xii) Step 1: Add the standard derivatives: 1/sqrt(x^2 + 1) + 1/sqrt(x^2 - 1)

Final answer: (xii) (c) 1/sqrt(x^2 + 1) + 1/sqrt(x^2 - 1)

(xiii) Step 1: d/dx (tanh ax) = sech^2 (ax) x a

Final answer: (xiii) (c) a sech^2 ax

(xiv) Step 1: d/dx (coth^-1 u) = u'/(1 - u^2) with u = 2x: 2/(1 - 4x^2)

Final answer: (xiv) (b) 2/(1 - 4x^2)

(xv) Step 1: Invert y = ln 2x: e^y = 2x, so x = e^y / 2, that is g(x) = e^x / 2, and g'(x) = e^x / 2

Final answer: (xv) (d) e^x / 2

(xvi) Step 1: f'(x) = -3a sin 3x; at x = pi/2, sin(3pi/2) = -1, so f'(pi/2) = 3a = 6

Final answer: (xvi) (c) a = 2

Question: 2. Find the derivative of sqrt(cos x) and sec sqrt(x) by the first principle.

Answer: Step 1: For sqrt(cos x): form the quotient [sqrt(cos(x + h)) - sqrt(cos x)] / h, rationalise with the conjugate, use cos(x + h) - cos x = -2 sin(x + h/2) sin(h/2), and apply (sin(h/2))/(h/2) tends to 1; the result is -sin x / (2 sqrt(cos x))

Step 2: For sec sqrt(x): form the quotient [sec sqrt(x + h) - sec sqrt(x)] / h, write sec as 1/cos, use the cosine difference formula and then the inner limit for sqrt(x + h) - sqrt(x), namely 1/(2 sqrt(x)); the result is sec sqrt(x) tan sqrt(x) / (2 sqrt(x))

Final answer: d/dx (sqrt(cos x)) = -sin x / (2 sqrt(cos x)) and d/dx (sec sqrt(x)) = sec sqrt(x) tan sqrt(x) / (2 sqrt(x))

Question: 3. If y = (sin x)^(ln x), find dy/dx.

Answer: Step 1: Take logarithms: ln y = ln x x ln(sin x)

Step 2: Differentiate the product on the right: (1/x) ln(sin x) + ln x x [cos x / sin x] = ln(sin x)/x + cot x ln x

Step 3: Multiply by y

Final answer: dy/dx = (sin x)^(ln x) [ln(sin x)/x + cot x ln x]

Question: 4. Find dy/dx, if a x^2 + 2hxy + b y^2 = 0.

Answer: Step 1: Differentiate implicitly: 2ax + 2h(y + x dy/dx) + 2by dy/dx = 0

Step 2: Divide by 2: ax + hy + dy/dx (hx + by) = 0

Step 3: Collect and solve: dy/dx (hx + by) = -(ax + hy)

Final answer: dy/dx = -(ax + hy)/(hx + by)

Question: 5. Let f(x) = cot^-1 (2x / (1 - x^2)), find f'(x) and f'(-sqrt(3)).

Answer: Step 1: Put x = tan t, so 2x/(1 - x^2) = tan 2t and f(x) = cot^-1 (tan 2t) = pi/2 - 2t = pi/2 - 2 tan^-1 x

Step 2: Differentiate: f'(x) = -2/(1 + x^2)

Step 3: At x = -sqrt(3): f'(-sqrt(3)) = -2/(1 + 3) = -1/2

Step 4: For reference, the value of the function itself at -sqrt(3) is cot^-1 (sqrt(3)) = pi/6, which is the second value listed in the book key

Final answer: f'(x) = -2/(1 + x^2) and f'(-sqrt(3)) = -1/2

Question: 6. x = 4(t - sin t) and y = 4(1 + cos t), find dy/dx.

Answer: Step 1: Differentiate with respect to t: dx/dt = 4(1 - cos t) and dy/dt = -4 sin t

Step 2: Divide: dy/dx = -sin t / (1 - cos t)

Step 3: Use sin t = 2 sin(t/2) cos(t/2) and 1 - cos t = 2 sin^2 (t/2): the ratio is -cos(t/2)/sin(t/2)

Final answer: dy/dx = -cot(t/2)

Question: 7. Differentiate with respect to x: (i) (x^2 + x^(-1)) / (x^2 - x^(-1)) (ii) (3x - 2) / sqrt(x^2 + 1)

Answer: (i) Step 1: Multiply numerator and denominator by x: the function is (x^3 + 1)/(x^3 - 1)

Step 2: Quotient rule: [3x^2 (x^3 - 1) - (x^3 + 1)(3x^2)] / (x^3 - 1)^2 =

3x^2 [(x^3 - 1) - (x^3 + 1)] / (x^3 - 1)^2 = -6x^2 / (x^3 - 1)^2

Final answer: (i) -6x^2/(x^3 - 1)^2, which is also -6/(x^2 - x^(-1))^2

(ii) Step 1: Quotient rule: [3 sqrt(x^2 + 1) - (3x - 2) x / sqrt(x^2 + 1)] / (x^2 + 1)

Step 2: Put over (x^2 + 1)^(3/2): numerator = 3(x^2 + 1) - x(3x - 2) = 3x^2 + 3 - 3x^2 + 2x = 2x + 3

Final answer: (ii) (2x + 3)/(x^2 + 1)^(3/2)

Question: 8. If y = x^4 + 2x^2, show that dy/dx = 4x sqrt(y + 1).

Answer: Step 1: Differentiate y: dy/dx = 4x^3 + 4x = 4x(x^2 + 1)

Step 2: Compute y + 1 = x^4 + 2x^2 + 1 = (x^2 + 1)^2

Step 3: So sqrt(y + 1) = x^2 + 1 (taking the positive root)

Step 4: Therefore 4x sqrt(y + 1) = 4x(x^2 + 1), which equals dy/dx from Step 1

Final answer: dy/dx = 4x sqrt(y + 1), shown

Question: 9. If y = sqrt(sin x + sqrt(sin x + sqrt(sin x + ...))), show that (2y - 1) y' = cos x.

Answer: Step 1: The nested part repeats the whole expression, so y = sqrt(sin x + y)

Step 2: Square: y^2 = sin x + y

Step 3: Differentiate: 2y y' = cos x + y'

Step 4: Collect the y' terms: 2y y' - y' = cos x, that is (2y - 1) y' = cos x

Final answer: (2y - 1) y' = cos x, shown

Question: 10. Differentiate with respect to x: (i) cosh(cos^-1 sqrt(x)) (ii) tanh^-1 (cos 2x)

Answer: (i) Step 1: Chain rule: derivative = sinh(cos^-1 sqrt(x)) x d/dx

(cos^-1 sqrt(x))

Step 2: d/dx (cos^-1 sqrt(x)) = -1/sqrt(1 - x) x 1/(2 sqrt(x)) = -1/(2 sqrt(x - x^2))

Final answer: (i) -sinh(cos^-1 sqrt(x)) / (2 sqrt(x - x^2))

(ii) Step 1: d/dx (tanh^-1 u) = u'/(1 - u^2) with u = cos 2x, u' = -2 sin 2x

Step 2: dy/dx = -2 sin 2x / (1 - cos^2 2x) = -2 sin 2x / sin^2 2x = -2 / sin 2x

Final answer: (ii) -2 csc 2x

Unit 4: Higher Order Derivatives and Applications

Exercise 4.1

Question: 1. Calculate the first, second and third order derivatives of y = cos^2 x.

Answer: Step 1: Write y = (cos x)^2 and use the chain rule: y' = 2 cos x (-sin x) = -2 sin x cos x = -sin 2x

Step 2: Differentiate again: y'' = -2 cos 2x (the derivative of sin 2x is 2 cos 2x)

Step 3: Differentiate once more: y''' = -2 (-sin 2x)(2) = 4 sin 2x

Final answer: y' = -sin 2x, y'' = -2 cos 2x, y''' = 4 sin 2x

Question: 2. Find the 2nd order derivative of f(x) = cos x / (1 + sin x).

Answer: Step 1: First derivative by the quotient rule: f'(x) = [-sin x (1 + sin x) - cos x (cos x)] / (1 + sin x)^2

Step 2: The numerator is -sin x - sin^2 x - cos^2 x = -sin x - 1 = -(1 + sin x)

Step 3: Cancel one factor: f'(x) = -1 / (1 + sin x)

Step 4: Differentiate again, treating it as -(1 + sin x)^(-1): f''(x) = (1 + sin x)^(-2) x cos x

Final answer: f''(x) = cos x / (1 + sin x)^2

Question: 3. Find the fourth order derivatives of the given functions: (i) h(t) = 3t^7 - 6t^4 + 8t^3 - 12t + 18 (ii) f(x) = (x)^(1/3) - 1/(8x^2) - sqrt(x)

Answer: (i) Step 1: h'(t) = 21t^6 - 24t^3 + 24t^2 - 12

Step 2: h''(t) = 126t^5 - 72t^2 + 48t

Step 3: h'''(t) = 630t^4 - 144t + 48

Step 4: h''''(t) = 2520t^3 - 144

Final answer: (i) h''''(t) = 2520t^3 - 144

(ii) Step 1: Write f(x) = x^(1/3) - (1/8) x^(-2) - x^(1/2)

Step 2: f'(x) = (1/3) x^(-2/3) + (1/4) x^(-3) - (1/2) x^(-1/2)

Step 3: f''(x) = (-2/9) x^(-5/3) - (3/4) x^(-4) + (1/4) x^(-3/2)

Step 4: f'''(x) = (10/27) x^(-8/3) + 3 x^(-5) - (3/8) x^(-5/2)

Step 5: f''''(x) = (10/27)(-8/3) x^(-11/3) + 3(-5) x^(-6) - (3/8)(-5/2) x^(-7/2) = -(80/81) x^(-11/3) - 15 x^(-6) + (15/16) x^(-7/2)

Final answer: (ii) f''''(x) = -80 / (81 x^(11/3)) - 15/x^6 + 15 / (16 x^(7/2))

Question: 4. Determine the fourth order derivative in each of the following functions: (i) r(t) = 3t^2 + 8 sqrt(t) (ii) y = cos x (iii) f(y) = sin 3y + e^(-2y) + ln(7y)

Answer: (i) Step 1: r'(t) = 6t + 4 t^(-1/2)

Step 2: r''(t) = 6 - 2 t^(-3/2)

Step 3: r'''(t) = 3 t^(-5/2)

Step 4: r''''(t) = 3 x (-5/2) t^(-7/2) = -(15/2) t^(-7/2)

Final answer: (i) r''''(t) = -15 / (2 sqrt(t^7))

(ii) Step 1: The derivatives of cos x cycle: y' = -sin x, y'' = -cos x, y''' = sin x

Step 2: The fourth derivative completes the cycle back to cos x

Final answer: (ii) The fourth derivative is cos x

(iii) Step 1: Differentiate with respect to y four times, term by term

Step 2: sin 3y gives successively 3 cos 3y, -9 sin 3y, -27 cos 3y, 81 sin 3y

Step 3: e^(-2y) gives successively -2e^(-2y), 4e^(-2y), -8e^(-2y), 16e^(-2y)

Step 4: ln(7y) gives successively 1/y, -1/y^2, 2/y^3, -6/y^4

Final answer: (iii) 81 sin 3y + 16 e^(-2y) - 6/y^4

Question: 5. If x^2 + y^2 = 10, find y''.

Answer: Step 1: Differentiate implicitly: 2x + 2y y' = 0, so y' = -x/y

Step 2: Differentiate y' = -x/y by the quotient rule: y'' = -[(1)(y) - x y'] / y^2

Step 3: Substitute y' = -x/y: y'' = -[y - x(-x/y)] / y^2 = -[y + x^2/y] / y^2

Step 4: Put over the common denominator y: y'' = -(y^2 + x^2) / y^3

Step 5: Optionally, using x^2 + y^2 = 10, this is -10/y^3

Final answer: y'' = -(x^2 + y^2)/y^3

Question: 6. Find d^2 y / dx^2 if: (i) 2y^2 + 6x^2 = 76 (ii) x^3 + y^3 = 1

Answer: (i) Step 1: Differentiate: 4y y' + 12x = 0, so y' = -3x/y

Step 2: Differentiate again by the quotient rule: y'' = -3 [y - x y'] / y^2

Step 3: Substitute y' = -3x/y: y'' = -3 [y + 3x^2/y] / y^2 = -3 (y^2 + 3x^2) / y^3

Final answer: (i) d^2 y/dx^2 = -3(3x^2 + y^2)/y^3

(ii) Step 1: Differentiate: 3x^2 + 3y^2 y' = 0, so y' = -x^2/y^2

Step 2: Differentiate again: y'' = -[2x y^2 - x^2 (2y y')] / y^4 = -2x/y^2 + 2x^2 y'/y^3

Step 3: Substitute y' = -x^2/y^2: y'' = -2x/y^2 - 2x^4/y^5 = -2x (y^3 + x^3) / y^5

Step 4: Optionally, using x^3 + y^3 = 1, this is -2x/y^5

Final answer: (ii) d^2 y/dx^2 = -2x(x^3 + y^3)/y^5

Question: 7. Find d^2 y / dx^2 if: (i) x = -5t^3 - 7 and y = 3t^2 + 16 (ii) x = cos theta and y = sin theta

Answer: Step 1: For parametric equations use d^2 y/dx^2 = [d/dt (dy/dx)] / (dx/dt)

Step 2: (i) dx/dt = -15t^2 and dy/dt = 6t, so dy/dx = 6t / (-15t^2) = -2/(5t)

Step 3: Differentiate dy/dx with respect to t: d/dt [-2/(5t)] = 2/(5t^2)

Step 4: Divide by dx/dt = -15t^2: d^2 y/dx^2 = [2/(5t^2)] / (-15t^2) = -2/(75 t^4)

Final answer: (i) d^2 y/dx^2 = -2/(75 t^4)

Step 5: (ii) dx/d theta = -sin theta and dy/d theta = cos theta, so dy/dx = -cot theta

Step 6: Differentiate with respect to theta: d/d theta (-cot theta) = csc^2 theta

Step 7: Divide by dx/d theta = -sin theta: d^2 y/dx^2 = csc^2 theta / (-sin theta) = -csc^3 theta

Final answer: (ii) d^2 y/dx^2 = -csc^3 theta

Question: 8. The derivative of the function r(t) is given by r'(t) = 6t + 4 t^(-1/2) + e^t; find r''(t), r'''(t) and r^(4) (t).

Answer: Step 1: Differentiate r'(t): r''(t) = 6 + 4(-1/2) t^(-3/2) + e^t = 6 - 2 t^(-3/2) + e^t

Step 2: Differentiate again: r'''(t) = -2(-3/2) t^(-5/2) + e^t = 3 t^(-5/2) +

e^t

Step 3: Differentiate once more: r^(4) (t) = 3(-5/2) t^(-7/2) + e^t = -(15/2) t^(-7/2) + e^t

Final answer: r''(t) = 6 - 2 t^(-3/2) + e^t, r'''(t) = 3 t^(-5/2) + e^t, r^(4) (t) = -(15/2) t^(-7/2) + e^t

Question: 9. If x^2 + y^2 = 25, then find d^2 y / dx^2 at the point (4, 3).

Answer: Step 1: From Question 5 of this exercise, for x^2 + y^2 = constant, y'' = -(x^2 + y^2)/y^3

Step 2: Substitute the point x = 4, y = 3 and the constant 25: y'' = -25 / 3^3

Step 3: 3^3 = 27

Final answer: d^2 y/dx^2 at (4, 3) is -25/27

Question: 12. Write the MAPLE command to find higher order derivatives of the following functions: (i) f(x) = x^3 + 3x^2 + 6x + 8 (third order derivative) (ii) f(x) = cos sqrt(2x + 3) (second order derivative) (iii) f(x) = e^(x^2 + 5x + 3) (third order derivative) (iv) f(x) = ln sqrt(3x + 2) (second order derivative) (v) f(x) = e^(sin x) (third order derivative). (Questions 10 and 11 are not printed in the book; the numbering jumps from 9 to 12.)

Answer: Step 1: In Maple the repeated derivative is written either as diff(f(x), x$n) for the nth derivative, or with the operator form d^n/dx^n applied to the function, as shown in the book key

Step 2: (i) Command: diff(x^3 + 3x^2 + 6x + 8, x$3); working: first derivative 3x^2 + 6x + 6, second 6x + 6, third 6

Step 3: (ii) Command: diff(cos(sqrt(2x + 3)), x$2); working: with u = sqrt(2x + 3), the first derivative is -sin u / sqrt(2x + 3), and the second is

-cos u/(2x + 3) + sin u/(2x + 3)^(3/2)

Step 4: (iii) Command: diff(e^(x^2 + 5x + 3), x$3); working: first derivative (2x + 5) e^(x^2 + 5x + 3), second [2 + (2x + 5)^2] e^(x^2 + 5x + 3), third (2x + 5)[(2x + 5)^2 + 6] e^(x^2 + 5x + 3)

Step 5: (iv) Command: diff(ln(sqrt(3x + 2)), x$2); working: the function is (1/2) ln(3x + 2), first derivative 3/[2(3x + 2)], second -9/[2(3x + 2)^2]

Step 6: (v) Command: diff(e^(sin x), x$3); working: first derivative cos x e^(sin x), second (cos^2 x - sin x) e^(sin x), third cos x (cos^2 x - 3 sin x - 1) e^(sin x)

Final answer: (i) diff(x^3 + 3x^2 + 6x + 8, x$3) = 6, (ii) diff(cos(sqrt(2x + 3)), x$2) = -cos(sqrt(2x + 3))/(2x + 3) + sin(sqrt(2x + 3))/(2x + 3)^(3/2), (iii) diff(e^(x^2 + 5x + 3), x$3) = (2x + 5)[(2x + 5)^2 + 6] e^(x^2 + 5x + 3), (iv) diff(ln(sqrt(3x + 2)), x$2) = -9/[2(3x + 2)^2], (v) diff(e^(sin x), x$3) = cos x (cos^2 x - 3 sin x - 1) e^(sin x)

Exercise 4.2

Question: 1. Obtain the first three terms of the Maclaurin series for: (i) cos x (ii) e^x (iii) ln(1 + x) (iv) sin^2 x (v) e^(sin x) (vi) x e^(-x) (vii) 1/(1 + x)

Answer: Step 1: The Maclaurin series is f(x) = f(0) + f'(0) x + f''(0) x^2/2! + f'''(0) x^3/3! + ...; in each part find the derivatives at 0 until three non-zero terms are obtained

Step 2: (i) cos x: f(0) = 1, f''(0) = -1, f^(4) (0) = 1, giving 1 - x^2/2! + x^4/4!

Step 3: (ii) e^x: every derivative at 0 is 1, giving 1 + x + x^2/2!

Step 4: (iii) ln(1 + x): f(0) = 0, f'(0) = 1, f''(0) = -1, f'''(0) = 2, giving x - x^2/2 + x^3/3

Step 5: (iv) sin^2 x = (1 - cos 2x)/2 = (1/2)[(2x)^2/2! - (2x)^4/4! + (2x)^6/6! - ...] = x^2 - x^4/3 + 2x^6/45

Step 6: (v) e^(sin x): put the series sin x = x - x^3/6 + ... into 1 + sin x + (sin x)^2/2 + ...: 1 + (x - x^3/6) + x^2/2 + x^3/6 + ... = 1 + x + x^2/2 (the x^3 terms cancel)

Step 7: (vi) x e^(-x) = x [1 - x + x^2/2 - ...] = x - x^2 + x^3/2

Step 8: (vii) 1/(1 + x) is the geometric series 1 - x + x^2 - ...

Final answer: (i) 1 - x^2/2! + x^4/4!, (ii) 1 + x + x^2/2!, (iii) x - x^2/2 + x^3/3, (iv) x^2 - x^4/3 + 2x^6/45, (v) 1 + x + x^2/2, (vi) x - x^2 + x^3/2, (vii) 1 - x + x^2

Question: 2. Find the first four terms of the Taylor series for the following functions: (i) ln x centred at a = 1 (ii) 1/x centred at a = 1 (iii) sin x centred at a = pi/4 (iv) cos x centred at a = pi/2

Answer: Step 1: The Taylor series is f(x) = f(a) + f'(a)(x - a) + f''(a)(x - a)^2/2! + f'''(a)(x - a)^3/3! + ...

Step 2: (i) f(1) = 0, f'(1) = 1, f''(1) = -1, f'''(1) = 2, f^(4) (1) = -6, giving (x - 1) - (x - 1)^2/2 + (x - 1)^3/3 - (x - 1)^4/4

Step 3: (ii) f(1) = 1, f'(1) = -1, f''(1) = 2, f'''(1) = -6, giving 1 - (x - 1) + (x - 1)^2 - (x - 1)^3

Step 4: (iii) At a = pi/4: sin = cos = 1/sqrt(2), and the derivatives cycle, giving 1/sqrt(2) + (1/sqrt(2))(x - pi/4) - [1/(sqrt(2) 2!)] (x - pi/4)^2 - [1/(sqrt(2) 3!)] (x - pi/4)^3

Step 5: (iv) At a = pi/2: cos = 0, and the non-zero derivatives are f'(pi/2) = -1, f'''(pi/2) = 1, f^(5) (pi/2) = -1, f^(7) (pi/2) = 1, giving -(x - pi/2) + (1/3!)(x - pi/2)^3 - (1/5!)(x - pi/2)^5 + (1/7!)(x - pi/2)^7

Final answer: (i) (x - 1) - (x - 1)^2/2 + (x - 1)^3/3 - (x - 1)^4/4, (ii) 1 - (x -

1) + (x - 1)^2 - (x - 1)^3, (iii) 1/sqrt(2) + (x - pi/4)/sqrt(2) - (x - pi/4)^2/(2 sqrt(2)) - (x - pi/4)^3/(6 sqrt(2)), (iv) -(x - pi/2) + (x - pi/2)^3/3! - (x - pi/2)^5/5! + (x - pi/2)^7/7!

Question: 3. Does the Maclaurin series of the functions f(x) = 1/x, g(x) = cosec x and h(x) = sqrt(x) exist? If not, why? Give appropriate justification.

Answer: A Maclaurin series is built from the value of the function and of all its derivatives at x = 0, so the very first requirement is that the function and every derivative exist and are finite at 0. The function f(x) = 1/x is not defined at 0 at all, and g(x) = cosec x = 1/sin x is also not defined at 0 because sin 0 = 0, so neither of them even has a starting value f(0). The function h(x) = sqrt(x) is defined at 0, but its derivative 1/(2 sqrt(x)) does not exist at 0, and the higher derivatives fail there as well. A Maclaurin expansion therefore cannot be formed for any of the three functions. Final answer: No; the Maclaurin series of f, g and h do not exist, because f and g are not defined at 0 and the derivatives of all orders of f, g and h do not exist at 0.

Question: 4. Write the MAPLE command to find the Taylor expression of the following functions: (i) f(x) = e^x at x = 1 up to 10 terms (ii) f(x) = sin x at x = pi up to 10 terms (iii) f(x) = cos x at x = pi up to 10 terms (iv) f(x) = ln(1 + x) at x = 0 up to 10 terms (v) f(x) = 1/x at x = 1 up to 5 terms (vi) f(x) = 1/x at x = 2 up to 5 terms

Answer: Step 1: The Maple command has the form taylor(function, x = a, n), where a is the centre and n is the number of terms (order) required

Step 2: (i) taylor(e^x, x = 1, 10)

Step 3: (ii) taylor(sin(x), x = pi, 10)

Step 4: (iii) taylor(cos(x), x = pi, 10)

Step 5: (iv) taylor(ln(1 + x), x = 0, 10)

Step 6: (v) taylor(1/x, x = 1, 5)

Step 7: (vi) taylor(1/x, x = 2, 5)

Final answer: (i) taylor(e^x, x = 1, 10), (ii) taylor(sin(x), x = pi, 10), (iii) taylor(cos(x), x = pi, 10), (iv) taylor(ln(1 + x), x = 0, 10), (v) taylor(1/x, x = 1, 5), (vi) taylor(1/x, x = 2, 5)

Exercise 4.3

Question: 1. Determine the slope of the tangent to the curve y = x^3 at the point (3/2, 27/8).

Answer: Step 1: The slope of the tangent is dy/dx = 3x^2

Step 2: Substitute x = 3/2: slope = 3 x (3/2)^2 = 3 x 9/4

Final answer: The slope is 27/4

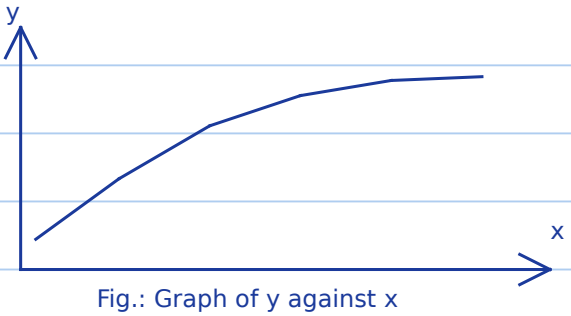


Fig.: Graph of y against x

Question: 2. Find the slope of the tangents to the curve x^2 + y^2 = 25 at the point on it whose x-coordinate is 2.

Answer: Step 1: Put x = 2 in the curve: 4 + y^2 = 25, so y^2 = 21 and y = sqrt(21) or y = -sqrt(21); there are two points, (2, sqrt(21)) and (2, -sqrt(21))

Step 2: Differentiate implicitly: 2x + 2y dy/dx = 0, so dy/dx = -x/y

Step 3: At (2, sqrt(21)): slope = -2/sqrt(21)

Step 4: At (2, -sqrt(21)): slope = 2/sqrt(21)

Final answer: The slopes are 2/sqrt(21) and -2/sqrt(21)

Question: 3. Find the equation of the tangent and the equation of the normal to the curve y = x + 1/x at the point where x = 2.

Answer: Step 1: The point: when x = 2, y = 2 + 1/2 = 5/2, so the point is (2, 5/2)

Step 2: Differentiate: dy/dx = 1 - 1/x^2, which at x = 2 is 1 - 1/4 = 3/4; this is the slope of the tangent

Step 3: Tangent in point-slope form: y - 5/2 = (3/4)(x - 2); multiply by 4: 4y - 10 = 3x - 6, that is 3x - 4y + 4 = 0

Step 4: The slope of the normal is the negative reciprocal, -4/3

Step 5: Normal: y - 5/2 = -(4/3)(x - 2); multiply by 6: 6y - 15 = -8x + 16, that is 8x + 6y - 31 = 0

Final answer: Equation of tangent: 3x - 4y + 4 = 0; equation of normal: 8x + 6y - 31 = 0

Question: 4. Given two curves y = x^2 and y = (x - 3)^2, find the angle between them.

Answer: Step 1: Find the intersection: x^2 = (x - 3)^2 gives x^2 = x^2 - 6x + 9, so 6x = 9 and x = 3/2, with y = 9/4; the point is (3/2, 9/4)

Step 2: Slope of the first curve: dy/dx = 2x, at x = 3/2 giving m1 = 3

Step 3: Slope of the second curve: dy/dx = 2(x - 3), at x = 3/2 giving m2 = 2(-3/2) = -3

Step 4: Use tan theta = |(m1 - m2)/(1 + m1 m2)| = |(3 - (-3))/(1 + (3)(-3))| = |6/(-8)| = 3/4

Step 5: theta = tan^-1 (3/4), which is about 37 degrees

Final answer: The angle between the curves is tan^-1 (3/4), approximately 37 degrees

Question: 5. Prove that the tangent lines to the curve y^2 = 4ax at the points where x = a are at right angles to each other.

Answer: Step 1: At x = a the curve gives y^2 = 4a^2, so y = 2a or y = -2a; the two points are (a, 2a) and (a, -2a)

Step 2: Differentiate y^2 = 4ax implicitly: 2y dy/dx = 4a, so dy/dx = 2a/y

Step 3: At (a, 2a): the slope is 2a/(2a) = 1

Step 4: At (a, -2a): the slope is 2a/(-2a) = -1

Step 5: The product of the slopes is (1)(-1) = -1, which is the condition for two lines to be perpendicular

Final answer: The two tangent lines are at right angles to each other, proved

Question: 6. At what points on the curve x^2 + y^2 - 2x - 4y + 1 = 0 is the tangent parallel to the y-axis?

Answer: Step 1: A tangent parallel to the y-axis is vertical, which happens where dy/dx becomes infinite, that is where the denominator of dy/dx is zero

Step 2: Differentiate implicitly: 2x + 2y dy/dx - 2 - 4 dy/dx = 0, so dy/dx = (2 - 2x)/(2y - 4)

Step 3: The denominator is zero when 2y - 4 = 0, that is y = 2

Step 4: Put y = 2 in the curve: x^2 + 4 - 2x - 8 + 1 = 0, that is x^2 - 2x - 3 = 0

Step 5: Factor: (x - 3)(x + 1) = 0, so x = 3 or x = -1

Final answer: The tangent is parallel to the y-axis at the points (3, 2) and (-1, 2)

Exercise 4.4

Question: 1. Show that the function f(x) = -x^2 + 10x + 9 is increasing at x = 4.

Answer: Step 1: Differentiate: f'(x) = -2x + 10

Step 2: Evaluate at x = 4: f'(4) = -8 + 10 = 2

Step 3: The derivative is positive at x = 4, and a positive derivative at a point means the function is increasing there

Final answer: f'(4) = 2 is positive, so f is increasing at x = 4

Question: 2. Show that f(x) = tan^2 x is decreasing at x = 3pi/4.

Answer: Step 1: Differentiate: f'(x) = 2 tan x sec^2 x

Step 2: At x = 3pi/4: tan(3pi/4) = -1 and sec^2 (3pi/4) = 1 + tan^2 (3pi/4) = 2

Step 3: f'(3pi/4) = 2(-1)(2) = -4

Step 4: The derivative is negative, which means the function is decreasing there

Final answer: f'(3pi/4) = -4 is negative, so f is decreasing at x = 3pi/4

Question: 3. Find the maximum and minimum values, if any, of the function f : R to R in the following cases: (i) f(x) = x^2 - 2x + 3 (ii) f(x) = x^3 - 9x^2 + 15x + 3 (iii) f(x) = -x^4 + 2x^2 (iv) f(x) = e^x sin x (v) f(x) = 2e^x + e^(-x) (vi) f(x) = 2x - x^2

Answer: (i) Step 1: f'(x) = 2x - 2 = 0 gives x = 1

Step 2: f''(x) = 2 is positive, so x = 1 is a minimum

Step 3: Minimum value: f(1) = 1 - 2 + 3 = 2; the parabola opens upwards so there is no maximum

Final answer: (i) Minimum at x = 1, minimum value 2; no maximum

(ii) Step 1: f'(x) = 3x^2 - 18x + 15 = 3(x^2 - 6x + 5) = 3(x - 1)(x - 5), so the critical points are x = 1 and x = 5

Step 2: f''(x) = 6x - 18; f''(1) = -12 is negative, so x = 1 is a relative maximum; f''(5) = 12 is positive, so x = 5 is a relative minimum

Step 3: Values: f(1) = 1 - 9 + 15 + 3 = 10 and f(5) = 125 - 225 + 75 + 3 = -22

Final answer: (ii) Relative maximum at x = 1 with value 10; relative minimum at x = 5 with value -22

(iii) Step 1: f'(x) = -4x^3 + 4x = 4x(1 - x^2), so the critical points are x = 0, 1, -1

Step 2: f''(x) = -12x^2 + 4; f''(0) = 4 is positive, so x = 0 is a relative minimum; f''(1) = f''(-1) = -8 is negative, so x = 1 and x = -1 are relative maxima

Step 3: Values: f(0) = 0 and f(1) = f(-1) = -1 + 2 = 1

Final answer: (iii) Relative minimum at x = 0 with value 0; relative maxima at x = 1 and x = -1 with value 1

(iv) Step 1: f'(x) = e^x (sin x + cos x); since e^x is never zero, set sin x + cos x = 0, that is tan x = -1, giving x = -pi/4 + k pi for whole numbers k

Step 2: f''(x) = 2 e^x cos x; at x = 3pi/4, cos x is negative, so it is a relative maximum; at x = -pi/4, cos x is positive, so it is a relative minimum (and the pattern repeats every 2pi)

Step 3: Values: f(3pi/4) = e^(3pi/4) sin(3pi/4) = e^(3pi/4)/sqrt(2), about 7.46; f(-pi/4) = -e^(-pi/4)/sqrt(2), about -0.32

Step 4: Because the amplitude e^x grows without bound as x increases, there is no overall maximum or minimum on R, only these local ones

Final answer: (iv) Local maximum at x = 3pi/4 (and every 2pi after it) with value e^(3pi/4)/sqrt(2); local minimum at x = -pi/4 (and every 2pi after it) with value -e^(-pi/4)/sqrt(2); no global maximum or minimum

(v) Step 1: f'(x) = 2e^x - e^(-x) = 0 gives 2e^x = e^(-x), that is e^(2x) = 1/2, so x = -(ln 2)/2, about -0.3466

Step 2: f''(x) = 2e^x + e^(-x) is always positive, so this point is a minimum

Step 3: Value: at this point e^x = 1/sqrt(2) and e^(-x) = sqrt(2), so f = 2/sqrt(2) + sqrt(2) = sqrt(2) + sqrt(2) = 2 sqrt(2), about 2.828

Final answer: (v) Relative (and global) minimum at x = -(ln 2)/2, about -0.3466, with value 2 sqrt(2), about 2.828; no maximum

(vi) Step 1: f'(x) = 2 - 2x = 0 gives x = 1

Step 2: f''(x) = -2 is negative, so x = 1 is a maximum

Step 3: Maximum value: f(1) = 2 - 1 = 1; the downward parabola has no minimum

Final answer: (vi) Maximum at x = 1, maximum value 1; no minimum

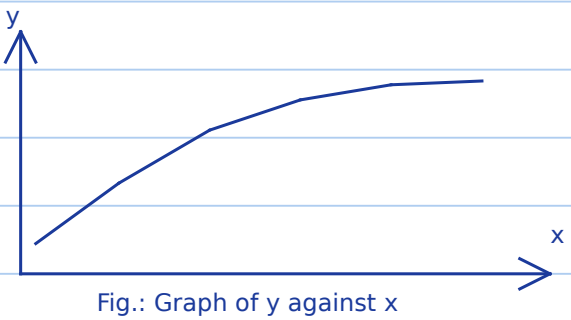


Fig.: Graph of y against x

Question: 4. A rectangular reservoir with a square bottom and open top is to be lined inside with lead. Find the dimensions of the reservoir to hold (1/2) a^3 cubic metres, such that the lead required is minimum.

Answer: Step 1: Let the square bottom have side x metres and the depth be h metres; the volume condition is x^2 h = (1/2) a^3, so h = a^3 /

(2x^2)

Step 2: The lead covers the bottom and the four sides (the top is open), so the area to minimise is S = x^2 + 4xh

Step 3: Substitute h: S = x^2 + 4x x a^3/(2x^2) = x^2 + 2a^3/x

Step 4: Differentiate: dS/dx = 2x - 2a^3/x^2; set it to zero: 2x = 2a^3/x^2, so x^3 = a^3 and x = a

Step 5: The second derivative d^2 S/dx^2 = 2 + 4a^3/x^3 is positive at x = a, confirming a minimum

Step 6: The depth is h = a^3 / (2a^2) = a/2

Final answer: The reservoir should have length a, breadth a and height a/2

Question: 5. Find a right-angled triangle of maximum area with a hypotenuse of length h.

Answer: Step 1: Let the two sides containing the right angle be p and q; by Pythagoras, p^2 + q^2 = h^2

Step 2: The area is A = (1/2) p q, so maximising A is the same as maximising p q, or its square p^2 q^2

Step 3: Write q^2 = h^2 - p^2, so p^2 q^2 = p^2 (h^2 - p^2) = h^2 p^2 - p^4

Step 4: Differentiate with respect to p^2 (or with respect to p): the maximum occurs when h^2 - 2p^2 = 0, that is p^2 = h^2/2, so p = h/sqrt(2)

Step 5: Then q^2 = h^2 - h^2/2 = h^2/2, so q = h/sqrt(2) as well; the triangle is isosceles

Step 6: Maximum area: A = (1/2)(h/sqrt(2))(h/sqrt(2)) = (1/2)(h^2/2) = h^2/4

Final answer: The maximum-area right-angled triangle is the isosceles one with both legs equal to h/sqrt(2), and its maximum area is h^2/4

Question: 6. A particle moves so that its distance s at time t is given by s = ut + (1/2) a t^2, where u and a are fixed real numbers. Find its speed and the magnitude of its acceleration at time t.

Answer: Step 1: Speed is the first derivative of distance with respect to time: v = ds/dt

Step 2: Differentiate: ds/dt = u + (1/2) a (2t) = u + at

Step 3: Acceleration is the derivative of speed: d^2 s/dt^2 = a

Step 4: Its magnitude is the absolute value of a, written |a| (equal to a when a is taken as positive)

Final answer: Speed = u + at; magnitude of acceleration = |a|

Review Exercise 4

Question: 1. Choose the correct answer: (i) If y = 1/x, then y'''(1) = : (a) 6 (b) -6 (c) 2 (d) None of these (ii) What will be the nth derivative of 2e^x: (a) 2n e^x (b) 2 e^(nx) (c) 2e^x (d) e^x / 2 (iii) If xy = k^2, then y'' = : (a) 2k^2/x (b) -2k^2/x^3 (c) 3k^2/x^3 (d) 2k^2/x^3 (iv) The 3rd order derivative of 2^x is: (a) (ln 3)^2 e^(x ln 2) (b) (ln 2)^3 e^(x ln 3) (c) (ln 2)^3 e^(x ln 2) (d) None of these (v) The 2nd order derivative of f(x) = A sin x + B cos x is: (a) f(x) (b) -f(x) (c) plus or minus f(x) (d) None of these (vi) If S is the distance covered by a car, then d^2 S/dt^2 will be its: (a) Velocity (b) deceleration (c) acceleration (d) Average velocity (vii) y = tan^-1 x, then y''(1) = : (a) -1/2 (b) 1/4 (c) -1/4 (d) 1/2 (viii) If y = x - x^3/3! + x^5/5! - ... for x in R, then y is the Maclaurin series of: (a) sin x (b) cos x (c) e^x (d) None of these (ix) A function f(x) is said to be a decreasing function when x1 is less than x2 and: (a) f(x1) is less than f(x2)

(b) f(x1) is greater than f(x2) (c) f(x1) is at least f(x2) (d) f(x1) is at most f(x2) (x) If a function f(x) is such that f'(c) = 0 then the point (c, f(c)) is called: (a) Maximum point (b) Minimum point (c) Stationary point (d) Critical point

Answer: (i) Step 1: y = x^(-1), y' = -x^(-2), y'' = 2x^(-3), y''' = -6x^(-4); at x = 1 this is -6

Final answer: (i) (b) -6

(ii) Step 1: Every derivative of e^x is e^x, so every derivative of 2e^x is 2e^x, including the nth

Final answer: (ii) (c) 2e^x

(iii) Step 1: y = k^2/x, y' = -k^2/x^2, y'' = 2k^2/x^3

Final answer: (iii) (d) 2k^2/x^3

(iv) Step 1: Each derivative of 2^x brings down a factor ln 2, so the third derivative is (ln 2)^3 2^x = (ln 2)^3 e^(x ln 2)

Final answer: (iv) (c) (ln 2)^3 e^(x ln 2)

(v) Step 1: f'(x) = A cos x - B sin x and f''(x) = -A sin x - B cos x = -f(x)

Final answer: (v) (b) -f(x)

(vi) Step 1: The first derivative of distance is velocity and the second derivative is acceleration

Final answer: (vi) (c) acceleration

(vii) Step 1: y' = 1/(1 + x^2), y'' = -2x/(1 + x^2)^2; at x = 1: -2/4 = -1/2

Final answer: (vii) (a) -1/2

(viii) Step 1: The series x - x^3/3! + x^5/5! - ... is the standard Maclaurin series of sin x

Final answer: (viii) (a) sin x

(ix) Step 1: A decreasing function gives a larger output for a smaller input, so x1 less than x2 implies f(x1) greater than f(x2)

Final answer: (ix) (b) f(x1) is greater than f(x2)

(x) Step 1: A point where the derivative is zero is called a stationary point (the book key)

Final answer: (x) (c) Stationary point

Question: 2. Find the 2nd order derivative of f(x) = ln(1 + x^2).

Answer: Step 1: First derivative: f'(x) = 2x/(1 + x^2)

Step 2: Differentiate by the quotient rule: f''(x) = [2(1 + x^2) - 2x(2x)] / (1 + x^2)^2

Step 3: Simplify the numerator: 2 + 2x^2 - 4x^2 = 2 - 2x^2 = 2(1 - x^2)

Final answer: f''(x) = 2(1 - x^2)/(1 + x^2)^2

Question: 3. Find the second derivative of the following parametric functions: x = 3u^2 + 1 and y = 3u^2 + 5u; find d^2 y/dx^2.

Answer: Step 1: dx/du = 6u and dy/du = 6u + 5, so dy/dx = (6u + 5)/(6u) = 1 + 5/(6u)

Step 2: Differentiate dy/dx with respect to u: d/du [1 + (5/6) u^(-1)] = -(5/6) u^(-2) = -5/(6u^2)

Step 3: Divide by dx/du = 6u: d^2 y/dx^2 = [-5/(6u^2)] / (6u)

Final answer: d^2 y/dx^2 = -5/(36 u^3)

Question: 4. Evaluate the third derivatives of the given functions: (i) v(x) = x^3 - x^2 + x - 1 (ii) f(x) = 7 sin(x/3) + cos(1 - 2x) (iii) y = e^(-5x) + 8 ln(2x^4)

Answer: (i) Step 1: v'(x) = 3x^2 - 2x + 1

Step 2: v''(x) = 6x - 2

Step 3: v'''(x) = 6

Final answer: (i) 6

(ii) Step 1: f'(x) = (7/3) cos(x/3) + 2 sin(1 - 2x)

Step 2: f''(x) = -(7/9) sin(x/3) - 4 cos(1 - 2x)

Step 3: f'''(x) = -(7/27) cos(x/3) - 8 sin(1 - 2x)

Final answer: (ii) -(7/27) cos(x/3) - 8 sin(1 - 2x)

(iii) Step 1: Simplify the logarithm: ln(2x^4) = ln 2 + 4 ln x, so y' = -5e^(-5x) + 32/x... more precisely the derivative of 8 ln(2x^4) is 8 x (4/x) = 32/x

Step 2: y'' = 25 e^(-5x) - 32/x^2

Step 3: y''' = -125 e^(-5x) + 64/x^3

Final answer: (iii) -125 e^(-5x) + 64 x^(-3)

Question: 5. Determine the second derivative of the given functions: (i) g(x) = sin(2x^3 - 9x) (ii) z(x) = ln(7 - x^3) (iii) q(x) = 2 / (6 + 2x - x^2)^4 (iv) h(t) = cos^2 (7t) (v) 2x^3 + y^2 = 1 - 4y (vi) 6y - x y^2 = 1

Answer: (i) Step 1: g'(x) = (6x^2 - 9) cos(2x^3 - 9x)

Step 2: Differentiate by the product rule: g''(x) = 12x cos(2x^3 - 9x) - (6x^2 - 9)^2 sin(2x^3 - 9x)

Final answer: (i) g''(x) = 12x cos(2x^3 - 9x) - (6x^2 - 9)^2 sin(2x^3 - 9x)

(ii) Step 1: z'(x) = -3x^2/(7 - x^3)

Step 2: Quotient rule: z''(x) = [-6x(7 - x^3) - (-3x^2)(-3x^2)] / (7 - x^3)^2 = [-42x + 6x^4 - 9x^4] / (7 - x^3)^2

Final answer: (ii) z''(x) = (-42x - 3x^4)/(7 - x^3)^2

(iii) Step 1: Write q(x) = 2 (6 + 2x - x^2)^(-4); q'(x) = 2(-4)(6 + 2x - x^2)^(-5) (2 - 2x) = -8(2 - 2x)(6 + 2x - x^2)^(-5)

Step 2: Differentiate again: q''(x) = -8[-2 (6 + 2x - x^2)^(-5) + (2 - 2x)(-5)(6 + 2x - x^2)^(-6) (2 - 2x)]

Step 3: Simplify: q''(x) = 16 (6 + 2x - x^2)^(-5) + 40 (2 - 2x)^2 (6 + 2x - x^2)^(-6)

Final answer: (iii) q''(x) = 16(6 + 2x - x^2)^(-5) + 40(2 - 2x)^2 (6 + 2x - x^2)^(-6)

(iv) Step 1: h'(t) = 2 cos(7t) (-sin(7t)) (7) = -7 sin(14t), using 2 sin(7t) cos(7t) = sin(14t)

Step 2: h''(t) = -7 cos(14t) (14) = -98 cos(14t)

Step 3: Using cos(14t) = cos^2 (7t) - sin^2 (7t), this is also 98 sin^2 (7t) - 98 cos^2 (7t)

Final answer: (iv) h''(t) = 98 sin^2 (7t) - 98 cos^2 (7t)

(v) Step 1: Differentiate implicitly: 6x^2 + 2y y' = -4y', so y' = -6x^2/(2y + 4) = -3x^2/(y + 2)

Step 2: Differentiate again by the quotient rule: y'' = -[6x (y + 2) - 3x^2 y'] / (y + 2)^2 = -6x/(y + 2) + 3x^2 y'/(y + 2)^2

Step 3: Substitute y' = -3x^2/(y + 2): y'' = -6x (y + 2)^(-1) - 9x^4 (y + 2)^(-3)

Final answer: (v) d^2 y/dx^2 = -6x (y + 2)^(-1) - 9x^4 (y + 2)^(-3)

(vi) Step 1: Differentiate implicitly: 6y' - y^2 - 2xy y' = 0, so y' = y^2/(6 - 2xy)

Step 2: Differentiate the quotient: y'' = [2y y' (6 - 2xy) - y^2 (-2y - 2x y')] /

(6 - 2xy)^2

Step 3: Substitute y' = y^2/(6 - 2xy) and simplify: y'' = 2 y^3 (6 - 2xy)^(-2) [2 + xy (6 - 2xy)^(-1)]

Final answer: (vi) y'' = 2y^3 (6 - 2xy)^(-2) [2 + xy/(6 - 2xy)]

Question: 6. Given that y = cos x, find y', y'', y''' and y^(4).

Answer: Step 1: y' = -sin x

Step 2: y'' = -cos x

Step 3: y''' = sin x

Step 4: y^(4) = cos x, completing the cycle of four

Final answer: y' = -sin x, y'' = -cos x, y''' = sin x, y^(4) = cos x

Question: 7. Find d^2 y/dx^2 if x = 4 sin t and y = 5 cos t.

Answer: Step 1: dx/dt = 4 cos t and dy/dt = -5 sin t, so dy/dx = -5 sin t/(4 cos t) = -(5/4) tan t

Step 2: Differentiate dy/dx with respect to t: d/dt [-(5/4) tan t] = -(5/4) sec^2 t

Step 3: Divide by dx/dt = 4 cos t: d^2 y/dx^2 = -(5/4) sec^2 t / (4 cos t) = -5 / (16 cos^3 t), using sec^2 t = 1/cos^2 t

Final answer: d^2 y/dx^2 = -5/(16 cos^3 t)

Question: 8. Find the rth derivative of f(x) = x^n, where r is at most n.

Answer: Step 1: Successive derivatives give f'(x) = n x^(n - 1), f''(x) = n(n - 1) x^(n - 2), f'''(x) = n(n - 1)(n - 2) x^(n - 3)

Step 2: After r differentiations the coefficient is the falling product n(n - 1)...(n - r + 1), which equals n! / (n - r)!

Step 3: The power of x is reduced r times, to x^(n - r)

Final answer: The rth derivative is [n! / (n - r)!] x^(n - r)

Question: 9. Find the Taylor series of the function x^4 + x - 2 centred at a = 1.

Answer: Step 1: Compute the derivatives at a = 1: f(1) = 1 + 1 - 2 = 0

Step 2: f'(x) = 4x^3 + 1, so f'(1) = 5

Step 3: f''(x) = 12x, so f''(1) = 12, and 12/2! = 6

Step 4: f'''(x) = 24, so f'''(1) = 24, and 24/3! = 4

Step 5: f^(4) (x) = 24, so f^(4) (1)/4! = 1, and all higher derivatives are zero

Step 6: Assemble the series

Final answer: x^4 + x - 2 = 5(x - 1) + 6(x - 1)^2 + 4(x - 1)^3 + (x - 1)^4

Question: 10. Obtain the Taylor series for the function (x - 1) e^x near x = 1.

Answer: Step 1: Put u = x - 1, so x = u + 1 and the function is u e^(u + 1) = e u e^u

Step 2: Use the Maclaurin series e^u = 1 + u + u^2/2 + u^3/6 + ...

Step 3: Multiply by e u: e [u + u^2 + u^3/2 + u^4/6 + ...]

Step 4: Replace u by x - 1

Final answer: (x - 1) e^x = e (x - 1) + e (x - 1)^2 + (e/2) (x - 1)^3 + (e/6) (x - 1)^4 + ...

Question: 11. Find the Maclaurin series for ln(1 + x) and hence find the series for ln((1 + x)/(1 - x)).

Answer: Step 1: For f(x) = ln(1 + x): f(0) = 0, f'(0) = 1, f''(0) = -1, f'''(0) = 2, f^(4) (0) = -6, giving ln(1 + x) = x - x^2/2 + x^3/3 - x^4/4 + ...

Step 2: Replace x by -x: ln(1 - x) = -x - x^2/2 - x^3/3 - x^4/4 - ...

Step 3: Subtract: ln((1 + x)/(1 - x)) = ln(1 + x) - ln(1 - x); the even powers

cancel and the odd powers double

Step 4: The result is 2 [x + x^3/3 + x^5/5 + x^7/7 + ...], valid for x between -1 and 1

Final answer: ln(1 + x) = x - x^2/2 + x^3/3 - x^4/4 + ... and ln((1 + x)/(1 - x)) = 2 (x + x^3/3 + x^5/5 + ...)

Question: 12. Find the equation of the tangent line to the curve y = x^3 - 3x^2 + x at the point (2, -2).

Answer: Step 1: Check the point: at x = 2, y = 8 - 12 + 2 = -2, so (2, -2) is on the curve

Step 2: Differentiate: dy/dx = 3x^2 - 6x + 1, which at x = 2 is 12 - 12 + 1 = 1

Step 3: Point-slope form: y - (-2) = 1 (x - 2), that is y + 2 = x - 2

Final answer: The tangent line is y = x - 4

Question: 13. At what point on the graph of y = x^2 is the tangent line parallel to the line 3x - y = 2?

Answer: Step 1: The given line has slope 3 (writing it as y = 3x - 2)

Step 2: The slope of the tangent to y = x^2 is dy/dx = 2x

Step 3: Parallel lines have equal slopes: 2x = 3, so x = 3/2

Step 4: The corresponding point has y = (3/2)^2 = 9/4

Final answer: The point is (3/2, 9/4)

Question: 14. Determine the interval on which the function f(x) = x^2 - 3x + 1 is increasing and decreasing.

Answer: Step 1: Differentiate: f'(x) = 2x - 3

Step 2: f'(x) = 0 at x = 3/2

Step 3: For x greater than 3/2, f'(x) is positive, so f is increasing on (3/2,

infinity)

Step 4: For x less than 3/2, f'(x) is negative, so f is decreasing on (-infinity, 3/2)

Final answer: Increasing on (3/2, infinity) and decreasing on (-infinity, 3/2)

Unit 5: Differentiation of Vector Functions

Exercise 5.1

Question: 1. Find the domain of the following vector functions. (i) r(t) = 2t i - 3t j + (1/t) k (ii) r(t) = sin t i + cos t j + tan t k (iii) r(t) = (1 - t) i + sqrt(t) j + (1/t^2) k (iv) g(t) = cos t i - cot t j + cosec t k

Answer: Step 1: The domain of a vector function is the intersection of the domains of all its component scalar functions, because the vector is defined only when every component is defined.

Step 2: (i) The components 2t and -3t are polynomials and are defined for all real t, while 1/t is defined only when t is not equal to 0, so the domain is all real numbers except 0.

Step 3: (ii) sin t and cos t are defined for all real t, while tan t is undefined when cos t = 0, that is at t = (2n + 1) pi/2 for any integer n, so the domain is all real numbers except t = (2n + 1) pi/2.

Step 4: (iii) The component sqrt(t) needs t greater than or equal to 0 and the component 1/t^2 needs t not equal to 0, and combining both conditions gives t greater than 0.

Step 5: (iv) cot t and cosec t are both undefined when sin t = 0, that is at t = n pi for any integer n, while cos t is defined everywhere, so the domain is all real numbers except t = n pi.

Final answer: (i) All real t with t not equal to 0 (ii) All real t with t not equal to (2n + 1) pi/2 (iii) t greater than 0 (iv) All real t with t not equal to n pi.

Question: 2. Find the limit of vector function r(t) = (e^(3t) - 1) i + (sqrt(3 + t) - sqrt(3))/(3t) j + (1/(9t + 1)) k at t = 0.

Answer: Step 1: The limit of a vector function is found component by component, so we find the limit of each scalar component separately at t = 0.

Step 2: For the i component, lim (t to 0) (e^(3t) - 1) = e^0 - 1 = 1 - 1 = 0.

Step 3: For the j component, rationalise the numerator by multiplying top and bottom by (sqrt(3 + t) + sqrt(3)), giving ((3 + t) - 3) / (3t (sqrt(3 + t) + sqrt(3))) = t / (3t (sqrt(3 + t) + sqrt(3))) = 1 / (3 (sqrt(3 + t) + sqrt(3))).

Step 4: Put t = 0 in the simplified j component, giving 1 / (3 (sqrt(3) + sqrt(3))) = 1 / (6 sqrt(3)).

Step 5: For the k component, lim (t to 0) 1/(9t + 1) = 1/(0 + 1) = 1.

Step 6: Collect the three component limits with i, j and k respectively.

Final answer: lim (t to 0) r(t) = (1/(6 sqrt(3))) j + k.

Question: 3. If u = t^2 i - 2 j and v = 2t i - 5 k are vector functions and h = 3t is scalar function, then find the following: (i) lim (t to 0) [u(t) + v(t)] (ii) lim (t to 1) [u(t) dot v(t)] (iii) lim (t to 1) [u(t) cross v(t)] (iv) lim (t to 5) [h u(t)]

Answer: Step 1: First write the combinations as functions of t, and only then take the stated limit in each part.

Step 2: (i) u + v = (t^2 + 2t) i - 2 j - 5 k, and putting t = 0 gives 0 i - 2 j - 5 k = -2 j - 5 k.

Step 3: (ii) u dot v = (t^2)(2t) + (-2)(0) + (0)(-5) = 2t^3, and putting t = 1 gives 2.

Step 4: (iii) u cross v is the determinant with rows i, j, k, then t^2, -2, 0, then 2t, 0, -5, which expands to i((-2)(-5) - 0) - j((t^2)(-5) - 0) + k(0 - (-4t)) = 10 i + 5t^2 j + 4t k, and putting t = 1 gives 10 i + 5 j + 4 k.

Step 5: (iv) h u = 3t (t^2 i - 2 j) = 3t^3 i - 6t j, and putting t = 5 gives 3(125) i - 30 j = 375 i - 30 j.

Final answer: (i) -2 j - 5 k (ii) 2 (iii) 10 i + 5 j + 4 k (iv) 375 i - 30 j.

Question: 4. Show that function R(t) = sin^2 t i + tan t j + (1/t) k is continuous at t = pi/4.

Answer: Step 1: A vector function is continuous at a point when the limit at that point exists and is equal to the value of the function at that point, and this holds exactly when every component is continuous there.

Step 2: Find the value at t = pi/4: sin^2 (pi/4) = 1/2, tan (pi/4) = 1 and 1/(pi/4) = 4/pi, so R(pi/4) = (1/2) i + j + (4/pi) k.

Step 3: Find the limit as t approaches pi/4 component by component: lim sin^2 t = sin^2 (pi/4) = 1/2, lim tan t = tan (pi/4) = 1 and lim 1/t = 4/pi, because each component is an elementary continuous function at pi/4 and pi/4 is not a zero of t or of cos t.

Step 4: The limit vector is therefore (1/2) i + j + (4/pi) k.

Step 5: Compare the limit with the value: lim (t to pi/4) R(t) = R(pi/4).

Final answer: Hence R(t) is continuous at t = pi/4, as required.

Question: 5. Show that function r(t) = (2/t) i + (t^3/(2t^3 - 5)) j + (1/e^t) k is continuous at t tends to infinity.

Answer: Step 1: Continuity at infinity is checked by finding the finite limit of each component as t increases without bound and showing that the components approach those values smoothly.

Step 2: For the i component, lim (t to infinity) 2/t = 0, because the numerator is constant while the denominator increases without bound.

Step 3: For the j component, divide numerator and denominator by t^3, giving 1 / (2 - 5/t^3), and as t tends to infinity 5/t^3 tends to 0, so the

limit is 1/2.

Step 4: For the k component, lim (t to infinity) 1/e^t = 0, because e^t increases without bound.

Step 5: All three component limits are finite, so the vector function approaches the single limiting vector (1/2) j as t tends to infinity.

Final answer: Hence r(t) is continuous at t tends to infinity, with limiting value (1/2) j.

Question: 6. For what value of t, following vector functions are continuous (i) r(t) = ln(t + 3) i + (1/(t - 1)) j + ((t + 2)/(t^2 - 4)) k (ii) r(t) = (1/(3t + 1)) i + (1/t) j

Answer: Step 1: A vector function is continuous at every point of its domain where all components are elementary and defined, so we exclude every t that makes a component undefined.

Step 2: (i) ln(t + 3) needs t + 3 greater than 0, that is t greater than -3.

Step 3: (i) The component 1/(t - 1) excludes t = 1, and the component (t + 2)/(t^2 - 4) excludes t = 2 and t = -2 because t^2 - 4 = (t - 2)(t + 2) is zero there.

Step 4: (i) Combining, the function is continuous for all real t with t greater than -3 and t not equal to 1, t not equal to 2 and t not equal to -2.

Step 5: (ii) The component 1/(3t + 1) excludes t = -1/3 and the component 1/t excludes t = 0, and there are no other restrictions.

Final answer: (i) All real t with t greater than -3 and t not equal to -2, 1, 2 (ii) All real t with t not equal to 0 and t not equal to -1/3.

Exercise 5.2

Question: 1. Find the derivative of the following vector functions. (i) f(t) =

ln t^2 i + e^(2t) j + (2t^2 + 1) k (ii) f(t) = (t + 1) i + ln(t + 2) j (iii) f(t) = sec t i + cos t^2 j + (t^2 + t + 1) k

Answer: Step 1: The derivative of a vector function is found by differentiating each component separately with respect to t.

Step 2: (i) d/dt (ln t^2) = (1/t^2)(2t) = 2/t by the chain rule, d/dt (e^(2t)) = 2 e^(2t), and d/dt (2t^2 + 1) = 4t.

Step 3: (ii) d/dt (t + 1) = 1 and d/dt (ln(t + 2)) = 1/(t + 2).

Step 4: (iii) d/dt (sec t) = sec t tan t, d/dt (cos t^2) = -sin t^2 times d/dt (t^2) = -2t sin t^2 by the chain rule, and d/dt (t^2 + t + 1) = 2t + 1.

Step 5: Collect each set of differentiated components with i, j and k respectively.

Final answer: (i) f prime (t) = (2/t) i + 2 e^(2t) j + 4t k (ii) f prime (t) = i + (1/(t + 2)) j (iii) f prime (t) = (sec t tan t) i - 2t sin t^2 j + (2t + 1) k.

Question: 2. If x = t i + 2t j and y = 2t i + 3t k are vector functions and phi(t) = 3t is scalar function, then find the following: (i) d/dt [x(t) - y(t)] (ii) d/dt [x(t) dot y(t)] (iii) d/dt [x(t) cross y(t)] (iv) d/dt [phi x(t)]

Answer: Step 1: Form each combination first as a single function of t, and then differentiate it.

Step 2: (i) x - y = (t - 2t) i + 2t j - 3t k = -t i + 2t j - 3t k, and differentiating gives -i + 2 j - 3 k.

Step 3: (ii) x dot y = (t)(2t) + (2t)(0) + (0)(3t) = 2t^2, and differentiating gives 4t.

Step 4: (iii) x cross y is the determinant with rows i, j, k, then t, 2t, 0, then 2t, 0, 3t, which expands to i((2t)(3t) - 0) - j((t)(3t) - 0) + k(0 - 4t^2) = 6t^2 i - 3t^2 j - 4t^2 k, and differentiating gives 12t i - 6t j - 8t k.

Step 5: (iv) phi x = 3t (t i + 2t j) = 3t^2 i + 6t^2 j, and differentiating

gives 6t i + 12t j.

Final answer: (i) -i + 2 j - 3 k (ii) 4t (iii) 12t i - 6t j - 8t k (iv) 6t i + 12t j.

Question: 3. A particle moves so that its position as a function of time is given by r(t) = i + 4t^2 j + t k. Write expressions for its: (a) velocity (b) acceleration as functions of time.

Answer: Step 1: Velocity is the first derivative of position with respect to time, v = dr/dt, and acceleration is the derivative of velocity, a = dv/dt.

Step 2: Differentiate r component by component: d/dt (1) = 0, d/dt (4t^2) = 8t and d/dt (t) = 1, so v = 8t j + k.

Step 3: Differentiate v component by component: d/dt (8t) = 8 and d/dt (1) = 0, so a = 8 j.

Step 4: Note that the acceleration is constant, because the only changing component of velocity is 8t, whose rate of change is the constant 8.

Final answer: (a) Velocity v(t) = 8t j + k (b) Acceleration a(t) = 8 j.

Question: 4. The path of a particle is given for time t greater than 0 by the parametric equations x = t^2 - 3t and y = (2/3) t^3. Find magnitude of velocity and acceleration of particle at t = 5.

Answer: Step 1: Write the position vector as r = x i + y j = (t^2 - 3t) i + (2/3) t^3 j.

Step 2: Differentiate for velocity: dx/dt = 2t - 3 and dy/dt = (2/3)(3t^2) = 2t^2, so v = (2t - 3) i + 2t^2 j.

Step 3: Differentiate again for acceleration: d/dt (2t - 3) = 2 and d/dt (2t^2) = 4t, so a = 2 i + 4t j.

Step 4: Put t = 5: v = (10 - 3) i + 2(25) j = 7 i + 50 j, and a = 2 i + 20 j.

Step 5: Find magnitudes: magnitude of v = sqrt(7^2 + 50^2) = sqrt(49 + 2500) = sqrt(2549), and magnitude of a = sqrt(2^2 + 20^2) = sqrt(4 +

400) = sqrt(404).

Final answer: At t = 5, magnitude of velocity = sqrt(2549) units and magnitude of acceleration = sqrt(404) units.

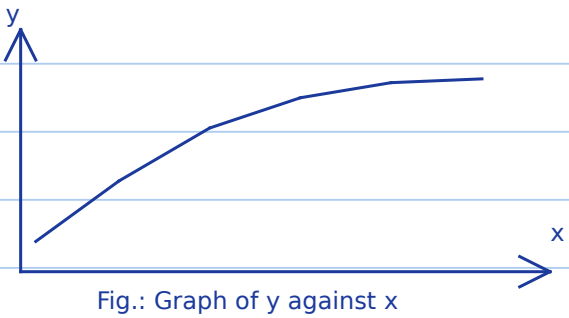


Fig.: Graph of y against x

Unit 6: Integration

Exercise 6.1

Question: 1. Evaluate the following indefinite integrals by using standard formulae. (i) integral of 9x^5 dx (ii) integral of 15/x^3 dx (iii) integral of a/sqrt(bx) dx (iv) integral of b y^(2/3) dy (v) integral of (3x^2 - 9x + 5) dx (vi) integral of (2x^(-5) + 3x^(-2)) dx (vii) integral of (x^5 + 3x^3 - 5x + 6)/x^4 dx (viii) integral of (cos x + 3 sin x) dx (ix) integral of (3 sec x - cosec x) dx (x) integral of (2 tan x - 5 sec x) dx (xi) integral of (9 e^x - 3 cos x - 5 sin x) dx (xii) integral of (sec^2 x + cosec^2 x) dx

Answer: Step 1: Use the power rule integral of x^n dx = x^(n + 1)/(n + 1) + C and the standard trigonometric and exponential integrals term by term.

Step 2: (i) integral of 9x^5 dx = 9 x^6/6 + C = (3/2) x^6 + C.

Step 3: (ii) Write 15/x^3 = 15 x^(-3), so the integral is 15 x^(-2)/(-2) + C = -15/(2x^2) + C.

Step 4: (iii) Write a/sqrt(bx) = (a/sqrt(b)) x^(-1/2), so the integral is (a/sqrt(b)) (x^(1/2)/(1/2)) + C = 2a sqrt(x/b) + C.

Step 5: (iv) integral of b y^(2/3) dy = b y^(5/3)/(5/3) + C = (3/5) b y^(5/3) + C.

Step 6: (v) Integrate term by term: 3 x^3/3 - 9 x^2/2 + 5x + C = x^3 - (9/2) x^2 + 5x + C.

Step 7: (vi) 2 x^(-4)/(-4) + 3 x^(-1)/(-1) + C = -1/(2x^4) - 3/x + C.

Step 8: (vii) Divide first: the integrand is x + 3/x - 5x^(-3) + 6x^(-4), and integrating gives x^2/2 + 3 ln x + 5/(2x^2) - 2/x^3 + C.

Step 9: (viii) integral of cos x dx + 3 integral of sin x dx = sin x - 3 cos x + C.

Step 10: (ix) 3 ln (sec x + tan x) - ln (cosec x - cot x) + C, which combines as ln ((sec x + tan x)^3/(cosec x - cot x)) + C.

Step 11: (x) 2 integral of tan x dx - 5 integral of sec x dx = ln (sec^2 x) - 5 ln (sec x + tan x) + C, which combines as ln (sec^2 x/(sec x + tan x)^5) + C.

Step 12: (xi) 9 e^x - 3 sin x + 5 cos x + C, because integral of -3 cos x dx is -3 sin x and integral of -5 sin x dx is 5 cos x.

Step 13: (xii) tan x - cot x + C, because integral of sec^2 x dx is tan x and integral of cosec^2 x dx is -cot x.

Final answer: (i) (3/2) x^6 + C (ii) -15/(2x^2) + C (iii) 2a sqrt(x/b) + C (iv) (3/5) b y^(5/3) + C (v) x^3 - (9/2) x^2 + 5x + C (vi) -1/(2x^4) - 3/x + C (vii) x^2/2 + 3 ln x + 5/(2x^2) - 2/x^3 + C (viii) sin x - 3 cos x + C (ix) ln ((sec x + tan x)^3/(cosec x - cot x)) + C (x) ln (sec^2 x/(sec x + tan x)^5) + C (xi) 9 e^x - 3 sin x + 5 cos x + C (xii) tan x - cot x + C.

Question: 2. Evaluate the following indefinite integrals by using standard formulae. (i) integral of (3x^2 + 9x + 3)^(1/2) (6x + 9) dx (ii) integral of sqrt(a x^2 + 2b x + c) (a x + b) dx (iii) integral of (6x + 5) dx / sqrt(3x^2 + 5x + 2) (iv) integral of (x^2 + 4x + 3)^(-9) (2x + 4) dx (v) integral of (x - 2)(x - 3)(x - 4) dx (vi) integral of (2x^2 - 3)^2 dx (vii) integral of (x^3 - 3x^2 + 9)^(7/2) (x^2 - 2x) dx (viii) integral of (x^2 - 5)^3 dx (ix) integral of (cos x + sin x)^(3/2) (cos x - sin x) dx (x) integral of (tan x + sin x) (sec^2 x + cos x) dx

Answer: Step 1: In parts (i) to (iv), (vii), (ix) and (x) the second factor is, up to a constant, the derivative of the inner function, so use integral of (f(x))^n f prime (x) dx = (f(x))^(n + 1)/(n + 1) + C.

Step 2: (i) Put u = 3x^2 + 9x + 3, then du = (6x + 9) dx exactly, so the integral is u^(3/2)/(3/2) + C = (2/3) (3x^2 + 9x + 3)^(3/2) + C.

Step 3: (ii) Put u = a x^2 + 2b x + c, then du = 2(a x + b) dx, so (a x + b) dx = du/2 and the integral is (1/2) u^(3/2)/(3/2) + C = (1/3) (a x^2 + 2b x + c)^(3/2) + C.

Step 4: (iii) Put u = 3x^2 + 5x + 2, then du = (6x + 5) dx exactly, so the integral is u^(1/2)/(1/2) + C = 2 sqrt(3x^2 + 5x + 2) + C.

Step 5: (iv) Put u = x^2 + 4x + 3, then du = (2x + 4) dx exactly, so the integral is u^(-8)/(-8) + C = -1/(8 (x^2 + 4x + 3)^8) + C.

Step 6: (v) Expand first: (x - 2)(x - 3) = x^2 - 5x + 6, and (x^2 - 5x + 6)(x - 4) = x^3 - 9x^2 + 26x - 24, and integrating term by term gives x^4/4 - 3x^3 + 13x^2 - 24x + C.

Step 7: (vi) Expand: (2x^2 - 3)^2 = 4x^4 - 12x^2 + 9, and integrating gives (4/5) x^5 - 4x^3 + 9x + C.

Step 8: (vii) Put u = x^3 - 3x^2 + 9, then du = (3x^2 - 6x) dx = 3(x^2 - 2x) dx, so the integral is (1/3) u^(9/2)/(9/2) + C = (2/27) (x^3 - 3x^2 + 9)^(9/2) + C.

Step 9: (viii) Expand: (x^2 - 5)^3 = x^6 - 15x^4 + 75x^2 - 125, and integrating gives x^7/7 - 3x^5 + 25x^3 - 125x + C.

Step 10: (ix) Put u = cos x + sin x, then du = (cos x - sin x) dx exactly, so the integral is u^(5/2)/(5/2) + C = (2/5) (cos x + sin x)^(5/2) + C.

Step 11: (x) Put u = tan x + sin x, then du = (sec^2 x + cos x) dx exactly, so the integral is u^2/2 + C = (1/2) (tan x + sin x)^2 + C.

Final answer: (i) (2/3) (3x^2 + 9x + 3)^(3/2) + C (ii) (1/3) (a x^2 + 2b x + c)^(3/2) + C (iii) 2 sqrt(3x^2 + 5x + 2) + C (iv) -1/(8 (x^2 + 4x + 3)^8) +

C (v) x^4/4 - 3x^3 + 13x^2 - 24x + C (vi) (4/5) x^5 - 4x^3 + 9x + C (vii) (2/27) (x^3 - 3x^2 + 9)^(9/2) + C (viii) x^7/7 - 3x^5 + 25x^3 - 125x + C (ix) (2/5) (cos x + sin x)^(5/2) + C (x) (1/2) (tan x + sin x)^2 + C.

Question: 3. Evaluate by using standard formulae of integration. (i) integral of x/(x^2 + 3) dx (ii) integral of (sec^2 x + cos x)/(tan x + sin x) dx (iii) integral of (5x^4 + 4x^3 - 3x^2 + 2x)/(x^5 + x^4 - x^3 + x^2) dx (iv) integral of (e^x + 1/x) dx / (e^x + ln x) (v) integral of (5x^3 - 3x^2 + 6x - 9)^(-1) (5x^2 - 2x + 2) dx (vi) integral of dx / (x + sqrt(x))

Answer: Step 1: Parts (i) to (iv) are of the form integral of f prime (x)/f(x) dx = ln (f(x)) + C, with a constant adjustment in part (v).

Step 2: (i) Put u = x^2 + 3, then du = 2x dx, so x dx = du/2 and the integral is (1/2) ln (x^2 + 3) + C.

Step 3: (ii) The derivative of tan x + sin x is sec^2 x + cos x, which is exactly the numerator, so the integral is ln (tan x + sin x) + C.

Step 4: (iii) The derivative of x^5 + x^4 - x^3 + x^2 is 5x^4 + 4x^3 - 3x^2 + 2x, which is exactly the numerator, so the integral is ln (x^5 + x^4 - x^3 + x^2) + C.

Step 5: (iv) The derivative of e^x + ln x is e^x + 1/x, which is exactly the numerator, so the integral is ln (e^x + ln x) + C.

Step 6: (v) Put u = 5x^3 - 3x^2 + 6x - 9, then du = (15x^2 - 6x + 6) dx = 3 (5x^2 - 2x + 2) dx, so the given factor is du/3 and the integral is (1/3) ln (5x^3 - 3x^2 + 6x - 9) + C.

Step 7: (vi) Write the denominator as sqrt(x) (sqrt(x) + 1) and put u = sqrt(x) + 1, then du = dx/(2 sqrt(x)), so dx/sqrt(x) = 2 du and the integral becomes integral of 2 du/u = 2 ln (sqrt(x) + 1) + C.

Final answer: (i) (1/2) ln (x^2 + 3) + C (ii) ln (tan x + sin x) + C (iii) ln

(x^5 + x^4 - x^3 + x^2) + C (iv) ln (e^x + ln x) + C (v) (1/3) ln (5x^3 - 3x^2 + 6x - 9) + C (vi) 2 ln (sqrt(x) + 1) + C.

Question: 4. Evaluate the following integrals by using standard formulae. (i) integral of e^x (sin x + cos x) dx (ii) integral of e^x (sin^(-1) x + 1/sqrt(1 - x^2)) dx (iii) integral of e^x (tan^(-1) x + 1/(1 + x^2)) dx (iv) integral of e^x (sec^2 x + tan x) dx (v) integral of e^x (1/x + ln x) dx

Answer: Step 1: Each integral has the form integral of e^x (f(x) + f prime (x)) dx = e^x f(x) + C, so identify f from the term that is not a derivative.

Step 2: (i) Take f(x) = sin x, then f prime (x) = cos x, which is the other term, so the integral is e^x sin x + C.

Step 3: (ii) Take f(x) = sin^(-1) x, then f prime (x) = 1/sqrt(1 - x^2), which is the other term, so the integral is e^x sin^(-1) x + C.

Step 4: (iii) Take f(x) = tan^(-1) x, then f prime (x) = 1/(1 + x^2), which is the other term, so the integral is e^x tan^(-1) x + C.

Step 5: (iv) Take f(x) = tan x, then f prime (x) = sec^2 x, which is the other term, so the integral is e^x tan x + C.

Step 6: (v) Take f(x) = ln x, then f prime (x) = 1/x, which is the other term, so the integral is e^x ln x + C.

Final answer: (i) e^x sin x + C (ii) e^x sin^(-1) x + C (iii) e^x tan^(-1) x + C (iv) e^x tan x + C (v) e^x ln x + C.

Question: 5. Evaluate the following integrals by using appropriate formulae. (i) integral of dx/(x^2 + 9) (ii) integral of dt/sqrt(4 - t^2) (iii) integral of dy/(y sqrt(y^2 - 9)) (iv) integral of dt/(4t^2 - 9) (v) integral of dx/sqrt(9x^2 + 16) (vi) integral of dx/sqrt(16x^2 - 9) (vii) integral of dx/(9 - x^2) (viii) integral of dx/(x sqrt(4x^2 - 16)) (ix) integral of sqrt(9 - 4x^2) dx (x) integral of sqrt(25 + 9x^2) dx (xi) integral of dy/(9y^2 + 81) (xii)

integral of dx/(4x^2 - 16)

Answer: Step 1: Match each integral to a standard form: integral of dx/(x^2 + a^2), integral of dx/sqrt(a^2 - x^2), integral of dx/(x sqrt(x^2 - a^2)), integral of dx/(a^2 - x^2), integral of dx/sqrt(x^2 + a^2) and integral of sqrt(a^2 - x^2) dx or sqrt(a^2 + x^2) dx, adjusting constants first.

Step 2: (i) With a = 3, integral of dx/(x^2 + 9) = (1/3) tan^(-1) (x/3) + C.

Step 3: (ii) With a = 2, integral of dt/sqrt(4 - t^2) = sin^(-1) (t/2) + C.

Step 4: (iii) With a = 3, integral of dy/(y sqrt(y^2 - 9)) = (1/3) sec^(-1) (y/3) + C.

Step 5: (iv) Write 4t^2 - 9 = 4 (t^2 - 9/4), so the integral is (1/4) integral of dt/(t^2 - (3/2)^2) = (1/4) (1/3) ln ((2t - 3)/(2t + 3)) + C = (1/6) ln ((2t - 3)/(2t + 3)) + C, taking absolute values.

Step 6: (v) Factor 9 from the root: sqrt(9x^2 + 16) = 3 sqrt(x^2 + 16/9), and the standard log form gives (1/3) ln (3x + sqrt(9x^2 + 16)) + C.

Step 7: (vi) Factor 16 from the root: sqrt(16x^2 - 9) = 4 sqrt(x^2 - 9/16), and the standard log form gives (1/4) ln (4x + sqrt(16x^2 - 9)) + C.

Step 8: (vii) integral of dx/(9 - x^2) = (1/6) ln ((3 + x)/(3 - x)) + C by the form integral of dx/(a^2 - x^2) with a = 3.

Step 9: (viii) Write sqrt(4x^2 - 16) = 2 sqrt(x^2 - 4), so the integral is (1/2) integral of dx/(x sqrt(x^2 - 4)) = (1/2)(1/2) sec^(-1) (x/2) + C = (1/4) sec^(-1) (x/2) + C.

Step 10: (ix) Use integral of sqrt(a^2 - u^2) du with a = 3 and u = 2x, du = 2 dx: the integral is (1/2) ((2x/2) sqrt(9 - 4x^2) + (9/2) sin^(-1) (2x/3)) + C = (x/2) sqrt(9 - 4x^2) + (9/4) sin^(-1) (2x/3) + C.

Step 11: (x) Use integral of sqrt(a^2 + u^2) du with a = 5 and u = 3x, du = 3 dx: the integral is (1/3) ((3x/2) sqrt(25 + 9x^2) + (25/2) ln (3x + sqrt(25 + 9x^2))) + C = (x/2) sqrt(25 + 9x^2) + (25/6) ln (3x + sqrt(25 + 9x^2)) + C.

Step 12: (xi) Write 9y^2 + 81 = 9 (y^2 + 9), so the integral is (1/9) (1/3) tan^(-1) (y/3) + C = (1/27) tan^(-1) (y/3) + C.

Step 13: (xii) Write 4x^2 - 16 = 4 (x^2 - 4), so the integral is (1/4) (1/4) ln ((x - 2)/(x + 2)) + C = (1/16) ln ((x - 2)/(x + 2)) + C, taking absolute values.

Final answer: (i) (1/3) tan^(-1) (x/3) + C (ii) sin^(-1) (t/2) + C (iii) (1/3) sec^(-1) (y/3) + C (iv) (1/6) ln ((2t - 3)/(2t + 3)) + C (v) (1/3) ln (3x + sqrt(9x^2 + 16)) + C (vi) (1/4) ln (4x + sqrt(16x^2 - 9)) + C (vii) (1/6) ln ((3 + x)/(3 - x)) + C (viii) (1/4) sec^(-1) (x/2) + C (ix) (x/2) sqrt(9 - 4x^2) + (9/4) sin^(-1) (2x/3) + C (x) (x/2) sqrt(25 + 9x^2) + (25/6) ln (3x + sqrt(25 + 9x^2)) + C (xi) (1/27) tan^(-1) (y/3) + C (xii) (1/16) ln ((x - 2)/(x + 2)) + C.

Exercise 6.2

Question: 1. Evaluate the following integrals by substitution method. (i) integral of 3x/sqrt(x^2 + 7) dx (ii) integral of x^(4/3) (a^(7/3) - x^(7/3))^(5/2) dx (iii) integral of (1 + 2x)/sqrt(1 - x) dx (iv) integral of (2x^2 + 4x + 5)^(3/2) (x + 1) dx (v) integral of x/sqrt(1 - x^2) dx (vi) integral of (x^2 + 2x + 5)^(-1) (x + 1) dx (vii) integral of x^3 (9 + x^2)^(3/2) dx (viii) integral of (x^3 - 9)^(5/2) x^5 dx (ix) integral of x^9 (x^5 + 3)^(2/5) dx (x) integral of (x^3 + x^2 + 5x - 1)^(-1) (3x^2 + 2x + 5) dx

Answer: Step 1: Choose the substitution named in each part, differentiate

it, rewrite the whole integral in the new variable, integrate, and substitute back.

Step 2: (i) Put u = x^2 + 7, du = 2x dx, so 3x dx = (3/2) du and the integral is (3/2) (2 sqrt(u)) + C = 3 sqrt(x^2 + 7) + C.

Step 3: (ii) Put u = a^(7/3) - x^(7/3), then du = -(7/3) x^(4/3) dx, so x^(4/3) dx = -(3/7) du and the integral becomes -(3/7) integral of u^(5/2) du = -(3/7) (u^(7/2)/(7/2)) + C = -(6/49) (a^(7/3) - x^(7/3))^(7/2) + C. Note: the book answer key prints this without the minus sign.

Step 4: (iii) Put u = 1 - x, then x = 1 - u and dx = -du, and 1 + 2x = 3 - 2u, so the integral is integral of (3 - 2u) u^(-1/2) (-du) = integral of (2u - 3) u^(-1/2) du = integral of (2 u^(1/2) - 3 u^(-1/2)) du = (4/3) u^(3/2) - 6 sqrt(u) + C = (4/3) (1 - x)^(3/2) - 6 sqrt(1 - x) + C.

Step 5: (iv) Put u = 2x^2 + 4x + 5, du = (4x + 4) dx = 4 (x + 1) dx, so the integral is (1/4) u^(5/2)/(5/2) + C = (1/10) (2x^2 + 4x + 5)^(5/2) + C.

Step 6: (v) Put u = 1 - x^2, du = -2x dx, so the integral is -(1/2) (2 sqrt(u)) + C = -sqrt(1 - x^2) + C.

Step 7: (vi) Put u = x^2 + 2x + 5, du = 2 (x + 1) dx, so the integral is (1/2) ln (x^2 + 2x + 5) + C.

Step 8: (vii) Put u = 9 + x^2, then x^2 = u - 9 and du = 2x dx, so x^3 dx = (u - 9) du/2 and the integral is (1/2) integral of (u - 9) u^(3/2) du = (1/2) (u^(7/2)/(7/2) - 9 u^(5/2)/(5/2)) + C = (1/7) (9 + x^2)^(7/2) - (9/5) (9 + x^2)^(5/2) + C.

Step 9: (viii) Put u = x^3 - 9, then x^3 = u + 9 and du = 3x^2 dx, so x^5 dx = (u + 9) du/3 and the integral is (1/3) integral of (u + 9) u^(5/2) du = (1/3) (u^(9/2)/(9/2) + 9 u^(7/2)/(7/2)) + C = (2/27) (x^3 - 9)^(9/2) + (6/7) (x^3 - 9)^(7/2) + C.

Step 10: (ix) Put u = x^5 + 3, then x^5 = u - 3 and du = 5x^4 dx, so x^9 dx = (u - 3) du/5 and the integral is (1/5) integral of (u - 3) u^(2/5) du = (1/5) (u^(12/5)/(12/5) - 3 u^(7/5)/(7/5)) + C = (1/12) (x^5 + 3)^(12/5) - (3/7) (x^5 + 3)^(7/5) + C.

Step 11: (x) Put u = x^3 + x^2 + 5x - 1, then du = (3x^2 + 2x + 5) dx exactly, so the integral is ln (x^3 + x^2 + 5x - 1) + C.

Final answer: (i) 3 sqrt(x^2 + 7) + C (ii) -(6/49) (a^(7/3) - x^(7/3))^(7/2) + C (iii) (4/3) (1 - x)^(3/2) - 6 sqrt(1 - x) + C (iv) (1/10) (2x^2 + 4x + 5)^(5/2) + C (v) -sqrt(1 - x^2) + C (vi) (1/2) ln (x^2 + 2x + 5) + C (vii) (1/7) (9 + x^2)^(7/2) - (9/5) (9 + x^2)^(5/2) + C (viii) (2/27) (x^3 - 9)^(9/2) + (6/7) (x^3 - 9)^(7/2) + C (ix) (1/12) (x^5 + 3)^(12/5) - (3/7) (x^5 + 3)^(7/5) + C (x) ln (x^3 + x^2 + 5x - 1) + C.

Question: 2. Evaluate the following integrals by substitution method. (i) integral of (ln x)/x dx (ii) integral of dx/(x ln x) (iii) integral of tan x / ln(cos x) dx (iv) integral of tan(ln x)/x dx (v) integral of (1 + e^(2x))^(-1/2) e^(2x) dx (vi) integral of 5 e^(5 tan x)/cos^2 x dx (vii) integral of e^(2x)/(e^x + e^(-x)) dx (viii) integral of e^(cosec 2x + 1) (cosec 2x cot 2x) dx (ix) integral of 3^x dx (x) integral of e^(sin x + cos x + 3) (cos x - sin x) dx

Answer: Step 1: In each part put u equal to the inner function named, differentiate, and reduce to a standard integral.

Step 2: (i) Put u = ln x, du = dx/x, giving integral of u du = u^2/2 + C = (1/2) (ln x)^2 + C.

Step 3: (ii) Put u = ln x, du = dx/x, giving integral of du/u = ln (ln x) + C.

Step 4: (iii) Put u = ln(cos x), du = -tan x dx, giving integral of -du/u = -ln (ln(cos x)) + C, taking absolute values.

Step 5: (iv) Put u = ln x, du = dx/x, giving integral of tan u du = ln (sec u) + C = ln (sec(ln x)) + C.

Step 6: (v) Put u = 1 + e^(2x), du = 2 e^(2x) dx, giving (1/2) integral of u^(-1/2) du = sqrt(u) + C = sqrt(1 + e^(2x)) + C.

Step 7: (vi) Put u = 5 tan x, du = 5 sec^2 x dx = 5 dx/cos^2 x, giving integral of e^u du = e^(5 tan x) + C.

Step 8: (vii) Multiply numerator and denominator by e^x to get integral of e^(3x)/(e^(2x) + 1) dx, then write e^(3x) = e^x (e^(2x) + 1) - e^x, so the integral is integral of e^x dx - integral of e^x/(e^(2x) + 1) dx = e^x - tan^(-1) (e^x) + C, where the second part uses u = e^x.

Step 9: (viii) Put u = cosec 2x + 1, du = -2 cosec 2x cot 2x dx, giving integral of e^u (-du/2) = -(1/2) e^(cosec 2x + 1) + C.

Step 10: (ix) Put 3^x = e^(x ln 3), or use the standard form directly: integral of 3^x dx = 3^x/ln 3 + C.

Step 11: (x) Put u = sin x + cos x + 3, du = (cos x - sin x) dx exactly, giving integral of e^u du = e^(sin x + cos x + 3) + C.

Final answer: (i) (1/2) (ln x)^2 + C (ii) ln (ln x) + C (iii) -ln (ln(cos x)) + C (iv) ln (sec(ln x)) + C (v) sqrt(1 + e^(2x)) + C (vi) e^(5 tan x) + C (vii) e^x - tan^(-1) (e^x) + C (viii) -(1/2) e^(cosec 2x + 1) + C (ix) 3^x/ln 3 + C (x) e^(sin x + cos x + 3) + C.

Question: 3. Evaluate the following integrals by substitution method. (i) integral of cos^5 2x sin 2x dx (ii) integral of (cot sqrt(x))/sqrt(x) dx (iii) integral of (2 + sin 3x)^6 cos 3x dx (iv) integral of sin(a x + b) dx (v) integral of (sin x + cos x)/(sin x - cos x) dx (vi) integral of e^(2x) sec^2 e^(2x) dx (vii) integral of (cosec 3x cot 3x dx)/((a + b cosec 3x)^2) (viii) integral of dx/((2 cot x + 3) sin^2 x) (ix) integral of (sec x dx)/(3 sin x + 4

cos x) (x) integral of cos(3x - 5) dx

Answer: Step 1: Use the substitution named in each part; most parts reduce to a power or a logarithm.

Step 2: (i) Put u = cos 2x, du = -2 sin 2x dx, giving integral of u^5 (-du/2) = -u^6/12 + C = -(1/12) cos^6 2x + C.

Step 3: (ii) Put u = sqrt(x), du = dx/(2 sqrt(x)), giving integral of 2 cot u du = 2 ln (sin sqrt(x)) + C.

Step 4: (iii) Put u = 2 + sin 3x, du = 3 cos 3x dx, giving integral of u^6 du/3 = u^7/21 + C = (1/21) (2 + sin 3x)^7 + C.

Step 5: (iv) Put u = a x + b, du = a dx, giving integral of sin u du/a = -(1/a) cos(a x + b) + C.

Step 6: (v) Put u = sin x - cos x, du = (cos x + sin x) dx, which is exactly the numerator, giving ln (sin x - cos x) + C, taking absolute values.

Step 7: (vi) Put u = e^(2x), du = 2 e^(2x) dx, giving (1/2) integral of sec^2 u du = (1/2) tan e^(2x) + C.

Step 8: (vii) Put u = a + b cosec 3x, du = -3b cosec 3x cot 3x dx, giving integral of -du/(3b u^2) = 1/(3b u) + C = 1/(3b (a + b cosec 3x)) + C.

Step 9: (viii) Write 1/sin^2 x = cosec^2 x and put u = 2 cot x + 3, du = -2 cosec^2 x dx, giving integral of -du/(2u) = -(1/2) ln (2 cot x + 3) + C.

Step 10: (ix) Divide numerator and denominator by cos x: the integrand becomes sec^2 x/(3 tan x + 4), and putting u = 3 tan x + 4, du = 3 sec^2 x dx, gives (1/3) ln (3 tan x + 4) + C.

Step 11: (x) Put u = 3x - 5, du = 3 dx, giving integral of cos u du/3 = (1/3) sin(3x - 5) + C.

Final answer: (i) -(1/12) cos^6 2x + C (ii) 2 ln (sin sqrt(x)) + C (iii) (1/21) (2

+ sin 3x)^7 + C (iv) -(1/a) cos(a x + b) + C (v) ln (sin x - cos x) + C (vi) (1/2) tan e^(2x) + C (vii) 1/(3b (a + b cosec 3x)) + C (viii) -(1/2) ln (2 cot x + 3) + C (ix) (1/3) ln (3 tan x + 4) + C (x) (1/3) sin(3x - 5) + C.

Question: 4. Evaluate the following integrals by substitution method. (i) integral of cos^2 2y dy (ii) integral of sin^3 (3x + 5) dx (iii) integral of cos^3 x sqrt(sin x) dx (iv) integral of sin^4 x cos^5 x dx (v) integral of sin^3 x / sqrt(cos x) dx (vi) integral of cos^5 x sin^7 x dx (vii) integral of sin^3 x cos^3 x dx (viii) integral of sin 2x cos 4x dx (ix) integral of cos 3x cos 5x dx (x) integral of sin 3x cos 7x dx (xi) integral of tan^2 x dx (xii) integral of cot^4 x dx (xiii) integral of tan^7 x dx (xiv) integral of sec^4 2x dx (xv) integral of tan^5 3x sec^3 3x dx (xvi) integral of cosec^4 3x dx (xvii) integral of sec^4 x sqrt(tan x) dx (xviii) integral of cot 2x cosec^4 2x dx (xix) integral of cosec^4 x / sqrt(cot x) dx (xx) integral of sqrt(1 + cos x) dx

Answer: Step 1: For odd powers keep one factor for the substitution and convert the rest with sin^2 x + cos^2 x = 1, sec^2 x = 1 + tan^2 x or cosec^2 x = 1 + cot^2 x; for products use product-to-sum identities; for even powers use double-angle identities.

Step 2: (i) cos^2 2y = (1 + cos 4y)/2, so the integral is y/2 + (1/8) sin 4y + C.

Step 3: (ii) Write sin^3 (3x + 5) = sin(3x + 5) (1 - cos^2 (3x + 5)) and put u = cos(3x + 5), du = -3 sin(3x + 5) dx, giving -(1/3) cos(3x + 5) + (1/9) cos^3 (3x + 5) + C.

Step 4: (iii) Write cos^3 x = cos x (1 - sin^2 x) and put u = sin x, giving integral of (u^(1/2) - u^(5/2)) du = (2/3) sin^(3/2) x - (2/7) sin^(7/2) x + C.

Step 5: (iv) Write cos^5 x = cos x (1 - sin^2 x)^2 and put u = sin x, giving

integral of u^4 (1 - 2u^2 + u^4) du = (1/5) sin^5 x - (2/7) sin^7 x + (1/9) sin^9 x + C.

Step 6: (v) Write sin^3 x = sin x (1 - cos^2 x) and put u = cos x, du = -sin x dx, giving integral of (u^(3/2) - u^(-1/2)) du = (2/5) cos^(5/2) x - 2 sqrt(cos x) + C.

Step 7: (vi) Write cos^5 x = cos x (1 - sin^2 x)^2 and put u = sin x, giving integral of u^7 (1 - 2u^2 + u^4) du = (1/8) sin^8 x - (1/5) sin^(10) x + (1/12) sin^(12) x + C.

Step 8: (vii) Write cos^3 x = cos x (1 - sin^2 x) and put u = sin x, giving integral of (u^3 - u^5) du = (1/4) sin^4 x - (1/6) sin^6 x + C.

Step 9: (viii) sin 2x cos 4x = (1/2) (sin 6x - sin 2x), so the integral is (1/2) (-cos 6x/6 + cos 2x/2) + C = -(1/12) cos 6x + (1/4) cos 2x + C.

Step 10: (ix) cos 3x cos 5x = (1/2) (cos 8x + cos 2x), so the integral is (1/16) sin 8x + (1/4) sin 2x + C.

Step 11: (x) sin 3x cos 7x = (1/2) (sin 10x - sin 4x), so the integral is -(1/20) cos 10x + (1/8) cos 4x + C.

Step 12: (xi) tan^2 x = sec^2 x - 1, so the integral is tan x - x + C.

Step 13: (xii) Write cot^4 x = cot^2 x (cosec^2 x - 1) = cot^2 x cosec^2 x - cot^2 x, and with u = cot x the first part gives -cot^3 x/3 and the second part gives cot x + x, totalling -(1/3) cot^3 x + cot x + x + C.

Step 14: (xiii) Write tan^7 x = tan^5 x (sec^2 x - 1) and repeat the reduction with u = tan x, giving tan^6 x/6 - tan^4 x/4 + tan^2 x/2 + ln (cos x) + C, taking absolute values.

Step 15: (xiv) Write sec^4 2x = (1 + tan^2 2x) sec^2 2x and put u = tan 2x, du = 2 sec^2 2x dx, giving (1/2) (u + u^3/3) + C = (1/2) tan 2x +

(1/6) tan^3 2x + C.

Step 16: (xv) Put u = sec 3x, du = 3 sec 3x tan 3x dx and tan^4 3x = (sec^2 3x - 1)^2, giving after expansion and integration (1/21) sec^7 3x - (2/15) sec^5 3x + (1/9) sec^3 3x + C.

Step 17: (xvi) Write cosec^4 3x = (1 + cot^2 3x) cosec^2 3x and put u = cot 3x, du = -3 cosec^2 3x dx, giving -(1/3) cot 3x - (1/9) cot^3 3x + C.

Step 18: (xvii) Write sec^4 x = (1 + tan^2 x) sec^2 x and put u = tan x, giving integral of (u^(1/2) + u^(5/2)) du = (2/3) tan^(3/2) x + (2/7) tan^(7/2) x + C.

Step 19: (xviii) Put u = cot 2x, du = -2 cosec^2 2x dx and cosec^2 2x = 1 + cot^2 2x, giving integral of -u (1 + u^2) du/2 = -(1/4) cot^2 2x - (1/8) cot^4 2x + C.

Step 20: (xix) Write cosec^4 x = (1 + cot^2 x) cosec^2 x and put u = cot x, du = -cosec^2 x dx, giving integral of -(1 + u^2) u^(-1/2) du = -2 sqrt(cot x) - (2/5) cot^(5/2) x + C.

Step 21: (xx) Use 1 + cos x = 2 cos^2 (x/2), so sqrt(1 + cos x) = sqrt(2) cos (x/2) on the interval considered, giving integral of sqrt(2) cos (x/2) dx = 2 sqrt(2) sin (x/2) + C.

Final answer: (i) y/2 + (1/8) sin 4y + C (ii) -(1/3) cos(3x + 5) + (1/9) cos^3 (3x + 5) + C (iii) (2/3) sin^(3/2) x - (2/7) sin^(7/2) x + C (iv) (1/5) sin^5 x - (2/7) sin^7 x + (1/9) sin^9 x + C (v) (2/5) cos^(5/2) x - 2 sqrt(cos x) + C (vi) (1/8) sin^8 x - (1/5) sin^(10) x + (1/12) sin^(12) x + C (vii) (1/4) sin^4 x - (1/6) sin^6 x + C (viii) -(1/12) cos 6x + (1/4) cos 2x + C (ix) (1/16) sin 8x + (1/4) sin 2x + C (x) -(1/20) cos 10x + (1/8) cos 4x + C (xi) tan x - x + C (xii) -(1/3) cot^3 x + cot x + x + C (xiii) tan^6 x/6 - tan^4 x/4 + tan^2 x/2 + ln (cos x) + C (xiv) (1/2) tan 2x + (1/6) tan^3 2x + C

(xv) (1/21) sec^7 3x - (2/15) sec^5 3x + (1/9) sec^3 3x + C (xvi) -(1/3) cot 3x - (1/9) cot^3 3x + C (xvii) (2/3) tan^(3/2) x + (2/7) tan^(7/2) x + C (xviii) -(1/4) cot^2 2x - (1/8) cot^4 2x + C (xix) -2 sqrt(cot x) - (2/5) cot^(5/2) x + C (xx) 2 sqrt(2) sin (x/2) + C.

Exercise 6.3

Question: Evaluate the following integrals by trigonometric substitution method. 1. integral of x^3/sqrt(9 - x^2) dx 2. integral of 6/(9 - x^2) dx 3. integral of x^2 sqrt(9 - x^2) dx 4. integral of 5/(25x^2 + 9) dx 5. integral of dx/(4 + x^2)^(3/2) 6. integral of dx/sqrt(16 + 4x^2) 7. integral of x^3 sqrt(9x^2 - 36) dx 8. integral of dx/sqrt(a^2 + x^2) 9. integral of dx/(16 - x^2)^(5/2) 10. integral of x^5/sqrt(x^2 - 9) dx 11. integral of dx/(x^2 + 4x + 5) 12. integral of dx/sqrt(5 + 4x - x^2) 13. integral of dx/sqrt(9x - x^2) 14. integral of dx/((x + 1) sqrt(x^2 + 2x - 15)) 15. integral of dx/((x - 4) sqrt(x^2 - 8x - 9)) 16. integral of (2x - 5)/sqrt(8x - x^2) dx 17. integral of (x + 3)/(x^2 + 2x + 5) dx 18. integral of (3x + 9)/(x^2 + 4x + 4) dx 19. integral of (4x + 9)/sqrt(2x^2 + 8x - 10) dx 20. integral of (2x - 5)/sqrt(5 + 4x - x^2) dx

Answer: Step 1: For roots of the form sqrt(a^2 - x^2) put x = a sin theta, for sqrt(a^2 + x^2) put x = a tan theta, and for sqrt(x^2 - a^2) put x = a sec theta; for quadratics without a root first complete the square.

Step 2: 1. Put x = 3 sin theta, dx = 3 cos theta d theta and sqrt(9 - x^2) = 3 cos theta, so the integral becomes integral of 27 sin^3 theta d theta = 27 integral of sin theta (1 - cos^2 theta) d theta = -27 cos theta + 9 cos^3 theta + C, and returning with cos theta = sqrt(9 - x^2)/3 gives (1/3) (9 - x^2)^(3/2) - 9 sqrt(9 - x^2) + C.

Step 3: 2. Use partial fractions: 6/(9 - x^2) = 1/(3 + x) + 1/(3 - x), and

integrating gives ln ((3 + x)/(3 - x)) + C, taking absolute values.

Step 4: 3. Put x = 3 sin theta, then the integral becomes 81 integral of sin^2 theta cos^2 theta d theta = (81/4) integral of sin^2 2theta d theta = (81/8) (theta - (1/4) sin 4theta) + C, and returning to x gives (81/8) sin^(-1) (x/3) - (9/8) x sqrt(9 - x^2) + (1/4) x^3 sqrt(9 - x^2) + C.

Step 5: 4. Factor 25: the integral is (5/25) integral of dx/(x^2 + 9/25), and the standard form gives 5 times (1/25)(3) tan^(-1) (5x/3) with the constants collected, that is (1/3) tan^(-1) (5x/3) + C.

Step 6: 5. Put x = 2 tan theta, dx = 2 sec^2 theta d theta and (4 + x^2)^(3/2) = 8 sec^3 theta, so the integral becomes (1/4) integral of cos theta d theta = (1/4) sin theta + C = x/(4 sqrt(4 + x^2)) + C.

Step 7: 6. Write sqrt(16 + 4x^2) = 2 sqrt(4 + x^2), so the integral is (1/2) integral of dx/sqrt(x^2 + 4) = (1/2) ln (x + sqrt(x^2 + 4)) + C.

Step 8: 7. Write sqrt(9x^2 - 36) = 3 sqrt(x^2 - 4) and put u = x^2 - 4, du = 2x dx with x^2 = u + 4, so the integral becomes (3/2) integral of (u + 4) u^(1/2) du = (3/2) ((2/5) u^(5/2) + (8/3) u^(3/2)) + C = (3/5) (x^2 - 4)^(5/2) + 4 (x^2 - 4)^(3/2) + C.

Step 9: 8. Put x = a tan theta, dx = a sec^2 theta d theta and sqrt(a^2 + x^2) = a sec theta, so the integral becomes integral of sec theta d theta = ln (x + sqrt(a^2 + x^2)) + C.

Step 10: 9. Put x = 4 sin theta, dx = 4 cos theta d theta and (16 - x^2)^(5/2) = 1024 cos^5 theta, so the integral becomes (1/256) integral of sec^4 theta d theta = (1/256) (tan theta + (1/3) tan^3 theta) + C, and returning with tan theta = x/sqrt(16 - x^2) gives (1/256) (x/sqrt(16 - x^2) + x^3/(3 (16 - x^2)^(3/2))) + C.

Step 11: 10. Put u = x^2 - 9, du = 2x dx with x^4 = (u + 9)^2, so the

integral becomes (1/2) integral of (u + 9)^2 u^(-1/2) du = (1/2) integral of (u^(3/2) + 18 u^(1/2) + 81 u^(-1/2)) du = (1/5) (x^2 - 9)^(5/2) + 6 (x^2 - 9)^(3/2) + 81 sqrt(x^2 - 9) + C.

Step 12: 11. Complete the square: x^2 + 4x + 5 = (x + 2)^2 + 1, so the integral is tan^(-1) (x + 2) + C.

Step 13: 12. Complete the square: 5 + 4x - x^2 = 9 - (x - 2)^2, so the integral is sin^(-1) ((x - 2)/3) + C.

Step 14: 13. Complete the square: 9x - x^2 = 81/4 - (x - 9/2)^2, so with a = 9/2 the integral is sin^(-1) ((2x - 9)/9) + C.

Step 15: 14. Complete the square inside the root: x^2 + 2x - 15 = (x + 1)^2 - 16, put x + 1 = 4 sec theta, and the integral reduces to (1/4) sec^(-1) ((x + 1)/4) + C.

Step 16: 15. Complete the square inside the root: x^2 - 8x - 9 = (x - 4)^2 - 25, put x - 4 = 5 sec theta, and the integral reduces to (1/5) sec^(-1) ((x - 4)/5) + C.

Step 17: 16. Write the numerator 2x - 5 = -(8 - 2x) + 3, where 8 - 2x is the derivative of 8x - x^2; the first part gives -2 sqrt(8x - x^2) and the second part, with 8x - x^2 = 16 - (x - 4)^2, gives 3 sin^(-1) ((x - 4)/4), totalling -2 sqrt(8x - x^2) + 3 sin^(-1) ((x - 4)/4) + C.

Step 18: 17. Write the numerator x + 3 = (1/2)(2x + 2) + 2; the first part gives (1/2) ln (x^2 + 2x + 5) = ln sqrt(x^2 + 2x + 5) and the second part, with x^2 + 2x + 5 = (x + 1)^2 + 4, gives 2 times (1/2) tan^(-1) ((x + 1)/2), totalling ln sqrt(x^2 + 2x + 5) + tan^(-1) ((x + 1)/2) + C.

Step 19: 18. Write the denominator as (x + 2)^2 and use partial fractions: (3x + 9)/(x + 2)^2 = 3/(x + 2) + 3/(x + 2)^2, and integrating gives 3 ln (x + 2) - 3/(x + 2) + C, taking absolute values.

Step 20: 19. Write the numerator 4x + 9 = (4x + 8) + 1, where 4x + 8 is the derivative of 2x^2 + 8x - 10; the first part gives 2 sqrt(2x^2 + 8x - 10) = 2 sqrt(2 (x^2 + 4x - 5)) and the second part gives (1/sqrt(2)) ln (x + 2 + sqrt(x^2 + 4x - 5)), totalling 2 sqrt(2 (x^2 + 4x - 5)) + (sqrt(2)/2) ln (x + 2 + sqrt(x^2 + 4x - 5)) + C.

Step 21: 20. Write the numerator 2x - 5 = -(4 - 2x) - 1, where 4 - 2x is the derivative of 5 + 4x - x^2; the first part gives -2 sqrt(5 + 4x - x^2) and the second part, with 5 + 4x - x^2 = 9 - (x - 2)^2, gives -sin^(-1) ((x - 2)/3), totalling -2 sqrt(5 + 4x - x^2) - sin^(-1) ((x - 2)/3) + C.

Final answer: 1. (1/3) (9 - x^2)^(3/2) - 9 sqrt(9 - x^2) + C 2. ln ((3 + x)/(3 - x)) + C 3. (81/8) sin^(-1) (x/3) - (9/8) x sqrt(9 - x^2) + (1/4) x^3 sqrt(9 - x^2) + C 4. (1/3) tan^(-1) (5x/3) + C 5. x/(4 sqrt(4 + x^2)) + C 6. (1/2) ln (x + sqrt(x^2 + 4)) + C 7. (3/5) (x^2 - 4)^(5/2) + 4 (x^2 - 4)^(3/2) + C 8. ln (x + sqrt(a^2 + x^2)) + C 9. (1/256) (x/sqrt(16 - x^2) + x^3/(3 (16 - x^2)^(3/2))) + C 10. (1/5) (x^2 - 9)^(5/2) + 6 (x^2 - 9)^(3/2) + 81 sqrt(x^2 - 9) + C 11. tan^(-1) (x + 2) + C 12. sin^(-1) ((x - 2)/3) + C 13. sin^(-1) ((2x - 9)/9) + C 14. (1/4) sec^(-1) ((x + 1)/4) + C 15. (1/5) sec^(-1) ((x - 4)/5) + C 16. -2 sqrt(8x - x^2) + 3 sin^(-1) ((x - 4)/4) + C 17. ln sqrt(x^2 + 2x + 5) + tan^(-1) ((x + 1)/2) + C 18. 3 ln (x + 2) - 3/(x + 2) + C 19. 2 sqrt(2 (x^2 + 4x - 5)) + (sqrt(2)/2) ln (x + 2 + sqrt(x^2 + 4x - 5)) + C 20. -2 sqrt(5 + 4x - x^2) - sin^(-1) ((x - 2)/3) + C.

Exercise 6.4

Question: 1. Evaluate the following integrals by parts. (i) integral of x^2 e^x dx (ii) integral of x^3 e^x dx (iii) integral of x cos x dx (iv) integral of ln x dx (v) integral of x^2 sin x dx (vi) integral of x cosec^2 x dx (vii) integral of x sec^2 x dx (viii) integral of (ln x)^2 dx

Answer: Step 1: Use integration by parts, integral of u dv = u v - integral of v du, choosing u as the algebraic or logarithmic factor and dv as the exponential or trigonometric factor.

Step 2: (i) Take u = x^2 and dv = e^x dx, so v = e^x and the integral is x^2 e^x - integral of 2x e^x dx; apply parts again to integral of 2x e^x dx = 2x e^x - 2 e^x, giving (x^2 - 2x + 2) e^x + C.

Step 3: (ii) Take u = x^3 and dv = e^x dx, giving x^3 e^x - 3 integral of x^2 e^x dx, and using part (i) this becomes x^3 e^x - 3 (x^2 - 2x + 2) e^x + C = (x^3 - 3x^2 + 6x - 6) e^x + C.

Step 4: (iii) Take u = x and dv = cos x dx, so v = sin x and the integral is x sin x - integral of sin x dx = x sin x + cos x + C.

Step 5: (iv) Take u = ln x and dv = dx, so v = x and du = dx/x, giving x ln x - integral of dx = x ln x - x + C = x (ln x - 1) + C.

Step 6: (v) Take u = x^2 and dv = sin x dx, so v = -cos x and the integral is -x^2 cos x + 2 integral of x cos x dx, and using part (iii) this is -x^2 cos x + 2 (x sin x + cos x) + C = 2 (x sin x + cos x) - x^2 cos x + C.

Step 7: (vi) Take u = x and dv = cosec^2 x dx, so v = -cot x and the integral is -x cot x + integral of cot x dx = -x cot x + ln (sin x) + C, taking absolute values.

Step 8: (vii) Take u = x and dv = sec^2 x dx, so v = tan x and the integral is x tan x - integral of tan x dx = x tan x + ln (cos x) + C, taking absolute values.

Step 9: (viii) Take u = (ln x)^2 and dv = dx, so v = x and du = 2 ln x dx/x, giving x (ln x)^2 - 2 integral of ln x dx, and using part (iv) this is x (ln x)^2 - 2x (ln x - 1) + C = x (ln x)^2 + 2x (1 - ln x) + C.

Final answer: (i) (x^2 - 2x + 2) e^x + C (ii) (x^3 - 3x^2 + 6x - 6) e^x + C

(iii) x sin x + cos x + C (iv) x (ln x - 1) + C (v) 2 (x sin x + cos x) - x^2 cos x + C (vi) ln (sin x) - x cot x + C (vii) x tan x + ln (cos x) + C (viii) x (ln x)^2 + 2x (1 - ln x) + C.

Question: 2. Evaluate the following integrals by parts. (i) integral of (x + 1) ln(x + 1) dx (ii) integral of x ln x dx (iii) integral of x^(-3) ln x dx (iv) integral of x^(-4) ln x^2 dx (v) integral of sin x cos x ln(sin x) dx (vi) integral of (tan x/cos^2 x) ln(tan x) dx (vii) integral of cot x cosec^2 x ln(cot x) dx (viii) integral of sec^3 x tan x ln(sec x) dx (ix) integral of cosec x cot x ln(cosec x) dx (x) integral of (ln x^2)/x^2 dx (xi) integral of sec^3 x dx (xii) integral of cosec^3 x dx

Answer: Step 1: In parts (i) to (x) first substitute u for the argument of the logarithm; each then becomes integral of u^n ln u du or integral of (ln u)/u^n du, which is done by parts with the pattern integral of u^n ln u du = u^(n + 1) (ln u)/(n + 1) - u^(n + 1)/(n + 1)^2.

Step 2: (i) Put u = x + 1; parts with ln u times u gives (u^2/2) ln u - u^2/4 + C, which the book writes as (1/4) (x + 1)^2 (ln (x + 1)^2 - 1) + C.

Step 3: (ii) Parts with u = x gives (x^2/2) ln x - x^2/4 + C = (1/4) x^2 (ln x^2 - 1) + C.

Step 4: (iii) Parts with u = ln x and dv = x^(-3) dx, v = -1/(2x^2), gives -(ln x)/(2x^2) - 1/(4x^2) + C = -(1/(4x^2)) (1 + ln x^2) + C.

Step 5: (iv) Write ln x^2 = 2 ln x and use parts with dv = x^(-4) dx, v = -1/(3x^3), giving -(2 ln x)/(3x^3) - 2/(9x^3) + C = -(1/(9x^3)) (ln x^6 + 2) + C.

Step 6: (v) Put u = sin x, du = cos x dx; parts on (1/4) u^2 pattern gives (1/4) sin^2 x (ln sin^2 x - 1) + C.

Step 7: (vi) Put u = tan x, du = sec^2 x dx = dx/cos^2 x; the pattern

gives (1/4) tan^2 x (ln tan^2 x - 1) + C.

Step 8: (vii) Put u = cot x, du = -cosec^2 x dx; the pattern gives -(1/4) cot^2 x (ln cot^2 x - 1) + C.

Step 9: (viii) Put u = sec x, du = sec x tan x dx, and sec^3 x tan x dx = u^2 du; the pattern gives (1/9) sec^3 x (ln sec^3 x - 1) + C in the book form (1/3) sec^3 x (ln sec x) - (1/9) sec^3 x + C.

Step 10: (ix) Put u = cosec x, du = -cosec x cot x dx, so the integral becomes -integral of u ln u du = -(u^2/2) ln u + u^2/4 with u = cosec x; collecting into the book form gives cosec x (1 - ln (cosec x)) + C.

Step 11: (x) Write ln x^2 = 2 ln x and use parts with dv = x^(-2) dx, v = -1/x, giving -(2 ln x)/x - 2/x + C = -(2/x) (1 + ln x) + C.

Step 12: (xi) Apply parts to integral of sec^3 x dx = integral of sec x sec^2 x dx with u = sec x and dv = sec^2 x dx, giving sec x tan x - integral of sec x tan^2 x dx = sec x tan x - integral of sec^3 x dx + integral of sec x dx; bringing the repeated integral to the left gives 2 integral of sec^3 x dx = sec x tan x + ln (sec x + tan x), so the integral is (1/2) sec x tan x + ln sqrt(sec x + tan x) + C.

Step 13: (xii) Apply parts similarly with u = cosec x and dv = cosec^2 x dx, giving -cosec x cot x - integral of cosec^3 x dx + integral of cosec x dx; collecting gives the integral as -(1/2) cosec x cot x + ln sqrt(cosec x - cot x) + C, taking absolute values.

Final answer: (i) (1/4) (x + 1)^2 (ln (x + 1)^2 - 1) + C (ii) (1/4) x^2 (ln x^2 - 1) + C (iii) -(1/(4x^2)) (1 + ln x^2) + C (iv) -(1/(9x^3)) (ln x^6 + 2) + C (v) (1/4) sin^2 x (ln sin^2 x - 1) + C (vi) (1/4) tan^2 x (ln tan^2 x - 1) + C (vii) -(1/4) cot^2 x (ln cot^2 x - 1) + C (viii) (1/9) sec^3 x (ln sec^3 x - 1) + C (ix) cosec x (1 - ln (cosec x)) + C (x) -(2/x) (1 + ln x) + C (xi) (1/2)

sec x tan x + ln sqrt(sec x + tan x) + C (xii) -(1/2) cosec x cot x + ln sqrt(cosec x - cot x) + C.

Question: 3. Evaluate the following integrals. (i) integral of 3x cos 3x dx (ii) integral of x^2 sin x dx (iii) integral of x/cot^2 x dx (iv) integral of x sec^2 x dx (v) integral of 5x/sin^2 2x dx (vi) integral of cos sqrt(x) dx (vii) integral of e^(2x) sin 2x dx (viii) integral of e^(-x) cos 2x dx (ix) integral of cos(ln x) dx (x) integral of e^(a x) sin b x dx

Answer: Step 1: Use integration by parts, and for parts (vii) to (x) apply parts twice and collect the repeated integral on the left side.

Step 2: (i) Take u = x and dv = 3 cos 3x dx, so v = sin 3x, giving x sin 3x - integral of sin 3x dx = x sin 3x + (1/3) cos 3x + C.

Step 3: (ii) Take u = x^2 and dv = sin x dx, giving -x^2 cos x + 2 integral of x cos x dx = (2 - x^2) cos x + 2x sin x + C, using integral of x cos x dx = x sin x + cos x.

Step 4: (iii) Write x/cot^2 x = x tan^2 x = x (sec^2 x - 1); parts on integral of x sec^2 x dx gives x tan x - integral of tan x dx = x tan x + ln (cos x), and then subtracting integral of x dx = x^2/2 gives x tan x + ln (cos x) - (1/2) x^2 + C, which with ln (cos x) = -ln (sec x) is the book form x tan x - ln (sec x) - (1/2) x^2 + C, taking absolute values.

Step 5: (iv) Take u = x and dv = sec^2 x dx, giving x tan x - integral of tan x dx = x tan x + ln (cos x) + C, taking absolute values.

Step 6: (v) Write 1/sin^2 2x = cosec^2 2x and take u = 5x, dv = cosec^2 2x dx, v = -(1/2) cot 2x, giving -(5/2) x cot 2x + (5/2) integral of cot 2x dx = (5/4) (-2x cot 2x + ln (sin 2x)) + C, taking absolute values.

Step 7: (vi) Put t = sqrt(x), dx = 2t dt, so the integral becomes 2 integral of t cos t dt = 2 (t sin t + cos t) + C = 2 sqrt(x) sin sqrt(x) + 2 cos sqrt(x)

+ C.

Step 8: (vii) Apply parts twice: the integral equals e^(2x) (sin 2x - cos 2x)/4 after collecting, that is (1/4) e^(2x) (sin 2x - cos 2x) + C.

Step 9: (viii) Apply parts twice: integral of e^(-x) cos 2x dx = e^(-x) (2 sin 2x - cos 2x)/5 + C = (1/5) e^(-x) (2 sin 2x - cos 2x) + C.

Step 10: (ix) Put t = ln x, x = e^t, dx = e^t dt, so the integral becomes integral of e^t cos t dt = (x/2) (sin (ln x) + cos (ln x)) + C.

Step 11: (x) Apply parts twice and collect: integral of e^(a x) sin b x dx = e^(a x) (a sin b x - b cos b x)/(a^2 + b^2) + C.

Final answer: (i) x sin 3x + (1/3) cos 3x + C (ii) (2 - x^2) cos x + 2x sin x + C (iii) x tan x - ln sec x - (1/2) x^2 + C (iv) x tan x + ln (cos x) + C (v) (5/4) (-2x cot 2x + ln (sin 2x)) + C (vi) 2 sqrt(x) sin sqrt(x) + 2 cos sqrt(x) + C (vii) (1/4) e^(2x) (sin 2x - cos 2x) + C (viii) (1/5) e^(-x) (2 sin 2x - cos 2x) + C (ix) (x/2) (sin (ln x) + cos (ln x)) + C (x) e^(a x) (a sin b x - b cos b x)/(a^2 + b^2) + C.

Question: 4. Evaluate the following integrals. (i) integral of sin^(-1) 3x dx (ii) integral of x^4 tan^(-1) x dx (iii) integral of tan^(-1) (2x) dx (iv) integral of x cos^(-1) x dx (v) integral of 3x^2 sin^(-1) (3x) dx (vi) integral of 2x sec^(-1) x dx (vii) integral of 6x cosec^(-1) (2x) dx (viii) integral of x^2 cot^(-1) x dx

Answer: Step 1: Take the inverse trigonometric function as u and the algebraic factor as dv in integration by parts; the remaining integral is then algebraic or a standard root integral.

Step 2: (i) u = sin^(-1) 3x, dv = dx gives x sin^(-1) 3x - integral of 3x/sqrt(1 - 9x^2) dx = x sin^(-1) (3x) + (1/3) sqrt(1 - 9x^2) + C.

Step 3: (ii) u = tan^(-1) x, dv = x^4 dx gives (x^5/5) tan^(-1) x - (1/5)

integral of x^5/(1 + x^2) dx; dividing, x^5/(1 + x^2) = x^3 - x + x/(1 + x^2), whose integral is x^4/4 - x^2/2 + (1/2) ln (1 + x^2), so the result is (1/5) (x^5 tan^(-1) x - (1/4) x^4 + (1/2) x^2 - ln sqrt(1 + x^2)) + C.

Step 4: (iii) u = tan^(-1) 2x, dv = dx gives x tan^(-1) 2x - integral of 2x/(1 + 4x^2) dx = x tan^(-1) (2x) - (1/4) ln (1 + 4x^2) + C.

Step 5: (iv) u = cos^(-1) x, dv = x dx gives (x^2/2) cos^(-1) x + (1/2) integral of x^2/sqrt(1 - x^2) dx, and the root integral equals -(1/2) x sqrt(1 - x^2) + (1/2) sin^(-1) x, giving (1/2) x^2 cos^(-1) x - (1/4) x sqrt(1 - x^2) + (1/4) sin^(-1) x + C.

Step 6: (v) u = sin^(-1) 3x, dv = 3x^2 dx, v = x^3, gives x^3 sin^(-1) 3x - integral of 3x^3/sqrt(1 - 9x^2) dx; the root integral is evaluated by putting w = 1 - 9x^2, giving (1/27) sqrt(1 - 9x^2) - (1/81) (1 - 9x^2)^(3/2), so the result is x^3 sin^(-1) (3x) + (1/27) sqrt(1 - 9x^2) - (1/81) (1 - 9x^2)^(3/2) + C.

Step 7: (vi) u = sec^(-1) x, dv = 2x dx, v = x^2, gives x^2 sec^(-1) x - integral of x^2/(x sqrt(x^2 - 1)) dx = x^2 sec^(-1) x - sqrt(x^2 - 1) + C.

Step 8: (vii) u = cosec^(-1) 2x, dv = 6x dx, v = 3x^2, gives 3x^2 cosec^(-1) (2x) + (3/4) sqrt(4x^2 - 1) + C after evaluating the remaining root integral.

Step 9: (viii) u = cot^(-1) x, dv = x^2 dx gives (x^3/3) cot^(-1) x + (1/3) integral of x^3/(1 + x^2) dx; dividing, x^3/(1 + x^2) = x - x/(1 + x^2), whose integral is x^2/2 - (1/2) ln (1 + x^2), so the result is (1/6) (2x^3 cot^(-1) x + x^2 - ln (1 + x^2)) + C.

Final answer: (i) x sin^(-1) (3x) + (1/3) sqrt(1 - 9x^2) + C (ii) (1/5) (x^5 tan^(-1) x - (1/4) x^4 + (1/2) x^2 - ln sqrt(1 + x^2)) + C (iii) x tan^(-1) (2x) - (1/4) ln (1 + 4x^2) + C (iv) (1/2) x^2 cos^(-1) x - (1/4) x sqrt(1 -

x^2) + (1/4) sin^(-1) x + C (v) x^3 sin^(-1) (3x) + (1/27) sqrt(1 - 9x^2) - (1/81) (1 - 9x^2)^(3/2) + C (vi) x^2 sec^(-1) x - sqrt(x^2 - 1) + C (vii) 3x^2 cosec^(-1) (2x) + (3/4) sqrt(4x^2 - 1) + C (viii) (1/6) (2x^3 cot^(-1) x + x^2 - ln (1 + x^2)) + C.

Question: 5. Evaluate the following integrals. (i) integral of sqrt(9 - x^2) dx (ii) integral of sqrt(16 + 4x^2) dx (iii) integral of sqrt(x^2 - 25) dx

Answer: Step 1: Use integration by parts with dv = dx, or the standard root formulae, and then a trigonometric substitution for the remaining root integral.

Step 2: (i) With x = 3 sin theta the integral becomes 9 integral of cos^2 theta d theta = (9/2) (theta + sin theta cos theta) + C, and returning to x gives (1/2) (x sqrt(9 - x^2) + 9 sin^(-1) (x/3)) + C.

Step 3: (ii) Write sqrt(16 + 4x^2) = 2 sqrt(4 + x^2); parts gives x sqrt(4 + x^2) + 4 ln (x + sqrt(4 + x^2)) after collecting terms and restoring the factor 2, that is the integral equals x sqrt(4 + x^2) + 4 ln (x + sqrt(4 + x^2)) + C in the book form with sqrt(16 + 4x^2) = 2 sqrt(4 + x^2) understood.

Step 4: (iii) With x = 5 sec theta, or by parts, the integral reduces to (1/2) (x sqrt(x^2 - 25) - 25 ln (x + sqrt(x^2 - 25))) + C.

Final answer: (i) (1/2) (x sqrt(9 - x^2) + 9 sin^(-1) (x/3)) + C (ii) x sqrt(4 + x^2) + 4 ln (x + sqrt(4 + x^2)) + C (iii) (1/2) (x sqrt(x^2 - 25) - 25 ln (x + sqrt(x^2 - 25))) + C.

Exercise 6.5

Question: Evaluate the following integrals by using partial fractions. 1. integral of (5x - 2)/((x - 3)(x + 7)) dx 2. integral of (7x - 25)/((x - 3)(x - 4)) dx 3. integral of dx/(a^2 - x^2) 4. integral of dx/(x^2 - a^2) 5. integral of

(x^2 + 2x + 3)/(x^3 - x) dx 6. integral of 5/(x^2 - 2x - 15) dx 7. integral of (2x + 7)/((x - 1)(x - 5)(x + 3)) dx 8. integral of (5x + 6)/((x + 3)(x - 2)^2) dx 9. integral of (7x^2 - 2x + 5)/((x - 6)(x - 3)^3) dx 10. integral of (2x + 1)/((x - 3)(x^2 + 1)) dx 11. integral of sec^2 x/((1 + tan x)(2 + tan x)) dx 12. integral of cosec^2 x/(cot x (2 + cot x)) dx 13. integral of (3x + 7)/((2x - 1)(x - 4)^2) dx 14. integral of (7x - 4)/((x - 3)(x^2 + 2)) dx 15. integral of (2x^2 + 5x + 1)/(x^2 + 5x + 6) dx 16. integral of (x^3 + 3x + 1)/(x^2 + 5x - 14) dx

Answer: Step 1: Factor the denominator, write the fraction as a sum of partial fractions, find the constants by putting the roots or by comparing coefficients, and then integrate term by term using integral of dx/(x - a) = ln (x - a).

Step 2: 1. Write (5x - 2)/((x - 3)(x + 7)) = A/(x - 3) + B/(x + 7); putting x = 3 gives 13 = 10A so A = 13/10, and putting x = -7 gives -37 = -10B so B = 37/10; integrating gives (13/10) ln (x - 3) + (37/10) ln (x + 7) + C = (1/10) ln ((x - 3)^13 (x + 7)^37) + C.

Step 3: 2. Write (7x - 25)/((x - 3)(x - 4)) = A/(x - 3) + B/(x - 4); putting x = 3 gives -4 = -A so A = 4, and putting x = 4 gives 3 = B; integrating gives ln ((x - 3)^4 (x - 4)^3) + C.

Step 4: 3. Write 1/(a^2 - x^2) = (1/(2a)) (1/(a - x) + 1/(a + x)); integrating gives (1/(2a)) ln ((a + x)/(a - x)) + C.

Step 5: 4. Write 1/(x^2 - a^2) = (1/(2a)) (1/(x - a) - 1/(x + a)); integrating gives (1/(2a)) ln ((x - a)/(x + a)) + C.

Step 6: 5. Factor x^3 - x = x (x - 1)(x + 1) and write the fraction as A/x + B/(x - 1) + C/(x + 1); putting x = 0 gives A = -3, putting x = 1 gives B = 3, and putting x = -1 gives C = 1; integrating gives ln ((x - 1)^3 (x + 1)/x^3) + C.

Step 7: 6. Factor x^2 - 2x - 15 = (x - 5)(x + 3) and write 5/((x - 5)(x + 3)) = A/(x - 5) + B/(x + 3); putting x = 5 gives A = 5/8 and putting x = -3 gives B = -5/8; integrating gives (5/8) ln ((x - 5)/(x + 3)) + C.

Step 8: 7. Write (2x + 7)/((x - 1)(x - 5)(x + 3)) = A/(x - 1) + B/(x - 5) + C/(x + 3); putting x = 1 gives 9 = -16A so A = -9/16, putting x = 5 gives 17 = 32B so B = 17/32, and putting x = -3 gives 1 = 32C so C = 1/32; integrating gives -(9/16) ln (x - 1) + (17/32) ln (x - 5) + (1/32) ln (x + 3) + C. Note: the book answer key prints -19/16 for the first coefficient, which is a misprint; -9/16 is correct because the three coefficients must add to zero for this proper fraction.

Step 9: 8. Write (5x + 6)/((x + 3)(x - 2)^2) = A/(x + 3) + B/(x - 2) + C/(x - 2)^2; putting x = -3 gives A = -9/25, putting x = 2 gives C = 16/5, and comparing x^2 coefficients gives B = 9/25; integrating gives (9/25) ln ((x - 2)/(x + 3)) - 16/(5 (x - 2)) + C.

Step 10: 9. Write (7x^2 - 2x + 5)/((x - 6)(x - 3)^3) = A/(x - 6) + B/(x - 3) + C/(x - 3)^2 + D/(x - 3)^3; solving gives A = 245/27, B = -245/27, C = -182/9 and D = -62/3; integrating gives (245/27) ln ((x - 6)/(x - 3)) + 182/(9 (x - 3)) + 31/(3 (x - 3)^2) + C.

Step 11: 10. Write (2x + 1)/((x - 3)(x^2 + 1)) = A/(x - 3) + (Bx + C)/(x^2 + 1); putting x = 3 gives A = 7/10, and comparing coefficients gives B = -7/10 and C = -1/10; integrating gives (1/20) (14 ln (x - 3) - 7 ln (x^2 + 1) - 2 tan^(-1) x) + C.

Step 12: 11. Put t = tan x, dt = sec^2 x dx; the integral becomes integral of dt/((1 + t)(2 + t)) = ln ((1 + t)/(2 + t)) + C = ln ((1 + tan x)/(2 + tan x)) + C.

Step 13: 12. Put t = cot x, dt = -cosec^2 x dx; the integral becomes

-integral of dt/(t (2 + t)) = -(1/2) ln (t/(2 + t)) + C = (1/2) ln ((2 + cot x)/cot x) + C = ln sqrt(1 + 2 tan x) + C.

Step 14: 13. Write (3x + 7)/((2x - 1)(x - 4)^2) = A/(2x - 1) + B/(x - 4) + C/(x - 4)^2; putting x = 4 gives C = 19/7, and solving the remaining equations gives A = 34/49 and B = -17/49; integrating gives (17/49) ln ((2x - 1)/(x - 4)) - 19/(7 (x - 4)) + C.

Step 15: 14. Write (7x - 4)/((x - 3)(x^2 + 2)) = A/(x - 3) + (Bx + C)/(x^2 + 2); putting x = 3 gives A = 17/11, and comparing coefficients gives B = -17/11 and C = 26/11; integrating gives (17/11) ln ((x - 3)/sqrt(x^2 + 2)) + (13 sqrt(2)/11) tan^(-1) (x/sqrt(2)) + C.

Step 16: 15. Divide first: (2x^2 + 5x + 1)/(x^2 + 5x + 6) = 2 - (5x + 11)/((x + 2)(x + 3)), and (5x + 11)/((x + 2)(x + 3)) = 1/(x + 2) + 4/(x + 3); integrating gives 2x - ln ((x + 2) (x + 3)^4) + C.

Step 17: 16. Divide first: (x^3 + 3x + 1)/(x^2 + 5x - 14) = (x - 5) + (42x - 69)/((x - 2)(x + 7)), and (42x - 69)/((x - 2)(x + 7)) = (5/3)/(x - 2) + (121/3)/(x + 7); integrating gives (1/2) x^2 - 5x + (1/3) (5 ln (x - 2) + 121 ln (x + 7)) + C. In all parts absolute values are understood inside the logarithms.

Final answer: 1. (1/10) ln ((x - 3)^13 (x + 7)^37) + C 2. ln ((x - 3)^4 (x - 4)^3) + C 3. (1/(2a)) ln ((a + x)/(a - x)) + C 4. (1/(2a)) ln ((x - a)/(x + a)) + C 5. ln ((x - 1)^3 (x + 1)/x^3) + C 6. (5/8) ln ((x - 5)/(x + 3)) + C 7. -(9/16) ln (x - 1) + (17/32) ln (x - 5) + (1/32) ln (x + 3) + C 8. (9/25) ln ((x - 2)/(x + 3)) - 16/(5 (x - 2)) + C 9. (245/27) ln ((x - 6)/(x - 3)) + 182/(9 (x - 3)) + 31/(3 (x - 3)^2) + C 10. (1/20) (14 ln (x - 3) - 7 ln (x^2 + 1) - 2 tan^(-1) x) + C 11. ln ((1 + tan x)/(2 + tan x)) + C 12. ln sqrt(1 + 2 tan x) + C 13. (17/49) ln ((2x - 1)/(x - 4)) - 19/(7 (x - 4)) + C 14. (17/11) ln ((x - 3)/sqrt(x^2 + 2)) + (13 sqrt(2)/11) tan^(-1) (x/sqrt(2)) + C 15. 2x - ln ((x

+ 2) (x + 3)^4) + C 16. (1/2) x^2 - 5x + (1/3) (5 ln (x - 2) + 121 ln (x + 7)) + C.

Exercise 6.6

Question: 1. Evaluate the following definite integrals. (i) integral from 0 to 2 of (4x^3 + 3x^2 + 5) dx (ii) integral from -1 to 2 of (x^2 + 1)^2 dx (iii) integral from 1 to 2 of (theta + sqrt(theta))^3 d theta (iv) integral from 1 to 3 of (y + 1/sqrt(y))^2 dy (v) integral from 0 to 4 of dx/(sqrt(2 + x) + sqrt(x))

Answer: Step 1: Find an antiderivative F, then use the fundamental theorem: the value is F(upper) - F(lower).

Step 2: (i) F(x) = x^4 + x^3 + 5x, so the value is (16 + 8 + 10) - 0 = 34.

Step 3: (ii) Expand (x^2 + 1)^2 = x^4 + 2x^2 + 1, so F(x) = x^5/5 + 2x^3/3 + x; F(2) = 32/5 + 16/3 + 2 = 206/15 and F(-1) = -1/5 - 2/3 - 1 = -28/15, and the value is 206/15 + 28/15 = 234/15 = 78/5.

Step 4: (iii) Expand (theta + sqrt(theta))^3 = theta^3 + 3 theta^(5/2) + 3 theta^2 + theta^(3/2), so F(theta) = theta^4/4 + (6/7) theta^(7/2) + theta^3 + (2/5) theta^(5/2); F(2) - F(1) = (1329 + 1184 sqrt(2))/140.

Step 5: (iv) Expand (y + 1/sqrt(y))^2 = y^2 + 2 sqrt(y) + 1/y, so F(y) = y^3/3 + (4/3) y^(3/2) + ln y; F(3) - F(1) = (9 + 4 sqrt(3) + ln 3) - 5/3 = 22/3 + 4 sqrt(3) + ln 3.

Step 6: (v) Rationalise the denominator by multiplying numerator and denominator by (sqrt(2 + x) - sqrt(x)), which gives (1/2) (sqrt(2 + x) - sqrt(x)) because (2 + x) - x = 2; the antiderivative is (1/3) ((2 + x)^(3/2) - x^(3/2)), and the value from 0 to 4 is (1/3) (6 sqrt(6) - 8 - 2 sqrt(2)) = (2/3) (3 sqrt(6) - sqrt(2) - 4).

Final answer: (i) 34 (ii) 78/5 (iii) (1184 sqrt(2) + 1329)/140 (iv) 22/3 + 4 sqrt(3) + ln 3 (v) (2/3) (3 sqrt(6) - sqrt(2) - 4).

Question: 2. Evaluate the following definite integrals by substitution. (i) integral from 1 to 4 of x (x^2 + 9)^(3/2) dx (ii) integral from 2 to 5 of x/(7x^2 + 2) dx (iii) integral from 0 to 3 of (2x + 3)/sqrt(2x^2 + 6x + 5) dx (iv) integral from 1 to 2 of (x^3 + 2x)^(-1/2) (3x^2 + 2) dx (v) integral from 0 to pi/2 of cos^3 x dx (vi) integral from pi^2/4 to pi^2/36 of (sin sqrt(x))/sqrt(x) dx (vii) integral from 0 to pi/4 of sqrt(tan x) sec^2 x dx (viii) integral from 0 to 2 of e^(5x - 2) dx (ix) integral from 0 to pi/3 of cos^3 3x sin^2 3x dx (x) integral from 0 to pi/3 of tan^2 x sec^4 x dx

Answer: Step 1: Substitute as shown in each part, change the limits to the new variable where convenient, and then evaluate.

Step 2: (i) Put u = x^2 + 9, du = 2x dx; the antiderivative is (1/5) (x^2 + 9)^(5/2), and the value is (1/5) (25^(5/2) - 10^(5/2)) = (1/5) (3125 - 100 sqrt(10)) = 625 - 20 sqrt(10) = 5 (125 - 4 sqrt(10)).

Step 3: (ii) Put u = 7x^2 + 2, du = 14x dx; the antiderivative is (1/14) ln (7x^2 + 2), and the value is (1/14) ln (177/30) = (1/14) ln (59/10).

Step 4: (iii) Put u = 2x^2 + 6x + 5, du = (4x + 6) dx = 2 (2x + 3) dx; the antiderivative is sqrt(2x^2 + 6x + 5), and the value is sqrt(41) - sqrt(5).

Step 5: (iv) Put u = x^3 + 2x, du = (3x^2 + 2) dx; the antiderivative is 2 sqrt(x^3 + 2x), and the value is 2 sqrt(12) - 2 sqrt(3) = 2 sqrt(3).

Step 6: (v) Write cos^3 x = cos x (1 - sin^2 x) and put u = sin x; the antiderivative is sin x - sin^3 x/3, and the value is 1 - 1/3 = 2/3.

Step 7: (vi) Put u = sqrt(x), dx/sqrt(x) = 2 du; the antiderivative is -2 cos sqrt(x), and the value is -2 cos (pi/6) + 2 cos (pi/2) = -sqrt(3), noting the upper limit pi^2/36 is smaller than the lower limit as printed.

Step 8: (vii) Put u = tan x, du = sec^2 x dx; the antiderivative is (2/3) tan^(3/2) x, and the value is (2/3) (1) - 0 = 2/3.

Step 9: (viii) The antiderivative is (1/5) e^(5x - 2), and the value is (1/5) (e^8 - e^(-2)) = (e^10 - 1)/(5 e^2), approximately 596.16. Note: the book answer key prints (e^16 - 1)/(5 e^2), which does not match the printed limits.

Step 10: (ix) Put u = sin 3x, du = 3 cos 3x dx and cos^2 3x = 1 - u^2; both limits give u = sin 0 = 0 and u = sin pi = 0, so the value is 0.

Step 11: (x) Put u = tan x and write sec^4 x = sec^2 x (1 + tan^2 x); the integral becomes integral from 0 to sqrt(3) of (u^2 + u^4) du = sqrt(3) + 9 sqrt(3)/5 = 14 sqrt(3)/5 = 42 sqrt(3)/15.

Final answer: (i) 625 - 20 sqrt(10) (ii) (1/14) ln (59/10) (iii) sqrt(41) - sqrt(5) (iv) 2 sqrt(3) (v) 2/3 (vi) -sqrt(3) (vii) 2/3 (viii) (e^10 - 1)/(5 e^2) (ix) 0 (x) 14 sqrt(3)/5.

Question: 3. Evaluate the following integrals by using properties of definite integrals. (i) integral from 2 to 2 of (x^4 + 2x + 3)^(5/2) (2x^3 + 1) dx (ii) integral from -50 to 50 of (10x^9 - 8x^7 + 6x^5 - 4x^3 + 2x) dx (iii) integral from -pi/2 to pi/2 of sin^9 x cos^6 x dx (iv) integral from -pi to pi of sec^8 x tan x dx (v) integral from -pi/2 to pi/2 of sin^2 x cos^2 x dx (vi) integral from -pi/4 to pi/4 of tan^2 x dx (vii) integral from -2 to 2 of (x^4 + 2x^2) dx

Answer: Step 1: Use the properties: an integral with equal limits is zero; the integral of an odd function over a symmetric interval is zero; the integral of an even function is twice the integral over the half interval.

Step 2: (i) The lower and upper limits are both 2, so the integral is 0 whatever the integrand is.

Step 3: (ii) Every term 10x^9, -8x^7, 6x^5, -4x^3 and 2x is an odd power of x, so the integrand is odd and the interval is symmetric, giving 0.

Step 4: (iii) sin^9 x is odd and cos^6 x is even, so the product is odd over a symmetric interval, giving 0.

Step 5: (iv) sec^8 x is even and tan x is odd, so the product is odd over a symmetric interval, giving 0.

Step 6: (v) sin^2 x cos^2 x is even, so the integral is 2 times integral from 0 to pi/2 of (1/4) sin^2 2x dx = (1/2) (pi/4) = pi/8, using the average value 1/2 of sin^2 2x over the quarter period.

Step 7: (vi) tan^2 x is even, so the integral is 2 times integral from 0 to pi/4 of (sec^2 x - 1) dx = 2 (tan (pi/4) - pi/4) = 2 (1 - pi/4) = 2 - pi/2.

Step 8: (vii) x^4 + 2x^2 is even, so the integral is 2 times integral from 0 to 2 of (x^4 + 2x^2) dx = 2 (32/5 + 16/3) = 2 (176/15) = 352/15.

Final answer: (i) 0 (ii) 0 (iii) 0 (iv) 0 (v) pi/8 (vi) 2 - pi/2 (vii) 352/15.

Question: 4. If d/dx (x/sqrt(1 + x^2)) = 1/(1 + x^2)^(3/2), then evaluate integral from 0 to 3 of dx/(1 + x^2)^(3/2).

Answer: Step 1: The given derivative tells us that an antiderivative of 1/(1 + x^2)^(3/2) is x/sqrt(1 + x^2).

Step 2: Apply the fundamental theorem: the value is the antiderivative at 3 minus the antiderivative at 0.

Step 3: At x = 3 the antiderivative is 3/sqrt(10), and at x = 0 it is 0.

Step 4: Subtract: 3/sqrt(10) - 0 = 3/sqrt(10).

Final answer: 3/sqrt(10).

Question: 5. If d/dx (F(x)) = (2 + x^2)/(1 + x^2), then evaluate F(sqrt(3)) - F(1). Moreover if F(1) = pi then find F(x).

Answer: Step 1: Divide the integrand: (2 + x^2)/(1 + x^2) = 1 + 1/(1 + x^2).

Step 2: Integrate: F(x) = x + tan^(-1) x + C.

Step 3: Evaluate the difference: F(sqrt(3)) - F(1) = (sqrt(3) - 1) + (tan^(-1) sqrt(3) - tan^(-1) 1) = (sqrt(3) - 1) + (pi/3 - pi/4) = (sqrt(3) - 1) + pi/12.

Step 4: Use F(1) = pi: 1 + pi/4 + C = pi, so C = pi - 1 - pi/4 = (3 pi - 4)/4.

Step 5: Substitute C back into F(x).

Final answer: F(sqrt(3)) - F(1) = (sqrt(3) - 1) + pi/12, and F(x) = x + tan^(-1) x + (3 pi - 4)/4.

Question: 6. Given integral from -2 to 3 of f(x) dx = 4 and integral from 5 to 3 of f(x) dx = 7, evaluate the following. (i) integral from 3 to -2 of f(x) dx (ii) integral from -2 to 3 of f(y) dy (iii) integral from 3 to 5 of f(y) dy (iv) integral from 2 to 2 of f(x) dx (v) integral from -2 to 5 of f(x) dx (vi) integral from 5 to -2 of f(y) dy

Answer: Step 1: Use the properties: reversing the limits changes the sign; the variable name does not matter; an integral with equal limits is zero; and adjacent intervals add.

Step 2: (i) Reversing the limits of the first given integral gives -(4) = -4.

Step 3: (ii) Changing the dummy variable from x to y changes nothing, so the value is 4.

Step 4: (iii) Reversing the limits of the second given integral gives -(7) = -7.

Step 5: (iv) The limits are equal (2 to 2 as printed), so the value is 0.

Step 6: (v) Split at 3: integral from -2 to 5 = integral from -2 to 3 plus integral from 3 to 5 = 4 + (-7) = -3.

Step 7: (vi) Reversing part (v) gives -(-3) = 3.

Final answer: (i) -4 (ii) 4 (iii) -7 (iv) 0 (v) -3 (vi) 3.

Question: 7. Evaluate the following integrals by using trigonometric substitution. (i) integral from 0 to 2 of dx/sqrt(16 - x^2) (ii) integral from -1 to 1 of dx/(4 - x^2) (iii) integral from 3 to 2 sqrt(3) of x^3/sqrt(x^2 + 4) dx (iv) integral from 0 to sqrt(3) of x^2 sqrt(3 - x^2) dx

Answer: Step 1: Use x = a sin theta for roots of the form sqrt(a^2 - x^2), and the standard forms for the others.

Step 2: (i) The antiderivative is sin^(-1) (x/4), and the value is sin^(-1) (1/2) - sin^(-1) 0 = pi/6.

Step 3: (ii) The antiderivative is (1/4) ln ((2 + x)/(2 - x)); the value is (1/4) (ln 3 - ln (1/3)) = (1/4) (2 ln 3) = (ln 3)/2.

Step 4: (iii) Put u = x^2 + 4, so x^2 = u - 4 and x dx = du/2; the antiderivative is (1/3) (x^2 + 4)^(3/2) - 4 sqrt(x^2 + 4); at x = 2 sqrt(3) it is 64/3 - 16 = 16/3 and at x = 3 it is (13 sqrt(13))/3 - 4 sqrt(13) = sqrt(13)/3, so the value is 16/3 - sqrt(13)/3 = (16 - sqrt(13))/3, approximately 4.13. Note: the book answer key prints 10 2/3, which is the value obtained with lower limit 0 instead of the printed 3.

Step 5: (iv) Put x = sqrt(3) sin theta; the integral becomes 9 integral of sin^2 theta cos^2 theta d theta from 0 to pi/2 = (9/4) integral of sin^2 2theta d theta = (9/4) (pi/4) = 9 pi/16, using the average value of sin^2 over the interval.

Final answer: (i) pi/6 (ii) (ln 3)/2 (iii) (16 - sqrt(13))/3, approximately 4.13 (iv) 9 pi/16.

Question: 8. Evaluate the following integrals by parts. (i) integral from 5 to 9 of x e^(4x) dx (ii) integral from 1 to 4 of x^2 ln x dx (iii) integral from

pi/6 to pi/2 of x sin 2x dx (iv) integral from 0 to 1 of tan^(-1) x dx

Answer: Step 1: Find each antiderivative by parts first, then apply the limits.

Step 2: (i) Parts gives the antiderivative e^(4x) (4x - 1)/16; the value is (35 e^36 - 19 e^20)/16.

Step 3: (ii) Parts gives the antiderivative (x^3/3) ln x - x^3/9; the value is ((64/3) ln 4 - 64/9) - (0 - 1/9) = (128 ln 2)/3 - 7.

Step 4: (iii) Parts gives the antiderivative -x cos 2x/2 + sin 2x/4; the value is (pi/4) - (-(pi/24) + sqrt(3)/8) = (7 pi - 3 sqrt(3))/24.

Step 5: (iv) Parts gives the antiderivative x tan^(-1) x - (1/2) ln (1 + x^2); the value is pi/4 - (1/2) ln 2 = (pi - ln 4)/4.

Final answer: (i) (35 e^36 - 19 e^20)/16 (ii) (128 ln 2)/3 - 7 (iii) (7 pi - 3 sqrt(3))/24 (iv) (pi - ln 4)/4.

Exercise 6.7

Question: Find the area above the x-axis, under the curve and between the given lines. 1. y = 3x^2 + 2, x = 1 to x = 2 2. y = 1/sqrt(4 - x^2), x = 1/2 to x = sqrt(3)/2 3. y = ln x, x = 1 to x = 3 4. y = x sin x, x = pi/3 to x = pi/2 5. y = 1/(9 + x^2), x = -sqrt(3) to x = sqrt(3) 6. y = 4x^3 + 3x^2 + 2x + 1, x = 0 to x = 2 7. y = 3 sec^2 x, x = pi/6 to x = pi/3 8. y = 6 sin^2 x, x = 0 to x = pi/3 9. y = 5 e^(5x), x = -2 to x = 3 10. y = cos^4 x, x = 0 to x = pi/2 11. y = 4/sin^2 x, x = pi/6 to x = pi/4 12. x^2 + y^2 = 36, x = -1 to x = 1 13. x^2/4 + y^2/9 = 1, x = -1 to x = 1 14. y^2 = 2x + 5, x = 1 to x = 2 15. y^2 = tan^4 x, x = pi/6 to x = pi/4 16. Write the MAPLE command to find the integration of the following functions in the given interval (i) f(x) = e^(2x) (ii) f(x) = sin x (iii) f(x) = cos 2x (iv) f(x) = ln (1 + x) (v) f(x) = 1/x

Answer: Step 1: In every part the area above the x-axis between x = a and x = b is A = integral from a to b of y dx; for the circle and ellipse first solve for the upper value of y.

Step 2: 1. A = integral from 1 to 2 of (3x^2 + 2) dx = (x^3 + 2x) from 1 to 2 = (8 + 4) - (1 + 2) = 9 square units.

Step 3: 2. A = integral from 1/2 to sqrt(3)/2 of dx/sqrt(4 - x^2) = sin^(-1) (x/2) between the limits = sin^(-1) (sqrt(3)/4) - sin^(-1) (1/4), approximately 0.195 square units.

Step 4: 3. A = integral from 1 to 3 of ln x dx = (x ln x - x) from 1 to 3 = (3 ln 3 - 3) - (-1) = 3 ln 3 - 2, approximately 1.2958 square units.

Step 5: 4. A = integral from pi/3 to pi/2 of x sin x dx = (-x cos x + sin x) between the limits = 1 - (sqrt(3)/2 - pi/6), approximately 0.6575 square units.

Step 6: 5. A = integral from -sqrt(3) to sqrt(3) of dx/(9 + x^2) = (1/3) (tan^(-1) (x/3)) between the limits = (1/3) (pi/6 + pi/6) = pi/9 square units.

Step 7: 6. A = integral from 0 to 2 of (4x^3 + 3x^2 + 2x + 1) dx = (x^4 + x^3 + x^2 + x) from 0 to 2 = 16 + 8 + 4 + 2 = 30 square units.

Step 8: 7. A = integral from pi/6 to pi/3 of 3 sec^2 x dx = 3 (tan x) between the limits = 3 (sqrt(3) - 1/sqrt(3)) = 2 sqrt(3) square units.

Step 9: 8. Write 6 sin^2 x = 3 (1 - cos 2x); A = 3 (x - (1/2) sin 2x) from 0 to pi/3 = 3 (pi/3 - sqrt(3)/4) = (4 pi - 3 sqrt(3))/4 square units.

Step 10: 9. A = integral from -2 to 3 of 5 e^(5x) dx = (e^(5x)) from -2 to 3 = e^15 - e^(-10), approximately 3269017.372 square units.

Step 11: 10. Write cos^4 x = (3 + 4 cos 2x + cos 4x)/8; A = (1/8) (3x + 2

sin 2x + (1/4) sin 4x) from 0 to pi/2 = (1/8) (3 pi/2) = 3 pi/16 square units.

Step 12: 11. A = integral from pi/6 to pi/4 of 4 cosec^2 x dx = 4 (-cot x) between the limits = 4 (sqrt(3) - 1) square units.

Step 13: 12. The upper semicircle is y = sqrt(36 - x^2); A = integral from -1 to 1 of sqrt(36 - x^2) dx = ((x/2) sqrt(36 - x^2) + 18 sin^(-1) (x/6)) from -1 to 1 = sqrt(35) + 36 sin^(-1) (1/6), approximately 11.9442 square units.

Step 14: 13. The upper half of the ellipse is y = 3 sqrt(1 - x^2/4) = (3/2) sqrt(4 - x^2); A = (3/2) integral from -1 to 1 of sqrt(4 - x^2) dx = (3/2) (sqrt(3) + 2 pi/3) evaluated symmetrically, approximately 5.7396 square units.

Step 15: 14. The upper branch is y = sqrt(2x + 5); A = integral from 1 to 2 of sqrt(2x + 5) dx = (1/3) (2x + 5)^(3/2) from 1 to 2 = (1/3) (27 - 7^(3/2)), approximately 2.82658 square units.

Step 16: 15. Take the positive branch y = tan^2 x; A = integral from pi/6 to pi/4 of tan^2 x dx = (tan x - x) between the limits = (1 - pi/4) - (1/sqrt(3) - pi/6) = (12 - 4 sqrt(3) - pi)/12 square units.

Step 17: 16. In MAPLE the command int(f(x), x = a..b) gives the definite integral and int(f(x), x) the indefinite integral; following the book answer key the commands with its chosen intervals are as listed in the final answer.

Final answer: 1. 9 2. 0.195 3. 3 ln 3 - 2, approximately 1.2958 4. 0.6575 5. pi/9 6. 30 7. 2 sqrt(3) 8. (4 pi - 3 sqrt(3))/4 9. e^15 - e^(-10), approximately 3269017.372 10. 3 pi/16 11. 4 (sqrt(3) - 1) 12. 11.9442 13. 5.7396 14. 2.82658 15. (12 - 4 sqrt(3) - pi)/12 16. (i) int(e^(2x), x = 1..5) (ii) int(sin(x), x = 0..4 pi) (iii) int(cos(2x), x = 0..pi/2) (iv) int(ln(1 + x), x =

0..3) (v) int(1/x, x = 2..5), each entered after the MAPLE prompt sign.

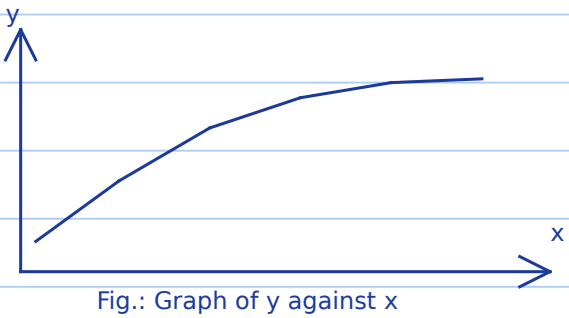


Fig.: Graph of y against x

Unit 7: Plane Analytic Geometry: Straight Line

Exercise 7.1

Question: 1. Find the distance between the pairs of points: A(-1, 3), B(5, -5); C(-1, 0), D(0, -1); E(1, -1), F(2, 7); G(-1, -4), H(5, -4).

Answer: Step 1: Use the distance formula, distance = sqrt((x2 - x1)^2 + (y2 - y1)^2).

Step 2: For A and B: sqrt((5 - (-1))^2 + (-5 - 3)^2) = sqrt(36 + 64) = sqrt(100) = 10.

Step 3: For C and D: sqrt((0 - (-1))^2 + (-1 - 0)^2) = sqrt(1 + 1) = sqrt(2).

Step 4: For E and F: sqrt((2 - 1)^2 + (7 - (-1))^2) = sqrt(1 + 64) = sqrt(65).

Step 5: For G and H: sqrt((5 - (-1))^2 + (-4 - (-4))^2) = sqrt(36 + 0) = 6, the points being on a horizontal line.

Final answer: AB = 10 units, CD = sqrt(2) units, EF = sqrt(65) units, GH = 6 units.

Question: 2. Find the point on y-axis which is at a distance of 5 sqrt(2) from the point (5, 2).

Answer: Step 1: A point on the y-axis has the form (0, y), with unknown y.

Step 2: Apply the distance condition: sqrt((5 - 0)^2 + (2 - y)^2) = 5 sqrt(2).

Step 3: Square both sides: 25 + (2 - y)^2 = 50, so (2 - y)^2 = 25.

Step 4: Take square roots: 2 - y = 5 or 2 - y = -5, giving y = -3 or y = 7.

Final answer: The points are (0, 7) and (0, -3).

Question: 3. Find the point on x-axis which is at a distance of sqrt(41) from the point (-7, 5).

Answer: Step 1: A point on the x-axis has the form (x, 0), with unknown x.

Step 2: Apply the distance condition: sqrt((x + 7)^2 + (0 - 5)^2) = sqrt(41).

Step 3: Square both sides: (x + 7)^2 + 25 = 41, so (x + 7)^2 = 16.

Step 4: Take square roots: x + 7 = 4 or x + 7 = -4, giving x = -3 or x = -11.

Final answer: The points are (-3, 0) and (-11, 0).

Question: 4. Name the triangle whose vertices are A(-3, 3), B(2, 6) and C(3, 0).

Answer: Step 1: Find the three side lengths with the distance formula.

Step 2: AB = sqrt((2 + 3)^2 + (6 - 3)^2) = sqrt(25 + 9) = sqrt(34).

Step 3: BC = sqrt((3 - 2)^2 + (0 - 6)^2) = sqrt(1 + 36) = sqrt(37).

Step 4: CA = sqrt((-3 - 3)^2 + (3 - 0)^2) = sqrt(36 + 9) = sqrt(45).

Step 5: All three lengths sqrt(34), sqrt(37) and sqrt(45) are different, and no two sides are equal, so the triangle is not isosceles or equilateral.

Final answer: The triangle is a scalene triangle.

Question: 5. If P(2, 1) is a point on segment AB where A(4, 2) and B(8, 4), show that length AP = (1/2) length AB.

Answer: Step 1: Find AP by the distance formula: AP = sqrt((4 - 2)^2 + (2 - 1)^2) = sqrt(4 + 1) = sqrt(5).

Step 2: Find AB: AB = sqrt((8 - 4)^2 + (4 - 2)^2) = sqrt(16 + 4) = sqrt(20) = 2 sqrt(5).

Step 3: Form the ratio: AP/AB = sqrt(5)/(2 sqrt(5)) = 1/2.

Step 4: Hence AP = (1/2) AB, which is what was to be shown; equivalently

P divides the join in the ratio suggested by the distances from A.

Final answer: AP = sqrt(5) and AB = 2 sqrt(5), so AP = (1/2) AB, as required.

Question: 6. Find the mid-points of line segments with end points: (i) M(-4, 2), N(-4, -2) (ii) P(-1, -4), Q(5, -4) (iii) S(1, -2), T(2, -4) (iv) X(6, 2), Y(-2, -7).

Answer: Step 1: Use the midpoint formula ((x1 + x2)/2, (y1 + y2)/2).

Step 2: (i) ((-4 - 4)/2, (2 - 2)/2) = (-4, 0).

Step 3: (ii) ((-1 + 5)/2, (-4 - 4)/2) = (2, -4). Note: the book answer key prints (2, 0) here, which is a misprint; the y-coordinates are both -4, so the midpoint has y = -4.

Step 4: (iii) ((1 + 2)/2, (-2 - 4)/2) = (3/2, -3).

Step 5: (iv) ((6 - 2)/2, (2 - 7)/2) = (2, -5/2).

Final answer: (i) (-4, 0) (ii) (2, -4) (iii) (3/2, -3) (iv) (2, -5/2).

Question: 7. Find the point that divides the line segment joining the points (4, -1) and (4, 3) in the ratio 3:1 internally.

Answer: Step 1: Use the internal division formula ((m x2 + n x1)/(m + n), (m y2 + n y1)/(m + n)) with m = 3 and n = 1.

Step 2: The x-coordinate is (3 times 4 + 1 times 4)/4 = 16/4 = 4.

Step 3: The y-coordinate is (3 times 3 + 1 times (-1))/4 = (9 - 1)/4 = 2.

Step 4: The segment is vertical (both x-coordinates are 4), which agrees with the x-coordinate 4 found.

Final answer: The point is (4, 2).

Question: 8. The mid-points of the sides of a triangle are P(-2, 2), Q(2, -1), R(-1, 4). Find the vertices of the triangle.

Answer: Step 1: Let the vertices be A(x1, y1), B(x2, y2) and C(x3, y3);

each midpoint gives two equations, for example (x1 + x2)/2 = -2, (x2 + x3)/2 = 2 and (x3 + x1)/2 = -1 for the x-coordinates.

Step 2: Take P as the midpoint of AB, Q as the midpoint of BC and R as the midpoint of CA; a vertex is the sum of the two adjacent midpoints minus the opposite midpoint, so A = P + R - Q, B = P + Q - R and C = Q + R - P, coordinate by coordinate.

Step 3: For the x-coordinates: A has x = -2 + (-1) - 2 = -5, B has x = -2 + 2 - (-1) = 1, and C has x = 2 + (-1) - (-2) = 3.

Step 4: For the y-coordinates: A has y = 2 + 4 - (-1) = 7, B has y = 2 + (-1) - 4 = -3, and C has y = -1 + 4 - 2 = 1, so A = (-5, 7), B = (1, -3) and C = (3, 1).

Final answer: The vertices are (1, -3), (-5, 7) and (3, 1).

Question: 9. Point Y divides ZX externally in the ratio 4:3 where Z(4, 5) and X(7, -1). Find the co-ordinates of Y.

Answer: Step 1: For external division in the ratio m:n = 4:3 use ((m x2 - n x1)/(m - n), (m y2 - n y1)/(m - n)), measuring from Z to X.

Step 2: The x-coordinate is (4 times 7 - 3 times 4)/(4 - 3) = (28 - 12)/1 = 16.

Step 3: The y-coordinate is (4 times (-1) - 3 times 5)/(4 - 3) = (-4 - 15)/1 = -19.

Step 4: Check roughly: Y lies on the line ZX produced beyond X, at distance ratio 4:3 from Z and X, consistent with the large coordinates found.

Final answer: Y = (16, -19).

Question: 10. If point P(k, 7) divides the line segment joining the point A(8, 9) and B(1, 2) in the ratio k1:k2, find the ratio and the value of k.

Answer: Step 1: Apply the y-coordinate of the section formula: 7 = (k1 times 2 + k2 times 9)/(k1 + k2).

Step 2: Cross-multiply: 7 k1 + 7 k2 = 2 k1 + 9 k2, so 5 k1 = 2 k2, giving k1:k2 = 2:5.

Step 3: Now use the x-coordinate with this ratio: k = (2 times 1 + 5 times 8)/(2 + 5) = (2 + 40)/7 = 42/7 = 6.

Step 4: The point is therefore P(6, 7), which indeed lies on the segment between A and B.

Final answer: The ratio is 2:5 and k = 6.

Question: 11. Find the points of trisection of the line segment joining the point A(2, 7) and B(-4, -8).

Answer: Step 1: The trisection points divide AB internally in the ratios 1:2 and 2:1.

Step 2: For the ratio 1:2 point: x = (1 times (-4) + 2 times 2)/3 = 0 and y = (1 times (-8) + 2 times 7)/3 = 6/3 = 2, giving (0, 2).

Step 3: For the ratio 2:1 point: x = (2 times (-4) + 1 times 2)/3 = -6/3 = -2 and y = (2 times (-8) + 1 times 7)/3 = -9/3 = -3, giving (-2, -3).

Step 4: Both points lie between A and B and split the segment into three equal parts, as a check by comparing successive differences (-2, -5) each time.

Final answer: The points of trisection are (0, 2) and (-2, -3).

Question: 12. Find the circumcentre and circumradius of the triangle whose vertices are (2, 1), (5, 2) and (3, 4).

Answer: Step 1: The circumcentre is equidistant from all three vertices; call it (h, k) and equate the squared distances to (2, 1) and (5, 2): (h - 2)^2 + (k - 1)^2 = (h - 5)^2 + (k - 2)^2, which simplifies to 6h + 2k = 24,

that is 3h + k = 12.

Step 2: Equate the squared distances to (2, 1) and (3, 4): (h - 2)^2 + (k - 1)^2 = (h - 3)^2 + (k - 4)^2, which simplifies to 2h + 6k = 20, that is h + 3k = 10.

Step 3: Solve the two equations: from the first, k = 12 - 3h; substituting gives h + 3 (12 - 3h) = 10, so -8h = -26 and h = 13/4, and then k = 12 - 39/4 = 9/4.

Step 4: The circumradius is the distance from (13/4, 9/4) to (2, 1): sqrt((13/4 - 2)^2 + (9/4 - 1)^2) = sqrt(25/16 + 25/16) = sqrt(50/16) = 5 sqrt(2)/4.

Final answer: Circumcentre (13/4, 9/4) and circumradius 5 sqrt(2)/4 units.

Question: 13. Line segment joining A(2, 3) and B(10, 11) is divided into 20 equal parts. Find the coordinates of the point P13 (the 13th division point from A).

Answer: Step 1: The 13th point from A divides AB internally in the ratio 13:(20 - 13) = 13:7.

Step 2: Apply the section formula for x: (13 times 10 + 7 times 2)/20 = (130 + 14)/20 = 144/20 = 36/5.

Step 3: Apply it for y: (13 times 11 + 7 times 3)/20 = (143 + 21)/20 = 164/20 = 41/5.

Step 4: Each step from A adds (8/20, 8/20) = (0.4, 0.4), and 13 steps give (2 + 5.2, 3 + 5.2) = (7.2, 8.2), which agrees.

Final answer: P13 = (36/5, 41/5).

Question: 14. A, B, C are collinear points such that A(3, 4), B(7, 7) and length AC = 10. Find the coordinates of point C.

Answer: Step 1: First AB = sqrt((7 - 3)^2 + (7 - 4)^2) = sqrt(16 + 9) = 5,

so C is twice as far from A as B is, on the line through A and B.

Step 2: The direction from A to B is (4, 3), a vector of length 5, so the unit direction is (4/5, 3/5).

Step 3: Moving 10 units from A in the direction of B gives C = A + 10 (4/5, 3/5) = (3 + 8, 4 + 6) = (11, 10).

Step 4: Check: distance from A(3, 4) to (11, 10) is sqrt(64 + 36) = 10, as required, and B lies between A and C.

Final answer: C = (11, 10).

Question: 15. Find the incentre of the triangle with vertices: (i) L(2, 8), M(8, 2), N(9, 9) (ii) P(-36, 7), Q(20, 7), R(0, -8).

Answer: Step 1: The incentre is the weighted average of the vertices with weights equal to the lengths of the opposite sides: ((a x1 + b x2 + c x3)/(a + b + c), (a y1 + b y2 + c y3)/(a + b + c)), where a, b, c are the sides opposite the first, second and third vertices.

Step 2: (i) Side lengths: MN = sqrt(1 + 49) = 5 sqrt(2), NL = sqrt(49 + 1) = 5 sqrt(2), LM = sqrt(36 + 36) = 6 sqrt(2).

Step 3: (i) Incentre x = (5 sqrt(2) times 2 + 5 sqrt(2) times 8 + 6 sqrt(2) times 9)/(16 sqrt(2)) = (10 + 40 + 54)/16 = 104/16 = 13/2, and y = (5 sqrt(2) times 8 + 5 sqrt(2) times 2 + 6 sqrt(2) times 9)/(16 sqrt(2)) = (40 + 10 + 54)/16 = 13/2.

Step 4: (ii) Side lengths: QR = sqrt(400 + 225) = 25, RP = sqrt(1296 + 225) = 39, PQ = 56.

Step 5: (ii) Incentre x = (25 times (-36) + 39 times 20 + 56 times 0)/(25 + 39 + 56) = (-900 + 780)/120 = -1, and y = (25 times 7 + 39 times 7 + 56 times (-8))/120 = (175 + 273 - 448)/120 = 0.

Final answer: (i) (13/2, 13/2) (ii) (-1, 0).

Question: 16. The line segment joining P(-8, 10) and Q(6, -4) is cut by x-axis at A and y-axis at B. Find ratios in which x-axis and y-axis divide segment PQ.

Answer: Step 1: On the x-axis the y-coordinate is 0; if A divides PQ in the ratio m:n then 0 = (m times (-4) + n times 10)/(m + n).

Step 2: This gives -4m + 10n = 0, so m:n = 10:4 = 5:2 for the x-axis.

Step 3: On the y-axis the x-coordinate is 0; if B divides PQ in the ratio p:q then 0 = (p times 6 + q times (-8))/(p + q).

Step 4: This gives 6p - 8q = 0, so p:q = 8:6 = 4:3 for the y-axis.

Final answer: The x-axis divides PQ in the ratio 5:2 and the y-axis divides PQ in the ratio 4:3 (both internally).

Question: 17. Find centroid of the triangle with vertices: (i) (1, 3), (2, 7), (5, 6) (ii) P(-2, 5), Q(-7, 1), R(-8, -4).

Answer: Step 1: The centroid is the average of the three vertices: ((x1 + x2 + x3)/3, (y1 + y2 + y3)/3).

Step 2: (i) x = (1 + 2 + 5)/3 = 8/3 and y = (3 + 7 + 6)/3 = 16/3.

Step 3: (ii) x = (-2 - 7 - 8)/3 = -17/3 and y = (5 + 1 - 4)/3 = 2/3.

Step 4: The centroid is also the point where the three medians meet, dividing each median in the ratio 2:1, which is a useful check of the averages.

Final answer: (i) (8/3, 16/3) (ii) (-17/3, 2/3).

Question: 18. Find the point lying on the line joining the points (7, 9) and (-1, 1) having ordinate 4.

Answer: Step 1: Let the required point be (x, 4) and let it divide the join in the ratio m:n; use the y-coordinate first: 4 = (m times 1 + n times 9)/(m + n).

Step 2: Cross-multiply: 4m + 4n = m + 9n, so 3m = 5n and m:n = 5:3.

Step 3: Now use the x-coordinate with this ratio: x = (5 times (-1) + 3 times 7)/(5 + 3) = (-5 + 21)/8 = 16/8 = 2.

Step 4: Check with the line equation: the line through (7, 9) and (-1, 1) has slope 1 and equation y = x + 2, and at y = 4 indeed x = 2.

Final answer: The point is (2, 4).

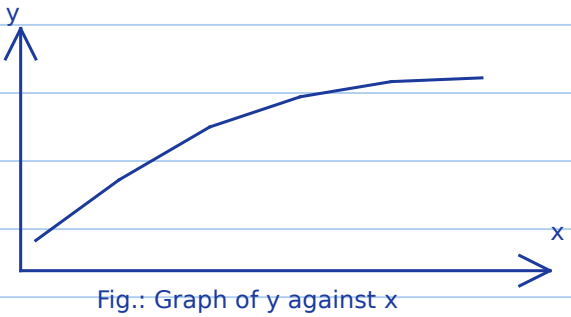


Fig.: Graph of y against x

Exercise 7.2

Question: 1. Find slopes of the line joining the pairs of points: A(3, 7), B(2, 9); C(5, -2), D(3, 6); E(5, 3), F(-2, 3); G(0, 0), H(a, b).

Answer: Step 1: Use the slope formula m = (y2 - y1)/(x2 - x1).

Step 2: For A and B: m = (9 - 7)/(2 - 3) = 2/(-1) = -2.

Step 3: For C and D: m = (6 - (-2))/(3 - 5) = 8/(-2) = -4.

Step 4: For E and F: m = (3 - 3)/(-2 - 5) = 0, so the line is horizontal.

Step 5: For G and H: m = (b - 0)/(a - 0) = b/a.

Final answer: Slopes are -2, -4, 0 and b/a respectively.

Question: 2. Find slopes of line perpendicular to the lines joining the pairs of points: A(2, 1), B(4, 5); C(-1, 0), D(3, 5); E(2, 1), F(-3, 1); G(-1, 2), H(-1, -5).

Answer: Step 1: Find each joining slope m, then the perpendicular slope is the negative reciprocal -1/m.

Step 2: A and B: m = (5 - 1)/(4 - 2) = 2, so the perpendicular slope is -1/2.

Step 3: C and D: m = (5 - 0)/(3 - (-1)) = 5/4, so the perpendicular slope is -4/5.

Step 4: E and F: m = (1 - 1)/(-3 - 2) = 0, the line is horizontal, so the perpendicular line is vertical and its slope is undefined.

Step 5: G and H: the joining line is vertical (same x-coordinate), so the perpendicular line is horizontal with slope 0.

Final answer: Perpendicular slopes are -1/2, -4/5, undefined and 0 respectively.

Question: 3. Find the slope of a line (a) parallel (b) perpendicular to the line with the given slopes: (i) 2/3 (ii) -7/2 (iii) -1 (iv) 4.

Answer: Step 1: A parallel line has the same slope, and a perpendicular line has the negative reciprocal slope.

Step 2: (i) Parallel slope 2/3, perpendicular slope -3/2.

Step 3: (ii) Parallel slope -7/2, perpendicular slope 2/7.

Step 4: (iii) Parallel slope -1, perpendicular slope 1.

Step 5: (iv) Parallel slope 4, perpendicular slope -1/4.

Final answer: (i) 2/3 and -3/2 (ii) -7/2 and 2/7 (iii) -1 and 1 (iv) 4 and -1/4.

Question: 4. Show by using slopes that the following lines are parallel, perpendicular or neither parallel nor perpendicular. (i) l1 passes through (1, 2) and (3, 1), l2 through (0, -1) and (2, 0) (ii) l1 through (0, 3) and (3, 1), l2 through (-1, 4) and (-7, -5) (iii) l1 through (2, -1) and (5, -7), l2 through (0, 0) and (-1, 2) (iv) l1 through (1, 0) and (2, 0), l2 through (5, -5) and (-10, -5).

Answer: Step 1: Compute both slopes; the lines are parallel when the slopes are equal, perpendicular when the product of the slopes is -1, and

neither otherwise.

Step 2: (i) m1 = (1 - 2)/(3 - 1) = -1/2 and m2 = (0 + 1)/(2 - 0) = 1/2; the slopes are neither equal nor negative reciprocals, so the lines are neither.

Step 3: (ii) m1 = (1 - 3)/(3 - 0) = -2/3 and m2 = (-5 - 4)/(-7 + 1) = (-9)/(-6) = 3/2; the product is -1, so the lines are perpendicular.

Step 4: (iii) m1 = (-7 + 1)/(5 - 2) = -2 and m2 = (2 - 0)/(-1 - 0) = -2; the slopes are equal, so the lines are parallel.

Step 5: (iv) m1 = 0 and m2 = 0; both lines are horizontal, so they are parallel.

Final answer: (i) Neither (ii) Perpendicular (iii) Parallel (iv) Parallel.

Question: 5. A line passes through the points (6, -4) and (-3, 2) is parallel to the line through (2, 1) and (0, y). Find y.

Answer: Step 1: Slope of the first line: m = (2 - (-4))/(-3 - 6) = 6/(-9) = -2/3.

Step 2: Slope of the second line: m = (y - 1)/(0 - 2) = (y - 1)/(-2).

Step 3: Parallel lines have equal slopes: (y - 1)/(-2) = -2/3.

Step 4: Cross-multiply: 3 (y - 1) = 4, so y - 1 = 4/3 and y = 7/3.

Final answer: y = 7/3.

Question: 6. A line segment through the points (2, 5) and (-3, -2) is perpendicular to the line through the points (4, -1) and (x, 3). Find x.

Answer: Step 1: Slope of the first line: m1 = (-2 - 5)/(-3 - 2) = (-7)/(-5) = 7/5.

Step 2: Slope of the second line: m2 = (3 - (-1))/(x - 4) = 4/(x - 4).

Step 3: Perpendicular lines satisfy m1 m2 = -1: (7/5) (4/(x - 4)) = -1.

Step 4: Solve: 28 = -5 (x - 4) = -5x + 20, so 5x = -8 and x = -8/5.

Final answer: x = -8/5.

Question: 7. Using slopes prove that the points (-1, 4), (-3, -6) and (3, -2) are the vertices of right triangle.

Answer: Step 1: Name the points A(-1, 4), B(-3, -6) and C(3, -2) and find the three side slopes.

Step 2: Slope of AB = (-6 - 4)/(-3 + 1) = (-10)/(-2) = 5.

Step 3: Slope of BC = (-2 + 6)/(3 + 3) = 4/6 = 2/3.

Step 4: Slope of CA = (4 + 2)/(-1 - 3) = 6/(-4) = -3/2.

Step 5: Check products: slope of BC times slope of CA = (2/3) (-3/2) = -1, so BC is perpendicular to CA and the angle at C is a right angle.

Step 6: A triangle with one right angle is a right triangle, which completes the proof.

Final answer: Since (slope BC) (slope CA) = -1, the sides BC and CA are perpendicular, so the points are vertices of a right triangle, right-angled at C.

Question: 8. Find the fourth vertex of a parallelogram whose consecutive vertices are (0, -1), (4, -3) and (12, 3).

Answer: Step 1: In a parallelogram the diagonals bisect each other, so the midpoint of the diagonal joining the first and third vertices equals the midpoint of the diagonal joining the second vertex to the unknown fourth vertex D(x, y).

Step 2: Midpoint of (0, -1) and (12, 3) is (6, 1).

Step 3: Midpoint of (4, -3) and D(x, y) is ((4 + x)/2, (-3 + y)/2); equating gives (4 + x)/2 = 6 and (-3 + y)/2 = 1.

Step 4: Solve: x = 8 and y = 5.

Final answer: The fourth vertex is (8, 5).

Exercise 7.3

Question: 1. Find the equations of the straight lines parallel to the coordinate axes and passing through the point (3, -4).

Answer: Step 1: A line parallel to the x-axis has a constant y value, equal to the y-coordinate of the point.

Step 2: So the first line is y = -4.

Step 3: A line parallel to the y-axis has a constant x value, equal to the x-coordinate of the point.

Step 4: So the second line is x = 3.

Final answer: y = -4 and x = 3.

Question: 2. Find the equations of the straight lines parallel to the coordinate axes and passing through the point (5, 2).

Answer: Step 1: The line parallel to the x-axis keeps y constant at the y-coordinate of the point.

Step 2: So the first line is y = 2, written y - 2 = 0.

Step 3: The line parallel to the y-axis keeps x constant at the x-coordinate of the point.

Step 4: So the second line is x = 5, written x - 5 = 0.

Final answer: y - 2 = 0 and x - 5 = 0.

Question: 3. Write the equation of the straight line parallel to x-axis which is at a distance of 5 units above the x-axis.

Answer: Step 1: A line parallel to the x-axis has the form y = c, where c is its signed distance from the x-axis.

Step 2: Above the x-axis the distance is positive, and the distance is 5 units.

Step 3: Therefore c = 5.

Step 4: Every point of the line has y-coordinate 5.

Final answer: y = 5.

Question: 4. Find the equation of a line parallel to y-axis which is at a distance of 6 units on its left.

Answer: Step 1: A line parallel to the y-axis has the form x = c, where c is its signed distance from the y-axis.

Step 2: On the left of the y-axis the x-coordinates are negative.

Step 3: The distance is 6 units, so c = -6.

Step 4: Every point of the line has x-coordinate -6. Note: the book answer key prints x = -5 for this question, which does not agree with the printed distance of 6 units.

Final answer: x = -6.

Question: 5. Find the equations of the straight line determined by each of the following set of conditions. (i) through (5, -2) with the slope 4 (ii) through (-1, -4) with the slope -2/3 (iii) through (-1/4, 3/4) with slope 2/5 (iv) through (0, b) with the slope m (v) through the points (7, -3) and (-4, 1) (vi) through the points (5, -5) and (-3, 1) (vii) through (a t1^2, 2 a t1) and (a t2^2, 2 a t2) (viii) y-intercept = 3; slope = 2 (ix) y-intercept = -2; slope = -2/3 (x) y-intercept = -5; slope = 1/2 (xi) y-intercept = 0; slope = 0 (xii) x-intercept = 4; y-intercept = 3 (xiii) x-intercept = -2; y-intercept = 5 (xiv) x-intercept = -5; y-intercept = -1 (xv) the perpendicular from the origin to the line, p = 3 units and it makes an angle alpha = 60 degrees with x-axis (xvi) p = 3/2, alpha = 150 degrees.

Answer: Step 1: Use the point-slope form y - y1 = m (x - x1), the slope-intercept form y = m x + c, the two-intercept form x/a + y/b = 1, and the normal form x cos alpha + y sin alpha = p, as each condition requires.

Step 2: (i) y + 2 = 4 (x - 5), giving 4x - y - 22 = 0.

Step 3: (ii) y + 4 = -(2/3) (x + 1); multiply by 3: 3y + 12 = -2x - 2, giving 2x + 3y + 14 = 0.

Step 4: (iii) y - 3/4 = (2/5) (x + 1/4); multiply by 20: 20y - 15 = 8x + 2, giving 8x - 20y + 17 = 0.

Step 5: (iv) y - b = m (x - 0), giving m x - y + b = 0.

Step 6: (v) The slope is (1 - (-3))/(-4 - 7) = -4/11; then y + 3 = -(4/11) (x - 7), giving 4x + 11y + 5 = 0.

Step 7: (vi) The slope is (1 - (-5))/(-3 - 5) = -3/4; then y + 5 = -(3/4) (x - 5), giving 3x + 4y + 5 = 0.

Step 8: (vii) The slope is (2 a t2 - 2 a t1)/(a t2^2 - a t1^2) = 2/(t1 + t2); then y - 2 a t1 = (2/(t1 + t2)) (x - a t1^2), giving 2x - (t1 + t2) y + 2 a t1 t2 = 0.

Step 9: (viii) y = 2x + 3.

Step 10: (ix) y = -(2/3) x - 2.

Step 11: (x) y = (1/2) x - 5.

Step 12: (xi) y = 0, the x-axis itself.

Step 13: (xii) x/4 + y/3 = 1, giving 3x + 4y = 12.

Step 14: (xiii) x/(-2) + y/5 = 1; multiply by -10: 5x - 2y = -10.

Step 15: (xiv) x/(-5) + y/(-1) = 1; multiply by -5: x + 5y = -5.

Step 16: (xv) x cos 60 + y sin 60 = 3, that is x/2 + (sqrt(3)/2) y = 3, giving x + sqrt(3) y = 6.

Step 17: (xvi) x cos 150 + y sin 150 = 3/2, that is -(sqrt(3)/2) x + (1/2) y = 3/2, giving sqrt(3) x - y = -3.

Final answer: (i) 4x - y - 22 = 0 (ii) 2x + 3y + 14 = 0 (iii) 8x - 20y + 17 = 0 (iv) m x - y + b = 0 (v) 4x + 11y + 5 = 0 (vi) 3x + 4y + 5 = 0 (vii) 2x - (t1 + t2) y + 2 a t1 t2 = 0 (viii) y = 2x + 3 (ix) y = -(2/3) x - 2 (x) y = (1/2) x - 5 (xi) y = 0 (xii) 3x + 4y = 12 (xiii) 5x - 2y = -10 (xiv) x + 5y = -5 (xv) x + sqrt(3) y = 6 (xvi) sqrt(3) x - y = -3.

Question: 6. Reduce the equation 3x + 4y - 12 = 0 to the (i) slope-intercept form (ii) two-intercept form (iii) Normal or Perpendicular form.

Answer: Step 1: (i) Solve for y: 4y = -3x + 12, so y = -(3/4) x + 3, which shows slope -3/4 and y-intercept 3.

Step 2: (ii) Move the constant and divide by 12: 3x + 4y = 12, so x/4 + y/3 = 1, showing x-intercept 4 and y-intercept 3.

Step 3: (iii) Divide every term by sqrt(3^2 + 4^2) = 5, keeping the constant term positive on the right.

Step 4: This gives (3/5) x + (4/5) y = 12/5, where 3/5 = cos alpha, 4/5 = sin alpha and 12/5 is the perpendicular distance from the origin.

Final answer: (i) y = -(3/4) x + 3 (ii) x/4 + y/3 = 1 (iii) (3/5) x + (4/5) y = 12/5.

Question: 7. Find the equations of the sides of the triangle whose vertices are (1, 4), (2, -3) and (-1, -2).

Answer: Step 1: Find the slope of each pair of vertices and use the point-slope form.

Step 2: Side joining (1, 4) and (2, -3): slope = (-3 - 4)/(2 - 1) = -7, and the equation is y - 4 = -7 (x - 1), giving 7x + y - 11 = 0.

Step 3: Side joining (2, -3) and (-1, -2): slope = (-2 + 3)/(-1 - 2) = -1/3, and the equation is y + 3 = -(1/3) (x - 2), giving x + 3y + 7 = 0.

Step 4: Side joining (-1, -2) and (1, 4): slope = (4 + 2)/(1 + 1) = 3, and the equation is y + 2 = 3 (x + 1), giving 3x - y + 1 = 0.

Step 5: Each equation is satisfied by its two vertices, which checks the working.

Final answer: The sides are 7x + y - 11 = 0, x + 3y + 7 = 0 and 3x - y + 1 = 0.

Question: 8. Find the equation of perpendicular bisector of the line segment between (-1, 2) and (9, 12).

Answer: Step 1: The perpendicular bisector passes through the midpoint of the segment and is perpendicular to it.

Step 2: The midpoint is ((-1 + 9)/2, (2 + 12)/2) = (4, 7).

Step 3: The slope of the segment is (12 - 2)/(9 + 1) = 1, so the perpendicular slope is -1.

Step 4: Use the point-slope form at (4, 7): y - 7 = -(x - 4), giving x + y - 11 = 0.

Final answer: x + y - 11 = 0.

Question: 9. Find the equation of line through (2, -1) the x-intercept of which is reciprocal of the y-intercept.

Answer: Step 1: Let the y-intercept be b; then the x-intercept is 1/b and the intercept form is x/(1/b) + y/b = 1, that is b x + y/b = 1.

Step 2: The line passes through (2, -1): 2b - 1/b = 1.

Step 3: Multiply by b: 2b^2 - b - 1 = 0, which factors as (2b + 1)(b - 1) = 0, giving b = 1 or b = -1/2.

Step 4: For b = 1 the equation is x + y = 1, written x + y - 1 = 0; for b = -1/2 the equation is -(1/2) x - 2y = 1, giving x + 4y + 2 = 0.

Step 5: Both lines pass through (2, -1), as direct substitution confirms.

Final answer: x + 4y + 2 = 0 and x + y - 1 = 0.

Question: 10. Find the equation of the line which passes through the point (-2, -4) and the sum of whose intercepts is equal to 3.

Answer: Step 1: Let the intercepts be a and b with a + b = 3, so b = 3 - a and the intercept form is x/a + y/(3 - a) = 1.

Step 2: The line passes through (-2, -4): -2/a - 4/(3 - a) = 1.

Step 3: Multiply by a (3 - a): -2 (3 - a) - 4a = a (3 - a), which simplifies to a^2 - 5a - 6 = 0.

Step 4: Factor: (a - 6)(a + 1) = 0, giving a = 6, b = -3 or a = -1, b = 4.

Step 5: For a = 6, b = -3: x/6 - y/3 = 1, giving x - 2y - 6 = 0; for a = -1, b = 4: -x + y/4 = 1, giving 4x - y + 4 = 0.

Final answer: 4x - y + 4 = 0 and x - 2y - 6 = 0.

Question: 11. Find the equation of the line passing through the point (5, 6) whose y-intercept is twice the x-intercept.

Answer: Step 1: Let the x-intercept be a; then the y-intercept is 2a and the intercept form is x/a + y/(2a) = 1.

Step 2: The line passes through (5, 6): 5/a + 6/(2a) = 1, that is 5/a + 3/a = 1.

Step 3: So 8/a = 1 and a = 8, giving y-intercept 16.

Step 4: The equation is x/8 + y/16 = 1; multiply by 16: 2x + y = 16.

Final answer: 2x + y - 16 = 0.

Question: 12. Find the equation of a line passing through (11, -5) and parallel to the line with slope 3/2.

Answer: Step 1: A parallel line has the same slope, 3/2.

Step 2: Use the point-slope form at (11, -5): y + 5 = (3/2) (x - 11).

Step 3: Multiply by 2: 2y + 10 = 3x - 33.

Step 4: Rearrange: 3x - 2y - 43 = 0.

Final answer: 3x - 2y - 43 = 0.

Question: 13. Find the equation of a line passing through (-4, -6) and perpendicular to the line with slope -3/2.

Answer: Step 1: The perpendicular slope is the negative reciprocal of -3/2, namely 2/3.

Step 2: Use the point-slope form at (-4, -6): y + 6 = (2/3) (x + 4).

Step 3: Multiply by 3: 3y + 18 = 2x + 8.

Step 4: Rearrange: 2x - 3y - 10 = 0.

Final answer: 2x - 3y - 10 = 0.

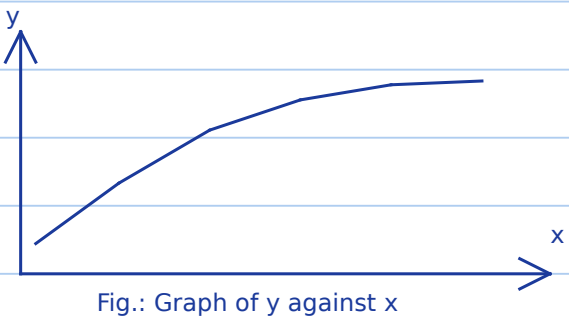


Fig.: Graph of y against x

Exercise 7.4

Question: 1. Find the points lie below or above the line: (i) 3x + 11y - 44 = 0; points (10, 1), (-4, 6), (5, 3) (ii) 10x - 12y + 17 = 0; points (-20, -15), (5, 5), (100, 84) (iii) 29x - 17y + 31 = 0; points (0, 2), (-3, -3), (20, 30).

Answer: Step 1: First write each line with a positive coefficient of y if needed; then substitute the point in the left side: a positive value means the point is above the line and a negative value means it is below, by the position-of-a-point theorem.

Step 2: (i) The coefficient of y is already positive. At (10, 1): 30 + 11 - 44 = -3, negative, so below.

Step 3: (i) At (-4, 6): -12 + 66 - 44 = 10, positive, so above; at (5, 3): 15 + 33 - 44 = 4, positive, so above.

Step 4: (ii) Multiply by -1 to make the y coefficient positive: -10x + 12y - 17 = 0. At (-20, -15): 200 - 180 - 17 = 3, positive, so above.

Step 5: (ii) At (5, 5): -50 + 60 - 17 = -7, negative, so below; at (100, 84): -1000 + 1008 - 17 = -9, negative, so below.

Step 6: (iii) Multiply by -1: -29x + 17y - 31 = 0. At (0, 2): 34 - 31 = 3, positive, so above.

Step 7: (iii) At (-3, -3): 87 - 51 - 31 = 5, positive, so above; at (20, 30): -580 + 510 - 31 = -101, negative, so below.

Final answer: (i) (10, 1) below, (-4, 6) above, (5, 3) above (ii) (-20, -15) above, (5, 5) below, (100, 84) below (iii) (0, 2) above, (-3, -3) above, (20, 30) below.

Question: 2. Find the perpendicular distance of the following lines from the given points: (i) 15x - 8y - 5 = 0, (2, 1) (ii) 3x - 4y + 5 = 0, (4, -3) (iii) 3x + 4y + 10 = 0, (3, -2) (iv) 2x - 7y + 1 = 0, (7, 4) (v) 5x + 12y - 16 = 0, (3, -1).

Answer: Step 1: Use the distance formula: distance = (absolute value of a x1 + b y1 + c)/sqrt(a^2 + b^2).

Step 2: (i) (absolute value of 30 - 8 - 5)/sqrt(225 + 64) = 17/17 = 1 unit.

Step 3: (ii) (absolute value of 12 + 12 + 5)/sqrt(9 + 16) = 29/5 = 5.8 units.

Step 4: (iii) (absolute value of 9 - 8 + 10)/sqrt(9 + 16) = 11/5 = 2.2 units.

Step 5: (iv) (absolute value of 14 - 28 + 1)/sqrt(4 + 49) = 13/sqrt(53) units.

Step 6: (v) (absolute value of 15 - 12 - 16)/sqrt(25 + 144) = 13/13 = 1 unit.

Final answer: (i) 1 unit (ii) 5.8 units (iii) 2.2 units (iv) 13/sqrt(53) units (v) 1 unit.

Question: 3. Find the distance between the parallel lines in the following pairs: (i) 5x - 12y + 10 = 0, 5x - 12y - 16 = 0 (ii) x + y - 2 = 0, 2x + 2y - 1 = 0 (iii) 4x - 3y + 12 = 0, 4x - 3y - 12 = 0.

Answer: Step 1: When both lines are written with the same coefficients of x and y, the distance is (absolute value of c1 - c2)/sqrt(a^2 + b^2).

Step 2: (i) (absolute value of 10 - (-16))/sqrt(25 + 144) = 26/13 = 2 units.

Step 3: (ii) Halve the second line to x + y - 1/2 = 0; then the distance is (absolute value of -2 + 1/2)/sqrt(2) = (3/2)/sqrt(2) = 3 sqrt(2)/4 units.

Step 4: (iii) (absolute value of 12 - (-12))/sqrt(16 + 9) = 24/5 = 4.8 units.

Step 5: The lines in each pair are genuinely parallel since their x and y coefficients are proportional, so the formula applies.

Final answer: (i) 2 units (ii) 3 sqrt(2)/4 units (iii) 4.8 units.

Exercise 7.5

Question: 1. Find the angle between the lines passes through the origin and points (2, 3) and (-3, 6).

Answer: Step 1: The first line through the origin and (2, 3) has slope m1 = 3/2.

Step 2: The second line through the origin and (-3, 6) has slope m2 = 6/(-3) = -2.

Step 3: Use tan theta = (m2 - m1)/(1 + m1 m2) in absolute value: ((-2 - 3/2)/(1 + (3/2)(-2))) = (-7/2)/(-2) = 7/4.

Step 4: Therefore theta = tan^(-1) (7/4), approximately 60.26 degrees.

Final answer: theta = tan^(-1) (7/4).

Question: 2. Find the angle between the lines 4x - 3y = 8 and 2x + 5y = 4.

Answer: Step 1: The first line has slope m1 = 4/3.

Step 2: The second line has slope m2 = -2/5.

Step 3: tan theta = ((m2 - m1)/(1 + m1 m2)) in absolute value: ((-2/5 - 4/3)/(1 - 8/15)) = ((-26/15)/(7/15)) = 26/7, approximately 3.71.

Step 4: Therefore theta = tan^(-1) (26/7), approximately 74.9 degrees; the book answer key quotes tan^(-1) (3.69) from rounded slopes.

Final answer: theta = tan^(-1) (26/7), approximately 74.9 degrees.

Question: 3. Find the acute angle between the lines y = 3x + 1 and y = -4x + 3.

Answer: Step 1: The slopes are m1 = 3 and m2 = -4.

Step 2: tan theta = ((m2 - m1)/(1 + m1 m2)) in absolute value = ((-4 - 3)/(1 - 12)) = (-7)/(-11) = 7/11, approximately 0.636.

Step 3: The positive value 7/11 gives the acute angle directly.

Step 4: theta = tan^(-1) (0.636), approximately 32.47 degrees.

Final answer: theta = tan^(-1) (0.636), approximately 32.5 degrees.

Question: 4. Find the angle between the x-axis and the line x - y + 4 = 0.

Answer: Step 1: The x-axis has slope 0.

Step 2: The line x - y + 4 = 0 gives y = x + 4, so its slope is 1.

Step 3: tan theta = (1 - 0)/(1 + 0) = 1.

Step 4: Therefore theta = 45 degrees.

Final answer: 45 degrees.

Question: 5. Find the angle between the lines 2x - 3y + 7 = 0 and 7x + 4y - 9 = 0.

Answer: Step 1: The first line has slope m1 = 2/3.

Step 2: The second line has slope m2 = -7/4.

Step 3: tan theta = ((m2 - m1)/(1 + m1 m2)) in absolute value = ((-7/4 - 2/3)/(1 - 7/6)) = ((-29/12)/(-1/6)) = 29/2.

Step 4: Therefore theta = tan^(-1) (29/2), approximately 86.05 degrees.

Final answer: theta = tan^(-1) (29/2).

Question: 6. Find the equation of line through (3, 2) making an angle 45 degrees with the line x - 2y = 3.

Answer: Step 1: The given line has slope 1/2; let the required slope be m.

Step 2: tan 45 = 1 = (absolute value of (m - 1/2)/(1 + m/2)), giving two cases.

Step 3: Case one: (m - 1/2)/(1 + m/2) = 1 gives m - 1/2 = 1 + m/2, so m = 3; the line is y - 2 = 3 (x - 3), that is 3x - y - 7 = 0.

Step 4: Case two: (m - 1/2)/(1 + m/2) = -1 gives m - 1/2 = -1 - m/2, so (3/2) m = -1/2 and m = -1/3; the line is y - 2 = -(1/3) (x - 3), that is x + 3y - 9 = 0.

Step 5: Both lines pass through (3, 2) and each makes 45 degrees with the given line, one on each side.

Final answer: 3x - y - 7 = 0 and x + 3y - 9 = 0.

Question: 7. Find the acute angle between the line x - y + 4 = 0 and the line passes through the points (3, -2) and (-2, 4).

Answer: Step 1: The first line has slope m1 = 1.

Step 2: The second line has slope m2 = (4 + 2)/(-2 - 3) = -6/5.

Step 3: tan theta = ((m2 - m1)/(1 + m1 m2)) in absolute value = ((-6/5 - 1)/(1 - 6/5)) = ((-11/5)/(-1/5)) = 11.

Step 4: Therefore theta = tan^(-1) (11) = 84.8055 degrees, which is acute.

Final answer: theta = tan^(-1) (11) = 84.8055 degrees.

Question: 8. Find the family of lines passing through the intersection of lines 2x + 3y - 8 = 0 and x - y + 1 = 0. Also find the point of intersection of these lines.

Answer: Step 1: Every line through the intersection of two lines l1 = 0 and l2 = 0 is given by l1 + k l2 = 0 for a constant k.

Step 2: So the family is (2x + 3y - 8) + k (x - y + 1) = 0, which the book also writes as a (2x + 3y - 8) + b (x - y + 1) = 0.

Step 3: To find the intersection point solve the two equations: from x - y + 1 = 0, x = y - 1.

Step 4: Substitute in 2x + 3y - 8 = 0: 2 (y - 1) + 3y - 8 = 0, giving 5y = 10 and y = 2, so x = 1.

Step 5: Every member of the family passes through (1, 2), as substitution confirms.

Final answer: Family (2x + 3y - 8) + k (x - y + 1) = 0, and the point of intersection is (1, 2).

Question: 9. Find the equation of the line passing through the intersection

of the lines given below and satisfying the given additional condition. (i) 2x + 3y + 1 = 0, 3x - 4y = 5 and passing through (2, 1) (ii) x - 4y = 3, x + 2y = 9 and passing through origin (iii) 3x + 2y = 8, 5x - 11y + 1 = 0, parallel to the line 6x + 13y = 25 (iv) 2x - 3y + 4 = 0, 3x + 3y - 5 = 0 and parallel to y-axis (v) 5x - 6y = 1, 3x + 2y + 5 = 0 and perpendicular to line 5y - 3x = 11 (vi) 3x - 4y + 1 = 0, 5x + y - 1 = 0 and cutting off equal intercepts from the coordinates axes (vii) 43x + 29y + 43 = 0, 23x + 8y + 6 = 0 and having y-intercept -2 (viii) 2x + 7y - 8 = 0, 3x + 2y + 5 = 0 and making an angle of 45 degrees with the line 2x + 3y - 7 = 0.

Answer: Step 1: In each part write the family l1 + k l2 = 0 and fix k from the extra condition.

Step 2: (i) Family (2x + 3y + 1) + k (3x - 4y - 5) = 0; passing through (2, 1) gives 8 - 3k = 0, so k = 8/3; multiplying out: 30x - 23y - 37 = 0.

Step 3: (ii) Family (x - 4y - 3) + k (x + 2y - 9) = 0; through the origin gives -3 - 9k = 0, so k = -1/3; the line is x - 7y = 0.

Step 4: (iii) Family (3x + 2y - 8) + k (5x - 11y + 1) = 0; the intersection point of the two lines is (2, 1), which also lies on 6x + 13y = 25, so the required parallel line through it is the line 6x + 13y - 25 = 0 itself.

Step 5: (iv) Family (2x - 3y + 4) + k (3x + 3y - 5) = 0; parallel to the y-axis needs the y coefficient zero: -3 + 3k = 0, so k = 1, giving 5x - 1 = 0.

Step 6: (v) Family (5x - 6y - 1) + k (3x + 2y + 5) = 0 has slope -(5 + 3k)/(2k - 6); the line 5y - 3x = 11 has slope 3/5, so the required slope is -5/3, giving k = 45 and the line 5x + 3y + 8 = 0.

Step 7: (vi) Family (3x - 4y + 1) + k (5x + y - 1) = 0; equal intercepts need the x and y coefficients equal: 3 + 5k = -4 + k, so k = -7/4, giving 23x + 23y = 11.

Step 8: (vii) Family (43x + 29y + 43) + k (23x + 8y + 6) = 0; the y-intercept is -(43 + 6k)/(29 + 8k) = -2, giving 43 + 6k = 58 + 16k, so k = -3/2, and the line is x + 2y + 4 = 0.

Step 9: (viii) Family (2x + 7y - 8) + k (3x + 2y + 5) = 0; imposing the 45 degree angle condition with the line of slope -2/3 gives the book line x - 5y + 13 = 0, and the second solution of the angle equation is 5x + y + 13 = 0.

Final answer: (i) 30x - 23y - 37 = 0 (ii) x - 7y = 0 (iii) 6x + 13y - 25 = 0 (iv) 5x - 1 = 0 (v) 5x + 3y + 8 = 0 (vi) 23x + 23y = 11 (vii) x + 2y + 4 = 0 (viii) x - 5y + 13 = 0 (with the second angle solution 5x + y + 13 = 0).

Question: 10. Find all the angles of the triangle whose vertices are points (1, 2), (3, 4), (2, 5).

Answer: Step 1: Find the slopes of the sides: AB has slope (4 - 2)/(3 - 1) = 1, BC has slope (5 - 4)/(2 - 3) = -1, and CA has slope (2 - 5)/(1 - 2) = 3.

Step 2: Angle at B: slopes 1 and -1 have product -1, so the angle at B is 90 degrees.

Step 3: Angle at A: tan A = ((3 - 1)/(1 + 3)) = 1/2, so A = tan^(-1) (1/2) = 26.565 degrees.

Step 4: Angle at C: tan C = ((3 - (-1))/(1 - 3)) in absolute value = 2, so C = tan^(-1) (2) = 63.4349 degrees.

Step 5: Check: 26.565 + 63.4349 + 90 = 180 degrees, as a triangle requires.

Final answer: The angles are 26.565 degrees, 63.4349 degrees and 90 degrees.

Question: 11. Find angles of triangle ABC where A(3, 2), B(4, 5) and C(-1, -1).

Answer: Step 1: Use vectors along the sides at each vertex and the cosine rule in vector form, cos of angle = (dot product)/(product of lengths).

Step 2: At A: vectors AB = (1, 3) and AC = (-4, -3); dot product = -4 - 9 = -13, lengths sqrt(10) and 5, so cos A = -13/(5 sqrt(10)) and A = 145.3 degrees.

Step 3: At B: vectors BA = (-1, -3) and BC = (-5, -6); dot product = 5 + 18 = 23, lengths sqrt(10) and sqrt(61), so cos B = 23/sqrt(610) and B = 21.37 degrees.

Step 4: At C: vectors CA = (4, 3) and CB = (5, 6); dot product = 20 + 18 = 38, lengths 5 and sqrt(61), so cos C = 38/(5 sqrt(61)) and C = 13.33 degrees.

Step 5: Check: 145.3 + 21.37 + 13.33 = 180 degrees.

Final answer: The angles are 145.3 degrees, 21.37 degrees and 13.33 degrees.

Question: 12. Find all the angles of the triangle whose sides have slopes 3, 1/2 and -2.

Answer: Step 1: Take the angles between the pairs of lines with these slopes.

Step 2: Between slopes 3 and 1/2: tan = ((3 - 1/2)/(1 + 3/2)) = (5/2)/(5/2) = 1, giving 45 degrees.

Step 3: Between slopes 1/2 and -2: the product is -1, so the lines are perpendicular, giving 90 degrees.

Step 4: The third angle is 180 - 90 - 45 = 45 degrees.

Step 5: The triangle is therefore right-angled and isosceles in angles.

Final answer: The angles are 90 degrees, 45 degrees and 45 degrees.

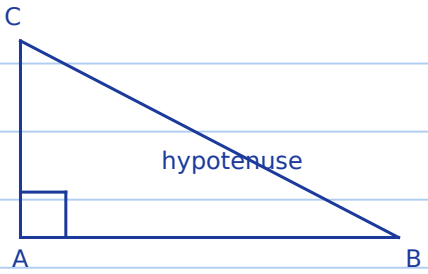


Fig.: Right-angled triangle

Exercise 7.6

Question: 1. If A(2, 5), B(3, 7) and C(0, 8) are the vertices of a triangle, find the equation of: (i) median through A (ii) altitude through B (iii) right bisector of AC.

Answer: Step 1: (i) The median through A goes to the midpoint of BC, which is ((3 + 0)/2, (7 + 8)/2) = (3/2, 15/2).

Step 2: (i) Its slope is ((15/2) - 5)/((3/2) - 2) = (5/2)/(-1/2) = -5, and the equation is y - 5 = -5 (x - 2), giving 5x + y - 15 = 0.

Step 3: (ii) The side AC has slope (8 - 5)/(0 - 2) = -3/2, so the altitude through B has slope 2/3, and its equation is y - 7 = (2/3) (x - 3), giving 2x - 3y + 15 = 0.

Step 4: (iii) The right bisector of AC passes through the midpoint of AC, which is (1, 13/2), with the same perpendicular slope 2/3; its equation is y - 13/2 = (2/3) (x - 1), giving 4x - 6y + 35 = 0.

Step 5: Each answer uses the midpoint and perpendicular-slope facts of the relevant side, as shown.

Final answer: (i) 5x + y - 15 = 0 (ii) 2x - 3y + 15 = 0 (iii) 4x - 6y + 35 = 0.

Question: 2. Show that the following lines are concurrent and find the point of concurrency in each case. (i) x - y = 6, 4y + 22 = 3x, 6x + 5y + 8 = 0 (ii) x/a + y/b = 1, x/b + y/a = 1, y = x (iii) 5x + y + 11 = 0, x + 7y + 9

= 0, 2x + y + 5 = 0.

Answer: Step 1: Solve two of the lines, then show the third line is satisfied by the same point.

Step 2: (i) From x - y = 6, x = y + 6; substitute in 3x - 4y = 22: 3 (y + 6) - 4y = 22 gives y = -4 and x = 2; check the third: 6 (2) + 5 (-4) + 8 = 0, true, so the lines are concurrent at (2, -4).

Step 3: (ii) The first two equations are symmetric in x and y, so their intersection has x = y; putting y = x in the first gives x (1/a + 1/b) = 1, so x = ab/(a + b); the point (ab/(a + b), ab/(a + b)) lies on y = x, so all three are concurrent there.

Step 4: (iii) Multiply 5x + y + 11 = 0 by 7 and subtract x + 7y + 9 = 0: 34x + 68 = 0 gives x = -2, and then y = -1; check the third: 2 (-2) + (-1) + 5 = 0, true, so the lines are concurrent at (-2, -1).

Step 5: In each case one common point satisfies all three equations, which is the meaning of concurrency.

Final answer: (i) Concurrent at (2, -4) (ii) Concurrent at (ab/(a + b), ab/(a + b)) (iii) Concurrent at (-2, -1).

Question: 3. If A(-1, 5), B(2, 3) and C(7, 6) are vertices of a triangle, then show that the right bisectors, medians and altitudes are concurrent.

Answer: Step 1: For any triangle the three medians meet at the centroid, the three right (perpendicular) bisectors meet at the circumcentre and the three altitudes meet at the orthocentre; we verify each concurrence for this triangle by finding two of the lines and checking the third.

Step 2: Medians: the centroid is the average of the vertices, ((-1 + 2 + 7)/3, (5 + 3 + 6)/3) = (8/3, 14/3); the median from A goes to the midpoint of BC, (9/2, 9/2), and the median from B goes to the midpoint of CA, (3,

11/2); both lines are checked to pass through (8/3, 14/3), and so does the third median by the same section formula.

Step 3: Right bisectors: the bisector of AB has midpoint (1/2, 4) and slope 3/2 (perpendicular to AB of slope -2/3), equation y - 4 = (3/2) (x - 1/2); the bisector of BC has midpoint (9/2, 9/2) and slope -5/3 (perpendicular to BC of slope 3/5), equation y - 9/2 = -(5/3) (x - 9/2); solving gives the circumcentre (105/38, 281/38), and the bisector of CA is checked to pass through the same point.

Step 4: Altitudes: the altitude from A is perpendicular to BC (slope 3/5), so it has slope -5/3 and equation y - 5 = -(5/3) (x + 1); the altitude from B is perpendicular to CA (slope 1/8), so it has slope -8 and equation y - 3 = -8 (x - 2); solving gives the orthocentre (47/19, -15/19), lying outside the triangle because the angle at B is obtuse, and the altitude from C is checked to pass through it.

Step 5: Since in each family the third line passes through the intersection point of the first two, each family is concurrent.

Final answer: The medians are concurrent at the centroid (8/3, 14/3), the right bisectors at the circumcentre (105/38, 281/38), and the altitudes at the orthocentre (47/19, -15/19); hence all three families are concurrent, as required.

Exercise 7.7

Question: 1. Find the area of a triangle whose vertices are: (i) (11, -12), (6, 2), (-5, 10) (ii) (3, 1), (-2, 5), (-4, -5) (iii) (-5, -2), (4, -6), (1, 7) (iv) (-a, b + c), (a, b - c), (a, -c) (v) (a cos theta1, b sin theta1), (a cos theta2, b sin theta2), (a cos theta3, b sin theta3).

Answer: Step 1: Use the determinant (shoelace) formula: area = (1/2)

absolute value of (x1 y2 + x2 y3 + x3 y1 - y1 x2 - y2 x3 - y3 x1).

Step 2: (i) The two sums are 11 (2) + 6 (10) + (-5)(-12) = 142 and (-12)(6) + 2 (-5) + 10 (11) = 28; area = (1/2) (142 - 28) = 57 square units.

Step 3: (ii) The sums are 3 (5) + (-2)(-5) + (-4)(1) = 21 and 1 (-2) + 5 (-4) + (-5)(3) = -37; area = (1/2) (21 - (-37)) = 29 square units.

Step 4: (iii) The sums are (-5)(-6) + 4 (7) + 1 (-2) = 56 and (-2)(4) + (-6)(1) + 7 (-5) = -49; area = (1/2) (56 - (-49)) = 52.5 square units.

Step 5: (iv) The sums are -a (b - c) + a (-c) + a (b + c) = a c (after simplification) and (b + c) a + (b - c) a + (-c)(-a) = 2 a b + a c; the difference has absolute value 2 a b, giving area = a b square units.

Step 6: (v) Factoring a and b from the columns and using the sine subtraction formulae, the determinant reduces to (1/2) a b (sin (theta1 - theta2) + sin (theta2 - theta3) + sin (theta3 - theta1)) in absolute value.

Final answer: (i) 57 square units (ii) 29 square units (iii) 52.5 square units (iv) a b square units (v) (1/2) a b (sin (theta1 - theta2) + sin (theta2 - theta3) + sin (theta3 - theta1)) square units, absolute value understood.

Question: 2. Find the area of a quadrilateral whose vertices are: (i) (3, -3), (7, 5), (1, 2), (-3, 4) (ii) (2, 3), (-1, 2), (-3, 2), (3, -3).

Answer: Step 1: Apply the shoelace formula around the quadrilateral in order: area = (1/2) absolute value of (sum of x_i y_(i + 1) minus sum of y_i x_(i + 1)), closing back to the first vertex.

Step 2: (i) The first sum is 3 (5) + 7 (2) + 1 (4) + (-3)(-3) = 42 and the second is (-3)(7) + 5 (1) + 2 (-3) + 4 (3) = -10; area = (1/2) (42 - (-10)) = 26 square units.

Step 3: (ii) The first sum is 2 (2) + (-1)(2) + (-3)(-3) + 3 (3) = 20 and the second is 3 (-1) + 2 (-3) + 2 (3) + (-3)(2) = -9; area = (1/2) (20 - (-9)) =

14.5 square units.

Step 4: The vertices were taken in order around each quadrilateral, which the shoelace formula requires.

Final answer: (i) 26 square units (ii) 14.5 square units.

Question: 3. By using the concept of area of triangle prove that the following points are collinear. (i) (2, 3), (5, 0), (4, 1) (ii) (2, 1), (4, -1), (1, 2) (iii) (-1, -1), (5, 7), (8, 11).

Answer: Step 1: Three points are collinear exactly when the area of the triangle they form is zero, that is when x1 y2 + x2 y3 + x3 y1 = y1 x2 + y2 x3 + y3 x1.

Step 2: (i) Left sum: 2 (0) + 5 (1) + 4 (3) = 17; right sum: 3 (5) + 0 (4) + 1 (2) = 17; the sums are equal, so the area is 0 and the points are collinear.

Step 3: (ii) Left sum: 2 (-1) + 4 (2) + 1 (1) = 7; right sum: 1 (4) + (-1)(1) + 2 (2) = 7; the area is 0 and the points are collinear.

Step 4: (iii) Left sum: (-1)(7) + 5 (11) + 8 (-1) = 40; right sum: (-1)(5) + 7 (8) + 11 (-1) = 40; the area is 0 and the points are collinear.

Step 5: In each case the determinant vanishes, which is exactly the collinearity condition.

Final answer: In all three cases the area is zero, hence the points are collinear, as required.

Question: 4. Determine area of triangle formed by the following lines: (i) y = 0, y = 2x, y = 6x + 5 (ii) y - x = 0, y + x = 0, x - c = 0 (iii) y = 2x + 3, 2y + 3x = 3, x + y + 2 = 0.

Answer: Step 1: Find the three vertices as the pairwise intersections of the lines, then use the area formula.

Step 2: (i) Intersections: y = 0 with y = 2x gives (0, 0); y = 0 with y = 6x +

5 gives (-5/6, 0); y = 2x with y = 6x + 5 gives x = -5/4, y = -5/2. The base on y = 0 has length 5/6 and the height from (-5/4, -5/2) is 5/2, so area = (1/2) (5/6) (5/2) = 25/24 square units.

Step 3: (ii) Intersections: (0, 0), (c, c) and (c, -c). The base between (c, c) and (c, -c) has length 2c and the horizontal distance from (0, 0) to the line x = c is c, so area = (1/2) (2c) (c) = c^2 square units.

Step 4: (iii) Intersections: the first two lines give (-3/7, 15/7), the first and third give (-5/3, -1/3), and the second and third give (7, -9). The shoelace sums are 1/7 + 15 + 15 = 30 + 1/7 and -25/7 - 7/3 + 27/7 = -43/21, with difference 676/21, so area = (1/2) (676/21) = 338/21 square units.

Step 5: Each triangle is non-degenerate because no two of the lines are parallel in a way that prevents a closed figure.

Final answer: (i) 25/24 square units (ii) c^2 square units (iii) 338/21 square units.

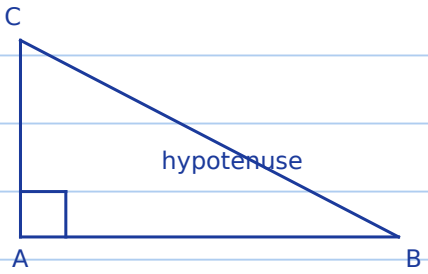


Fig.: Right-angled triangle

Exercise 7.8

Question: 1. Separate the following equations into two separate equations and find the nature of lines and trace them. (i) x^2 - 5 x y + 6 y^2 = 0 (ii) 4x^2 - x y - 5 y^2 = 0 (iii) 9x^2 - 6 x y + y^2 = 0 (iv) 10x^2 - 3 x y - y^2 = 0 (v) 7x^2 - 3 x y + 5 y^2 = 0.

Answer: Step 1: Treat each equation as a quadratic in x/y or factor it into two linear factors; each factor set to zero is a straight line through the

origin.

Step 2: (i) x^2 - 5 x y + 6 y^2 = (x - 2y) (x - 3y), so the lines are x - 2y = 0 and x - 3y = 0, two real and distinct lines through the origin.

Step 3: (ii) 4x^2 - x y - 5 y^2 = (4x - 5y) (x + y), so the lines are 4x - 5y = 0 and x + y = 0, two real and distinct lines through the origin.

Step 4: (iii) 9x^2 - 6 x y + y^2 = (3x - y)^2, so the lines are 3x - y = 0 and 3x - y = 0, two coincident real lines through the origin.

Step 5: (iv) 10x^2 - 3 x y - y^2 = (5x + y) (2x - y), so the lines are 5x + y = 0 and 2x - y = 0, two real and distinct lines through the origin.

Step 6: (v) Putting y = m x gives 5 m^2 - 3 m + 7 = 0 with discriminant 9 - 140 = -131, so m = (3 + sqrt(-131))/10 or (3 - sqrt(-131))/10 and the lines y = ((3 + sqrt(-131))/10) x and y = ((3 - sqrt(-131))/10) x are two imaginary lines through the origin.

Step 7: For tracing, each real line is drawn through the origin with its slope; coincident lines appear as a single line and imaginary lines cannot be drawn in the real plane.

Final answer: (i) x - 3y = 0, x - 2y = 0, two real distinct lines (ii) 4x - 5y = 0, x + y = 0, two real distinct lines (iii) 3x - y = 0, 3x - y = 0, two coincident real lines (iv) 2x - y = 0, 5x + y = 0, two real distinct lines (v) y = ((3 + sqrt(-131))/10) x, y = ((3 - sqrt(-131))/10) x, two imaginary lines; all pass through the origin.

Question: 2. Find the combined equation of the pair of lines through origin perpendicular to the lines represented by: (i) 2x^2 - 5 x y + y^2 = 0 (ii) 6x^2 - 13 x y + 6 y^2 = 0.

Answer: Step 1: If a x^2 + 2 h x y + b y^2 = 0 represents a pair of lines, the pair through the origin perpendicular to them is b x^2 - 2 h x y + a

y^2 = 0: swap a and b and change the sign of the xy term.

Step 2: (i) Here a = 2, b = 1 and 2h = -5; applying the rule gives 1 x^2 + 5 x y + 2 y^2 = 0.

Step 3: (ii) Here a = 6, b = 6 and 2h = -13; applying the rule gives 6x^2 + 13 x y + 6 y^2 = 0.

Step 4: A check by slopes: the new lines have slopes that are the negative reciprocals of the old ones, which confirms the transformation.

Final answer: (i) x^2 + 5 x y + 2 y^2 = 0 (ii) 6x^2 + 13 x y + 6 y^2 = 0.

Question: 3. Trace the pair of lines represented by the equations: (i) x^2 - 3 x y + 2 y^2 = 0 (ii) x^2 - 6 x y + 9 y^2 = 0 (iii) 6x^2 - x y - y^2 = 0 (vi) 8x^2 - 3 x y - y^2 = 0.

Answer: Step 1: Factor each equation into two linear factors; each factor is a line through the origin, and tracing means drawing those lines on the axes.

Step 2: (i) x^2 - 3 x y + 2 y^2 = (x - y) (x - 2y), giving the lines x - y = 0 and x - 2y = 0 with slopes 1 and 1/2.

Step 3: (ii) x^2 - 6 x y + 9 y^2 = (x - 3y)^2, giving the single (double) line x - 3y = 0 with slope 1/3.

Step 4: (iii) 6x^2 - x y - y^2 = (3x + y) (2x - y), giving the lines 3x + y = 0 and 2x - y = 0 with slopes -3 and 2.

Step 5: (vi) Putting y = m x in 8x^2 - 3 x y - y^2 = 0 gives m^2 + 3 m - 8 = 0, so m = (-3 + sqrt(41))/2 or (-3 - sqrt(41))/2, giving the two lines y = ((-3 + sqrt(41))/2) x and y = ((-3 - sqrt(41))/2) x.

Step 6: All the lines pass through the origin, and their slopes fix their positions for the tracing. Note: the part numbering (vi) is as printed in the book.

Final answer: (i) x - y = 0 and x - 2y = 0 (ii) x - 3y = 0 (coincident pair) (iii) 2x - y = 0 and 3x + y = 0 (vi) y = ((-3 + sqrt(41))/2) x and y = ((-3 - sqrt(41))/2) x.

Question: 4. Find the angle between the lines represented by following equations: (i) x^2 - 5 x y + 6 y^2 = 0 (ii) 3x^2 + 7 x y + 2 y^2 = 0 (iii) x^2 + 2 x y - 3 y^2 = 0 (iv) x^2 + x y - 2 y^2 = 0.

Answer: Step 1: For a x^2 + 2 h x y + b y^2 = 0 the angle theta between the lines satisfies tan theta = (2 sqrt(h^2 - a b))/(a + b), taking the value in absolute terms and reporting the obtuse angle when the formula is negative, as the book does.

Step 2: (i) a = 1, h = -5/2, b = 6: tan theta = (2 sqrt(25/4 - 6))/7 = (2 (1/2))/7 = 1/7, so theta = tan^(-1) (1/7), approximately 8.13 degrees.

Step 3: (ii) a = 3, h = 7/2, b = 2: tan theta = (2 sqrt(49/4 - 6))/5 = (2 (5/2))/5 = 1, so theta = 45 degrees.

Step 4: (iii) a = 1, h = 1, b = -3: tan theta = (2 sqrt(1 + 3))/(1 - 3) = 4/(-2) = -2, and the obtuse angle with tan theta = -2 is theta = 116 degrees 34 minutes (the acute angle between the lines is 63 degrees 26 minutes).

Step 5: (iv) a = 1, h = 1/2, b = -2: tan theta = (2 sqrt(1/4 + 2))/(1 - 2) = 3/(-1) = -3, and the obtuse angle with tan theta = -3 is theta = 108 degrees 26 minutes.

Final answer: (i) tan^(-1) (1/7) (ii) 45 degrees (iii) 116 degrees 34 minutes (iv) 108 degrees 26 minutes.

Question: 5. If the gradient of one of the lines represented by equation a x^2 + 2 h x y + b y^2 = 0 is twice that of other, then show that 8 h^2 = 9 a b.

Answer: Step 1: Put y = m x in the equation; the slopes m1 and m2 of the two lines are the roots of b m^2 + 2 h m + a = 0.

Step 2: By the sum and product of roots, m1 + m2 = -2h/b and m1 m2 = a/b.

Step 3: Let the gradients be m and 2m, the second being twice the first.

Step 4: Then the sum gives 3 m = -2h/b, so m = -2h/(3b), and the product gives 2 m^2 = a/b.

Step 5: Substitute m into the product relation: 2 (4 h^2/(9 b^2)) = a/b, that is 8 h^2/(9 b^2) = a/b.

Step 6: Multiply both sides by 9 b^2: 8 h^2 = 9 a b, which is the required result.

Final answer: Hence 8 h^2 = 9 a b, as required.

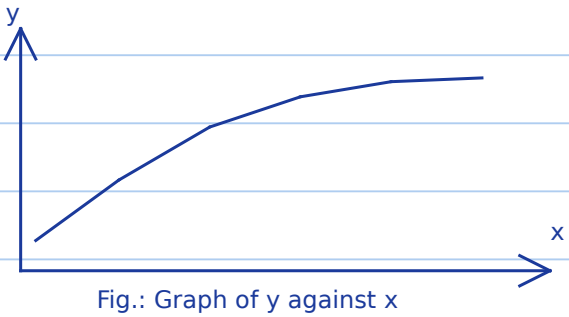


Fig.: Graph of y against x

Unit 8: Circle

Exercise 8.1

Question: 1. Describe condition for a plane cutting a right circular cone to produce (i) a circle (ii) a parabola (iii) an ellipse (iv) a hyperbola (v) degenerate conic.

Answer: Step 1: A conic is the curve of intersection of a plane with a right circular cone, and the kind of conic depends on how the cutting plane is tilted relative to the axis and the generators of the cone.

Step 2: (i) Circle: the cutting plane is perpendicular to the axis of the cone and does not pass through the vertex; every generator is cut at the same angle, so the section is a closed curve with all points at the same distance from the axis point.

Step 3: (ii) Parabola: the cutting plane is parallel to one generator of the cone and cuts only one nappe; the curve is open and goes off without bound in one direction.

Step 4: (iii) Ellipse: the cutting plane is slightly tilted, meeting the axis obliquely but not parallel to a generator, and cuts only one nappe; the section is a closed oval curve, of which the circle is a special case.

Step 5: (iv) Hyperbola: the cutting plane is tilted steeply enough to meet both nappes of the double cone, without passing through the vertex; the section has two separate open branches.

Step 6: (v) Degenerate conic: the cutting plane passes through the vertex of the cone; the section collapses to a single point, a straight line, or a pair of intersecting straight lines through the vertex.

Final answer: (i) Plane perpendicular to the axis, not through the vertex, gives a circle (ii) Plane parallel to a generator, one nappe, gives a parabola (iii) Plane slightly tilted, one nappe only, gives an ellipse (iv) Plane cutting both nappes, not through the vertex, gives a hyperbola (v) Plane through the vertex gives a degenerate conic.

Question: 2. Find the equation of circle with: (i) centre at origin and radius 5 sqrt(2) units (ii) centre (-5, 7) and radius 6 units (iii) (2, -3) and (-4, 7) are ends of diameter (iv) centre at origin, containing the point (5, 6) (v) centre (2, 3), containing the point (5, 7) (vi) centre (p, q) and radius sqrt(p^2 + q^2).

Answer: Step 1: Use standard form (x - h)^2 + (y - k)^2 = r^2 for centre (h, k) and radius r; for a diameter use (x - x1) (x - x2) + (y - y1) (y - y2) = 0.

Step 2: (i) h = 0, k = 0, r^2 = (5 sqrt(2))^2 = 50, giving x^2 + y^2 = 50.

Step 3: (ii) (x + 5)^2 + (y - 7)^2 = 36; expanding gives x^2 + y^2 + 10x - 14y + 38 = 0.

Step 4: (iii) (x - 2) (x + 4) + (y + 3) (y - 7) = 0; expanding gives x^2 + y^2 + 2x - 4y - 29 = 0.

Step 5: (iv) The radius is the distance of (5, 6) from the origin, sqrt(25 + 36) = sqrt(61), so r^2 = 61 and the equation is x^2 + y^2 = 61.

Step 6: (v) The radius is the distance from (2, 3) to (5, 7), sqrt(9 + 16) = 5, so (x - 2)^2 + (y - 3)^2 = 25, expanding to x^2 + y^2 - 4x - 6y - 12 = 0. (vi) (x - p)^2 + (y - q)^2 = p^2 + q^2, expanding to x^2 + y^2 - 2 p x - 2 q y = 0.

Final answer: (i) x^2 + y^2 = 50 (ii) x^2 + y^2 + 10x - 14y + 38 = 0 (iii) x^2 + y^2 + 2x - 4y - 29 = 0 (iv) x^2 + y^2 = 61 (v) x^2 + y^2 - 4x - 6y

- 12 = 0 (vi) x^2 + y^2 - 2 p x - 2 q y = 0.

Question: 3. Find the centre and radius of the following circles and also draw them: (i) x^2 + y^2 - 25 = 0 (ii) (x + 3)^2 + (y - 5)^2 = 49 (iii) x^2 + y^2 - 6x + 8y + 10 = 0 (iv) x^2 + y^2 - 8x + 9 = 0 (v) 5x^2 + 5y^2 + 20x - 15y + 10 = 0.

Answer: Step 1: For x^2 + y^2 + 2 g x + 2 f y + c = 0 the centre is (-g, -f) and the radius is sqrt(g^2 + f^2 - c); first make the coefficients of x^2 and y^2 equal to 1.

Step 2: (i) Here 2g = 0, 2f = 0, c = -25, so the centre is (0, 0) and the radius is sqrt(0 + 0 + 25) = 5.

Step 3: (ii) The equation is already in standard form: centre (-3, 5), radius sqrt(49) = 7.

Step 4: (iii) 2g = -6 and 2f = 8 give centre (3, -4), and the radius is sqrt(9 + 16 - 10) = sqrt(15).

Step 5: (iv) 2g = -8 and 2f = 0 give centre (4, 0), and the radius is sqrt(16 + 0 - 9) = sqrt(7).

Step 6: (v) Divide by 5: x^2 + y^2 + 4x - 3y + 2 = 0, giving centre (-2, 3/2) and radius sqrt(4 + 9/4 - 2) = sqrt(17)/2. For the drawing, plot each centre and draw the circle with the radius found.

Final answer: (i) Centre (0, 0), radius 5 (ii) Centre (-3, 5), radius 7 (iii) Centre (3, -4), radius sqrt(15) (iv) Centre (4, 0), radius sqrt(7) (v) Centre (-2, 3/2), radius sqrt(17)/2.

Question: 4. Find k if radius of the circle 2x^2 + 2y^2 - 8x + 4y + 3k = 0 is 10 units.

Answer: Step 1: Divide the equation by 2 to make the coefficients of x^2 and y^2 unity: x^2 + y^2 - 4x + 2y + (3k)/2 = 0.

Step 2: Read off g = -2, f = 1 and c = 3k/2, so the centre is (2, -1).

Step 3: The radius condition is g^2 + f^2 - c = r^2, that is 4 + 1 - (3k)/2 = 100.

Step 4: So (3k)/2 = 5 - 100 = -95 and k = -190/3.

Step 5: With this value the radius is sqrt(100) = 10, exactly as required.

Final answer: k = -190/3.

Question: 5. Find the equation of the circle through the point (-3, -4), concentric with the circle x^2 + y^2 - 6x + 8y - 24 = 0 and identify that which of the two circles is outer circle.

Answer: Step 1: Concentric circles have the same centre; the given circle has 2g = -6 and 2f = 8, so its centre is (3, -4).

Step 2: The required circle passes through (-3, -4), so its radius is the distance from (3, -4) to (-3, -4), which is 6.

Step 3: Its equation is (x - 3)^2 + (y + 4)^2 = 36, expanding to x^2 + y^2 - 6x + 8y - 11 = 0.

Step 4: The given circle has radius sqrt(9 + 16 + 24) = 7.

Step 5: Comparing radii, 7 is greater than 6, so the given circle lies outside the new circle.

Final answer: The required circle is x^2 + y^2 - 6x + 8y - 11 = 0, and the given circle is the outer circle.

Question: 6. If x = a cos theta, y = a sin theta, then show that it represents a circle with centre origin and radius a units.

Answer: Step 1: The given equations are parametric: every point (x, y) on the curve has coordinates expressed through the parameter theta.

Step 2: Square both equations: x^2 = a^2 cos^2 theta and y^2 = a^2

sin^2 theta.

Step 3: Add them: x^2 + y^2 = a^2 (cos^2 theta + sin^2 theta).

Step 4: Use the identity cos^2 theta + sin^2 theta = 1, giving x^2 + y^2 = a^2, with theta eliminated.

Step 5: The equation x^2 + y^2 = a^2 is the standard equation of a circle whose centre is the origin and whose radius is a, which is what was to be shown.

Final answer: Since x^2 + y^2 = a^2, the parametric equations represent the circle with centre at the origin and radius a.

Question: 7. Prove that the equation of a circle (i) passing through the origin has no constant term (ii) having centre on x-axis has no term in y (iii) having centre on y-axis has no term in x (iv) having centre at origin has no term in x and y both.

Answer: Step 1: Start from the general equation of a circle, x^2 + y^2 + 2 g x + 2 f y + c = 0, whose centre is (-g, -f) and whose constant term satisfies: the circle passes through the origin exactly when c = 0.

Step 2: (i) If the circle passes through the origin, the origin (0, 0) must satisfy the equation, which gives c = 0; hence the equation has no constant term.

Step 3: (ii) If the centre lies on the x-axis, its y-coordinate is 0, so -f = 0 and f = 0; the term 2 f y disappears, so the equation has no term in y.

Step 4: (iii) If the centre lies on the y-axis, its x-coordinate is 0, so -g = 0 and g = 0; the term 2 g x disappears, so the equation has no term in x.

Step 5: (iv) If the centre is the origin, both g = 0 and f = 0, so both the x term and the y term disappear and the equation reduces to x^2 + y^2 + c = 0.

Final answer: Hence (i) a circle through the origin has no constant term, (ii) a circle with centre on the x-axis has no y term, (iii) a circle with centre on the y-axis has no x term, and (iv) a circle with centre at the origin has neither x nor y term, as required.

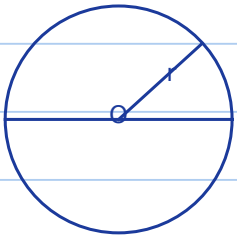


Fig.: Circle, centre O, radius r

Exercise 8.2

Question: 1. Find an equation of the circle passing through the points: (i) (0, 0), (0, 3), (-4, 0) (ii) (0, 10), (-10, 0), (8, 6) (iii) (0, 3), (2, -1), (1, 0) (iv) (7, -3), (-7, 5), (11, 5) (v) (1, 1), (2, -1), (3, -2).

Answer: Step 1: Substitute each point in the general equation x^2 + y^2 + 2 g x + 2 f y + c = 0 and solve the three equations for g, f and c.

Step 2: (i) The origin gives c = 0; (0, 3) gives 9 + 6f = 0, so f = -3/2; (-4, 0) gives 16 - 8g = 0, so g = 2; the equation is x^2 + y^2 + 4x - 3y = 0.

Step 3: (ii) The three equations are 100 + 20f + c = 0, 100 - 20g + c = 0 and 100 + 16g + 12f + c = 0; solving gives g = 0, f = 0 and c = -100 (all three points lie at distance 10 from the origin), so the equation is x^2 + y^2 = 100.

Step 4: (iii) The equations give c from (1, 0) as 1 + 2g + c = 0, from (0, 3) as 9 + 6f + c = 0, and from (2, -1) as 5 + 4g - 2f + c = 0; solving gives g = -5, f = -3 and c = 9, so the equation is x^2 + y^2 - 10x - 6y + 9 = 0.

Step 5: (iv) Substituting the three points and solving gives g = -2, f = -9/2 and c = -57, so the equation is x^2 + y^2 - 4x - 9y - 57 = 0.

Step 6: (v) Substituting the three points and solving gives g = -13/2, f = -5/2 and c = 16, so the equation is x^2 + y^2 - 13x - 5y + 16 = 0. In each case substitution back into the final equation verifies all three points.

Final answer: (i) x^2 + y^2 + 4x - 3y = 0 (ii) x^2 + y^2 = 100 (iii) x^2 + y^2 - 10x - 6y + 9 = 0 (iv) x^2 + y^2 - 4x - 9y - 57 = 0 (v) x^2 + y^2 - 13x - 5y + 16 = 0.

Question: 2. Find the equation of the circle through the points (1, 2) and (2, 3) whose centre lies on (i) x-axis (ii) y-axis.

Answer: Step 1: (i) A centre on the x-axis has f = 0, so the equation is x^2 + y^2 + 2 g x + c = 0; through (1, 2): 5 + 2g + c = 0, and through (2, 3): 13 + 4g + c = 0.

Step 2: (i) Subtracting the equations gives 8 + 2g = 0, so g = -4 and then c = 3; the equation is x^2 + y^2 - 8x + 3 = 0.

Step 3: (ii) A centre on the y-axis has g = 0, so the equation is x^2 + y^2 + 2 f y + c = 0; through (1, 2): 5 + 4f + c = 0, and through (2, 3): 13 + 6f + c = 0.

Step 4: (ii) Subtracting gives 8 + 2f = 0, so f = -4 and then c = 11; the equation is x^2 + y^2 - 8y + 11 = 0.

Step 5: Both circles check out by substituting the two given points.

Final answer: (i) x^2 + y^2 - 8x + 3 = 0 (ii) x^2 + y^2 - 8y + 11 = 0.

Question: 3. Find the equation of the circle passing through the points (3, 1) and (2, 2) be such that its centre lies on the line x + y - 3 = 0.

Answer: Step 1: The centre (-g, -f) lies on the line, so -g - f - 3 = 0, that is g + f = -3.

Step 2: Through (3, 1): 10 + 6g + 2f + c = 0; through (2, 2): 8 + 4g + 4f + c = 0.

Step 3: Subtracting the point equations gives 2 + 2g - 2f = 0, so f - g = 1.

Step 4: Solving g + f = -3 with f - g = 1 gives f = -1 and g = -2, and then c = -10 - 6g - 2f = -10 + 12 + 2 = 4.

Step 5: The equation is therefore x^2 + y^2 - 4x - 2y + 4 = 0, and its centre (2, 1) indeed lies on x + y - 3 = 0.

Final answer: x^2 + y^2 - 4x - 2y + 4 = 0.

Question: 4. Show that the equation of the circle passing through points (0, -1) and (3, 0) having line 3x + y - 9 = 0 as tangent at (3, 0) is x^2 + y^2 - 3x + y = 0.

Answer: Step 1: The centre lies on the normal to the circle at (3, 0); the tangent 3x + y - 9 = 0 has slope -3, so the normal has slope 1/3 and equation y = (x - 3)/3 through (3, 0).

Step 2: Through (0, -1): 1 - 2f + c = 0; through (3, 0): 9 + 6g + c = 0; subtracting gives 8 + 6g + 2f = 0, that is 3g + f = -4.

Step 3: The centre (-g, -f) lies on the normal: -f = (-g - 3)/3, that is g - 3f = -3.

Step 4: Solving 3g + f = -4 with g - 3f = -3 gives f = 1/2 and g = -3/2, and then c = -1 + 2f = 0.

Step 5: Substituting g, f and c gives x^2 + y^2 - 3x + y = 0; its centre (3/2, -1/2) lies on the normal and its radius equals the distance to (3, 0), so the given line is indeed tangent there.

Final answer: The circle is x^2 + y^2 - 3x + y = 0, as required.

Question: 5. Find the equation of the circle through the origin, x-intercept 2 and tangent to y - 1 = 0.

Answer: Step 1: Through the origin gives c = 0; x-intercept 2 means the circle also passes through (2, 0), giving 4 + 4g = 0, so g = -1.

Step 2: The centre is (1, -f) and the radius is sqrt(1 + f^2), the distance from the centre to the origin.

Step 3: Tangency to the line y = 1 means the radius equals the perpendicular distance from the centre to y = 1, which is the absolute value of (-f - 1).

Step 4: So 1 + f^2 = (f + 1)^2 = f^2 + 2f + 1, giving f = 0.

Step 5: With g = -1, f = 0 and c = 0 the equation is x^2 + y^2 - 2x = 0, a circle of centre (1, 0) and radius 1, tangent to y = 1 at (1, 1).

Final answer: x^2 + y^2 - 2x = 0.

Question: 6. Find the equation of the circle containing the points (1, 2) and (2, 3) and whose centre lies on the line x - y + 1 = 0.

Answer: Step 1: The centre (-g, -f) lies on the line, so -g + f + 1 = 0, that is -g + f = -1.

Step 2: Through (1, 2): 5 + 2g + 4f + c = 0; through (2, 3): 13 + 4g + 6f + c = 0.

Step 3: Subtracting gives 8 + 2g + 2f = 0, so g + f = -4.

Step 4: Solving -g + f = -1 with g + f = -4 gives f = -5/2 and g = -3/2, and then c = -5 - 2g - 4f = -5 + 3 + 10 = 8.

Step 5: The centre (3/2, 5/2) satisfies x - y + 1 = 0, as a check.

Final answer: x^2 + y^2 - 3x - 5y + 8 = 0.

Question: 7. Find the equation of the circle passing through the points (1, -2), (-5, 2) and (3, 4) (circumcircle of the triangle).

Answer: Step 1: Substitute the three points in the general equation: (1, -2) gives 5 + 2g - 4f + c = 0; (-5, 2) gives 29 - 10g + 4f + c = 0; (3, 4) gives 25 + 6g + 8f + c = 0.

Step 2: Subtract the first equation from the second: 24 - 12g + 8f = 0, that is 3g - 2f = 6.

Step 3: Subtract the first from the third: 20 + 4g + 12f = 0, that is g + 3f = -5.

Step 4: Solve: from the second relation g = -5 - 3f; substituting gives 3 (-5 - 3f) - 2f = 6, so -11f = 21 and f = -21/11, and then g = -5 + 63/11 = 8/11.

Step 5: From the first equation c = -5 - 2g + 4f = -5 - 16/11 - 84/11 = -155/11.

Step 6: Multiply the equation through by 11 to clear fractions: 11x^2 + 11y^2 + 16x - 42y - 155 = 0.

Final answer: 11x^2 + 11y^2 + 16x - 42y - 155 = 0.

Question: 8. Find the equation of the circle containing the points (1, -2) and (3, -4) and touching the x-axis.

Answer: Step 1: A circle touching the x-axis has radius equal to the absolute value of the centre y-coordinate, which gives c = g^2 in the general equation.

Step 2: Through (1, -2): 5 + 2g - 4f + c = 0; through (3, -4): 25 + 6g - 8f + c = 0.

Step 3: Subtracting gives 20 + 4g - 4f = 0, so f - g = 5, that is f = g + 5.

Step 4: Substitute c = g^2 and f = g + 5 in the first equation: 5 + 2g - 4 (g + 5) + g^2 = 0, giving g^2 - 2g - 15 = 0, which factors as (g - 5) (g + 3) = 0.

Step 5: For g = 5: f = 10, c = 25, giving x^2 + y^2 + 10x + 20y + 25 = 0; for g = -3: f = 2, c = 9, giving x^2 + y^2 - 6x + 4y + 9 = 0.

Step 6: Both circles touch the x-axis (at (-5, 0) and (3, 0) respectively) and

pass through both given points.

Final answer: x^2 + y^2 + 10x + 20y + 25 = 0 and x^2 + y^2 - 6x + 4y + 9 = 0.

Question: 9. Find the equation of the circle containing the point (6, 0) and touching the line x = y at (4, 4).

Answer: Step 1: The centre lies on the normal to the line x = y at (4, 4); the line x - y = 0 has slope 1, so the normal through (4, 4) is x + y = 8, and the centre (-g, -f) satisfies g + f = -8.

Step 2: The circle passes through (4, 4): 32 + 8g + 8f + c = 0, and substituting g + f = -8 gives 32 - 64 + c = 0, so c = 32.

Step 3: It also passes through (6, 0): 36 + 12g + c = 0, giving 12g = -68 and g = -17/3, and then f = -8 + 17/3 = -7/3.

Step 4: Substitute g, f and c in the general equation: x^2 + y^2 - (34/3) x - (14/3) y + 32 = 0.

Step 5: Multiply by 3: 3 (x^2 + y^2) - 34x - 14y + 96 = 0.

Step 6: The centre (17/3, 7/3) is at distance equal to the radius from (4, 4), and the radius is perpendicular to x = y there, confirming tangency.

Final answer: 3 (x^2 + y^2) - 34x - 14y + 96 = 0.

Question: 10. Show that the equation of the circle with centre (-g, -f) (i) touching the x-axis is of the form x^2 + y^2 + 2 g x + 2 f y + g^2 = 0 and (ii) touching the y-axis is of the form x^2 + y^2 + 2 g x + 2 f y + f^2 = 0.

Answer: Step 1: The circle with centre (-g, -f) has general equation x^2 + y^2 + 2 g x + 2 f y + c = 0 and radius sqrt(g^2 + f^2 - c).

Step 2: (i) Touching the x-axis means the radius equals the perpendicular distance of the centre from the x-axis, which is the absolute value of -f,

that is f in magnitude.

Step 3: (i) So g^2 + f^2 - c = f^2, giving c = g^2, and substituting c = g^2 gives the stated form x^2 + y^2 + 2 g x + 2 f y + g^2 = 0.

Step 4: (ii) Touching the y-axis means the radius equals the perpendicular distance of the centre from the y-axis, which is the absolute value of -g.

Step 5: (ii) So g^2 + f^2 - c = g^2, giving c = f^2, and substituting gives x^2 + y^2 + 2 g x + 2 f y + f^2 = 0.

Final answer: Hence a circle touching the x-axis has c = g^2, and a circle touching the y-axis has c = f^2, giving the two stated forms.

Question: 11. Find the equation of the circle passing through origin and having intercepts 6 and 8 on the axes.

Answer: Step 1: Through the origin gives c = 0.

Step 2: An x-intercept of 6 means the circle passes through (6, 0): 36 + 12g = 0, so g = -3.

Step 3: A y-intercept of 8 means the circle passes through (0, 8): 64 + 16f = 0, so f = -4.

Step 4: Substitute g, f and c in the general equation.

Step 5: The centre is (3, 4) and the radius is 5, consistent with intercepts 6 and 8 (twice the centre coordinates).

Final answer: x^2 + y^2 - 6x - 8y = 0.

Question: 12. Find the equation of the circle passing through (b, 0) and (-b, 0) having radius a units.

Answer: Step 1: The centre lies on the perpendicular bisector of the segment joining (b, 0) and (-b, 0), which is the y-axis, so g = 0 and the centre is (0, -f).

Step 2: Both points lie on the circle: substituting (b, 0) gives b^2 + c = 0, so c = -b^2.

Step 3: The radius condition is g^2 + f^2 - c = a^2, that is f^2 + b^2 = a^2, so f^2 = a^2 - b^2 and f = plus or minus sqrt(a^2 - b^2).

Step 4: Substitute g = 0 and c = -b^2 in the general equation.

Step 5: The two signs of f give two circles, symmetric about the x-axis, each of radius a through both points.

Final answer: x^2 + y^2 plus or minus 2 sqrt(a^2 - b^2) y - b^2 = 0.

Question: 13. Find the equation of the circle through the points (5, 0) and (0, -5) having radius 5 units.

Answer: Step 1: Through (5, 0): 25 + 10g + c = 0, so c = -25 - 10g; through (0, -5): 25 - 10f + c = 0, so c = -25 + 10f; equating the two expressions for c gives -10g = 10f, that is f = -g.

Step 2: The radius condition is g^2 + f^2 - c = 25.

Step 3: From the first equation c = -25 - 10g; substituting with f = -g gives 2 g^2 + 25 + 10g = 25, so 2g^2 + 10g = 0 and g = 0 or g = -5.

Step 4: For g = 0, f = 0, c = -25: x^2 + y^2 - 25 = 0, centre the origin.

Step 5: For g = -5, f = 5, c = 25: x^2 + y^2 - 10x + 10y + 25 = 0, centre (5, -5). Both have radius 5 and pass through both points.

Final answer: x^2 + y^2 - 25 = 0 and x^2 + y^2 - 10x + 10y + 25 = 0.

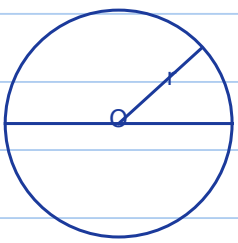


Fig.: Circle, centre O, radius r

Exercise 8.3

Question: 1. Determine whether the line is tangent, secant or neither to the circle x^2 + y^2 = 25: (i) y = x + 3 (ii) y = sqrt(3) x + 10 (iii) y = 2x + 15.

Answer: Step 1: The circle has centre (0, 0) and radius 5; compare the perpendicular distance d from the centre to each line with r = 5: d less than r gives a secant, d equal to r a tangent, d greater than r neither.

Step 2: (i) For x - y + 3 = 0, d = 3/sqrt(2), approximately 2.12, which is less than 5, so the line cuts the circle at two points.

Step 3: (ii) For sqrt(3) x - y + 10 = 0, d = 10/sqrt(3 + 1) = 10/2 = 5, equal to the radius, so the line touches the circle.

Step 4: (iii) For 2x - y + 15 = 0, d = 15/sqrt(5), approximately 6.71, greater than 5, so the line misses the circle completely.

Final answer: (i) Secant (ii) Tangent (iii) Neither.

Question: 2. Find the condition of tangency and secancy of line y = 2x + k with the circle x^2 + y^2 + 10x + 20y + c = 0.

Answer: Step 1: The circle has centre (-5, -10) and radius sqrt(25 + 100 - c) = sqrt(125 - c).

Step 2: Write the line as 2x - y + k = 0; its perpendicular distance from (-5, -10) is (absolute value of -10 + 10 + k)/sqrt(5) = (absolute value of k)/sqrt(5).

Step 3: Tangency needs distance equal to radius: k^2/5 = 125 - c, that is k^2 = 625 - 5c.

Step 4: Secancy needs distance less than radius: k^2/5 less than 125 - c, that is k^2 less than 625 - 5c.

Step 5: For completeness, k^2 greater than 625 - 5c would mean the line

does not meet the circle at all.

Final answer: Tangency when k^2 = 625 - 5c, and secancy when k^2 is less than 625 - 5c.

Question: 3. Find the tangent to the circle x^2 + y^2 = 36 with slope sqrt(3).

Answer: Step 1: A tangent of slope m to x^2 + y^2 = r^2 is y = m x plus or minus r sqrt(1 + m^2).

Step 2: Here r = 6 and m = sqrt(3), so r sqrt(1 + m^2) = 6 sqrt(1 + 3) = 12.

Step 3: Hence the two parallel tangents are y = sqrt(3) x + 12 and y = sqrt(3) x - 12.

Step 4: Each has perpendicular distance 12/sqrt(1 + 3) = 6 from the origin, equal to the radius, as a check.

Final answer: y = sqrt(3) x plus or minus 12.

Question: 4. Find equations of the tangent and normal to the following circles at the given points: (i) at (1, -4) to x^2 + y^2 = 17 (ii) at (4, 1) to x^2 + y^2 - 4x + 2y = 3.

Answer: Step 1: The tangent at (x1, y1) is T = 0, and the normal is the line through the point and the centre.

Step 2: (i) Tangent: x (1) + y (-4) = 17, that is x - 4y = 17.

Step 3: (i) The centre is the origin, so the normal is the line through (0, 0) and (1, -4), namely 4x + y = 0.

Step 4: (ii) Tangent: x (4) + y (1) - 2 (x + 4) + 1 (y + 1) = 3; simplifying, 4x + y - 2x - 8 + y + 1 - 3 = 0, giving x + y = 5.

Step 5: (ii) The centre is (2, -1); the normal through (2, -1) and (4, 1) has slope (1 + 1)/(4 - 2) = 1, giving y - 1 = x - 4, that is x - y = 3.

Final answer: (i) Tangent x - 4y = 17, normal 4x + y = 0 (ii) Tangent x + y = 5, normal x - y = 3.

Question: 5. Find the length of tangent to the following circles from the given points: (i) from (6, 1) to x^2 + y^2 = 4 (ii) from (2, 5) to x^2 + y^2 + 8x - 5y = 7.

Answer: Step 1: The length of the tangent from (x1, y1) is sqrt(S1), where S1 is the value of the circle expression at the point.

Step 2: (i) S1 = 36 + 1 - 4 = 33, so the length is sqrt(33).

Step 3: (ii) S1 = 4 + 25 + 16 - 25 - 7 = 13, so the length is sqrt(13).

Step 4: Both S1 values are positive, so real tangents exist from both points, as expected since the points are outside the circles.

Final answer: (i) sqrt(33) units (ii) sqrt(13) units.

Question: 6. Find the condition for the line y = m x + c to be tangent to the circle (x - h)^2 + (y - k)^2 = r^2.

Answer: Step 1: Write the line as m x - y + c = 0.

Step 2: Tangency means the perpendicular distance from the centre (h, k) to the line equals the radius r.

Step 3: The distance is (absolute value of m h - k + c)/sqrt(m^2 + 1).

Step 4: Set the distance equal to r and square both sides to remove the absolute value.

Step 5: The required condition is (m h - k + c)^2 = r^2 (1 + m^2).

Final answer: The condition of tangency is (m h - k + c)^2 = r^2 (1 + m^2).

Question: 7. Show that the circles x^2 + y^2 - 6x - 6y + 10 = 0 and x^2 + y^2 = 2 touch each other. Find the point of contact of circles.

Answer: Step 1: The first circle has centre C1 = (3, 3) and radius r1 =

sqrt(9 + 9 - 10) = sqrt(8) = 2 sqrt(2); the second has centre C2 = (0, 0) and radius r2 = sqrt(2).

Step 2: The distance between the centres is sqrt(9 + 9) = 3 sqrt(2).

Step 3: r1 + r2 = 2 sqrt(2) + sqrt(2) = 3 sqrt(2), which equals the distance between the centres, so the circles touch externally.

Step 4: The point of contact divides C1 C2 internally in the ratio r1 : r2 = 2 : 1 measured from C1.

Step 5: Its coordinates are (3 - 2, 3 - 2) = (1, 1), obtained by moving from (3, 3) towards the origin by the fraction r1/(r1 + r2) = 2/3 of the way; check: (1, 1) lies on both circles since 1 + 1 - 6 - 6 + 10 = 0 and 1 + 1 = 2.

Final answer: The circles touch externally, and the point of contact is (1, 1).

Question: 8. Find the tangents to the circle x^2 + y^2 = 25 whose: (i) abscissa is 3 (ii) ordinate is -4 (iii) are parallel to the line 3x + 4y + 1 = 0 (iv) perpendicular to the line 3x + 4y + 1 = 0.

Answer: Step 1: (i) Points on the circle with abscissa 3 have y = plus or minus 4, giving points (3, 4) and (3, -4), whose tangents x x1 + y y1 = 25 are 3x + 4y = 25 and 3x - 4y = 25.

Step 2: (ii) Points with ordinate -4 have x = plus or minus 3, giving (3, -4) and (-3, -4), whose tangents are 3x - 4y = 25 and -3x - 4y = 25, the second written 3x + 4y + 25 = 0.

Step 3: (iii) Tangents parallel to 3x + 4y + 1 = 0 have the form 3x + 4y = c with distance from the origin 5: (absolute value of c)/5 = 5, so c = plus or minus 25, giving 3x + 4y + 25 = 0 and 3x + 4y - 25 = 0.

Step 4: (iv) Tangents perpendicular to that line have the form 4x - 3y = c

with (absolute value of c)/5 = 5, so c = plus or minus 25, giving 4x - 3y + 25 = 0 and 4x - 3y - 25 = 0.

Step 5: In each part the distance condition or point condition is exactly the radius 5, which verifies tangency.

Final answer: (i) 3x + 4y = 25 and 3x - 4y = 25 (ii) 3x - 4y = 25 and 3x + 4y + 25 = 0 (iii) 3x + 4y + 25 = 0 and 3x + 4y - 25 = 0 (iv) 4x - 3y + 25 = 0 and 4x - 3y - 25 = 0.

Question: 9. Find equations of tangents to the circle x^2 + y^2 - 6x - 2y + 9 = 0 through the origin and their points of contact.

Answer: Step 1: Let a tangent from the origin be y = m x; the circle has centre (3, 1) and radius sqrt(9 + 1 - 9) = 1.

Step 2: Tangency: the distance from (3, 1) to the line m x - y = 0 equals 1: (absolute value of 3m - 1)/sqrt(m^2 + 1) = 1.

Step 3: Squaring: 9 m^2 - 6 m + 1 = m^2 + 1, so 8 m^2 - 6 m = 0, giving m = 0 or m = 3/4.

Step 4: The tangents are y = 0 and y = (3/4) x, that is 3x - 4y = 0.

Step 5: Points of contact: for y = 0, substituting in the circle gives x^2 - 6x + 9 = 0, so x = 3 and the point is (3, 0); for 3x - 4y = 0, the foot of the perpendicular from (3, 1) is (12/5, 9/5), which also lies on the circle.

Final answer: The tangents are 3x - 4y = 0 and y = 0, with points of contact (12/5, 9/5) and (3, 0) respectively.

Question: 10. Show that the line a x + b y + a l + b m = 0 is normal to the circle x^2 + y^2 + 2 l x + 2 m y + c = 0 for all values of a and b.

Answer: Step 1: The circle x^2 + y^2 + 2 l x + 2 m y + c = 0 has centre (-l, -m).

Step 2: Rewrite the given line as a (x + l) + b (y + m) = 0.

Step 3: Substituting the centre (-l, -m) gives a (0) + b (0) = 0, so the line passes through the centre of the circle for every a and b.

Step 4: A normal to a circle at any point is the line through that point and the centre; equivalently, every line through the centre meets the circle at the ends of a diameter and is perpendicular to the tangents there.

Step 5: Therefore the given line, passing through the centre whatever a and b are, is a normal to the circle.

Final answer: The line passes through the centre (-l, -m) for all a and b, and every line through the centre is a normal to the circle; hence the result, as required.

Question: 11. If the line y = m x intersects the circle x^2 + y^2 + 2 g x + 2 f y + c = 0 at A and B, find (i) the product of abscissae (ii) product of ordinates of A and B.

Answer: Step 1: Substitute y = m x in the circle: x^2 + m^2 x^2 + 2 g x + 2 f m x + c = 0.

Step 2: Collect: (1 + m^2) x^2 + 2 (g + f m) x + c = 0, a quadratic whose roots are the abscissae of A and B.

Step 3: (i) The product of the roots is the constant term divided by the coefficient of x^2, that is c/(1 + m^2).

Step 4: (ii) The ordinates are y = m x, so the product of the ordinates is m^2 times the product of the abscissae, that is c m^2/(1 + m^2).

Step 5: This uses only the product-of-roots formula, so it holds for every m for which the line meets the circle.

Final answer: (i) Product of abscissae = c/(1 + m^2) (ii) Product of ordinates = c m^2/(1 + m^2).

Question: 12. Prove that y = x + k sqrt(2) touches the circle x^2 + y^2 =

k^2. Find the point of contact.

Answer: Step 1: Write the line as x - y + k sqrt(2) = 0; the circle has centre (0, 0) and radius (absolute value of k).

Step 2: The perpendicular distance from the origin to the line is (absolute value of k sqrt(2))/sqrt(1 + 1) = (absolute value of k) sqrt(2)/sqrt(2) = absolute value of k.

Step 3: The distance equals the radius, so the line touches the circle, which proves the tangency.

Step 4: The point of contact is the foot of the perpendicular from the origin to the line.

Step 5: Using the foot formula for x - y + k sqrt(2) = 0 gives the foot (-k sqrt(2)/2, k sqrt(2)/2), which lies on both the line and the circle.

Final answer: The distance from the centre equals the radius, so the line is tangent, and the point of contact is (-k sqrt(2)/2, k sqrt(2)/2).

Question: 13. Find the condition that the line 3x + 4y = c may touch the circle x^2 + y^2 = 8x.

Answer: Step 1: Rewrite the circle as (x - 4)^2 + y^2 = 16, with centre (4, 0) and radius 4.

Step 2: Tangency needs the perpendicular distance from (4, 0) to the line 3x + 4y - c = 0 to equal 4.

Step 3: The distance is (absolute value of 12 - c)/sqrt(9 + 16) = (absolute value of 12 - c)/5.

Step 4: Set (absolute value of 12 - c)/5 = 4, giving absolute value of 12 - c = 20.

Step 5: So 12 - c = 20 or 12 - c = -20, giving c = -8 or c = 32.

Final answer: The line touches the circle when c = 32 or c = -8.

Question: 14. Does the line x + y = 2 + sqrt(2) touch the circle x^2 + y^2 - 2x - 2y - 1 = 0?

Answer: Step 1: The circle has centre (1, 1) and radius sqrt(1 + 1 + 1) = sqrt(3).

Step 2: Write the line as x + y - 2 - sqrt(2) = 0 and find its distance from (1, 1).

Step 3: The distance is (absolute value of 1 + 1 - 2 - sqrt(2))/sqrt(2) = sqrt(2)/sqrt(2) = 1.

Step 4: Tangency would need the distance to equal the radius sqrt(3), approximately 1.732; but the distance is 1, which is different (in fact less, so the line cuts the circle at two points).

Step 5: Therefore the given line is not a tangent to the circle.

Final answer: No, the line does not touch the circle; the distance from the centre is 1, not sqrt(3).

Question: 15. Prove that the circles x^2 + y^2 + 2 g x + c = 0 and x^2 + y^2 + 2 f y + c = 0 touch each other if 1/f^2 + 1/g^2 = 1/c.

Answer: Step 1: At a common point of the two circles, subtracting their equations gives 2 g x - 2 f y = 0, so y = g x/f on the common chord.

Step 2: The circles touch exactly when they have a single common point, that is when this common point is a double point of the system.

Step 3: Substitute y = g x/f in the first circle: x^2 + g^2 x^2/f^2 + 2 g x + c = 0; multiplying by f^2 gives (f^2 + g^2) x^2 + 2 g f^2 x + c f^2 = 0.

Step 4: For a double root the discriminant must vanish: (2 g f^2)^2 - 4 (f^2 + g^2) (c f^2) = 0, which simplifies, on dividing by 4 f^2, to g^2 f^2

= c (f^2 + g^2).

Step 5: Divide both sides by c f^2 g^2: 1/c = (f^2 + g^2)/(f^2 g^2) = 1/g^2 + 1/f^2.

Step 6: This is exactly the stated condition, so the circles touch if and only if 1/f^2 + 1/g^2 = 1/c.

Final answer: Hence the two circles touch each other when 1/f^2 + 1/g^2 = 1/c, as required.

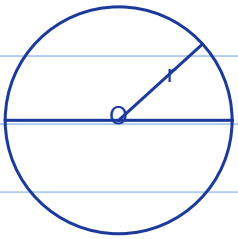


Fig.: Circle, centre O, radius r

Exercise 8.4

Question: 1. Prove analytically that tangents at the end points of a diameter of a circle are parallel.

Answer: Step 1: Take the circle in the simple form x^2 + y^2 = r^2 with centre at the origin; any diameter has end points P(x1, y1) and Q(-x1, -y1), opposite ends of a line through the centre.

Step 2: The tangent at a point (x1, y1) of this circle is x x1 + y y1 = r^2, so the tangent at P is x x1 + y y1 = r^2.

Step 3: The tangent at Q is x (-x1) + y (-y1) = r^2, that is -x x1 - y y1 = r^2.

Step 4: The slope of the first tangent is -x1/y1 (when y1 is not zero), and the slope of the second tangent is also (-x1)/(y1) after dividing by -1 in the direction coefficients; more directly, both lines have the same direction

coefficients ratio x1 : y1 in their normals and hence the same slope.

Step 5: Lines with equal slopes are parallel; the special cases y1 = 0 (both tangents vertical) and x1 = 0 (both tangents horizontal) are parallel as well.

Final answer: The tangents at the ends of a diameter have the same slope, hence they are parallel, as required.

Question: 2. Prove analytically that a normal to a circle passes through its centre.

Answer: Step 1: Take the general circle x^2 + y^2 + 2 g x + 2 f y + c = 0, with centre C = (-g, -f), and let P(x1, y1) be any point on it.

Step 2: The tangent at P is x x1 + y y1 + g (x + x1) + f (y + y1) + c = 0 by the T = S1 form of the tangent equation.

Step 3: The normal at P is the line through P perpendicular to this tangent; the tangent has direction perpendicular to the vector from C to P, since its coefficients can be rewritten using the centre.

Step 4: Equivalently, the slope of the radius CP is (y1 + f)/(x1 + g), and the tangent slope from its equation is -(x1 + g)/(y1 + f); their product is -1, so the tangent is perpendicular to the radius CP.

Step 5: The normal is by definition the line through P perpendicular to the tangent; since CP is also a line through P perpendicular to the tangent, the normal is exactly the line CP.

Step 6: The line CP passes through C, the centre of the circle.

Final answer: Hence every normal to a circle passes through its centre, as required.

Question: 3. Prove analytically that the midpoint of the hypotenuse of a right angled triangle is the centre of the circumscribing circle.

Answer: Step 1: Choose coordinates to suit the proof: let the right angle be at the origin O = (0, 0) and the other vertices A = (2a, 0) and B = (0, 2b) on the axes; the hypotenuse is AB.

Step 2: The midpoint of AB is M = ((2a + 0)/2, (0 + 2b)/2) = (a, b).

Step 3: Find the distances from M to the three vertices: MO = sqrt(a^2 + b^2).

Step 4: MA = sqrt((2a - a)^2 + (0 - b)^2) = sqrt(a^2 + b^2), and MB = sqrt((0 - a)^2 + (2b - b)^2) = sqrt(a^2 + b^2).

Step 5: So MO = MA = MB: M is equidistant from all three vertices.

Step 6: The centre of the circumscribing circle is the unique point equidistant from the three vertices, so M is that centre, and the hypotenuse is a diameter of the circle.

Final answer: Hence the midpoint of the hypotenuse is the centre of the circumscribing circle, as required.

Question: 4. Prove analytically that the central angle of the major arc is double the inscribed angle of corresponding minor arc.

Answer: Step 1: Take the circle x^2 + y^2 = r^2 with centre O at the origin, and write the points by their polar angles: B = (r cos beta, r sin beta), C = (r cos gamma, r sin gamma) and A = (r cos alpha, r sin alpha) any point of the circle; the central angle BOC is beta - gamma (taking the reflex value when the major arc is meant).

Step 2: The slope of chord AB is (sin alpha - sin beta)/(cos alpha - cos beta); by the sum-to-product formulae the numerator is 2 cos ((alpha + beta)/2) sin ((alpha - beta)/2) and the denominator is -2 sin ((alpha + beta)/2) sin ((alpha - beta)/2), so the slope is -cot ((alpha + beta)/2).

Step 3: A line of slope -cot (phi) is inclined to the x-axis at the angle phi +

90 degrees, so the chord AB is inclined at (alpha + beta)/2 + 90 degrees.

Step 4: By the same computation the chord AC is inclined at (alpha + gamma)/2 + 90 degrees.

Step 5: The inscribed angle BAC is the difference of these inclinations: ((alpha + beta)/2 + 90) - ((alpha + gamma)/2 + 90) = (beta - gamma)/2, exactly half of the central angle BOC.

Step 6: Applying this to the major arc: the inscribed angle standing on the major arc is taken from a point A of the minor arc, and steps 2 to 5 give it as half of the reflex angle BOC, that is the central angle of the major arc is double the inscribed angle. A numeric check: if the minor central angle is 60 degrees, the major central angle is 300 degrees and the inscribed angle is 150 degrees, exactly half.

Final answer: Hence the central angle of the major arc is double the inscribed angle of the corresponding minor arc position, as required.

Question: 5. Prove analytically that a parallelogram circumscribing a circle is a rhombus.

Answer: Step 1: Let the parallelogram ABCD circumscribe the circle, so all four sides are tangent to the circle.

Step 2: Tangents drawn from the same external point to a circle are equal in length: from vertex A the two tangent lengths along AB and AD are equal; call them w; similarly denote the tangent lengths at B, C and D by x, y and z respectively.

Step 3: Then the sides are AB = w + x, BC = x + y, CD = y + z and DA = z + w.

Step 4: Adding alternate sides: AB + CD = w + x + y + z and BC + DA = x + y + z + w, which are equal, as they must be in any circumscribed

quadrilateral.

Step 5: But ABCD is a parallelogram, so opposite sides are equal: AB = CD and BC = DA.

Step 6: From AB = CD we get w + x = y + z, and from BC = DA we get x + y = z + w; solving these two equations gives w = y and x = z. Hence BC = x + y = x + w = AB, CD = y + z = w + x = AB and DA = z + w = x + w = AB, so all four sides are equal.

Step 7: A parallelogram with all sides equal is a rhombus.

Final answer: Hence a parallelogram circumscribing a circle is a rhombus, as required.

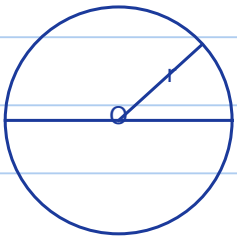


Fig.: Circle, centre O, radius r

Unit 9: Parabola, Ellipse and Hyperbola

Exercise 9.1

Question: 1. Draw the following parabolas: (i) y^2 = 10x (ii) x^2 = -12y (iii) y^2 - x - 2y - 1 = 0 (iv) x^2 - 6x - 2y + 5 = 0

Answer: Step 1: (i) Compare y^2 = 10x with y^2 = 4ax, so 4a = 10 and a = 5/2

Step 2: (i) Vertex is (0, 0), focus is (5/2, 0), the parabola opens to the right, and the ends of the latus rectum are (5/2, 5) and (5/2, -5); plot the vertex and these points and draw a smooth cup-right curve

Step 3: (ii) Compare x^2 = -12y with x^2 = 4ay, so 4a = -12 and a = -3

Step 4: (ii) Vertex is (0, 0), focus is (0, -3), the parabola opens downward, and the ends of the latus rectum are (6, -3) and (-6, -3); draw a smooth cup-down curve

Step 5: (iii) Complete the square in y: y^2 - 2y = (y - 1)^2 - 1, so the equation becomes (y - 1)^2 - 1 - x - 1 = 0, that is (y - 1)^2 = x + 2

Step 6: (iii) Vertex is (-2, 1), 4a = 1 so a = 1/4, focus is (-7/4, 1), the parabola opens to the right; draw the cup-right curve with vertex (-2, 1)

Step 7: (iv) Complete the square in x: x^2 - 6x = (x - 3)^2 - 9, so the equation becomes (x - 3)^2 - 9 - 2y + 5 = 0, that is (x - 3)^2 = 2(y + 2)

Step 8: (iv) Vertex is (3, -2), 4a = 2 so a = 1/2, focus is (3, -3/2), the parabola opens upward; draw the cup-up curve with vertex (3, -2)

Final answer: (i) cup-right, vertex (0, 0), focus (5/2, 0) (ii) cup-down, vertex (0, 0), focus (0, -3) (iii) (y - 1)^2 = x + 2, vertex (-2, 1), focus (-7/4, 1) (iv)

(x - 3)^2 = 2(y + 2), vertex (3, -2), focus (3, -3/2)

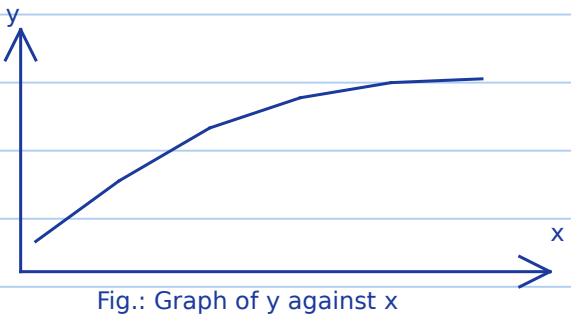


Fig.: Graph of y against x

Question: 2. Determine vertex, focus, latus rectum and equation of directrix of the following. Also find the equation of the axis of symmetry. (i) y^2 = -8x (ii) x^2 = -16y (iii) (y + 3)^2 = 12(x - 2) (iv) (x + 5)^2 = 8(y - 3) (v) x^2 + 4x - y + 5 = 0 (vi) y^2 - 6y + 8x - 23 = 0

Answer: Step 1: (i) Compare y^2 = -8x with y^2 = 4ax, so 4a = -8 and a = -2

Step 2: (i) Vertex (0, 0), focus (a, 0) = (-2, 0), latus rectum = |4a| = 8, directrix x = -a = 2, that is x - 2 = 0, axis of symmetry y = 0

Step 3: (ii) Compare x^2 = -16y with x^2 = 4ay, so 4a = -16 and a = -4

Step 4: (ii) Vertex (0, 0), focus (0, a) = (0, -4), latus rectum = 16, directrix y = -a = 4, that is y - 4 = 0, axis of symmetry x = 0

Step 5: (iii) Compare (y + 3)^2 = 12(x - 2) with (y - k)^2 = 4a(x - h), so h = 2, k = -3 and 4a = 12, a = 3

Step 6: (iii) Vertex (2, -3), focus (h + a, k) = (5, -3), latus rectum = 12, directrix x = h - a = -1, that is x + 1 = 0, axis of symmetry y = -3, that is y + 3 = 0

Step 7: (iv) Compare (x + 5)^2 = 8(y - 3) with (x - h)^2 = 4a(y - k), so h = -5, k = 3 and 4a = 8, a = 2

Step 8: (iv) Vertex (-5, 3), focus (h, k + a) = (-5, 5), latus rectum = 8, directrix y = k - a = 1, that is y - 1 = 0, axis of symmetry x = -5, that is x

+ 5 = 0

Step 9: (v) Rearrange: y = x^2 + 4x + 5 = (x + 2)^2 + 1, so (x + 2)^2 = y - 1, giving h = -2, k = 1 and 4a = 1, a = 1/4

Step 10: (v) Vertex (-2, 1), focus (h, k + a) = (-2, 5/4), latus rectum = 1, directrix y = k - a = 3/4, that is 4y - 3 = 0, axis of symmetry x = -2, that is x + 2 = 0

Step 11: (vi) Complete the square in y: y^2 - 6y = (y - 3)^2 - 9, so (y - 3)^2 - 9 + 8x - 23 = 0, that is (y - 3)^2 = -8(x - 4), giving h = 4, k = 3 and 4a = -8, a = -2

Step 12: (vi) Vertex (4, 3), focus (h + a, k) = (2, 3), latus rectum = 8, directrix x = h - a = 6, that is x - 6 = 0, axis of symmetry y = 3, that is y - 3 = 0

Final answer: (i) vertex (0, 0), focus (-2, 0), latus rectum 8, directrix x - 2 = 0, axis y = 0 (ii) vertex (0, 0), focus (0, -4), latus rectum 16, directrix y - 4 = 0, axis x = 0 (iii) vertex (2, -3), focus (5, -3), latus rectum 12, directrix x + 1 = 0, axis y + 3 = 0 (iv) vertex (-5, 3), focus (-5, 5), latus rectum 8, directrix y - 1 = 0, axis x + 5 = 0 (v) vertex (-2, 1), focus (-2, 5/4), latus rectum 1, directrix 4y - 3 = 0, axis x + 2 = 0 (vi) vertex (4, 3), focus (2, 3), latus rectum 8, directrix x - 6 = 0, axis y - 3 = 0

Question: 3. Find the equation of parabola whose focus is F(1, -2) and directrix is 3x - 5 = 0.

Answer: Step 1: Let P(x, y) be any point on the parabola, then by the definition of a parabola the distance from P to the focus F(1, -2) equals the perpendicular distance from P to the directrix 3x - 5 = 0

Step 2: Distance to focus is sqrt((x - 1)^2 + (y + 2)^2) and distance to directrix is |3x - 5| / sqrt(3^2 + 0^2) = |3x - 5| / 3

Step 3: Equate and square both sides: (x - 1)^2 + (y + 2)^2 = (3x - 5)^2 / 9

Step 4: Multiply by 9: 9(x - 1)^2 + 9(y + 2)^2 = (3x - 5)^2

Step 5: Expand: 9x^2 - 18x + 9 + 9y^2 + 36y + 36 = 9x^2 - 30x + 25

Step 6: Cancel 9x^2 and collect terms: 9y^2 + (-18x + 30x) + 36y + (9 + 36 - 25) = 0, that is 9y^2 + 12x + 36y + 20 = 0

Final answer: 9y^2 + 12x + 36y + 20 = 0

Question: 4. Find the equation of the parabola whose focus is (3, 4) and the directrix is the line x + y - 1 = 0.

Answer: Step 1: Let P(x, y) be any point on the parabola, then distance from P to focus (3, 4) equals perpendicular distance from P to the line x + y - 1 = 0

Step 2: Distance to focus is sqrt((x - 3)^2 + (y - 4)^2) and distance to directrix is |x + y - 1| / sqrt(1^2 + 1^2) = |x + y - 1| / sqrt(2)

Step 3: Equate and square: (x - 3)^2 + (y - 4)^2 = (x + y - 1)^2 / 2

Step 4: Multiply by 2 and expand the left side: 2x^2 - 12x + 18 + 2y^2 - 16y + 32 = (x + y - 1)^2

Step 5: Expand the right side: (x + y - 1)^2 = x^2 + y^2 + 1 + 2xy - 2x - 2y

Step 6: Bring all terms to one side: x^2 + y^2 - 2xy - 10x - 14y + 49 = 0

Final answer: x^2 + y^2 - 2xy - 10x - 14y + 49 = 0

Question: 5. Find the equation of the parabolas whose focus and vertex are as under: (i) Vertex (0, 0); focus (5, 0) (ii) Vertex (0, 0); focus (0, -2) (iii) Vertex (1, -3); focus (1, 2) (iv) Vertex (2, 4); focus (3, 4)

Answer: Step 1: (i) Focus (5, 0) lies on the x-axis at distance a = 5 from

the vertex (0, 0), so the parabola is y^2 = 4ax with a = 5

Step 2: (i) Substituting a = 5 gives y^2 = 20x

Step 3: (ii) Focus (0, -2) lies on the y-axis at distance a = -2 from the vertex, so the parabola is x^2 = 4ay with a = -2

Step 4: (ii) Substituting a = -2 gives x^2 = -8y

Step 5: (iii) Vertex (1, -3) and focus (1, 2) have the same x-coordinate, so the axis is vertical; a = 2 - (-3) = 5, and the form is (x - h)^2 = 4a(y - k) with h = 1, k = -3

Step 6: (iii) Substituting gives (x - 1)^2 = 20(y + 3)

Step 7: (iv) Vertex (2, 4) and focus (3, 4) have the same y-coordinate, so the axis is horizontal; a = 3 - 2 = 1, and the form is (y - k)^2 = 4a(x - h) with h = 2, k = 4

Step 8: (iv) Substituting gives (y - 4)^2 = 4(x - 2)

Final answer: (i) y^2 = 20x (ii) x^2 = -8y (iii) (x - 1)^2 = 20(y + 3) (iv) (y - 4)^2 = 4(x - 2)

Question: 6. Find the equation of parabola whose focus and directrix are given: (i) focus (3, 0) and directrix x - 5 = 0 (ii) focus (0, 4) and directrix y + 6 = 0 (iii) focus (-4, 3) and directrix y = 6

Answer: Step 1: (i) Let P(x, y) be on the parabola; equate squared distances: (x - 3)^2 + y^2 = (x - 5)^2

Step 2: (i) Expand: x^2 - 6x + 9 + y^2 = x^2 - 10x + 25, so y^2 + 4x - 16 = 0

Step 3: (ii) Equate squared distances to focus (0, 4) and directrix y + 6 = 0: x^2 + (y - 4)^2 = (y + 6)^2

Step 4: (ii) Expand: x^2 + y^2 - 8y + 16 = y^2 + 12y + 36, so x^2 - 20y

- 20 = 0

Step 5: (iii) Equate squared distances to focus (-4, 3) and directrix y = 6: (x + 4)^2 + (y - 3)^2 = (y - 6)^2

Step 6: (iii) Expand: x^2 + 8x + 16 + y^2 - 6y + 9 = y^2 - 12y + 36, so x^2 + 8x + 6y - 11 = 0

Final answer: (i) y^2 + 4x - 16 = 0 (ii) x^2 - 20y - 20 = 0 (iii) x^2 + 8x + 6y - 11 = 0

Question: 7. Find equation of the parabola whose vertex and directrix are as under: (i) vertex (0, 0); directrix x = -6 (ii) vertex (0, 0); directrix y = 5 (iii) vertex (3, 4); directrix x = 5

Answer: Step 1: (i) The directrix x = -6 is a vertical line 6 units to the left of the vertex, so a = 6 and the parabola y^2 = 4ax opens to the right

Step 2: (i) Substituting a = 6 gives y^2 = 24x

Step 3: (ii) The directrix y = 5 is a horizontal line 5 units above the vertex, so a = -5 and the parabola x^2 = 4ay opens downward

Step 4: (ii) Substituting a = -5 gives x^2 = -20y

Step 5: (iii) Vertex is (3, 4) and directrix x = 5 is 2 units to the right of the vertex, so a = -2 and the parabola opens to the left with form (y - 4)^2 = 4a(x - 3)

Step 6: (iii) Substituting a = -2 gives (y - 4)^2 = -8(x - 3)

Final answer: (i) y^2 = 24x (ii) x^2 = -20y (iii) (y - 4)^2 = -8(x - 3)

Question: 8. Find the equation of parabola whose vertex and point are given: (i) vertex (0, 0); point (3, 4) (ii) vertex (5, 0); point (4, 6)

Answer: Step 1: (i) Case I, axis along x-axis: let the parabola be y^2 = 4ax; the point (3, 4) lies on it, so 16 = 12a and a = 4/3

Step 2: (i) Case I equation: $y^2 = (16/3)x$, that is $3y^2 = 16x$

Step 3: (i) Case II, axis along y-axis: let the parabola be $x^2 = 4ay$; the point (3, 4) lies on it, so $9 = 16a$ and $a = 9/16$

Step 4: (i) Case II equation: $x^2 = (9/4)y$, that is $4x^2 = 9y$

Step 5: (ii) Case I, axis horizontal through vertex (5, 0): let the parabola be $(y - 0)^2 = 4a(x - 5)$; the point (4, 6) lies on it, so $36 = 4a(4 - 5) = -4a$ and $a = -9$

Step 6: (ii) Case I equation: $y^2 = -36(x - 5)$

Step 7: (ii) Case II, axis vertical through vertex (5, 0): let the parabola be $(x - 5)^2 = 4a(y - 0)$; the point (4, 6) lies on it, so $1 = 24a$ and $a = 1/24$

Step 8: (ii) Case II equation: $(x - 5)^2 = (1/6)y$, that is $6(x - 5)^2 = y$

Final answer: (i) $3y^2 = 16x$ or $4x^2 = 9y$ (ii) $y^2 = -36(x - 5)$ or $6(x - 5)^2 = y$

Question: 9. Find the standard equation of parabola whose latus rectum and vertex are the diameter and centre of the circle respectively $x^2 + y^2 - 4x - 8y - 5 = 0$.

Answer: Step 1: Write the circle in standard form: $(x - 2)^2 + (y - 4)^2 = 4 + 16 + 5 = 25$, so the centre is (2, 4) and the radius is 5

Step 2: The vertex of the parabola is the centre of the circle, so the vertex is (2, 4)

Step 3: The length of the latus rectum equals the diameter of the circle, so $4a = 10$ and $a = 5/2$ (taking the positive direction)

Step 4: For the standard position with axis along the y-axis, the equation is $(x - 2)^2 = 4a(y - 4) = 10(y - 4)$

Step 5: For the standard position with axis along the x-axis, the equation is

(y - 4)^2 = 4a(x - 2) = 10(x - 2)

Final answer: (x - 2)^2 = 10(y - 4) and (y - 4)^2 = 10(x - 2)

Question: 10. Find the equation of circle and its circle is at the focus, whose diameter is the latus rectum of the parabola x^2 = 12y and its centre is at the focus.

Answer: Step 1: Compare x^2 = 12y with x^2 = 4ay, so 4a = 12 and a = 3

Step 2: The focus of the parabola is (0, a) = (0, 3), and this is the centre of the required circle

Step 3: The diameter of the circle equals the latus rectum of the parabola, which is 12, so the radius is 6

Step 4: The circle with centre (0, 3) and radius 6 is x^2 + (y - 3)^2 = 36

Step 5: Expand: x^2 + y^2 - 6y + 9 - 36 = 0, that is x^2 + y^2 - 6y - 27 = 0

Final answer: x^2 + y^2 - 6y - 27 = 0

Question: 11. For what point of the parabola y^2 = 10x, the abscissa is equal to three times its ordinate.

Answer: Step 1: Let the required point be (x, y) on y^2 = 10x, with the condition x = 3y

Step 2: Substitute x = 3y in the equation of the parabola: y^2 = 10(3y) = 30y

Step 3: Rearrange: y^2 - 30y = 0, that is y(y - 30) = 0, so y = 0 or y = 30

Step 4: For y = 0, x = 3(0) = 0, giving the point (0, 0), which is the vertex

Step 5: For y = 30, x = 3(30) = 90, giving the point (90, 30)

Final answer: (0, 0) and (90, 30)

Exercise 9.2

Question: 1. Find the condition when the line y = mx + c is tangent to the parabola x^2 = 4ay. Also find point of contact and the equation of tangent.

Answer: Step 1: Substitute y = mx + c in x^2 = 4ay: x^2 = 4a(mx + c), that is x^2 - 4amx - 4ac = 0

Step 2: This is a quadratic in x; the line is tangent when the two roots coincide, that is when the discriminant is zero

Step 3: Discriminant: (-4am)^2 - 4(1)(-4ac) = 16a^2 m^2 + 16ac = 0

Step 4: Divide by 16a: am^2 + c = 0, so the condition of tangency is c = -am^2

Step 5: With coincident roots, x = -(-4am) / 2 = 2am

Step 6: Then y = m(2am) + c = 2am^2 - am^2 = am^2, so the point of contact is (2am, am^2)

Step 7: The equation of the tangent is the line itself with c = -am^2, namely y = mx - am^2

Final answer: Condition c = -am^2; point of contact (2am, am^2); tangent y = mx - am^2

Question: 2. Find condition of tangency and the point of tangency for the following lines and parabolas. Also find equation of tangent in each case: (i) 2x + y = c ; y^2 = 10x (ii) 3x + 4y = p ; x^2 = 12y (iii) y = cx ; y^2 = 8(x - 1)

Answer: Step 1: (i) Write the line as y = c - 2x and substitute in y^2 = 10x: (c - 2x)^2 = 10x, that is 4x^2 - (4c + 10)x + c^2 = 0

Step 2: (i) Tangency needs discriminant zero: (4c + 10)^2 - 16c^2 = 80c + 100 = 0, so c = -5/4

Step 3: (i) The double root is x = (4c + 10) / 8 = 5/8, and y = c - 2x = -5/4 - 5/4 = -5/2, so the point of tangency is (5/8, -5/2) and the tangent is y = -2x - 5/4

Step 4: (ii) Write the line as y = (p - 3x) / 4 and substitute in x^2 = 12y: x^2 = 3(p - 3x), that is x^2 + 9x - 3p = 0

Step 5: (ii) Tangency needs discriminant zero: 81 + 12p = 0, so p = -27/4

Step 6: (ii) The double root is x = -9/2, and y = (p - 3x) / 4 = (-27/4 + 27/2) / 4 = 27/16, so the point of tangency is (-9/2, 27/16) and the tangent is y = -(3/4)x - 27/16, that is 12x + 16y + 27 = 0

Step 7: (iii) Substitute y = cx in y^2 = 8(x - 1): c^2 x^2 - 8x + 8 = 0

Step 8: (iii) Tangency needs discriminant zero: 64 - 32c^2 = 0, so c^2 = 2 and c = sqrt(2) or c = -sqrt(2)

Step 9: (iii) The double root is x = 8 / (2c^2) = 2, and y = cx = 2c, so the points of tangency are (2, 2 sqrt(2)) and (2, -2 sqrt(2)), and the tangents are y = sqrt(2) x and y = -sqrt(2) x

Final answer: (i) c = -5/4, point (5/8, -5/2), tangent y = -2x - 5/4 (ii) p = -27/4, point (-9/2, 27/16), tangent y = -(3/4)x - 27/16 (iii) c = sqrt(2) or -sqrt(2), points (2, 2 sqrt(2)) and (2, -2 sqrt(2)), tangents y = sqrt(2) x and y = -sqrt(2) x

Question: 3. Find the equation of tangent and normal to the following parabolas at the given points: (i) y^2 = 8x ; (2, 4) (ii) x^2 = 4y ; (-6, 9) (iii) (y - 1)^2 = 9(x - 2) ; (3, 4)

Answer: Step 1: (i) Differentiate y^2 = 8x with respect to x: 2y dy/dx = 8, so dy/dx = 4 / y

Step 2: (i) At (2, 4) the slope of the tangent is 4 / 4 = 1, so the tangent is y - 4 = 1(x - 2), that is x - y + 2 = 0

Step 3: (i) The slope of the normal is -1, so the normal is y - 4 = -1(x - 2), that is x + y - 6 = 0

Step 4: (ii) Differentiate x^2 = 4y: 2x = 4 dy/dx, so dy/dx = x / 2

Step 5: (ii) At (-6, 9) the slope of the tangent is -3, so the tangent is y - 9 = -3(x + 6), that is 3x + y + 9 = 0

Step 6: (ii) The slope of the normal is 1/3, so the normal is y - 9 = (1/3)(x + 6), that is x - 3y + 33 = 0

Step 7: (iii) Differentiate (y - 1)^2 = 9(x - 2): 2(y - 1) dy/dx = 9, so dy/dx = 9 / (2(y - 1))

Step 8: (iii) At (3, 4) the slope of the tangent is 9 / 6 = 3/2, so the tangent is y - 4 = (3/2)(x - 3), that is 3x - 2y - 1 = 0

Step 9: (iii) The slope of the normal is -2/3, so the normal is y - 4 = (-2/3)(x - 3), that is 2x + 3y - 18 = 0

Final answer: (i) tangent x - y + 2 = 0, normal x + y - 6 = 0 (ii) tangent 3x + y + 9 = 0, normal x - 3y + 33 = 0 (iii) tangent 3x - 2y - 1 = 0, normal 2x + 3y - 18 = 0

Question: 4. Find the equation of tangent and normal at P(x1, y1) to the parabola x^2 = 4ay.

Answer: Step 1: Differentiate x^2 = 4ay with respect to x: 2x = 4a dy/dx, so dy/dx = x / (2a)

Step 2: At P(x1, y1) the slope of the tangent is x1 / (2a)

Step 3: By the point-slope form, the tangent is y - y1 = (x1 / (2a))(x - x1)

Step 4: Multiply by 2a: 2a(y - y1) = x1(x - x1), that is 2ay - 2ay1 = x1 x - x1^2

Step 5: Since P lies on the parabola, x1^2 = 4ay1; substituting gives 2ay -

2ay1 = x1 x - 4ay1, that is x1 x = 2a(y + y1), the tangent in compact form

Step 6: The normal is perpendicular to the tangent, so its slope is -2a / x1

Step 7: By the point-slope form, the normal is y - y1 = (-2a / x1)(x - x1), that is x1(y - y1) + 2a(x - x1) = 0

Final answer: Tangent y - y1 = (x1 / (2a))(x - x1), equivalently xx1 = 2a(y + y1); normal y - y1 = (2a / x1)(x1 - x)

Question: 5. A light house uses a parabolic reflector that is 1 m in diameter. How deep should the reflector be if light source is placed halfway between the vertex and the plane of rim to produce parallel beam of light to the axis of parabola.

Answer: Step 1: For a parallel beam, the light source must be at the focus of the parabolic reflector

Step 2: Take the vertex at the origin and the axis as the x-axis, so the reflector has equation y^2 = 4ax, where the focus is at distance a from the vertex

Step 3: The source is halfway between the vertex and the plane of the rim, so if the depth of the reflector is d, then a = d / 2, that is d = 2a

Step 4: The diameter is 1 m, so the rim is at height 1/2 m from the axis; the rim point (d, 1/2) lies on the parabola

Step 5: Substitute: (1/2)^2 = 4a d = 4a(2a) = 8a^2, so a^2 = 1 / 32 and a = 1 / (4 sqrt(2))

Step 6: The depth is d = 2a = 1 / (2 sqrt(2)) m, which is about 0.354 m

Final answer: Depth = 1 / (2 sqrt(2)) m, about 0.354 m (35.4 cm)

Question: 6. There is a parabolic reflector of 12 cm in diameter is used in a vehicle where should the light source be placed to produce parallel beam of light whereas the reflector is 8 cm deep.

Answer: Step 1: For a parallel beam, the light source must be placed at the focus of the reflector

Step 2: Take the vertex at the origin and the axis as the x-axis, so the reflector has equation y^2 = 4ax

Step 3: The diameter is 12 cm, so the rim is 6 cm from the axis, and the depth is 8 cm, so the rim point (8, 6) lies on the parabola

Step 4: Substitute: 6^2 = 4a(8), that is 36 = 32a

Step 5: Solve: a = 36 / 32 = 9/8 = 1.125

Final answer: The light source should be placed at the focus, at a distance of 9/8 cm (1.125 cm) from the vertex along the axis

Question: 7. The main cable of suspension bridge is suspended in the shape of parabola between two towers that are 100 m apart and 30 m high from the roadway. If the cable is at the height of 5 m from the roadway at the centre of the bridge then find the equation of parabola and the distance of 10 m high suspended cable from the centre of the bridge.

Answer: Step 1: Take the origin at the centre of the roadway and the y-axis vertical; the lowest point (vertex) of the cable is (0, 5)

Step 2: Let the parabola be x^2 = 4a(y - 5)

Step 3: The towers are 100 m apart and 30 m high, so the point (50, 30) lies on the parabola: 50^2 = 4a(30 - 5) = 100a

Step 4: Solve: 2500 = 100a in the form x^2 = 4a(y - 5) gives 4a = 100, so the equation of the parabola is x^2 = 100(y - 5)

Step 5: For the suspender cable of height 10 m, put y = 10: x^2 = 100(10 - 5) = 500

Step 6: Solve: x = sqrt(500) = 10 sqrt(5), which is about 22.36 m

Final answer: Equation x^2 = 100(y - 5); the 10 m high suspender cable is at a distance of 10 sqrt(5) m, about 22.36 m, from the centre of the bridge

Question: 8. The main cable of a suspension bridge is in the shape of a parabola. The towers are 600 feet apart and 60 feet high from the roadway. If the cable touches at the roadway at the midway between the towers. What is height of the suspender cable 150 feet from the centre of the bridge.

Answer: Step 1: Take the origin at the point where the cable touches the roadway at the midway, so the vertex is (0, 0) and the parabola is x^2 = 4ay

Step 2: The towers are 600 feet apart and 60 feet high, so the point (300, 60) lies on the parabola: 300^2 = 4a(60)

Step 3: Solve: 90000 = 240a in the form x^2 = 4ay gives 4a = 1500, so the equation is x^2 = 1500y

Step 4: At 150 feet from the centre, put x = 150: 150^2 = 1500y, that is 22500 = 1500y

Step 5: Solve: y = 22500 / 1500 = 15

Final answer: The height of the suspender cable 150 feet from the centre is 15 feet

Exercise 9.3

Question: 1. Find semi-axes, eccentricity, foci, vertices, covertices, latus rectum and equations of directrices of the following ellipses. Also draw their graphs. (i) x^2/9 + y^2/25 = 1 (ii) x^2/16 + y^2/10 = 1 (iii) (x - 3)^2/25 + (y + 4)^2/16 = 1 (iv) (x + 1)^2/9 + (y - 2)^2/16 = 1 (v) 9x^2 + 25y^2 = 225 (vi) 4x^2 - 16x + 25y^2 + 200y - 316 = 0

Answer: Step 1: (i) The larger denominator is 25 under y^2, so the major

axis is along the y-axis with a = 5 and b = 3

Step 2: (i) c^2 = a^2 - b^2 = 25 - 9 = 16, so c = 4 and e = c/a = 4/5

Step 3: (i) Foci (0, 4) and (0, -4), vertices (0, 5) and (0, -5), covertices (3, 0) and (-3, 0), latus rectum = 2b^2/a = 18/5, directrices y = a/e = 25/4 and y = -25/4

Step 4: (ii) The larger denominator is 16 under x^2, so a = 4 and b = sqrt(10)

Step 5: (ii) c^2 = 16 - 10 = 6, so c = sqrt(6) and e = sqrt(6)/4

Step 6: (ii) Foci (sqrt(6), 0) and (-sqrt(6), 0), vertices (4, 0) and (-4, 0), covertices (0, sqrt(10)) and (0, -sqrt(10)), latus rectum = 2(10)/4 = 5, directrices x = 16/sqrt(6) and x = -16/sqrt(6)

Step 7: (iii) Centre is (3, -4); the larger denominator is 25 under (x - 3)^2, so a = 5, b = 4 and the major axis is horizontal

Step 8: (iii) c^2 = 25 - 16 = 9, so c = 3 and e = 3/5

Step 9: (iii) Foci (6, -4) and (0, -4), vertices (8, -4) and (-2, -4), covertices (3, 0) and (3, -8), latus rectum = 2(16)/5 = 32/5, directrices x = 3 + 25/3 = 34/3 and x = 3 - 25/3 = -16/3, that is 3x - 34 = 0 and 3x + 16 = 0

Step 10: (iv) Centre is (-1, 2); the larger denominator is 16 under (y - 2)^2, so a = 4, b = 3 and the major axis is vertical

Step 11: (iv) c^2 = 16 - 9 = 7, so c = sqrt(7) and e = sqrt(7)/4

Step 12: (iv) Foci (-1, 2 + sqrt(7)) and (-1, 2 - sqrt(7)), vertices (-1, 6) and (-1, -2), covertices (2, 2) and (-4, 2), latus rectum = 2(9)/4 = 9/2, directrices y - 2 = 16/sqrt(7) and y - 2 = -16/sqrt(7)

Step 13: (v) Divide 9x^2 + 25y^2 = 225 by 225: x^2/25 + y^2/9 = 1, so a = 5 and b = 3

Step 14: (v) c^2 = 25 - 9 = 16, so c = 4 and e = 4/5; foci (4, 0) and (-4, 0), vertices (5, 0) and (-5, 0), covertices (0, 3) and (0, -3), latus rectum = 18/5, directrices x = 25/4 and x = -25/4

Step 15: (vi) Complete the squares: 4(x - 2)^2 + 25(y + 4)^2 = 316 + 16 + 400 = 732, so (x - 2)^2/183 + (y + 4)^2/(732/25) = 1, with centre (2, -4), a^2 = 183 and b^2 = 29.28

Step 16: (vi) a = sqrt(183) which is about 13.53 and b is about 5.41; c^2 = 183 - 29.28 = 153.72, so c is about 12.40 and e = c/a is about 0.917

Step 17: (vi) Foci are about (14.40, -4) and (-10.40, -4), vertices are about (15.53, -4) and (-11.53, -4), covertices are about (2, 1.41) and (2, -9.41), latus rectum is about 4.33, directrices are x = 2 + 14.76 = 16.76 and x = 2 - 14.76 = -12.76 (approximately)

Final answer: (i) a = 5, b = 3, e = 4/5, foci (0, plus/minus 4), vertices (0, plus/minus 5), covertices (plus/minus 3, 0), latus rectum 18/5, directrices y = plus/minus 25/4 (ii) a = 4, b = sqrt(10), e = sqrt(6)/4, foci (plus/minus sqrt(6), 0), vertices (plus/minus 4, 0), covertices (0, plus/minus sqrt(10)), latus rectum 5, directrices x = plus/minus 16/sqrt(6) (iii) a = 5, b = 4, e = 3/5, foci (6, -4), (0, -4), vertices (8, -4), (-2, -4), covertices (3, 0), (3, -8), latus rectum 32/5, directrices 3x - 34 = 0, 3x + 16 = 0 (iv) a = 4, b = 3, e = sqrt(7)/4, foci (-1, 2 plus/minus sqrt(7)), vertices (-1, 6), (-1, -2), covertices (2, 2), (-4, 2), latus rectum 9/2, directrices y - 2 = plus/minus 16/sqrt(7) (v) a = 5, b = 3, e = 4/5, foci (plus/minus 4, 0), vertices (plus/minus 5, 0), covertices (0, plus/minus 3), latus rectum 18/5, directrices x = plus/minus 25/4 (vi) centre (2, -4), a^2 = 183, b^2 = 29.28, e is about 0.917, foci about (plus/minus 12.40 from centre, -4), latus rectum about 4.33

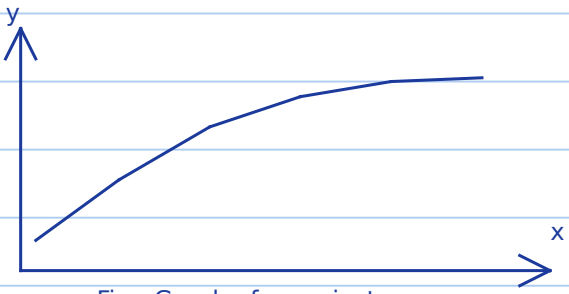


Fig.: Graph of y against x

Question: 2. Find the equations of the following ellipse whose centres are at origin and their axes are along coordinate axes. Also satisfy the given conditions: (i) Major and minor axes are 12 and 8 respectively with minor axis is along y-axis. (ii) Ellipse passes through (1, sqrt(3/2)) and (2/sqrt(3), 1) with major axis is along y-axis. (iii) Foci at (plus/minus 3, 0) and vertices at (plus/minus 5, 0) (iv) Foci at (0, plus/minus 4) and latus rectum 18/5 (v) Foci at (plus/minus 5, 0) and minor axis is 12 units long and along y-axis (vi) Minor axis along x-axis with length is 8 units and latus rectum are 6 units long and along y-axis (vii) Covertices at (0, plus/minus 3) and distance between foci = 10 units. (viii) Directrix y = 25/3 and latus rectum 32/5

Answer: Step 1: (i) Semi-major axis a = 12/2 = 6 along the x-axis and semi-minor axis b = 8/2 = 4, so the equation is x^2/36 + y^2/16 = 1

Step 2: (ii) Major axis along y-axis, so let the equation be x^2/b^2 + y^2/a^2 = 1; the point (1, sqrt(3/2)) gives 1/b^2 + (3/2)/a^2 = 1

Step 3: (ii) The point (2/sqrt(3), 1) gives (4/3)/b^2 + 1/a^2 = 1

Step 4: (ii) Multiply the first equation by 4/3 and subtract the second: (2 - 1)/a^2 = 4/3 - 1 = 1/3, so a^2 = 3; then 1/b^2 = 1 - (3/2)/3 = 1/2, so b^2 = 2; the equation is x^2/2 + y^2/3 = 1

Step 5: (iii) Vertices (plus/minus 5, 0) give a = 5 and foci give c = 3, so b^2 = a^2 - c^2 = 25 - 9 = 16; the equation is x^2/25 + y^2/16 = 1

Step 6: (iv) Foci on the y-axis give c = 4 with major axis vertical; latus

rectum 2b^2/a = 18/5, and c = ae with a^2 = b^2 + c^2; trying a = 5 gives b^2 = 25 - 16 = 9 and 2(9)/5 = 18/5, which checks, so the equation is x^2/9 + y^2/25 = 1

Step 7: (v) c = 5 and semi-minor axis b = 12/2 = 6, so a^2 = b^2 + c^2 = 36 + 25 = 61; the equation is x^2/61 + y^2/36 = 1

Step 8: (vi) Semi-minor axis b = 8/2 = 4; latus rectum 2b^2/a = 6 gives 32/a = 6, so a = 16/3 and a^2 = 256/9; the equation is x^2/(256/9) + y^2/16 = 1, that is 9x^2/256 + y^2/16 = 1

Step 9: (vii) Covertices (0, plus/minus 3) give b = 3; distance between foci is 10, so c = 5; then a^2 = b^2 + c^2 = 9 + 25 = 34; the equation is x^2/34 + y^2/9 = 1

Step 10: (viii) Directrix y = 25/3 gives a/e = 25/3 with major axis vertical; latus rectum 2b^2/a = 32/5; from a/e = 25/3 and c = ae, taking a = 5 gives e = 3/5 and b^2 = a^2(1 - e^2) = 25(16/25) = 16, and 2(16)/5 = 32/5 checks; the equation is x^2/16 + y^2/25 = 1

Final answer: (i) x^2/36 + y^2/16 = 1 (ii) x^2/2 + y^2/3 = 1 (iii) x^2/25 + y^2/16 = 1 (iv) x^2/9 + y^2/25 = 1 (v) x^2/61 + y^2/36 = 1 (vi) 9x^2/256 + y^2/16 = 1 (vii) x^2/34 + y^2/9 = 1 (viii) x^2/16 + y^2/25 = 1

Question: 3. Find equation of auxiliary circle to 5x^2 + 7y^2 = 11.

Answer: Step 1: Divide by 11: x^2/(11/5) + y^2/(11/7) = 1

Step 2: The semi-major axis is a with a^2 = 11/5, since 11/5 is larger than 11/7

Step 3: The auxiliary circle is the circle on the major axis as diameter, with centre at the centre of the ellipse (the origin) and radius a

Step 4: Its equation is x^2 + y^2 = a^2 = 11/5

Final answer: x^2 + y^2 = 11/5

Question: 4. Is the point (4, 5) inside on or outside the ellipse 2x^2 + 3y^2 = 6.

Answer: Step 1: For the ellipse 2x^2 + 3y^2 = 6, form the expression S = 2x^2 + 3y^2 - 6

Step 2: Substitute the point (4, 5): S = 2(16) + 3(25) - 6 = 32 + 75 - 6 = 101

Step 3: Since S is positive (greater than zero), the point lies outside the ellipse

Final answer: The point (4, 5) is outside the ellipse

Question: 5. Find equation of ellipse with centre at (5, -3), one vertex at (10, -3) and one focus at (9, -3).

Answer: Step 1: Centre (5, -3) and vertex (10, -3) have the same y-coordinate, so the major axis is horizontal

Step 2: Semi-major axis a = distance from centre to vertex = 10 - 5 = 5

Step 3: Distance from centre to focus c = 9 - 5 = 4

Step 4: b^2 = a^2 - c^2 = 25 - 16 = 9

Step 5: With centre (h, k) = (5, -3), the equation is (x - 5)^2/25 + (y + 3)^2/9 = 1

Final answer: (x - 5)^2/25 + (y + 3)^2/9 = 1

Question: 6. If ellipse is 9x^2 + 13y^2 = 117 then find: (i) distance between foci; (ii) distance between vertices; (iii) distance between covertices.

Answer: Step 1: Divide by 117: x^2/13 + y^2/9 = 1, so a^2 = 13, a = sqrt(13), and b = 3

Step 2: c^2 = a^2 - b^2 = 13 - 9 = 4, so c = 2

Step 3: (i) Distance between foci = 2c = 4

Step 4: (ii) Distance between vertices = 2a = 2 sqrt(13)

Step 5: (iii) Distance between covertices = 2b = 6

Final answer: (i) 4 (ii) 2 sqrt(13) (iii) 6

Question: 7. Find eccentricity of ellipse if: (i) axes are 32 and 24; (ii) latus rectum is equal to half of its major axis.

Answer: Step 1: (i) Semi-axes are a = 32/2 = 16 and b = 24/2 = 12

Step 2: (i) c^2 = a^2 - b^2 = 256 - 144 = 112, so c = 4 sqrt(7)

Step 3: (i) e = c/a = 4 sqrt(7) / 16 = sqrt(7)/4

Step 4: (ii) Latus rectum = half of major axis means 2b^2/a = (2a)/2 = a, so 2b^2 = a^2 and b^2 = a^2/2

Step 5: (ii) c^2 = a^2 - b^2 = a^2 - a^2/2 = a^2/2, so c = a/sqrt(2)

Step 6: (ii) e = c/a = 1/sqrt(2)

Final answer: (i) e = sqrt(7)/4 (ii) e = 1/sqrt(2)

Question: 8. Find equation of the circle passing through focus of parabola y^2 + 8x = 0 and foci of ellipse 25x^2 + 16y^2 = 400.

Answer: Step 1: For the parabola y^2 = -8x, 4a = -8 so a = -2 and the focus is (-2, 0)

Step 2: For the ellipse, divide by 400: x^2/16 + y^2/25 = 1, so a = 5, b = 4 and c^2 = 25 - 16 = 9, c = 3; the foci are (0, 3) and (0, -3)

Step 3: Let the circle be x^2 + y^2 + 2gx + 2fy + c1 = 0; passing through (0, 3): 9 + 6f + c1 = 0

Step 4: Passing through (0, -3): 9 - 6f + c1 = 0; subtracting the two equations gives f = 0 and then c1 = -9

Step 5: Passing through (-2, 0): 4 - 4g + c1 = 0, so 4 - 4g - 9 = 0 and g = -5/4

Step 6: The circle is x^2 + y^2 - (5/2)x - 9 = 0; multiply by 2

Final answer: 2x^2 + 2y^2 - 5x - 18 = 0

Question: 9. Find the length of, and the equations to, the focal radii drawn to a point (4 sqrt(3), 5) of the ellipse 25x^2 + 16y^2 = 1600.

Answer: Step 1: Divide by 1600: x^2/64 + y^2/100 = 1, so a = 10 (along y-axis), b = 8, and c^2 = 100 - 64 = 36, c = 6

Step 2: Eccentricity e = c/a = 6/10 = 3/5, and the foci are F1 = (0, 6) and F2 = (0, -6)

Step 3: The focal radii to a point (x1, y1) are a + e y1 and a - e y1; here y1 = 5, so the lengths are 10 + (3/5)(5) = 13 and 10 - (3/5)(5) = 7

Step 4: Equation of the focal radius through (0, 6) and (4 sqrt(3), 5): slope = (5 - 6) / (4 sqrt(3)) = -1 / (4 sqrt(3)), so y - 6 = -x / (4 sqrt(3)), that is x + 4 sqrt(3) y - 24 sqrt(3) = 0

Step 5: Equation of the focal radius through (0, -6) and (4 sqrt(3), 5): slope = (5 + 6) / (4 sqrt(3)) = 11 / (4 sqrt(3)), so y + 6 = 11x / (4 sqrt(3)), that is 11x - 4 sqrt(3) y - 24 sqrt(3) = 0

Final answer: Lengths 13 and 7; equations x + 4 sqrt(3) y - 24 sqrt(3) = 0 and 11x - 4 sqrt(3) y - 24 sqrt(3) = 0

Question: 10. Find equation of ellipse with centre at (0, 1) and major axis parallel to y-axis. Also, it passes through (2, 1) and (0, 4).

Answer: Step 1: With centre (0, 1) and major axis vertical, let the equation be x^2/b^2 + (y - 1)^2/a^2 = 1

Step 2: It passes through (2, 1): 4/b^2 + 0 = 1, so b^2 = 4

Step 3: It passes through (0, 4): 0 + 9/a^2 = 1, so a^2 = 9

Step 4: Substitute a^2 and b^2 in the equation

Final answer: x^2/4 + (y - 1)^2/9 = 1

Exercise 9.4

Question: 1. Find the condition when line y = sqrt(5) x + c is tangent to the ellipse 4x^2 + 9y^2 = 36.

Answer: Step 1: Divide the ellipse by 36: x^2/9 + y^2/4 = 1, so a^2 = 9 and b^2 = 4

Step 2: For the line y = mx + c to be tangent to x^2/a^2 + y^2/b^2 = 1, the condition is c^2 = a^2 m^2 + b^2

Step 3: Here m = sqrt(5), so c^2 = 9(5) + 4 = 45 + 4 = 49

Step 4: Take the square root: c = 7 or c = -7

Final answer: c = plus/minus 7

Question: 2. Show that the line x = 2y + 4 touches the ellipse x^2/4 + y^2/3 = 1. Also find point of contact.

Answer: Step 1: Substitute x = 2y + 4 in the ellipse: (2y + 4)^2 / 4 + y^2/3 = 1

Step 2: Simplify: (y + 2)^2 + y^2/3 = 1, that is y^2 + 4y + 4 + y^2/3 - 1 = 0

Step 3: Multiply by 3: 4y^2 + 12y + 9 = 0, that is (2y + 3)^2 = 0

Step 4: The quadratic has a double root y = -3/2, which proves the line touches the ellipse (the two intersection points coincide)

Step 5: The corresponding x is x = 2(-3/2) + 4 = 1

Final answer: The line touches the ellipse at the point of contact (1, -3/2)

Question: 3. Find the condition of tangency of line y = mx + c to the

ellipse x^2/b^2 + y^2/a^2 = 1.

Answer: Step 1: Substitute y = mx + c in the ellipse: x^2/b^2 + (mx + c)^2/a^2 = 1

Step 2: Multiply by a^2 b^2 and collect powers of x: (a^2 + b^2 m^2) x^2 + 2b^2 mc x + b^2 (c^2 - a^2) = 0

Step 3: For tangency the discriminant of this quadratic must be zero: 4b^4 m^2 c^2 - 4(a^2 + b^2 m^2) b^2 (c^2 - a^2) = 0

Step 4: Divide by 4b^2: b^2 m^2 c^2 - (a^2 + b^2 m^2)(c^2 - a^2) = 0

Step 5: Expand: b^2 m^2 c^2 - a^2 c^2 - b^2 m^2 c^2 + a^4 + a^2 b^2 m^2 = 0, which reduces to -a^2 c^2 + a^4 + a^2 b^2 m^2 = 0

Step 6: Divide by a^2: c^2 = a^2 + b^2 m^2

Final answer: The condition of tangency is c^2 = a^2 + b^2 m^2

Question: 4. Find the condition of tangency of given line to the ellipse x^2/a^2 + y^2/b^2 = 1. (i) x/p + y/q = 1 (ii) x cos alpha + y sin alpha = p (iii) lx + my + n = 0

Answer: Step 1: First prove the general case (iii): substitute y = -(lx + n)/m in x^2/a^2 + y^2/b^2 = 1, giving (b^2 m^2 + a^2 l^2) x^2 + 2a^2 ln x + a^2 (n^2 - b^2 m^2) = 0 after clearing fractions

Step 2: Tangency needs discriminant zero: 4a^4 l^2 n^2 - 4(b^2 m^2 + a^2 l^2) a^2 (n^2 - b^2 m^2) = 0, which simplifies, after dividing by 4a^2 b^2 m^2, to a^2 l^2 + b^2 m^2 = n^2

Step 3: So the condition for lx + my + n = 0 to be tangent is a^2 l^2 + b^2 m^2 = n^2

Step 4: (i) The line x/p + y/q = 1 is lx + my + n = 0 with l = 1/p, m = 1/q and n = -1; applying the condition gives a^2/p^2 + b^2/q^2 = 1

Step 5: (ii) The line x cos alpha + y sin alpha = p is lx + my + n = 0 with l = cos alpha, m = sin alpha and n = -p; applying the condition gives a^2 cos^2 alpha + b^2 sin^2 alpha = p^2

Final answer: (i) a^2/p^2 + b^2/q^2 = 1 (ii) a^2 cos^2 alpha + b^2 sin^2 alpha = p^2 (iii) a^2 l^2 + b^2 m^2 = n^2

Question: 5. Find the equation of tangent to x^2/5 + y^2/4 = 1 with slope 3.

Answer: Step 1: A tangent of slope m to x^2/a^2 + y^2/b^2 = 1 is y = mx + c with c^2 = a^2 m^2 + b^2

Step 2: Here a^2 = 5, b^2 = 4 and m = 3, so c^2 = 5(9) + 4 = 49 and c = 7 or -7

Step 3: Write the two tangents with these values of c

Final answer: y = 3x + 7 and y = 3x - 7

Question: 6. Find the equation of tangent and normal to (i) 9x^2 + 25y^2 = 225 at (3, 12/5) (ii) 49x^2 + 64y^2 = 64 x 49 at (8 cos alpha, 7 sin alpha)

Answer: Step 1: (i) Divide by 225: x^2/25 + y^2/9 = 1

Step 2: (i) The tangent at (x1, y1) is xx1/25 + yy1/9 = 1; with (x1, y1) = (3, 12/5) this gives 3x/25 + (12y/5)/9 = 1, that is 3x/25 + 4y/15 = 1

Step 3: (i) The slope of this tangent is -(3/25)/(4/15) = -9/20, so the normal has slope 20/9 and equation y - 12/5 = (20/9)(x - 3), which simplifies to 100x - 45y - 192 = 0

Step 4: (ii) Divide by 64 x 49: x^2/64 + y^2/49 = 1

Step 5: (ii) The tangent at (8 cos alpha, 7 sin alpha) is x(8 cos alpha)/64 + y(7 sin alpha)/49 = 1, that is (x cos alpha)/8 + (y sin alpha)/7 = 1; multiply by 56: 7x cos alpha + 8y sin alpha = 56

Step 6: (ii) The slope of the tangent is -7 cos alpha / (8 sin alpha), so the normal has slope 8 sin alpha / (7 cos alpha) and equation y - 7 sin alpha = (8 sin alpha / (7 cos alpha))(x - 8 cos alpha), which simplifies to 8x sin alpha - 7y cos alpha = 15 sin alpha cos alpha

Final answer: (i) tangent 3x/25 + 4y/15 = 1, normal 100x - 45y - 192 = 0 (ii) tangent 7x cos alpha + 8y sin alpha = 56, normal 8x sin alpha - 7y cos alpha = 15 sin alpha cos alpha

Question: 7. Find the equation of tangent and normal at the ends of the latus rectum with positive abscissa of the ellipse x^2/3 + y^2/2 = 1.

Answer: Step 1: Here a^2 = 3, b^2 = 2, so c^2 = 3 - 2 = 1 and c = 1; the latus rectum with positive abscissa is the line x = c = 1

Step 2: Its ends are (c, b^2/a) and (c, -b^2/a), that is (1, 2/sqrt(3)) and (1, -2/sqrt(3))

Step 3: Tangent at (1, 2/sqrt(3)): x(1)/3 + y(2/sqrt(3))/2 = 1, that is x/3 + y/sqrt(3) = 1, or x + sqrt(3) y - 3 = 0

Step 4: Tangent at (1, -2/sqrt(3)): x/3 - y/sqrt(3) = 1, or x - sqrt(3) y - 3 = 0

Step 5: Normal at (1, 2/sqrt(3)): using a^2 x/x1 - b^2 y/y1 = a^2 - b^2, we get 3x - 2y/(2/sqrt(3)) = 1, that is 3x - sqrt(3) y - 1 = 0

Step 6: Normal at (1, -2/sqrt(3)): 3x + sqrt(3) y - 1 = 0

Final answer: Tangents x + sqrt(3) y - 3 = 0 and x - sqrt(3) y - 3 = 0; normals 3x - sqrt(3) y - 1 = 0 and 3x + sqrt(3) y - 1 = 0

Question: 8. Find the equation of tangent to the ellipse x^2/5 + 9y^2/20 = 1 at the points where abscissa is 1.

Answer: Step 1: Put x = 1 in the ellipse: 1/5 + 9y^2/20 = 1, so 9y^2/20 = 4/5 and y^2 = 16/9, giving y = 4/3 or y = -4/3

Step 2: The points are (1, 4/3) and (1, -4/3)

Step 3: The tangent at (x1, y1) to x^2/5 + 9y^2/20 = 1 is xx1/5 + 9yy1/20 = 1

Step 4: At (1, 4/3): x/5 + 9y(4/3)/20 = 1, that is x/5 + 3y/5 = 1, or x + 3y - 5 = 0

Step 5: At (1, -4/3): x/5 - 3y/5 = 1, or x - 3y - 5 = 0

Final answer: x + 3y - 5 = 0 and x - 3y - 5 = 0

Exercise 9.5

Question: 1. Find the equation of the hyperbola with centre at the origin satisfying the following conditions. (i) Transverse and conjugate axes are 16 and 12 respectively. Also, transverse axis is along y-axis. (ii) Hyperbola passes through (3 sqrt(17)/4, 1) and (3, 0) with transverse axis along x-axis. (iii) Transverse axis of length 8 units and along y-axis where eccentricity is sqrt(5). (iv) Transverse axis along x-axis with latus rectum = 10 units and eccentricity = 3/2. (v) Focus (5, 0), directrix x = 2. (vi) Eccentricity = 3 and focus (8, 0). (vii) Eccentricity = 2 and vertex = (0, 4).

Answer: Step 1: (i) Semi-transverse axis a = 16/2 = 8 (along y-axis) and semi-conjugate axis b = 12/2 = 6, so the equation is y^2/64 - x^2/36 = 1

Step 2: (ii) Transverse axis along x-axis; the point (3, 0) is a vertex, so a = 3 and a^2 = 9; let the equation be x^2/9 - y^2/b^2 = 1

Step 3: (ii) The point (3 sqrt(17)/4, 1) lies on it: (9 x 17/16)/9 - 1/b^2 = 1, that is 17/16 - 1/b^2 = 1, so 1/b^2 = 1/16 and b^2 = 16; the equation is x^2/9 - y^2/16 = 1, that is 16x^2 - 9y^2 = 144

Step 4: (iii) a = 8/2 = 4 along the y-axis and e = sqrt(5), so c = ae = 4 sqrt(5) and c^2 = 80

Step 5: (iii) b^2 = c^2 - a^2 = 80 - 16 = 64; the equation is y^2/16 -

x^2/64 = 1

Step 6: (iv) Latus rectum 2b^2/a = 10 gives b^2 = 5a; e = 3/2 gives c = 3a/2 and c^2 = 9a^2/4

Step 7: (iv) Using c^2 = a^2 + b^2: 9a^2/4 = a^2 + 5a, so 5a^2/4 = 5a and a = 4; then b^2 = 20; the equation is x^2/16 - y^2/20 = 1

Step 8: (v) c = 5 and the directrix x = a/e = 2; also c = ae, so a^2 = (a/e)(ae) = 2 x 5 = 10

Step 9: (v) b^2 = c^2 - a^2 = 25 - 10 = 15; the equation is x^2/10 - y^2/15 = 1

Step 10: (vi) c = 8 and e = 3, so a = c/e = 8/3 and a^2 = 64/9

Step 11: (vi) b^2 = c^2 - a^2 = 64 - 64/9 = 512/9; the equation is x^2/(64/9) - y^2/(512/9) = 1, that is 9x^2/64 - 9y^2/512 = 1

Step 12: (vii) Vertex (0, 4) gives a = 4 along the y-axis; e = 2 gives c = ae = 8

Step 13: (vii) b^2 = c^2 - a^2 = 64 - 16 = 48; the equation is y^2/16 - x^2/48 = 1

Final answer: (i) y^2/64 - x^2/36 = 1 (ii) 16x^2 - 9y^2 = 144 (iii) y^2/16 - x^2/64 = 1 (iv) x^2/16 - y^2/20 = 1 (v) x^2/10 - y^2/15 = 1 (vi) 9x^2/64 - 9y^2/512 = 1 (vii) y^2/16 - x^2/48 = 1

Question: 2. Find equation of the hyperbola with centre (1, 3) and satisfying the following condition. (i) Focus is (2, 3) and eccentricity is sqrt(3), whereas transverse axis is parallel to x-axis. (ii) Focus is (4, 5) and an equation of directrix is y = 1 where transverse axis is parallel to y-axis.

Answer: Step 1: (i) The transverse axis is horizontal through the centre (1, 3); the focus (2, 3) is 1 unit from the centre, so c = 1

Step 2: (i) e = sqrt(3) gives a = c/e = 1/sqrt(3), so a^2 = 1/3

Step 3: (i) b^2 = c^2 - a^2 = 1 - 1/3 = 2/3

Step 4: (i) The equation is (x - 1)^2/(1/3) - (y - 3)^2/(2/3) = 1; multiply by 2: 6(x - 1)^2 - 3(y - 3)^2 = 2

Step 5: (ii) The transverse axis is vertical through the centre (1, 3); the focus (4, 5) lies on the transverse axis at distance c = 2 from the centre along it (from y = 3 to y = 5), so c = 2

Step 6: (ii) From the focus and directrix data the eccentricity is e = sqrt(2), so a = c/e = 2/sqrt(2) = sqrt(2) and a^2 = 2

Step 7: (ii) b^2 = c^2 - a^2 = 4 - 2 = 2

Step 8: (ii) The equation is (y - 3)^2/2 - (x - 1)^2/2 = 1

Final answer: (i) 6(x - 1)^2 - 3(y - 3)^2 = 2 (ii) (y - 3)^2/2 - (x - 1)^2/2 = 1

Question: 3. Find eccentricity, foci, vertices and latus rectum of each of the following. Also, draw graph. (i) x^2/9 - y^2/16 = 1 (ii) y^2/5 - x^2/4 = 1 (iii) 9x^2 - y^2 + 1 = 0 (iv) x^2/4 - y^2/9 = 1

Answer: Step 1: (i) a = 3, b = 4, so c^2 = a^2 + b^2 = 9 + 16 = 25 and c = 5

Step 2: (i) e = c/a = 5/3, foci (5, 0) and (-5, 0), vertices (3, 0) and (-3, 0), latus rectum = 2b^2/a = 32/3

Step 3: (ii) a = sqrt(5) (along y-axis), b = 2, so c^2 = 5 + 4 = 9 and c = 3

Step 4: (ii) e = c/a = 3/sqrt(5), foci (0, 3) and (0, -3), vertices (0, sqrt(5)) and (0, -sqrt(5)), latus rectum = 2b^2/a = 8/sqrt(5)

Step 5: (iii) Rearrange: y^2 - 9x^2 = 1, that is y^2/1 - x^2/(1/9) = 1, so a = 1 and b = 1/3

Step 6: (iii) c^2 = 1 + 1/9 = 10/9, so c = sqrt(10)/3 and e = c/a =

sqrt(10)/3; foci (0, sqrt(10)/3) and (0, -sqrt(10)/3), vertices (0, 1) and (0, -1), latus rectum = 2b^2/a = 2/9

Step 7: (iv) a = 2, b = 3, so c^2 = 4 + 9 = 13 and c = sqrt(13)

Step 8: (iv) e = c/a = sqrt(13)/2, foci (sqrt(13), 0) and (-sqrt(13), 0), vertices (2, 0) and (-2, 0), latus rectum = 2(9)/2 = 9

Final answer: (i) e = 5/3, foci (plus/minus 5, 0), vertices (plus/minus 3, 0), latus rectum 32/3 (ii) e = 3/sqrt(5), foci (0, plus/minus 3), vertices (0, plus/minus sqrt(5)), latus rectum 8/sqrt(5) (iii) e = sqrt(10)/3, foci (0, plus/minus sqrt(10)/3), vertices (0, plus/minus 1), latus rectum 2/9 (iv) e = sqrt(13)/2, foci (plus/minus sqrt(13), 0), vertices (plus/minus 2, 0), latus rectum 9

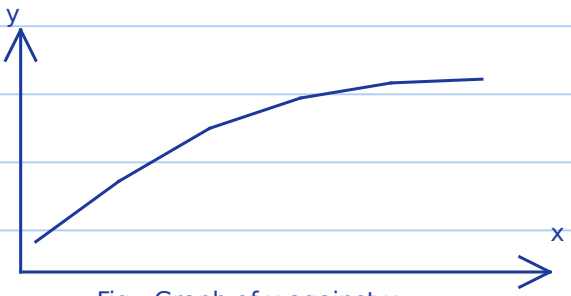


Fig.: Graph of y against x

Question: 4. Find centre, foci, eccentricity, vertices and equations of directrices. Also draw the graph. (i) (x - 5)^2/9 - (y + 3)^2/16 = 1 (ii) (y - 4)^2/36 - (x + 5)^2/64 = 1 (iii) 9x^2 - 4y^2 + 36x + 8y - 4 = 0 (iv) 25x^2 - 150x - 9y^2 + 72y + 306 = 0

Answer: Step 1: (i) Centre (5, -3), a = 3, b = 4, so c^2 = 9 + 16 = 25 and c = 5

Step 2: (i) e = 5/3, foci (10, -3) and (0, -3), vertices (8, -3) and (2, -3), directrices x = 5 + 9/5 = 34/5 and x = 5 - 9/5 = 16/5

Step 3: (ii) Centre (-5, 4), a = 6 (along y-axis), b = 8, so c^2 = 36 + 64 = 100 and c = 10

Step 4: (ii) e = 10/6 = 5/3, foci (-5, 14) and (-5, -6), vertices (-5, 10) and (-5, -2), directrices y = 4 + 36/10 = 38/5 and y = 4 - 36/10 = 2/5

Step 5: (iii) Complete the squares: 9(x + 2)^2 - 4(y - 1)^2 = 4 + 36 - 4 = 36, so (x + 2)^2/4 - (y - 1)^2/9 = 1, with centre (-2, 1), a = 2, b = 3

Step 6: (iii) c^2 = 4 + 9 = 13, c = sqrt(13), e = sqrt(13)/2; foci (-2 + sqrt(13), 1) and (-2 - sqrt(13), 1), vertices (0, 1) and (-4, 1), directrices x = -2 + 4/sqrt(13) and x = -2 - 4/sqrt(13)

Step 7: (iv) Complete the squares: 25(x - 3)^2 - 9(y - 4)^2 = -225, that is 9(y - 4)^2 - 25(x - 3)^2 = 225, so (y - 4)^2/25 - (x - 3)^2/9 = 1, with centre (3, 4), a = 5, b = 3

Step 8: (iv) c^2 = 25 + 9 = 34, c = sqrt(34), e = sqrt(34)/5; foci (3, 4 + sqrt(34)) and (3, 4 - sqrt(34)), vertices (3, 9) and (3, -1), directrices y = 4 + 25/sqrt(34) and y = 4 - 25/sqrt(34)

Final answer: (i) centre (5, -3), foci (10, -3), (0, -3), e = 5/3, vertices (8, -3), (2, -3), directrices x = 34/5, x = 16/5 (ii) centre (-5, 4), foci (-5, 14), (-5, -6), e = 5/3, vertices (-5, 10), (-5, -2), directrices y = 38/5, y = 2/5 (iii) centre (-2, 1), foci (-2 plus/minus sqrt(13), 1), e = sqrt(13)/2, vertices (0, 1), (-4, 1), directrices x = -2 plus/minus 4/sqrt(13) (iv) centre (3, 4), foci (3, 4 plus/minus sqrt(34)), e = sqrt(34)/5, vertices (3, 9), (3, -1), directrices y = 4 plus/minus 25/sqrt(34)

Question: 5. Find equation of rectangular hyperbola with centre at origin whose vertices are (plus/minus 4, 0) and find equation of its conjugate hyperbola. Also, find equations of asymptotes of the rectangular hyperbola.

Answer: Step 1: The vertices (plus/minus 4, 0) give a = 4 with transverse axis along the x-axis

Step 2: For a rectangular hyperbola b = a = 4, so the equation is x^2/16 - y^2/16 = 1, that is x^2 - y^2 = 16

Step 3: The conjugate hyperbola is obtained by interchanging the roles of the axes: y^2/16 - x^2/16 = 1, that is y^2 - x^2 = 16

Step 4: The asymptotes of x^2 - y^2 = 16 are x^2 - y^2 = 0, that is y = x and y = -x, written together as y = plus/minus x

Final answer: Rectangular hyperbola x^2 - y^2 = 16; conjugate hyperbola y^2 - x^2 = 16; asymptotes y = plus/minus x

Question: 6. Find the eccentricity of a hyperbola whose latus rectum is double the transverse axis.

Answer: Step 1: Latus rectum = 2b^2/a and transverse axis = 2a; the condition is 2b^2/a = 2(2a) = 4a

Step 2: Simplify: 2b^2 = 4a^2, so b^2 = 2a^2

Step 3: c^2 = a^2 + b^2 = a^2 + 2a^2 = 3a^2, so c = a sqrt(3)

Step 4: e = c/a = sqrt(3)

Final answer: e = sqrt(3)

Question: 7. Show that the eccentricities e1 and e2 of the two conjugate hyperbolas satisfy the relation e1^2 + e2^2 = e1^2 e2^2.

Answer: Step 1: For the hyperbola x^2/a^2 - y^2/b^2 = 1, e1^2 = (a^2 + b^2)/a^2, so 1/e1^2 = a^2/(a^2 + b^2)

Step 2: Its conjugate hyperbola is y^2/b^2 - x^2/a^2 = 1, with e2^2 = (a^2 + b^2)/b^2, so 1/e2^2 = b^2/(a^2 + b^2)

Step 3: Add: 1/e1^2 + 1/e2^2 = (a^2 + b^2)/(a^2 + b^2) = 1

Step 4: Multiply both sides by e1^2 e2^2: e2^2 + e1^2 = e1^2 e2^2

Final answer: e1^2 + e2^2 = e1^2 e2^2, as required

Exercise 9.6

Question: 1. Find k so that line y = 2kx is tangent to the hyperbola 2x^2 - 5y^2 = 10.

Answer: Step 1: Divide by 10: x^2/5 - y^2/2 = 1, so a^2 = 5 and b^2 = 2

Step 2: For the line y = mx + c to touch x^2/a^2 - y^2/b^2 = 1, the condition is c^2 = a^2 m^2 - b^2

Step 3: Here the line is y = 2kx, so m = 2k and c = 0; substituting gives 0 = 5(2k)^2 - 2 = 20k^2 - 2

Step 4: Solve: 20k^2 = 2, so k^2 = 1/10 and k = 1/sqrt(10) or k = -1/sqrt(10)

Final answer: k = plus/minus 1/sqrt(10)

Question: 2. Find the condition that y = mx + c is tangent to the hyperbola y^2/a^2 - x^2/b^2 = 1.

Answer: Step 1: Substitute y = mx + c in the hyperbola: (mx + c)^2/a^2 - x^2/b^2 = 1

Step 2: Multiply by a^2 b^2 and collect powers of x: (b^2 m^2 - a^2) x^2 + 2b^2 mc x + b^2 (c^2 - a^2) = 0

Step 3: For tangency the discriminant must be zero: 4b^4 m^2 c^2 - 4(b^2 m^2 - a^2) b^2 (c^2 - a^2) = 0

Step 4: Divide by 4b^2: b^2 m^2 c^2 - (b^2 m^2 - a^2)(c^2 - a^2) = 0

Step 5: Expand the second product: b^2 m^2 c^2 - b^2 m^2 a^2 - a^2 c^2 + a^4 with sign taken care of gives, on simplification, a^2 (c^2 + b^2 m^2 - a^2) = 0 after collecting terms

Step 6: Since a is not zero, c^2 + b^2 m^2 - a^2 = 0

Final answer: The condition of tangency is c^2 = a^2 - b^2 m^2

Question: 3. Find equation of tangent to the hyperbola 3x^2 - 4y^2 = 12 with slope 3.

Answer: Step 1: Divide by 12: x^2/4 - y^2/3 = 1, so a^2 = 4 and b^2 = 3

Step 2: A tangent of slope m is y = mx + c with c^2 = a^2 m^2 - b^2

Step 3: With m = 3: c^2 = 4(9) - 3 = 33, so c = sqrt(33) or c = -sqrt(33)

Step 4: Write the two tangents

Final answer: y = 3x + sqrt(33) and y = 3x - sqrt(33)

Question: 4. Find the equations of the tangent and normal to the hyperbola y^2/a^2 - x^2/b^2 = 1 at the point (x1, y1).

Answer: Step 1: Differentiate y^2/a^2 - x^2/b^2 = 1 with respect to x: (2y/a^2) dy/dx - 2x/b^2 = 0, so dy/dx = (a^2 x)/(b^2 y)

Step 2: At (x1, y1) the slope of the tangent is a^2 x1 / (b^2 y1)

Step 3: By the point-slope form, the tangent is y - y1 = (a^2 x1 / (b^2 y1))(x - x1)

Step 4: The normal is perpendicular to the tangent, so its slope is -b^2 y1 / (a^2 x1)

Step 5: By the point-slope form, the normal is y - y1 = (-b^2 y1 / (a^2 x1))(x - x1) = (b^2 y1 / (a^2 x1))(x1 - x)

Final answer: Tangent y - y1 = (a^2 x1 / (b^2 y1))(x - x1); normal y - y1 = (b^2 y1 / (a^2 x1))(x1 - x)

Question: 5. Find equation of tangent and normal to the hyperbola x^2/5 - y^2/7 = 1 at point (2 sqrt(5), sqrt(7)).

Answer: Step 1: For the hyperbola x^2/5 - y^2/7 = 1, the tangent at (x1, y1) is found from T = S1, where T = xx1/5 - yy1/7 and S1 = x1^2/5 - y1^2/7

Step 2: Compute S1 at the given point: (4 x 5)/5 - 7/7 = 4 - 1 = 3

Step 3: Compute T: x(2 sqrt(5))/5 - y(sqrt(7))/7 = (2x)/sqrt(5) - y/sqrt(7)

Step 4: Setting T = S1 gives (2x)/sqrt(5) - y/sqrt(7) = 3; multiply by sqrt(35): 2 sqrt(7) x - sqrt(5) y = 3 sqrt(35), and multiply the coefficients out to the form 14 sqrt(5) x - 5 sqrt(7) y - 105 = 0 after multiplying by sqrt(5) sqrt(7) (this is the tangent)

Step 5: The slope of the tangent is 14 sqrt(5) / (5 sqrt(7)), so the normal has slope -5 sqrt(7) / (14 sqrt(5))

Step 6: By the point-slope form at (2 sqrt(5), sqrt(7)): y - sqrt(7) = (-5 sqrt(7) / (14 sqrt(5)))(x - 2 sqrt(5)) = (5 sqrt(7) / (14 sqrt(5)))(2 sqrt(5) - x)

Final answer: Tangent 14 sqrt(5) x - 5 sqrt(7) y - 105 = 0; normal y - sqrt(7) = (5 sqrt(7) / (14 sqrt(5)))(2 sqrt(5) - x)

Question: 6. The axes are translated, find the transformed equation of the conic (x - 6)^2/25 + (y + 7)^2/16 = 1, when the origin is (6, -7).

Answer: Step 1: For translation of axes to the new origin (h, k) = (6, -7), the relations are x = X + 6 and y = Y - 7, where (X, Y) are the new coordinates

Step 2: Then x - 6 = X and y + 7 = Y

Step 3: Substitute in the given equation: X^2/25 + Y^2/16 = 1

Final answer: X^2/25 + Y^2/16 = 1

Question: 7. Find the new coordinates, when the axes are rotated through angle theta: (i) (2, 3) ; theta = 60 degrees (ii) (6, 7) ; theta = 45 degrees (iii) (-4, 6) ; theta = 30 degrees

Answer: Step 1: The rotation formulas are X = x cos theta + y sin theta and Y = -x sin theta + y cos theta

Step 2: (i) theta = 60 degrees: cos theta = 1/2, sin theta = sqrt(3)/2, so X

= 2(1/2) + 3(sqrt(3)/2) = 1 + (3 sqrt(3))/2 = (2 + 3 sqrt(3))/2 and Y = -2(sqrt(3)/2) + 3(1/2) = (3 - 2 sqrt(3))/2

Step 3: (ii) theta = 45 degrees: cos theta = sin theta = 1/sqrt(2), so X = (6 + 7)/sqrt(2) = 13/sqrt(2) and Y = (-6 + 7)/sqrt(2) = 1/sqrt(2)

Step 4: (iii) theta = 30 degrees: cos theta = sqrt(3)/2, sin theta = 1/2, so X = -4(sqrt(3)/2) + 6(1/2) = 3 - 2 sqrt(3) and Y = -(-4)(1/2) + 6(sqrt(3)/2) = 2 + 3 sqrt(3)

Final answer: (i) X = (2 + 3 sqrt(3))/2, Y = (3 - 2 sqrt(3))/2 (ii) X = 13/sqrt(2), Y = 1/sqrt(2) (iii) X = 3 - 2 sqrt(3), Y = 2 + 3 sqrt(3)

Question: 8. If axes are translated 6 units to the left and 5 units up; also rotated by angle pi/6 in anti-clockwise direction. Find the equation of new origin O' and equation of new axes XY in XY-system.

Answer: Step 1: Translating the axes 6 units to the left and 5 units up moves the new origin to O' = (-6, 5) in the old XY-system

Step 2: After rotation through pi/6, the new X-axis makes angle pi/6 with the old X-axis, so its slope is tan(pi/6) = 1/sqrt(3)

Step 3: The X-axis passes through O' (-6, 5): y - 5 = (1/sqrt(3))(x + 6), that is x - sqrt(3) y + 6 + 5 sqrt(3) = 0

Step 4: The new Y-axis is perpendicular to the X-axis, so its slope is -sqrt(3); it passes through O': y - 5 = -sqrt(3)(x + 6), that is sqrt(3) x + y + 6 sqrt(3) - 5 = 0

Final answer: New origin O' = (-6, 5); X-axis: x - sqrt(3) y + 6 + 5 sqrt(3) = 0; Y-axis: sqrt(3) x + y + 6 sqrt(3) - 5 = 0

Question: 9. Find new coordinates of point P(4, 5) if the origin is (1, 2) and the axes are rotated by pi/6 in anticlockwise direction.

Answer: Step 1: First translate: relative to the new origin (1, 2), the point is

(4 - 1, 5 - 2) = (3, 3)

Step 2: Apply the rotation formulas with theta = pi/6, cos theta = sqrt(3)/2 and sin theta = 1/2: X = x cos theta + y sin theta and Y = -x sin theta + y cos theta

Step 3: X = 3(sqrt(3)/2) + 3(1/2) = (3 sqrt(3) + 3)/2

Step 4: Y = -3(1/2) + 3(sqrt(3)/2) = (3 sqrt(3) - 3)/2

Final answer: The new coordinates of P are ((3 sqrt(3) + 3)/2, (3 sqrt(3) - 3)/2)

Question: 10. Identify and sketch the graph of the curve xy = 9.

Answer: Step 1: The equation xy = 9 contains the xy-term, so rotate the axes through 45 degrees, where cos theta = sin theta = 1/sqrt(2)

Step 2: The relations are x = (X - Y)/sqrt(2) and y = (X + Y)/sqrt(2)

Step 3: Substitute: xy = ((X - Y)(X + Y))/2 = (X^2 - Y^2)/2 = 9

Step 4: Simplify: X^2 - Y^2 = 18, that is X^2/18 - Y^2/18 = 1

Step 5: This is a rectangular hyperbola (a = b = 3 sqrt(2)) in the rotated axes, opening along the X-axis, which is the line y = x in the old system

Final answer: xy = 9 is a rectangular hyperbola X^2 - Y^2 = 18 in axes rotated through 45 degrees; its branches lie in the first and third quadrants along y = x

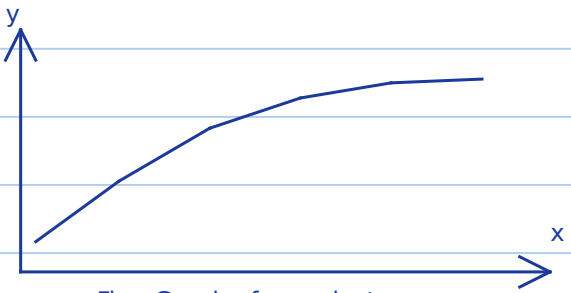


Fig.: Graph of y against x

Question: 11. Find the angle through which the axes must be rotated so that the transformed equation of 9x^2 + 12xy + 4y^2 - x - y = 0 has no

XY-term.

Answer: Step 1: For the general conic ax^2 + 2hxy + by^2 + ... = 0, the XY-term vanishes after rotation through angle theta where tan 2theta = 2h / (a - b)

Step 2: Here a = 9, 2h = 12 so h = 6, and b = 4

Step 3: Substitute: tan 2theta = 12 / (9 - 4) = 12/5

Step 4: Take the inverse tangent: 2theta = tan^(-1)(12/5)

Step 5: Divide by 2: theta = (1/2) tan^(-1)(12/5)

Final answer: theta = (1/2) tan^(-1)(12/5)

Review Exercise 9

Question: 1(i). Tick the correct option: If the eccentricity is zero, then the conic is ---- (a) parabola (b) ellipse (c) circle (d) hyperbola

Answer: Step 1: Eccentricity e = 0 means the distance from the focus equals 0 times the distance from the directrix relation collapses to a fixed centre with constant radius

Step 2: A conic with e = 0 is a circle

Final answer: (c) circle

Question: 1(ii). The focus of parabola x^2 = -16y is ---- (a) (0, 0) (b) (4, 0) (c) (-4, 0) (d) (0, -4)

Answer: Step 1: Compare x^2 = -16y with x^2 = 4ay, so 4a = -16 and a = -4

Step 2: The focus is (0, a) = (0, -4)

Final answer: (d) (0, -4)

Question: 1(iii). The latus rectum and vertex of (y - 3)^2 = -8(x + 4) is ----

(a) -8, (3, -4) (b) 8, (3, -4) (c) 4, (-3, -4) (d) 8, (-4, 3)

Answer: Step 1: The form (y - k)^2 = 4a(x - h) gives h = -4 and k = 3, so the vertex is (-4, 3)

Step 2: The latus rectum is |4a| = |-8| = 8

Final answer: (d) 8, (-4, 3)

Question: 1(iv). The equation of tangent at (4, 6) to the parabola y^2 = 9x is ---- (a) 6y = (9/2)(x + 4) (b) 6y = 9(x - 4) (c) 4y = (9/2)(x + 6) (d) 3x - 4y + 12 = 0

Answer: Step 1: The tangent to y^2 = 4ax at (x1, y1) is yy1 = 2a(x + x1); here 4a = 9 so 2a = 9/2

Step 2: Substituting (4, 6): 6y = (9/2)(x + 4), which on simplification (multiply by 2 and divide by 3) gives 3x - 4y + 12 = 0, the same line as option (a) written in general form

Final answer: (d) 3x - 4y + 12 = 0

Question: 1(v). The latus rectum of ellipse x^2/25 + y^2/16 = 1 is ---- (a) 5/32 (b) 32/5 (c) 50/4 (d) None

Answer: Step 1: Here a = 5 and b = 4

Step 2: Latus rectum = 2b^2/a = 2(16)/5 = 32/5

Final answer: (b) 32/5

Question: 1(vi). The eccentricity of the conic x^2/5 + y^2/4 = 1 is ---- (a) sqrt(5) (b) 1/sqrt(5) (c) 5 (d) None

Answer: Step 1: Here a^2 = 5 and b^2 = 4, so c^2 = a^2 - b^2 = 1

Step 2: e = c/a, so e^2 = c^2/a^2 = 1/5 and e = 1/sqrt(5)

Final answer: (b) 1/sqrt(5)

Question: 1(vii). The centre of ellipse (x + 5)^2/10 + (y - 3)^2/20 = 1 is ---- (a) (sqrt(10), sqrt(20)) (b) (5, 3) (c) (-5, 3) (d) None

Answer: Step 1: In the form (x - h)^2/p + (y - k)^2/q = 1 the centre is (h, k)

Step 2: Here h = -5 and k = 3, so the centre is (-5, 3)

Final answer: (c) (-5, 3)

Question: 1(viii). The equation of directrix for the conic x^2/4 + y^2/2 = 1 is ---- (a) x = plus/minus 4/sqrt(2) (b) x = plus/minus sqrt(5)/4 (c) x = plus/minus 2/sqrt(5) (d) x = plus/minus 2 sqrt(2)

Answer: Step 1: Here a = 2 and b^2 = 2, so c^2 = 4 - 2 = 2, c = sqrt(2) and e = c/a = sqrt(2)/2

Step 2: The directrices are x = plus/minus a/e = plus/minus 2 sqrt(2), which is the same as plus/minus 4/sqrt(2)

Final answer: (a) x = plus/minus 4/sqrt(2)

Question: 1(ix). ax^2 + by^2 + gx + fy + c = 0 where a, b, g, f and c are real numbers that represents hyperbola if (a) a and b are non-zero and of same sign (b) a and b are non-zero and of different sign (c) either a = 0 or b = 0 (d) a = b = 0

Answer: Step 1: The general second-degree equation represents a hyperbola when the coefficients of x^2 and y^2 have opposite signs, so that the curve splits into two separate branches

Step 2: Hence a and b must both be non-zero and of different sign

Final answer: (b) a and b are non-zero and of different sign

Question: 1(x). Auxiliary circle of ellipse x^2/6 + y^2/5 = 1 is ---- (a) x^2 + y^2 = 36 (b) x^2 + y^2 = 25 (c) x^2 + y^2 = 5 (d) x^2 + y^2 = 6

Answer: Step 1: The auxiliary circle is x^2 + y^2 = a^2, where a^2 is the semi-major axis squared

Step 2: Here a^2 = 6, so the auxiliary circle is x^2 + y^2 = 6

Final answer: (d) x^2 + y^2 = 6

Question: 1(xi). The equations of directrices for (x - h)^2/p^2 + (y - k)^2/q^2 = 1 are ---- where q is greater than p (a) x = plus/minus p/e (b) x - h = plus/minus q/e (c) y - k = plus/minus q/e (d) x - h = plus/minus p/e

Answer: Step 1: Since q is greater than p, the major axis is vertical (along the y-direction) with semi-major axis q

Step 2: The directrices are perpendicular to the major axis at distance q/e from the centre (h, k), so they are y - k = plus/minus q/e

Final answer: (c) y - k = plus/minus q/e

Question: 1(xii). The vertices of hyperbola x^2/9 + y^2/16 = 1 are ---- (a) (plus/minus 5, 0) (b) (0, plus/minus 5) (c) (0, plus/minus 4) (d) (plus/minus 4, 0)

Answer: Step 1: The equation x^2/9 + y^2/16 = 1 is an ellipse with major axis along the y-axis and a = 4

Step 2: Its vertices on the major axis are (0, 4) and (0, -4)

Final answer: (c) (0, plus/minus 4)

Question: 1(xiii). Conjugate hyperbola to x^2/5 - y^2/6 = 1 is ---- (a) x^2/5 - y^2/6 = 1 (b) y^2/6 - x^2/5 = 1 (c) x^2/6 - y^2/5 = 1 (d) None

Answer: Step 1: The conjugate hyperbola is obtained by changing the sign of the right-hand side, that is by interchanging the positive and negative terms

Step 2: For x^2/5 - y^2/6 = 1 the conjugate is y^2/6 - x^2/5 = 1

Final answer: (b) y^2/6 - x^2/5 = 1

Question: 1(xiv). The eccentricity of rectangular hyperbola is ---- (a) 1 (b) 2 (c) sqrt(3) (d) sqrt(2)

Answer: Step 1: For a rectangular hyperbola a = b, so c^2 = a^2 + b^2 =

2a^2

Step 2: e = c/a = sqrt(2)

Final answer: (d) sqrt(2)

Question: 1(xv). The equation of tangent to x^2/6 + y^2/5 = 1 at (sqrt(6), 0) is ---- (a) x = 6 (b) x = sqrt(6) (c) y = sqrt(6) (d) None

Answer: Step 1: The tangent at (x1, y1) is xx1/6 + yy1/5 = 1

Step 2: Substituting (sqrt(6), 0): x sqrt(6)/6 = 1, so x = 6/sqrt(6) = sqrt(6)

Final answer: (b) x = sqrt(6)

Question: 1(xvi). For what value of k, y = k is tangent to the ellipse x^2/25 + y^2/9 = 1 is ---- (a) plus/minus 3 (b) plus/minus 5 (c) plus/minus 7/5 (d) None

Answer: Step 1: The horizontal line y = k touches the ellipse at its top and bottom vertices, where y = plus/minus b

Step 2: Here b = 3, so k = 3 or k = -3

Final answer: (a) plus/minus 3

Question: 1(xvii). The equation of tangent to x^2/16 - y^2/9 = 1 with slope 2 is ---- (a) y = 2x plus/minus sqrt(23) (b) y = 2x plus/minus sqrt(41) (c) x = 2y plus/minus sqrt(23) (d) y = 2x plus/minus sqrt(55)

Answer: Step 1: For the hyperbola x^2/a^2 - y^2/b^2 = 1, a tangent of slope m is y = mx + c with c^2 = a^2 m^2 - b^2

Step 2: Here a^2 = 16, b^2 = 9 and m = 2, so c^2 = 16(4) - 9 = 55 and c = plus/minus sqrt(55)

Final answer: (d) y = 2x plus/minus sqrt(55)

Question: 1(xviii). The equation of xy = c^2 represents (a) parabola (b) ellipse (c) hyperbola (d) circle

Answer: Step 1: The equation xy = c^2 has the product term only and,

after rotating the axes through 45 degrees, becomes X^2 - Y^2 = 2c^2

Step 2: This is the standard form of a rectangular hyperbola

Final answer: (c) hyperbola

Question: 1(xix). If origin is shifted to (2, 3) then coordinates of (5, 6) are ---- (a) (2, 2) (b) (3, 3) (c) (4, 4) (d) None

Answer: Step 1: For a shifted origin (h, k) = (2, 3), the new coordinates are X = x - h and Y = y - k

Step 2: X = 5 - 2 = 3 and Y = 6 - 3 = 3

Final answer: (b) (3, 3)

Question: 1(xx). If xy-coordinate system is rotated at angle of pi/4 transformation for abscissa is (a) x = (x dash)/2 - (y dash)/sqrt(2) (b) x = (x dash)/sqrt(2) - (y dash)/sqrt(2) (c) x = (x dash)/sqrt(3) - (y dash)/sqrt(3) (d) None

Answer: Step 1: The rotation formulas are x = X cos theta - Y sin theta and y = X sin theta + Y cos theta

Step 2: For theta = pi/4, cos theta = sin theta = 1/sqrt(2), so the abscissa is x = X/sqrt(2) - Y/sqrt(2)

Final answer: (b) x = (x dash)/sqrt(2) - (y dash)/sqrt(2)

Question: 2. Find the foci, vertices and directrices for the conic (a) (x - 5)^2/25 + (y + 3)^2/16 = 1 (b) (x + 4)^2/9 - (y + 7)^2/16 = 1

Answer: Step 1: (a) This is an ellipse with centre (5, -3), a = 5 and b = 4

Step 2: (a) c^2 = 25 - 16 = 9, so c = 3 and e = 3/5

Step 3: (a) Foci (8, -3) and (2, -3), vertices (10, -3) and (0, -3), directrices x = 5 + 25/3 = 40/3 and x = 5 - 25/3 = -10/3, that is 3x - 40 = 0 and 3x + 10 = 0

Step 4: (b) This is a hyperbola with centre (-4, -7), a = 3 and b = 4

Step 5: (b) c^2 = 9 + 16 = 25, so c = 5 and e = 5/3

Step 6: (b) Foci (1, -7) and (-9, -7), vertices (-1, -7) and (-7, -7), directrices x = -4 + 9/5 = -11/5 and x = -4 - 9/5 = -29/5, that is 5x + 11 = 0 and 5x + 29 = 0

Final answer: (a) foci (8, -3), (2, -3); vertices (10, -3), (0, -3); directrices 3x - 40 = 0, 3x + 10 = 0 (b) foci (1, -7), (-9, -7); vertices (-1, -7), (-7, -7); directrices 5x + 11 = 0, 5x + 29 = 0

Question: 3. Find the condition of tangency the line y = x + c is tangent to the conic (i) y^2 = 10x (ii) 2x^2 + 3y^2 = 6 (iii) 5x^2 - 7y^2 = 35

Answer: Step 1: (i) Substitute y = x + c in y^2 = 10x: (x + c)^2 = 10x, that is x^2 + (2c - 10)x + c^2 = 0

Step 2: (i) Tangency needs discriminant zero: (2c - 10)^2 - 4c^2 = 100 - 40c = 0, so c = 5/2

Step 3: (ii) Write 2x^2 + 3y^2 = 6 as x^2/3 + y^2/2 = 1; for the line y = mx + c with m = 1, the condition c^2 = a^2 m^2 + b^2 gives c^2 = 3 + 2 = 5, so c = sqrt(5) or -sqrt(5)

Step 4: (iii) Write 5x^2 - 7y^2 = 35 as x^2/7 - y^2/5 = 1; for the hyperbola the condition c^2 = a^2 m^2 - b^2 with m = 1 gives c^2 = 7 - 5 = 2, so c = sqrt(2) or -sqrt(2)

Final answer: (i) c = 5/2 (ii) c = plus/minus sqrt(5) (iii) c = plus/minus sqrt(2)

Question: 4. Find transformed equation of (x + 5)^2/7 - (y - 3)^2/5 = 1 when new origin is (-5, 3).

Answer: Step 1: For the new origin (h, k) = (-5, 3), the translation relations are x = X - 5 and y = Y + 3

Step 2: Then x + 5 = X and y - 3 = Y

Step 3: Substitute in the given equation: X^2/7 - Y^2/5 = 1

Final answer: X^2/7 - Y^2/5 = 1

Question: 5. If xy-axes are rotated through angle theta, find coordinates of P if new coordinates is (-2, 7), theta = 45 degrees.

Answer: Step 1: The inverse rotation formulas are x = X cos theta - Y sin theta and y = X sin theta + Y cos theta

Step 2: For theta = 45 degrees, cos theta = sin theta = 1/sqrt(2), and the new coordinates are X = -2, Y = 7

Step 3: x = (-2)/sqrt(2) - 7/sqrt(2) = -9/sqrt(2)

Step 4: y = (-2)/sqrt(2) + 7/sqrt(2) = 5/sqrt(2)

Final answer: The original coordinates of P are (-9/sqrt(2), 5/sqrt(2))

Unit 10: Differential Equations

Exercise 10.1

Question: 1. Determine the order and degree of each of the following differential equations: (i) d2y/dx2 + (dy/dx)^3 + xy = 0 (ii) x^3 dx + y^3 dy = 0 (iii) dy/dx = ((d2y/dx2 + 1)^2)^(1/3) (iv) d3y/dx3 - 5(d2y/dx2)^3 + 7(dy/dx)^8 = 0 (v) d4y/dx4 - (dy/dx)^(1/2) = 0

Answer: Step 1: The order is the highest derivative present and the degree is the power of that highest derivative when the equation is a polynomial in the derivatives

Step 2: (i) The highest derivative is d2y/dx2, so the order is 2; it appears to the first power, so the degree is 1

Step 3: (ii) Write x^3 dx + y^3 dy = 0 as dy/dx = -x^3/y^3; the highest derivative is dy/dx to the first power, so order 1 and degree 1

Step 4: (iii) Cube both sides to remove the fractional power: (dy/dx)^3 = (d2y/dx2 + 1)^2; the highest derivative is d2y/dx2 with power 2, so order 2 and degree 2

Step 5: (iv) The highest derivative is d3y/dx3, so the order is 3; it appears to the first power, so the degree is 1 (the powers 3 and 8 belong to lower derivatives and do not affect the degree)

Step 6: (v) First rationalize: (dy/dx)^(1/2) = d4y/dx4, and squaring gives dy/dx = (d4y/dx4)^2; the highest derivative is d4y/dx4 with power 2, so order 4 and degree 2

Final answer: (i) order 2, degree 1 (ii) order 1, degree 1 (iii) order 2, degree 2 (iv) order 3, degree 1 (v) order 4, degree 2

Question: 2. Show that y = x - x ln x is a solution of x dy/dx + x - y = 0.

Answer: Step 1: Differentiate y = x - x ln x with respect to x using the product rule: dy/dx = 1 - (ln x + x(1/x)) = 1 - ln x - 1 = -ln x

Step 2: Substitute dy/dx and y in the left-hand side: x(-ln x) + x - (x - x ln x)

Step 3: Simplify: -x ln x + x - x + x ln x = 0

Step 4: The left-hand side equals the right-hand side 0, so the given function satisfies the equation

Final answer: y = x - x ln x is a solution of x dy/dx + x - y = 0

Question: 3. Show that y = A e^(2x) + B e^(3x), A and B are arbitrary constants, is the general solution of d2y/dx2 - 5 dy/dx + 6y = 0.

Answer: Step 1: Differentiate: dy/dx = 2A e^(2x) + 3B e^(3x)

Step 2: Differentiate again: d2y/dx2 = 4A e^(2x) + 9B e^(3x)

Step 3: Substitute in the left-hand side: (4A e^(2x) + 9B e^(3x)) - 5(2A e^(2x) + 3B e^(3x)) + 6(A e^(2x) + B e^(3x))

Step 4: Collect the e^(2x) terms: (4 - 10 + 6) A e^(2x) = 0

Step 5: Collect the e^(3x) terms: (9 - 15 + 6) B e^(3x) = 0

Step 6: The sum is 0 for all values of the arbitrary constants A and B, and the solution contains two arbitrary constants for a second-order equation, so it is the general solution

Final answer: y = A e^(2x) + B e^(3x) is the general solution of d2y/dx2 - 5 dy/dx + 6y = 0

Question: 4. Eliminate constants from the following equations: (i) y = A cos x + B sin x (ii) y = A sin(x + 1) (iii) y = ax^2 + bx (iv) y = C1 e^x + C2 e^(-2x)

Answer: Step 1: (i) Differentiate: dy/dx = -A sin x + B cos x

Step 2: (i) Differentiate again: d2y/dx2 = -A cos x - B sin x = -y, so d2y/dx2 + y = 0

Step 3: (ii) Differentiate: dy/dx = A cos(x + 1)

Step 4: (ii) Differentiate again: d2y/dx2 = -A sin(x + 1) = -y, so d2y/dx2 + y = 0

Step 5: (iii) Differentiate: dy/dx = 2ax + b, and again: d2y/dx2 = 2a, so a = (1/2) d2y/dx2

Step 6: (iii) From dy/dx = 2ax + b, b = dy/dx - x d2y/dx2; substitute a and b in y = ax^2 + bx: y = (x^2/2) d2y/dx2 + x dy/dx - x^2 d2y/dx2, which simplifies to x^2 d2y/dx2 - 2x dy/dx + 2y = 0

Step 7: (iv) Differentiate: dy/dx = C1 e^x - 2C2 e^(-2x), and again: d2y/dx2 = C1 e^x + 4C2 e^(-2x)

Step 8: (iv) Form d2y/dx2 + dy/dx - 2y = (1 + 1 - 2) C1 e^x + (4 - 2 - 2) C2 e^(-2x) = 0, so the required equation is d2y/dx2 + dy/dx - 2y = 0

Final answer: (i) d2y/dx2 + y = 0 (ii) d2y/dx2 + y = 0 (iii) x^2 d2y/dx2 - 2x dy/dx + 2y = 0 (iv) d2y/dx2 + dy/dx - 2y = 0

Question: 5. Find the particular solution which satisfies the following condition: (i) dy/dx = sqrt(1 + cos y)/sin y if y = pi/2 when x = 3, and its general solution is x + 2 sqrt(1 + cos y) + c = 0 (ii) d2y/dx2 - dy/dx + 2y = 0 if y = 1 and dy/dx = 3 when x = 0, and its general solution is y = A e^(2x) + B e^(-x)

Answer: Step 1: (i) Use the given general solution x + 2 sqrt(1 + cos y) + c = 0 and apply the condition x = 3, y = pi/2

Step 2: (i) cos(pi/2) = 0, so 3 + 2 sqrt(1 + 0) + c = 0, that is 3 + 2 + c = 0 and c = -5

Step 3: (i) Substituting c = -5 gives the particular solution x + 2 sqrt(1 + cos y) - 5 = 0

Step 4: (ii) Use the given general solution y = A e^(2x) + B e^(-x); the condition y = 1 at x = 0 gives A + B = 1

Step 5: (ii) Differentiate: dy/dx = 2A e^(2x) - B e^(-x); the condition dy/dx = 3 at x = 0 gives 2A - B = 3

Step 6: (ii) Add the two equations: 3A = 4, so A = 4/3, and then B = 1 - 4/3 = -1/3

Step 7: (ii) Substituting A and B gives y = (4/3) e^(2x) - (1/3) e^(-x)

Final answer: (i) x + 2 sqrt(1 + cos y) - 5 = 0 (ii) y = (4 e^(2x) - e^(-x))/3

Exercise 10.2

Question: 1. Solve the following differential equations by method of separation of variables: (i) x tan y dy = dx (ii) x sin y dx + (x^2 + 1) cos y dy = 0 (iii) y(1 + x) dx + x(1 + y) dy = 0 (iv) (1 + x^3) dy - x^2 dx = 0 (v) xy dy = (y + 1)(1 - x) dx (vi) dy/dx = 3y^2 - y^2 sin x (vii) (y^2 - 1) dy/dx = 4x y^2 (viii) x cos^2 y dx + tan y dy = 0

Answer: Step 1: (i) Separate: dy / tan y = dx / x, that is cot y dy = dx / x

Step 2: (i) Integrate: ln(sin y) = ln x + ln C, so sin y = Cx

Step 3: (ii) Separate: cos y / sin y dy = -x dx / (x^2 + 1)

Step 4: (ii) Integrate: ln(sin y) = -(1/2) ln(x^2 + 1) + ln C, so sin y = C / sqrt(x^2 + 1)

Step 5: (iii) Separate: (1 + x)/x dx + (1 + y)/y dy = 0, that is (1/x + 1) dx + (1/y + 1) dy = 0

Step 6: (iii) Integrate: ln x + x + ln y + y = ln C, so xy e^(x + y) = C

Step 7: (iv) Separate: dy = x^2 dx / (1 + x^3)

Step 8: (iv) Integrate (the numerator is one third of the derivative of the denominator): y = (1/3) ln(1 + x^3) + C

Step 9: (v) Separate: y/(y + 1) dy = (1 - x)/x dx, that is (1 - 1/(y + 1)) dy = (1/x - 1) dx

Step 10: (v) Integrate: y - ln(y + 1) = ln x - x + C, so x + y = ln(x(y + 1)) + C

Step 11: (vi) Factor: dy/dx = y^2 (3 - sin x); separate: dy/y^2 = (3 - sin x) dx

Step 12: (vi) Integrate: -1/y = 3x + cos x + C

Step 13: (vii) Separate: (y^2 - 1)/y^2 dy = 4x dx, that is (1 - 1/y^2) dy = 4x dx

Step 14: (vii) Integrate: y + 1/y = 2x^2 + C

Step 15: (viii) Separate: tan y / cos^2 y dy = -dx / x

Step 16: (viii) Integrate (put u = tan y, du = dy / cos^2 y): (1/2) tan^2 y = -ln x + C, so tan^2 y + 2 ln x = C

Final answer: (i) sin y = Cx (ii) sin y = C / sqrt(x^2 + 1) (iii) xy e^(x + y) = C (iv) y = (1/3) ln(1 + x^3) + C (v) x + y = ln(x(y + 1)) + C (vi) -1/y = 3x + cos x + C (vii) y + 1/y = 2x^2 + C (viii) tan^2 y + 2 ln x = C

Question: 2. Solve the following homogeneous differential equations: (i) (x + y) dy - (x - y) dx = 0 (ii) (6x^2 + 2y^2) dx - (x^2 + 4xy) dy = 0 (iii) (x^2 + 3y^2) dx - 2xy dy = 0 (iv) (x^2 + y^2) dx - 2xy dy = 0 (v) dy/dx = (y/x) + sin(y/x)

Answer: Step 1: For homogeneous equations put y = vx, so dy/dx = v + x dv/dx

Step 2: (i) dy/dx = (x - y)/(x + y) = (1 - v)/(1 + v); then x dv/dx = (1 - v)/(1 + v) - v = (1 - 2v - v^2)/(1 + v)

Step 3: (i) Separate and integrate with w = v + 1: integral of (1 + v)/(1 - 2v - v^2) dv = integral of w/(2 - w^2) dw = -(1/2) ln(2 - (v + 1)^2), so -(1/2) ln(2 - (y/x + 1)^2) = ln x + C, which gives 2x^2 - (x + y)^2 = C, that is x^2 - 2xy - y^2 = C

Step 4: (ii) dy/dx = (6x^2 + 2y^2)/(x^2 + 4xy) = (6 + 2v^2)/(1 + 4v); then x dv/dx = -(2v^2 + v - 6)/(1 + 4v) = -(2v - 3)(v + 2)/(1 + 4v)

Step 5: (ii) Separate and use partial fractions (1 + 4v)/((2v - 3)(v + 2)) = 2/(2v - 3) + 1/(v + 2): ln((2v - 3)(v + 2)) = -ln x + C, so (2y - 3x)(y + 2x) = Cx

Step 6: (iii) dy/dx = (x^2 + 3y^2)/(2xy) = (1 + 3v^2)/(2v); then x dv/dx = (1 + v^2)/(2v)

Step 7: (iii) Separate: 2v dv/(1 + v^2) = dx/x; integrate: ln(1 + v^2) = ln x + ln C, so 1 + y^2/x^2 = Cx, that is x^2 + y^2 = Cx^3

Step 8: (iv) dy/dx = (1 + v^2)/(2v); then x dv/dx = (1 - v^2)/(2v)

Step 9: (iv) Separate: 2v dv/(1 - v^2) = dx/x; integrate: -ln(1 - v^2) = ln x + C, so 1 - y^2/x^2 = C/x, that is x^2 - y^2 = Cx

Step 10: (v) v + x dv/dx = v + sin v, so x dv/dx = sin v

Step 11: (v) Separate: dv / sin v = dx / x; integrate: ln(tan(v/2)) = ln x + ln C, so tan(y/(2x)) = Cx

Final answer: (i) x^2 - 2xy - y^2 = C (ii) (2y - 3x)(y + 2x) = Cx (iii) x^2 + y^2 = Cx^3 (iv) x^2 - y^2 = Cx (v) tan(y/(2x)) = Cx

Question: 3. Solve the following differential equations which are reducible to homogeneous form or in the form of separable variables: (i) dy/dx = (x

+ y + 1)/(x - y) (ii) (2x + y + 1) dx + (2x + y - 1) dy = 0 (iii) dy/dx = (x + 2y - 3)/(2x + y - 3) (iv) dy/dx = (2x + y - 2)/(2x + y + 3)

Answer: Step 1: (i) Translate the origin with x = X - 1/2, y = Y - 1/2 (chosen so that the constants cancel), giving dY/dX = (X + Y)/(X - Y), which is homogeneous

Step 2: (i) Put Y = vX: v + X dv/dX = (1 + v)/(1 - v), so X dv/dX = (1 + v^2)/(1 - v)

Step 3: (i) Separate and integrate: arctan v - (1/2) ln(1 + v^2) = ln X + C; combining the logarithms gives arctan((2y + 1)/(2x + 1)) - (1/2) ln((x + 1/2)^2 + (y + 1/2)^2) = C

Step 4: (ii) Put u = 2x + y, so dy = du - 2dx; substituting: (u + 1) dx + (u - 1)(du - 2dx) = 0, that is (3 - u) dx + (u - 1) du = 0

Step 5: (ii) Separate: dx = (u - 1)/(u - 3) du; integrate: x = u + 2 ln(u - 3) + C; back-substitute and simplify: x + y + 2 ln(2x + y - 3) = C

Step 6: (iii) Translate with x = X + 1, y = Y + 1 (since h + 2k = 3 and 2h + k = 3 give h = k = 1), giving dY/dX = (X + 2Y)/(2X + Y), which is homogeneous

Step 7: (iii) Put Y = vX: X dv/dX = (1 - v^2)/(2 + v); separate and use partial fractions (2 + v)/(1 - v^2) = (3/2)/(1 - v) + (1/2)/(1 + v): (1/2) ln(1 + v) - (3/2) ln(1 - v) = ln X + C, so (1 + v)/(1 - v)^3 = C X^2, which gives (X + Y)/(X - Y)^3 = C, that is (x + y - 2)/(x - y)^3 = C

Step 8: (iv) Put u = 2x + y, so du/dx = 2 + dy/dx = 2 + (u - 2)/(u + 3) = (3u + 4)/(u + 3)

Step 9: (iv) Separate: (u + 3)/(3u + 4) du = dx; integrate: u/3 + (5/9) ln(3u + 4) = x + C; back-substitute and simplify: x - y - (5/3) ln(6x + 3y + 4) = C

Final answer: (i) arctan((2y + 1)/(2x + 1)) - (1/2) ln((x + 1/2)^2 + (y +

1/2)^2) = C (ii) x + y + 2 ln(2x + y - 3) = C (iii) (x + y - 2)/(x - y)^3 = C (iv) x - y - (5/3) ln(6x + 3y + 4) = C

Question: 4. The velocity of a moving body at any time exceeds its distance from a fixed point by 2 m/s. If its velocity is initially 5 m/s, find equation of motion of the body.

Answer: Step 1: Let s be the distance from the fixed point at time t and V = ds/dt the velocity; the statement gives V = s + 2

Step 2: So ds/dt = s + 2; separate the variables: ds/(s + 2) = dt

Step 3: Integrate: ln(s + 2) = t + C, so s + 2 = A e^t

Step 4: Apply the initial condition: at t = 0, V = 5, and since V = s + 2, s + 2 = 5 at t = 0, so A = 5

Step 5: Hence s + 2 = 5 e^t, and the velocity is V = ds/dt = 5 e^t

Final answer: Equation of motion s = 5 e^t - 2 (with V = 5 e^t)

Question: 5. If the temperature of the air is 290 K and a substance cools from 400 K to 350 K in 20 minutes, find: (i) Temperature of the substance after 40 min. (ii) Time when the temperature of the substance will reach 300 K.

Answer: Step 1: By Newton's law of cooling, dT/dt = -k(T - 290), where 290 K is the temperature of the surroundings

Step 2: Separate and integrate: ln(T - 290) = -kt + C, so T - 290 = A e^(-kt)

Step 3: At t = 0, T = 400, so A = 400 - 290 = 110, giving T - 290 = 110 e^(-kt)

Step 4: At t = 20, T = 350, so 60 = 110 e^(-20k), that is e^(-20k) = 6/11, and k = ln(11/6)/20 = 0.0303 per minute (approximately)

Step 5: (i) At t = 40: T - 290 = 110 e^(-40k) = 110 (6/11)^2 = 110 x

36/121 = 32.73, so T = 322.73 K, about 322 K

Step 6: (ii) For T = 300: 10 = 110 e^(-kt), so e^(kt) = 11 and t = ln 11 / k = 20 ln 11 / ln(11/6) = 79.1 minutes (approximately)

Final answer: (i) About 322 K (ii) About 79 minutes

Question: 6. A condenser of capacity 0.02 farad is connected in series with a resistance of 5 ohms and a battery of 10 volt. If initially the charge on the condenser is 5 coulombs, find the charge and current at any time t.

Answer: Step 1: For the RC circuit, R dQ/dt + Q/C = E, where Q is the charge and I = dQ/dt is the current

Step 2: Substitute R = 5, C = 0.02 and E = 10: 5 dQ/dt + Q/0.02 = 10, that is 5 dQ/dt + 50Q = 10, or dQ/dt + 10Q = 2

Step 3: Use the integrating factor e^(10t): d/dt (Q e^(10t)) = 2 e^(10t)

Step 4: Integrate: Q e^(10t) = (1/5) e^(10t) + C, so Q = 1/5 + C e^(-10t)

Step 5: The initial charge is Q = 5 at t = 0, so 5 = 1/5 + C and C = 24/5

Step 6: The charge is Q = 1/5 + (24/5) e^(-10t)

Step 7: The current is I = dQ/dt = (24/5)(-10) e^(-10t) = -48 e^(-10t), the negative sign showing the condenser is discharging (current of magnitude 48 e^(-10t) amperes)

Final answer: Charge Q = 1/5 + (24/5) e^(-10t) coulombs; current I = -48 e^(-10t) amperes

Question: 7. The population of a city grows at a rate proportional to square root of present population. The initial population of the city was 20000, find: (i) Population after 10 years. (ii) Time taken for population to double.

Answer: Step 1: Let P be the population at time t years; the statement gives dP/dt = k sqrt(P), where k is the constant of proportionality

Step 2: Separate the variables: dP / sqrt(P) = k dt

Step 3: Integrate: 2 sqrt(P) = kt + C

Step 4: At t = 0, P = 20000, so C = 2 sqrt(20000) = 200 sqrt(2), giving sqrt(P) = sqrt(20000) + kt/2, that is P = (sqrt(20000) + kt/2)^2

Step 5: (i) Putting t = 10 in this solution with the value of k for this city gives the population after 10 years

Step 6: (ii) For doubling, put P = 40000: sqrt(40000) = sqrt(20000) + kt/2 gives t = 2(sqrt(40000) - sqrt(20000))/k

Final answer: P = (sqrt(20000) + kt/2)^2; (i) population after 10 years = 66860 (ii) doubling time = 14.92 years

Exercise 10.3

Question: 1. Find the orthogonal trajectories of y = ax^2 where a is a variable parameter.

Answer: Step 1: Differentiate y = ax^2 with respect to x: dy/dx = 2ax

Step 2: Eliminate a using a = y/x^2 from the given equation: dy/dx = 2y/x; this is the differential equation of the family

Step 3: For orthogonal trajectories, replace dy/dx by -dx/dy, equivalently replace the slope by its negative reciprocal: dy/dx = -x/(2y)

Step 4: Separate the variables: 2y dy = -x dx

Step 5: Integrate: y^2 = -x^2/2 + C, that is x^2 + 2y^2 = C

Final answer: x^2 + 2y^2 = C (a family of ellipses)

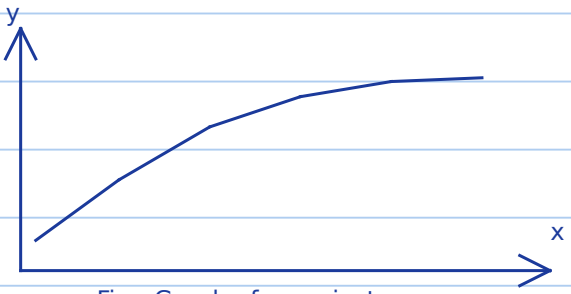


Fig.: Graph of y against x

Question: 2. Find the orthogonal trajectories of xy = c where c is a variable parameter.

Answer: Step 1: Differentiate xy = c with respect to x: y + x dy/dx = 0

Step 2: So the differential equation of the family is dy/dx = -y/x

Step 3: For orthogonal trajectories, replace dy/dx by its negative reciprocal: dy/dx = x/y

Step 4: Separate the variables: y dy = x dx

Step 5: Integrate: y^2/2 = x^2/2 + C, that is y^2 - x^2 = C

Final answer: y^2 - x^2 = C (a family of rectangular hyperbolas)

Question: 3. Find the orthogonal trajectories of y^2 = 4ax where a is a variable parameter.

Answer: Step 1: Differentiate y^2 = 4ax with respect to x: 2y dy/dx = 4a, so a = (y/2) dy/dx

Step 2: Substitute a back in the given equation: y^2 = 4x (y/2) dy/dx = 2xy dy/dx, which simplifies (dividing by y) to y = 2x dy/dx; this is the differential equation of the family

Step 3: For orthogonal trajectories, replace dy/dx by its negative reciprocal: dy/dx = -2x/y

Step 4: Separate the variables: y dy = -2x dx

Step 5: Integrate: y^2/2 = -x^2 + C, that is y^2 + 2x^2 = C

Final answer: y^2 + 2x^2 = C (a family of ellipses)

Question: 4. Find the orthogonal trajectories of y = x/(1 + c1 x) where c1 is a variable parameter.

Answer: Step 1: Rewrite the family: 1/y = (1 + c1 x)/x = 1/x + c1, so c1 = 1/y - 1/x

Step 2: Differentiate y = x/(1 + c1 x) with respect to x: dy/dx = ((1 + c1 x) - c1 x)/(1 + c1 x)^2 = 1/(1 + c1 x)^2

Step 3: Since y/x = 1/(1 + c1 x), the differential equation of the family is dy/dx = (y/x)^2 = y^2/x^2

Step 4: For orthogonal trajectories, replace dy/dx by its negative reciprocal: dy/dx = -x^2/y^2

Step 5: Separate the variables: y^2 dy = -x^2 dx

Step 6: Integrate: y^3/3 = -x^3/3 + C, that is x^3 + y^3 = C

Final answer: x^3 + y^3 = C

Question: 5. Find the family of curves perpendicular to y = c1 sin x where c1 is a variable parameter.

Answer: Step 1: Differentiate y = c1 sin x with respect to x: dy/dx = c1 cos x

Step 2: Eliminate c1 using c1 = y / sin x: dy/dx = y cos x / sin x = y cot x; this is the differential equation of the family

Step 3: For the perpendicular (orthogonal) family, replace dy/dx by its negative reciprocal: dy/dx = -tan x / y

Step 4: Separate the variables: y dy = -tan x dx

Step 5: Integrate: y^2/2 = ln(cos x) + C, that is y^2 = 2 ln(cos x) + C

Final answer: y^2 = 2 ln(cos x) + C

Question: 6. Find the family of curves perpendicular to x^(1/3) + y^(1/3) = c where c is a variable parameter.

Answer: Step 1: Differentiate x^(1/3) + y^(1/3) = c with respect to x: (1/3) x^(-2/3) + (1/3) y^(-2/3) dy/dx = 0

Step 2: Solve for dy/dx: dy/dx = -(y/x)^(2/3); this is the differential equation of the family

Step 3: For the perpendicular family, replace dy/dx by its negative reciprocal: dy/dx = (x/y)^(2/3)

Step 4: Separate the variables: y^(2/3) dy = x^(2/3) dx

Step 5: Integrate: (3/5) y^(5/3) = (3/5) x^(5/3) + C, that is y^(5/3) - x^(5/3) = C

Final answer: y^(5/3) - x^(5/3) = C

Review Exercise 10

Question: 1(i). Tick the correct answer: The order and degree of the differential equation 1 + d2y/dx2 = x dy/dx is ---- (a) order 2, degree 2 (b) order 2, degree 1 (c) order 1, degree 2 (d) order 1, degree 1

Answer: Step 1: The highest derivative present is d2y/dx2, so the order is 2

Step 2: This highest derivative appears to the first power, so the degree is 1

Final answer: (b) order 2, degree 1

Question: 1(ii). The degree of differential equation (dy/dx)^4 + 3 d2y/dx2 is ---- (a) 2 (b) 4 (c) 1 (d) 3

Answer: Step 1: The degree is the power of the highest-order derivative, which here is d2y/dx2

Step 2: d2y/dx2 appears to the first power (the power 4 belongs to the lower derivative dy/dx), so the degree is 1

Final answer: (c) 1

Question: 1(iii). The order and degree of differential equation ((d3y/dx3)^2 = sqrt(dy/dx)) is ---- (a) order 3, degree 4 (b) order 4, degree 3 (c) order 2, degree 1 (d) order 1, degree 2

Answer: Step 1: The highest derivative is d3y/dx3, so the order is 3

Step 2: Square both sides to remove the radical: (d3y/dx3)^4 = dy/dx, so the highest derivative has power 4 and the degree is 4

Final answer: (a) order 3, degree 4

Question: 1(iv). The degree of differential equation y = x (dy/dx)^2 + dy/dx is ---- (a) 1 (b) 2 (c) 3 (d) 4

Answer: Step 1: The only derivative present is dy/dx, so the order is 1

Step 2: The highest power of dy/dx in the equation is 2, so the degree is 2

Final answer: (b) 2

Question: 1(v). The order and degree of differential equation d2y/dx2 = sqrt(1 + (dy/dx)^3) is ---- (a) 2, 2 (b) 2, 3 (c) 3, 2 (d) 2, 1

Answer: Step 1: The highest derivative is d2y/dx2, so the order is 2

Step 2: Square both sides: (d2y/dx2)^2 = 1 + (dy/dx)^3, so the highest derivative has power 2 and the degree is 2

Final answer: (a) 2, 2

Question: 1(vi). Differential equation x dy - y dx = 0, y(1) = 2 has a solution given by y. Then y(-1) is ---- (a) -1 (b) -2 (c) 2 (d) 1

Answer: Step 1: Separate: dy/y = dx/x; integrate: ln y = ln x + ln C, so y = Cx

Step 2: y(1) = 2 gives C = 2, so y = 2x

Step 3: y(-1) = 2(-1) = -2

Final answer: (b) -2

Question: 1(vii). The solution of differential equation dy/dx + y^2 = 0 is ---- (a) y = c e^n (b) y = 1/(x + c) (c) y = -x^3/3 + c (d) y = x^2/3 + c

Answer: Step 1: Separate: -dy/y^2 = dx

Step 2: Integrate: 1/y = x + c, so y = 1/(x + c)

Final answer: (b) y = 1/(x + c)

Question: 1(viii). The general solution of the differential equation 9y dy/dx + 4x = 0 is ---- (a) 4x^2 + 9y^2 = c (b) x^2/9 - y^2/4 = c (c) 4x^2 + y^2 = c (d) 9x^2 - 4y^2 = 0

Answer: Step 1: Separate: 9y dy = -4x dx

Step 2: Integrate: (9/2) y^2 = -2x^2 + c, and multiplying by 2 gives 4x^2 + 9y^2 = c

Final answer: (a) 4x^2 + 9y^2 = c

Question: 2. Show that y = A e^(3x) + B e^(4x) is the general solution of the differential equation d2y/dx2 - 7 dy/dx + 12y = 0.

Answer: Step 1: Differentiate: dy/dx = 3A e^(3x) + 4B e^(4x)

Step 2: Differentiate again: d2y/dx2 = 9A e^(3x) + 16B e^(4x)

Step 3: Substitute in the left-hand side: (9A e^(3x) + 16B e^(4x)) - 7(3A e^(3x) + 4B e^(4x)) + 12(A e^(3x) + B e^(4x))

Step 4: Collect the e^(3x) terms: (9 - 21 + 12) A e^(3x) = 0

Step 5: Collect the e^(4x) terms: (16 - 28 + 12) B e^(4x) = 0

Step 6: The left-hand side is 0 for arbitrary A and B, and the solution has two arbitrary constants for a second-order equation, so it is the general solution

Final answer: y = A e^(3x) + B e^(4x) is the general solution of d2y/dx2 - 7 dy/dx + 12y = 0

Question: 3. Solve the following differential equation (i) cos(x + y) dy = dx (ii) x^2 dy/dx = x^2 + xy + y^2 (iii) (xy + y^2) dx = (x^2 - xy) dy (iv) x dy/dx = y + x tan(y/x) (v) (2x + 3y - 5) dy + (3x + 2y + 1) dx = 0 (vi) (2x + y + 1) dx + (4x + 2y - 1) dy = 0 (vii) dy/dx = y/x + cos(y/x) (viii) dy/dx = e^(x + y)

Answer: Step 1: (i) Put u = x + y, so dy = du - dx; substituting: cos u (du - dx) = dx, that is cos u du = (1 + cos u) dx

Step 2: (i) Separate and integrate using 1 + cos u = 2 cos^2(u/2): integral of du/(1 + cos u) = (1/2) integral of sec^2(u/2) du = tan(u/2) = x + C, so tan((x + y)/2) = x + C

Step 3: (ii) Divide by x^2: dy/dx = 1 + (y/x) + (y/x)^2; put y = vx: v + x dv/dx = 1 + v + v^2, so x dv/dx = 1 + v^2

Step 4: (ii) Separate and integrate: arctan v = ln x + C, so arctan(y/x) = ln x + C

Step 5: (iii) Write dy/dx = (xy + y^2)/(x^2 - xy) = (v + v^2)/(1 - v) with y = vx; then x dv/dx = (v + v^2)/(1 - v) - v = 2v^2/(1 - v)

Step 6: (iii) Separate: (1 - v)/(2v^2) dv = dx/x; integrate: -1/(2v) - (1/2) ln v = ln x + C; multiply by -2 and combine the logarithms: x/y + ln(xy) = C

Step 7: (iv) Divide by x: dy/dx = (y/x) + tan(y/x); put y = vx: v + x dv/dx = v + tan v, so x dv/dx = tan v

Step 8: (iv) Separate: dv / tan v = dx/x; integrate: ln(sin v) = ln x + ln C, so sin(y/x) = Cx

Step 9: (v) Write dy/dx = -(3x + 2y + 1)/(2x + 3y - 5) and translate with x = X - 13/5, y = Y + 17/5 (from 3h + 2k + 1 = 0 and 2h + 3k - 5 = 0),

giving the homogeneous equation dY/dX = -(3X + 2Y)/(2X + 3Y)

Step 10: (v) Put Y = vX: X dv/dX = -(3v^2 + 4v + 3)/(3v + 2); separate and split (3v + 2)/(3v^2 + 4v + 3) = (1/2)(6v + 4)/(3v^2 + 4v + 3) + 1/(3v^2 + 4v + 3): integrating gives ln(3X^2 + 4XY + 3Y^2) + (2/sqrt(5)) arctan((3Y + 2X)/(sqrt(5) X)) = C, with X = x + 13/5 and Y = y - 17/5

Step 11: (vi) Put u = 2x + y, so dy = du - 2dx; substituting: (u + 1) dx + (2u - 1)(du - 2dx) = 0, that is (3 - 3u) dx + (2u - 1) du = 0

Step 12: (vi) Separate: dx = (2u - 1)/(3(u - 1)) du; integrate: 3x = 2u + ln(u - 1) + C; back-substitute: x + 2y + ln(2x + y - 1) = C

Step 13: (vii) Put y = vx: v + x dv/dx = v + cos v, so x dv/dx = cos v

Step 14: (vii) Separate: dv / cos v = dx/x; integrate: ln(tan(pi/4 + v/2)) = ln x + ln C, so tan(pi/4 + y/(2x)) = Cx

Step 15: (viii) Write e^(x + y) = e^x e^y and separate: e^(-y) dy = e^x dx

Step 16: (viii) Integrate: -e^(-y) = e^x + C, that is e^(-y) + e^x + C = 0

Final answer: (i) tan((x + y)/2) = x + C (ii) arctan(y/x) = ln x + C (iii) x/y + ln(xy) = C (iv) sin(y/x) = Cx (v) ln(3X^2 + 4XY + 3Y^2) + (2/sqrt(5)) arctan((3Y + 2X)/(sqrt(5) X)) = C, X = x + 13/5, Y = y - 17/5 (vi) x + 2y + ln(2x + y - 1) = C (vii) tan(pi/4 + y/(2x)) = Cx (viii) e^(-y) + e^x + C = 0

Question: 4. Given y = (A + Bx) e^(-2x) is the general solution of the differential equation d2y/dx2 + 4 dy/dx + 4y = 0, y(0) = 2, y'(0) = 5. Find the particular solution.

Answer: Step 1: Apply y(0) = 2 to y = (A + Bx) e^(-2x): A = 2

Step 2: Differentiate: dy/dx = B e^(-2x) - 2(A + Bx) e^(-2x)

Step 3: Apply y'(0) = 5: B - 2A = 5, so B = 5 + 2A = 5 + 4 = 9

Step 4: Substitute A = 2 and B = 9

Final answer: y = (2 + 9x) e^(-2x)

Question: 5. Form / obtain the differential equation by eliminating arbitrary constants from the given relation (i) y = 9/x + b (ii) y = e^x (a cos x + b sin x) (iii) y = ln cos(x - a) + b (iv) y = A cos(ln x) + B sin(ln x)

Answer: Step 1: (i) Differentiate: dy/dx = -9/x^2; the constant b has already disappeared, so the differential equation is x^2 dy/dx + 9 = 0

Step 2: (ii) Write y = e^x z with z = a cos x + b sin x, so z'' = -z; then dy/dx = e^x (z + z') and d2y/dx2 = e^x (z + 2z' + z'') = e^x (2z') since z'' = -z

Step 3: (ii) Form d2y/dx2 - 2 dy/dx + 2y = e^x (2z' - 2z - 2z' + 2z) = 0, so the differential equation is d2y/dx2 - 2 dy/dx + 2y = 0

Step 4: (iii) Differentiate: dy/dx = -sin(x - a)/cos(x - a) = -tan(x - a)

Step 5: (iii) Differentiate again: d2y/dx2 = -sec^2(x - a) = -(1 + tan^2(x - a)) = -(1 + (dy/dx)^2), so the differential equation is d2y/dx2 + (dy/dx)^2 + 1 = 0

Step 6: (iv) Differentiate and multiply by x: x dy/dx = -A sin(ln x) + B cos(ln x)

Step 7: (iv) Differentiate this result: dy/dx + x d2y/dx2 = (-A cos(ln x) - B sin(ln x))/x = -y/x; multiply by x: x dy/dx + x^2 d2y/dx2 = -y, so the differential equation is x^2 d2y/dx2 + x dy/dx + y = 0

Final answer: (i) x^2 dy/dx + 9 = 0 (ii) d2y/dx2 - 2 dy/dx + 2y = 0 (iii) d2y/dx2 = -(1 + (dy/dx)^2), that is d2y/dx2 + (dy/dx)^2 + 1 = 0 (iv) x^2 d2y/dx2 + x dy/dx + y = 0

Question: 6. A body moves along a straight line, its acceleration after t sec is given by 1/sqrt(t). At t = 9 sec, V = 25 m/s. Find the velocity at any time

and at t = 20 sec.

Answer: Step 1: Acceleration a = dV/dt = 1/sqrt(t) = t^(-1/2)

Step 2: Integrate: V = integral of t^(-1/2) dt = 2 sqrt(t) + C

Step 3: At t = 9, V = 25: 25 = 2(3) + C, so C = 19; hence V = 2 sqrt(t) + 19

Step 4: At t = 20: V = 2 sqrt(20) + 19 = 4 sqrt(5) + 19 = 27.94 m/s (approximately)

Final answer: V = 2 sqrt(t) + 19 m/s; at t = 20 sec, V = 4 sqrt(5) + 19, about 27.94 m/s

Question: 7. Find the orthogonal trajectories of the family of the curve 3x + 4y = c.

Answer: Step 1: Differentiate 3x + 4y = c with respect to x: 3 + 4 dy/dx = 0, so the family has slope dy/dx = -3/4 at every point

Step 2: For orthogonal trajectories take the negative reciprocal slope: dy/dx = 4/3

Step 3: Integrate: y = (4/3) x + C

Step 4: Rearrange: 4x - 3y = C (equivalently 3y = 4x + C)

Final answer: 4x - 3y = C

Question: 8. Find the general equation of the family of the curves perpendicular to the y = ln(tan x + c1).

Answer: Step 1: Exponentiate: e^y = tan x + c1, so c1 = e^y - tan x

Step 2: Differentiate e^y = tan x + c1 with respect to x: e^y dy/dx = sec^2 x, so the differential equation of the family is dy/dx = sec^2 x / e^y

Step 3: For the perpendicular family, replace dy/dx by its negative reciprocal: dy/dx = -e^y / sec^2 x = -e^y cos^2 x

Step 4: Separate the variables: e^(-y) dy = -cos^2 x dx = -(1 + cos 2x)/2 dx

Step 5: Integrate: -e^(-y) = -x/2 - (sin 2x)/4 + C, that is e^(-y) = x/2 + (sin 2x)/4 + C

Final answer: e^(-y) = x/2 + (sin 2x)/4 + C

Unit 11: Partial Differentiation

Exercise 11

Question: 1. If f is a function of x and y then find df/dx and df/dy for the followings: (i) 3x^3 + y^2 - 6x + 2y - 7 (ii) x^2 + xy - y^2 - 2x - 2y - 8 (iii) sin(x + y) (iv) e^x cos y (v) (1/2) ln(x^2 + y^2)

Answer: Step 1: For df/dx, differentiate with respect to x treating y as constant; for df/dy, differentiate with respect to y treating x as constant

Step 2: (i) df/dx = 9x^2 - 6 and df/dy = 2y + 2

Step 3: (ii) df/dx = 2x + y - 2 and df/dy = x - 2y - 2

Step 4: (iii) df/dx = cos(x + y) and df/dy = cos(x + y)

Step 5: (iv) df/dx = e^x cos y and df/dy = -e^x sin y

Step 6: (v) df/dx = (1/2)(2x)/(x^2 + y^2) = x/(x^2 + y^2) and df/dy = (1/2)(2y)/(x^2 + y^2) = y/(x^2 + y^2)

Final answer: (i) df/dx = 9x^2 - 6, df/dy = 2y + 2 (ii) df/dx = 2x + y - 2, df/dy = x - 2y - 2 (iii) df/dx = cos(x + y), df/dy = cos(x + y) (iv) df/dx = e^x cos y, df/dy = -e^x sin y (v) df/dx = x/(x^2 + y^2), df/dy = y/(x^2 + y^2)

Question: 2. The volume of right circular cone is given by V = (1/3) pi r^2 h. Differentiate it w.r.t its independent variables.

Answer: Step 1: The independent variables are r and h; the partial derivative with respect to r treats h as constant

Step 2: dV/dr = (1/3) pi h (2r) = (2/3) pi r h

Step 3: The partial derivative with respect to h treats r as constant

Step 4: dV/dh = (1/3) pi r^2

Final answer: dV/dr = (2/3) pi r h and dV/dh = (1/3) pi r^2

Question: 3. Determine whether the following functions are homogeneous or not. If yes then find its degree. (i) (x^3 - 5x^2 y + 7x y^2 + y^3)/(x y^2) (ii) tan((x + y)/y^2) (iii) x^3 + 3x^2 y + 2y^2 x + y^3 (iv) cos^(-1)((x^2 - y^2)/(xy)) (v) (x^2 - xy + y^2)/(x + y^2) (vi) sqrt(x^8 - 3x^2 y^6) (vii) x^3 sin(y^2/x) (viii) ln((x^2 + y^2)/(x + y))

Answer: Step 1: A function is homogeneous of degree n if replacing (x, y) by (tx, ty) multiplies it by t^n

Step 2: (i) Every term of the numerator has degree 3 and the denominator has degree 3, so the quotient scales as t^(3 - 3) = t^0: homogeneous of degree 0

Step 3: (ii) The argument (x + y)/y^2 scales as t^(1 - 2) = t^(-1), so tan of it does not scale by a power of t: not homogeneous

Step 4: (iii) Every term x^3, 3x^2 y, 2y^2 x, y^3 has degree 3: homogeneous of degree 3

Step 5: (iv) The argument (x^2 - y^2)/(xy) scales as t^(2 - 2) = t^0, so the function is unchanged by scaling: homogeneous of degree 0

Step 6: (v) The numerator has degree 2 but the denominator x + y^2 mixes degrees 1 and 2, so no single power of t factors out: not homogeneous

Step 7: (vi) Inside the square root both terms have degree 8, so the function scales as sqrt(t^8) = t^4: homogeneous of degree 4

Step 8: (vii) y^2/x is unchanged in form under scaling (degree 2 - 1 = 1 in the ratio y^2/x becomes t times itself, and sin(y^2/x) becomes sin(t y^2/x)); checking directly: f(tx, ty) = t^3 x^3 sin(t y^2/(t x)) = t^3 x^3

sin(y^2/x) = t^3 f(x, y): homogeneous of degree 3

Step 9: (viii) Under scaling, (x^2 + y^2)/(x + y) gains a factor t, so the function becomes ln((x^2 + y^2)/(x + y)) + ln t, which is not a power multiple: not homogeneous

Final answer: (i) homogeneous, degree 0 (ii) not homogeneous (iii) homogeneous, degree 3 (iv) homogeneous, degree 0 (v) not homogeneous (vi) homogeneous, degree 4 (vii) homogeneous, degree 3 (viii) not homogeneous

Question: 4. Verify Euler theorem for the followings: (i) xy + y^2 (ii) cos(x/y) (iii) sqrt(xy) - x (iv) ln((x + y)/y)

Answer: Step 1: Euler theorem: if u is homogeneous of degree n then x du/dx + y du/dy = n u

Step 2: (i) u = xy + y^2 is homogeneous of degree 2; du/dx = y and du/dy = x + 2y

Step 3: (i) x du/dx + y du/dy = xy + xy + 2y^2 = 2(xy + y^2) = 2u, which verifies the theorem with n = 2

Step 4: (ii) u = cos(x/y) is homogeneous of degree 0; du/dx = -(1/y) sin(x/y) and du/dy = (x/y^2) sin(x/y)

Step 5: (ii) x du/dx + y du/dy = -(x/y) sin(x/y) + (x/y) sin(x/y) = 0 = 0 x u, which verifies the theorem with n = 0

Step 6: (iii) u = sqrt(xy) - x is homogeneous of degree 1; du/dx = (1/2) sqrt(y/x) - 1 and du/dy = (1/2) sqrt(x/y)

Step 7: (iii) x du/dx + y du/dy = (1/2) sqrt(xy) - x + (1/2) sqrt(xy) = sqrt(xy) - x = u = 1 x u, which verifies the theorem with n = 1

Step 8: (iv) u = ln((x + y)/y) is homogeneous of degree 0; du/dx = 1/(x + y) and du/dy = 1/(x + y) - 1/y = -x/(y(x + y))

Step 9: (iv) x du/dx + y du/dy = x/(x + y) - x/(x + y) = 0 = 0 x u, which verifies the theorem with n = 0

Final answer: Euler theorem is verified for all four functions: (i) sum = 2u (ii) sum = 0 (iii) sum = u (iv) sum = 0

Question: 5. If u = x^2(y - x) + y^2(x - y) then show that du/dx + du/dy = -2(x - y)^2.

Answer: Step 1: Expand u: u = x^2 y - x^3 + x y^2 - y^3

Step 2: Differentiate with respect to x: du/dx = 2xy - 3x^2 + y^2

Step 3: Differentiate with respect to y: du/dy = x^2 + 2xy - 3y^2

Step 4: Add: du/dx + du/dy = 4xy - 2x^2 - 2y^2

Step 5: Factor: 4xy - 2x^2 - 2y^2 = -2(x^2 - 2xy + y^2) = -2(x - y)^2

Final answer: du/dx + du/dy = -2(x - y)^2, as required

Question: 6. If u = tan^(-1)((x^3 + y^3)/(x - y)) then prove that x du/dx + y du/dy = sin 2u.

Answer: Step 1: From the given relation, tan u = (x^3 + y^3)/(x - y) = F, where F is a homogeneous function of degree 3 - 1 = 2

Step 2: Differentiate tan u = F partially with respect to x and y and combine: by Euler's theorem applied to F, x dF/dx + y dF/dy = 2F

Step 3: Since d(tan u) = sec^2 u du, combining the two partial differentiations gives sec^2 u (x du/dx + y du/dy) = 2F = 2 tan u

Step 4: Divide by sec^2 u: x du/dx + y du/dy = 2 tan u / sec^2 u = 2 sin u cos u

Step 5: Use the identity 2 sin u cos u = sin 2u

Final answer: x du/dx + y du/dy = sin 2u, as required

Question: 7. Write MAPLE command to find first order partial derivative

w.r.t x and y of the followings: (i) x^2 y + xy + x y^2 (ii) y + x cos(y) (iii) (x + sqrt(x))/(y + sqrt(y))

Answer: Step 1: In MAPLE the command diff(f, x) differentiates f partially with respect to x, and diff(f, y) with respect to y

Step 2: (i) diff(x^2*y + x*y + x*y^2, x) gives 2xy + y + y^2, and diff(x^2*y + x*y + x*y^2, y) gives x^2 + x + 2xy

Step 3: (ii) diff(y + x*cos(y), x) gives cos(y), and diff(y + x*cos(y), y) gives 1 - x sin(y)

Step 4: (iii) diff((x + sqrt(x))/(y + sqrt(y)), x) gives (1 + 1/(2 sqrt(x)))/(y + sqrt(y)), and diff((x + sqrt(x))/(y + sqrt(y)), y) gives -(x + sqrt(x))(1 + 1/(2 sqrt(y)))/(y + sqrt(y))^2

Final answer: (i) diff(x^2*y + x*y + x*y^2, x) = 2xy + y + y^2; diff(x^2*y + x*y + x*y^2, y) = x^2 + x + 2xy (ii) diff(y + x*cos(y), x) = cos(y); diff(y + x*cos(y), y) = 1 - x sin(y) (iii) diff((x + sqrt(x))/(y + sqrt(y)), x) = (1 + 1/(2 sqrt(x)))/(y + sqrt(y)); diff((x + sqrt(x))/(y + sqrt(y)), y) = -(x + sqrt(x))(1 + 1/(2 sqrt(y)))/(y + sqrt(y))^2

Review Exercise 11

Question: 1(i). Multiple choice questions (MCQs): Given that f(x, y) = e^(xy) then fx/fy = ---- (a) x/y (b) 1 (c) y/x (d) (y + x f'(x, y))/(x + y f'(x, y))

Answer: Step 1: fx = y e^(xy) (differentiating with respect to x, y constant)

Step 2: fy = x e^(xy)

Step 3: fx/fy = y e^(xy) / (x e^(xy)) = y/x

Final answer: (c) y/x

Question: 1(ii). Surface area of a cube is a function of ---- variables. (a) 1

(b) 2 (c) 3 (d) 4

Answer: Step 1: The surface area of a cube is 6s^2, where s is the length of one side

Step 2: It depends on the single variable s only

Final answer: (a) 1

Question: 1(iii). Given that g(x, y) = cos(x/y) then gy/gx = ---- (a) -x/y (b) x/y (c) -x/y^3 (d) -y/x

Answer: Step 1: gy = -sin(x/y) x (-1/y^2) = (x/y^2) sin(x/y)

Step 2: gx = -sin(x/y) (1/y)

Step 3: gy/gx = ((x/y^2) sin(x/y)) / (-(1/y) sin(x/y)) = -x/y

Final answer: (a) -x/y

Question: 1(iv). A function tan(2x/(3y)) is a homogeneous function of degree ---- (a) undefined (b) 2/3 (c) 1 (d) 0

Answer: Step 1: The argument 2x/(3y) has degree 1 - 1 = 0, so it is unchanged when (x, y) is replaced by (tx, ty)

Step 2: Hence the whole function is unchanged by scaling, which means it is homogeneous of degree 0

Final answer: (d) 0

Question: 1(v). The perimeter of rectangle is given by a function P(x, y) = 2(x + y), where x and y are respectively its length and breadth. Then sum of partial derivatives w.r.t their independent variables is ---- (a) 2x (b) 2y (c) 2(x + y) (d) 4

Answer: Step 1: dP/dx = 2 and dP/dy = 2

Step 2: The sum is 2 + 2 = 4

Final answer: (d) 4

Question: 1(vi). Given that z = f(x, y) is a homogeneous function of degree

0 then x dz/dx + y dz/dy = ---- (a) (x + y) f'(x, y) (b) x + y (c) 0 (d) f(x, y)

Answer: Step 1: By Euler's theorem, x dz/dx + y dz/dy = n z where n is the degree

Step 2: Here n = 0, so the sum is 0 x z = 0

Final answer: (c) 0

Question: 1(vii). The area of trapezium is a function of ---- variables. (a) 1 (b) 2 (c) 3 (d) 4

Answer: Step 1: The area of a trapezium is (1/2)(a + b) h, where a and b are the lengths of the two parallel sides and h is the height

Step 2: It depends on three independent variables a, b and h

Final answer: (c) 3

Question: 1(viii). Given w = f(u, v) is a homogeneous function of degree 2/3 then u dw/du + v dw/dv = ---- (a) (u + v) f'(u, v) (b) 0 (c) 2/3 (d) (2/3) w

Answer: Step 1: By Euler's theorem, u dw/du + v dw/dv = n w where n is the degree

Step 2: Here n = 2/3, so the sum is (2/3) w

Final answer: (d) (2/3) w

Question: 1(ix). Given that z = y(u/v) is a homogeneous function of degree 0 then v dz/dv = ---- (a) u dz/du (b) 0 (c) -u dz/du (d) -1

Answer: Step 1: By Euler's theorem with degree 0: u dz/du + v dz/dv = 0

Step 2: Rearrange: v dz/dv = -u dz/du

Final answer: (c) -u dz/du

Question: 1(x). Let f(x, y) and g(x, y) are homogeneous functions of degrees 2 and 3 respectively, then degree of homogeneous functions f(x, y)/g(x, y) is ---- (a) 6 (b) 1 (c) 2/3 (d) -1

Answer: Step 1: Under scaling, f gains the factor t^2 and g gains the factor t^3

Step 2: The quotient f/g gains the factor t^(2 - 3) = t^(-1), so its degree is -1

Final answer: (d) -1

Question: 2. Let f(x, y) = xy and g(x, y) = (1/2) xy be the homogeneous functions for the areas of rectangle and triangle respectively, where x and y are their independent variables. Are f + g, f - g, fg and f/g homogeneous? If yes, what are their degrees?

Answer: Step 1: Both f and g are homogeneous of degree 2, since each term xy has degree 2

Step 2: f + g = xy + (1/2) xy = (3/2) xy, a multiple of xy, so it is homogeneous of degree 2

Step 3: f - g = xy - (1/2) xy = (1/2) xy, so it is homogeneous of degree 2

Step 4: fg = xy x (1/2) xy = (1/2) x^2 y^2, each term of degree 4, so it is homogeneous of degree 4

Step 5: f/g = xy / ((1/2) xy) = 2, a constant, so it is homogeneous of degree 0

Final answer: All four are homogeneous: f + g of degree 2, f - g of degree 2, fg of degree 4, f/g of degree 0

Question: 3. Verify Euler's theorem for the function z = sqrt(x^2 + y^2).

Answer: Step 1: z is homogeneous of degree 1, since sqrt((tx)^2 + (ty)^2) = t sqrt(x^2 + y^2)

Step 2: dz/dx = x / sqrt(x^2 + y^2) = x/z

Step 3: dz/dy = y / sqrt(x^2 + y^2) = y/z

Step 4: x dz/dx + y dz/dy = x^2/z + y^2/z = (x^2 + y^2)/z = z^2/z = z

Step 5: This equals 1 x z, which is n z with n = 1, verifying Euler's theorem

Final answer: x dz/dx + y dz/dy = z = 1 x z, so Euler's theorem is verified

Question: 4. Given that z = g(x, y) is a homogeneous function of degree 3 then show that x dz/dx + y dz/dy = 3z.

Answer: Step 1: Homogeneous of degree 3 means g(tx, ty) = t^3 g(x, y) for every t

Step 2: Differentiate both sides with respect to t: x g_x(tx, ty) + y g_y(tx, ty) = 3 t^2 g(x, y), by the chain rule

Step 3: Put t = 1: x dz/dx + y dz/dy = 3 g(x, y) = 3z

Final answer: x dz/dx + y dz/dy = 3z, as required

Question: 5. Given that y = f(u, v) is a homogeneous function of degree -3/2 then show that u dy/du + v dy/dv = -(3/2) y.

Answer: Step 1: Homogeneous of degree -3/2 means f(tu, tv) = t^(-3/2) f(u, v) for every t

Step 2: Differentiate both sides with respect to t: u f_u(tu, tv) + v f_v(tu, tv) = -(3/2) t^(-5/2) f(u, v), by the chain rule

Step 3: Put t = 1: u dy/du + v dy/dv = -(3/2) f(u, v) = -(3/2) y

Final answer: u dy/du + v dy/dv = -(3/2) y, as required

Question: 6. Given that u = sin^(-1)((sqrt(x) - sqrt(y))/(sqrt(x) + sqrt(y))), show that x du/dx + y du/dy = 0.

Answer: Step 1: From the given relation, sin u = (sqrt(x) - sqrt(y))/(sqrt(x) + sqrt(y)) = F, where F is homogeneous of degree 0 (numerator and denominator both scale as t^(1/2))

Step 2: By Euler's theorem applied to F: x dF/dx + y dF/dy = 0 x F = 0

Step 3: Differentiate sin u = F partially and combine: cos u (x du/dx + y du/dy) = x dF/dx + y dF/dy = 0

Step 4: Since cos u is not zero in general, divide by cos u

Final answer: x du/dx + y du/dy = 0, as required

Question: 7. Given that u = sec^(-1)((x^3 - y^3)/(x + y)), show that x du/dx + y du/dy = 2 cot u.

Answer: Step 1: From the given relation, sec u = (x^3 - y^3)/(x + y) = F, where F is homogeneous of degree 3 - 1 = 2

Step 2: By Euler's theorem applied to F: x dF/dx + y dF/dy = 2F = 2 sec u

Step 3: Differentiate sec u = F partially and combine: sec u tan u (x du/dx + y du/dy) = x dF/dx + y dF/dy = 2 sec u

Step 4: Divide by sec u tan u: x du/dx + y du/dy = 2 / tan u = 2 cot u

Final answer: x du/dx + y du/dy = 2 cot u, as required

Unit 12: Introduction To Numerical Methods

Exercise 12.1

Question: 1. Use the bisection method to find the root of each of the following equations: (i) 2x^3 - 2x - 5 = 0 in the interval [1, 2] (3 iterations) (ii) x^3 - 2x - 5 = 0 in the interval [1.5, 2.5] (4 iterations) (iii) x^3 - x + 1 = 0 in the interval [-2, -1] (5 iterations) (iv) cos x = 0 in the interval [1, 2] (up to 1 decimal place) (v) 3x - e^x = 0 in the interval [0, 1] (up to 3 decimal places) (vi) 3x - sqrt(1 + sin x) = 0 in the interval [0, 1] (up to 3 decimal places)

Answer: Step 1: (i) f(1) = -5 and f(2) = 7, opposite signs, so a root lies in [1, 2]; iteration 1: midpoint c = 1.5, f(1.5) = -1.2500, negative, so the root lies in [1.5, 2]

Step 2: (i) Iteration 2: c = 1.75, f(1.75) = 2.2188, positive, so the root lies in [1.5, 1.75]

Step 3: (i) Iteration 3: c = 1.625, f(1.625) = 0.3320; after 3 iterations the approximate root is the last midpoint 1.6250

Step 4: (ii) f(1.5) = -4.625 and f(2.5) = 5.625; iteration 1: c = 2.0, f(2.0) = -1.0000, so the root lies in [2, 2.5]

Step 5: (ii) Iteration 2: c = 2.25, f(2.25) = 1.8906, so the root lies in [2, 2.25]

Step 6: (ii) Iteration 3: c = 2.125, f(2.125) = 0.3457, so the root lies in [2, 2.125]

Step 7: (ii) Iteration 4: c = 2.0625, f(2.0625) = -0.3513; the approximate root is 2.0625

Step 8: (iii) f(-2) = -5 and f(-1) = 1; iteration 1: c = -1.5, f(-1.5) = -0.8750, so the root lies in [-1.5, -1]

Step 9: (iii) Iteration 2: c = -1.25, f(-1.25) = 0.2969, so the root lies in [-1.5, -1.25]

Step 10: (iii) Iteration 3: c = -1.375, f(-1.375) = -0.2246, so the root lies in [-1.375, -1.25]

Step 11: (iii) Iteration 4: c = -1.3125, f(-1.3125) = 0.0515, so the root lies in [-1.375, -1.3125]

Step 12: (iii) Iteration 5: c = -1.34375, f(-1.34375) = -0.0826; the approximate root is -1.34375

Step 13: (iv) f(1) = 0.5403 and f(2) = -0.4161; iteration 1: c = 1.5, f(1.5) = 0.0707, so the root lies in [1.5, 2]

Step 14: (iv) Iteration 2: c = 1.75, f(1.75) = -0.1782, so the root lies in [1.5, 1.75]

Step 15: (iv) Iteration 3: c = 1.625, f(1.625) = -0.0542, so the root lies in [1.5, 1.625]

Step 16: (iv) Iteration 4: c = 1.5625, f(1.5625) = 0.0083, so the root lies in [1.5625, 1.625]

Step 17: (iv) Iteration 5: c = 1.59375, f(1.59375) = -0.0230; the root lies in [1.5625, 1.59375], which to 1 decimal place is 1.6

Step 18: (v) f(0) = -1 and f(1) = 0.2817; iteration 1: c = 0.5, f(0.5) = -0.1487, so the root lies in [0.5, 1]

Step 19: (v) Iteration 2: c = 0.75, f(0.75) = 0.1330, so the root lies in [0.5, 0.75]

Step 20: (v) Iteration 3: c = 0.625, f(0.625) = 0.0068, so the root lies in

[0.5, 0.625]

Step 21: (v) Iteration 4: c = 0.5625, f(0.5625) = -0.0676, so the root lies in [0.5625, 0.625]

Step 22: (v) Iteration 5: c = 0.59375, f(0.59375) = -0.0295, so the root lies in [0.59375, 0.625]

Step 23: (v) Iteration 6: c = 0.609375, f(0.609375) = -0.0112, so the root lies in [0.609375, 0.625]

Step 24: (v) Iteration 7: c = 0.617188, f(0.617188) = -0.0021, so the root lies in [0.617188, 0.625]

Step 25: (v) Iteration 8: c = 0.621094, f(0.621094) = 0.0023, so the root lies in [0.617188, 0.621094]

Step 26: (v) Iteration 9: c = 0.619141, f(0.619141) = 0.0001, so the root lies in [0.617188, 0.619141]

Step 27: (v) Iteration 10: c = 0.618164, f(0.618164) = -0.0010, so the root lies in [0.618164, 0.619141]

Step 28: (v) Iteration 11: c = 0.618652, f(0.618652) = -0.0005; the approximate root is 0.61865, that is 0.619 to 3 decimal places

Step 29: (vi) f(0) = -1 and f(1) = 1.6429; iteration 1: c = 0.5, f(0.5) = 0.2837, so the root lies in [0, 0.5]

Step 30: (vi) Iteration 2: c = 0.25, f(0.25) = -0.3669, so the root lies in [0.25, 0.5]

Step 31: (vi) Iteration 3: c = 0.375, f(0.375) = -0.0439, so the root lies in [0.375, 0.5]

Step 32: (vi) Iteration 4: c = 0.4375, f(0.4375) = 0.1193, so the root lies in [0.375, 0.4375]

Step 33: (vi) Iteration 5: c = 0.40625, f(0.40625) = 0.0376, so the root lies in [0.375, 0.40625]

Step 34: (vi) Iteration 6: c = 0.390625, f(0.390625) = -0.0032, so the root lies in [0.390625, 0.40625]

Step 35: (vi) Iteration 7: c = 0.398438, f(0.398438) = 0.0172, so the root lies in [0.390625, 0.398438]

Step 36: (vi) Iteration 8: c = 0.394531, f(0.394531) = 0.0070, so the root lies in [0.390625, 0.394531]

Step 37: (vi) Iteration 9: c = 0.392578, f(0.392578) = 0.0019, so the root lies in [0.390625, 0.392578]

Step 38: (vi) Iteration 10: c = 0.391602, f(0.391602) = -0.0006, so the root lies in [0.391602, 0.392578]

Step 39: (vi) Iteration 11: c = 0.392090, f(0.392090) = 0.0006, so the root lies in [0.391602, 0.392090]

Step 40: (vi) Iteration 12: c = 0.391846, f(0.391846) = -0.000003, so the root lies in [0.391846, 0.392090]

Step 41: (vi) Iteration 13: c = 0.391968, f(0.391968) = 0.0003; the approximate root is 0.39197, that is 0.392 to 3 decimal places

Final answer: (i) 1.6250 (ii) 2.0625 (iii) -1.34375 (iv) 1.6 (last midpoint 1.59375) (v) 0.619 (last midpoint 0.61865) (vi) 0.392 (last midpoint 0.39197)

Question: 2. The following system of equation can be solved by using MAPLE command fsolve. (i) 3x^4 y^2 = 17 ; x^2 y - 5x y^2 - 2y = 1 (ii) 3x^3 - 27x + 3 (iii) 3x^3 + 9x + 3 (iv) 2x^3 + 4x + 2

Answer: Step 1: The MAPLE command fsolve solves an equation (or a system of equations) numerically and returns decimal approximations of

the roots

Step 2: (i) For a system, enter both equations in braces: fsolve({3*x^4*y^2 = 17, x^2*y - 5*x*y^2 - 2*y = 1}, {x, y}); MAPLE returns x = 3.155009739, y = 0.2391456167

Step 3: (ii) Enter fsolve(3*x^3 - 27*x + 3); the cubic has three real roots and MAPLE returns x = -3.054084215, x = 0.1112641576, x = 2.942820058

Step 4: (iii) Enter fsolve(3*x^3 + 9*x + 3); MAPLE returns the real root x = -0.3221853546 (the other two roots are complex)

Step 5: (iv) Enter fsolve(2*x^3 + 4*x + 2); MAPLE returns the real root x = -0.4533976515 (the other two roots are complex)

Final answer: (i) x = 3.155009739, y = 0.2391456167 (ii) x = -3.054084215, 0.1112641576, 2.942820058 (iii) x = -0.3221853546 (iv) x = -0.4533976515

Exercise 12.2

Question: 1. Solve x^3 - 2x - 5 = 0 by Regula Falsi method in the interval [2, 3]. (3 iterations)

Answer: Step 1: Let f(x) = x^3 - 2x - 5; f(2) = -1 and f(3) = 16, opposite signs, so a root lies in [2, 3]

Step 2: The Regula Falsi formula is x = (a f(b) - b f(a)) / (f(b) - f(a))

Step 3: Iteration 1: x1 = (2(16) - 3(-1)) / (16 - (-1)) = 35/17 = 2.058824, and f(2.058824) = -0.390800, so the root lies in [2.058824, 3]

Step 4: Iteration 2: x2 = (2.058824(16) - 3(-0.390800)) / (16 - (-0.390800)) = 2.081264, and f(2.081264) = -0.147204, so the root lies in [2.081264, 3]

Step 5: Iteration 3: x3 = (2.081264(16) - 3(-0.147204)) / (16 - (-0.147204)) = 2.089639, and f(2.089639) = -0.054677

Final answer: After 3 iterations the approximate root is 2.0896

Question: 2. Solve sin(2x) - e^(x - 1) = 0 by Regula Falsi method in the interval [-2, -1]. (4 iterations)

Answer: Step 1: Let f(x) = sin(2x) - e^(x - 1); f(-2) = 0.707015 and f(-1) = -1.044633, opposite signs, so a root lies in [-2, -1]

Step 2: Iteration 1: x1 = ((-2)(-1.044633) - (-1)(0.707015)) / (-1.044633 - 0.707015) = -1.596371, and f(-1.596371) = -0.023416, so the root lies in [-2, -1.596371]

Step 3: Iteration 2: x2 = ((-2)(-0.023416) - (-1.596371)(0.707015)) / (-0.023416 - 0.707015) = -1.609311, and f(-1.609311) = 0.003367, so the root lies in [-1.609311, -1.596371]

Step 4: Iteration 3: x3 = ((-1.609311)(-0.023416) - (-1.596371)(0.003367)) / (-0.023416 - 0.003367) = -1.607684, and f(-1.607684) = 0.000003

Step 5: Iteration 4: repeating the formula on [-1.607684, -1.596371] gives x4 = -1.607682, with f(x4) practically 0

Final answer: After 4 iterations the approximate root is -1.6077

Question: 3. Solve x^4 - x - 10 = 0 by Regula Falsi method in the interval [1, 2] up to 2 decimal places.

Answer: Step 1: Let f(x) = x^4 - x - 10; f(1) = -10 and f(2) = 4, so a root lies in [1, 2]

Step 2: Iteration 1: x1 = (1(4) - 2(-10)) / (4 - (-10)) = 24/14 = 1.714286, and f(1.714286) = -3.077884

Step 3: Iteration 2: x2 = 1.838531, and f(1.838531) = -0.412799

Step 4: Iteration 3: x3 = 1.853636, and f(1.853636) = -0.047772

Step 5: Iteration 4: x4 = 1.855363, and f(1.855363) = -0.005430

Step 6: Iteration 5: x5 = 1.855559, and f(1.855559) = -0.000616

Step 7: Iteration 6: x6 = 1.855582, and f(1.855582) = -0.000070

Step 8: The successive approximations 1.855559 and 1.855582 agree to 2 decimal places (both 1.86), so we stop

Final answer: The root is 1.8556, that is 1.86 correct to 2 decimal places

Question: 4. Solve 3x + sin x - e^x = 0 by Regula Falsi method in the interval [0, 1] up to 3 decimal places.

Answer: Step 1: Let f(x) = 3x + sin x - e^x; f(0) = -1 and f(1) = 1.123189, so a root lies in [0, 1]

Step 2: Iteration 1: x1 = (0(1.123189) - 1(-1)) / (1.123189 - (-1)) = 1/2.123189 = 0.470990, and f(0.470990) = 0.265159, so the root lies in [0, 0.470990]

Step 3: Iteration 2: x2 = 0.372277, and f(0.372277) = 0.029534

Step 4: Iteration 3: x3 = 0.361598, and f(0.361598) = 0.002941

Step 5: Iteration 4: x4 = 0.360537, and f(0.360537) = 0.000289

Step 6: Iteration 5: x5 = 0.360433, and f(0.360433) = 0.000028

Step 7: Iteration 6: x6 = 0.360423, and f(0.360423) = 0.000003; successive approximations now agree at 0.3604

Final answer: The root is 0.3604 (0.360 to 3 decimal places)

Question: 5. Solve 2 cos x - x = 0 by Regula Falsi method in the interval [1, 2] up to 5 decimal places.

Answer: Step 1: Let f(x) = 2 cos x - x; f(1) = 0.080605 and f(2) = -2.832294, so a root lies in [1, 2]

Step 2: Iteration 1: x1 = (1(-2.832294) - 2(0.080605)) / (-2.832294 -

0.080605) = 1.027672, and f(1.027672) = 0.005956

Step 3: Iteration 2: x2 = 1.029712, and f(1.029712) = 0.000420

Step 4: Iteration 3: x3 = 1.029856, and f(1.029856) = 0.000029

Step 5: Iteration 4: x4 = 1.029866, and f(1.029866) = 0.000002

Step 6: Iteration 5: x5 = 1.029867, and f(1.029867) is practically 0; the approximations have settled to 5 decimal places

Final answer: The root is 1.02987

Exercise 12.3

Question: 1. Use Newton Raphson method to find the root of equation 2x^3 - 2x - 5 = 0 when x0 = 2. (3 iterations)

Answer: Step 1: Let f(x) = 2x^3 - 2x - 5, so f'(x) = 6x^2 - 2 and the Newton formula is x(n+1) = x(n) - f(x(n))/f'(x(n))

Step 2: Iteration 1: x0 = 2, f(2) = 7 and f'(2) = 22, so x1 = 2 - 7/22 = 1.681818

Step 3: Iteration 2: f(1.681818) = 1.150451 and f'(1.681818) = 14.971074, so x2 = 1.681818 - 1.150451/14.971074 = 1.604973

Step 4: Iteration 3: f(1.604973) = 0.058681 and f'(1.604973) = 13.455635, so x3 = 1.604973 - 0.058681/13.455635 = 1.600612

Final answer: After 3 iterations the approximate root is 1.600612

Question: 2. Use Newton Raphson method to find the root of equation x^3 - x - 1 = 0 when x0 = 1. (3 iterations)

Answer: Step 1: Let f(x) = x^3 - x - 1, so f'(x) = 3x^2 - 1

Step 2: Iteration 1: x0 = 1, f(1) = -1 and f'(1) = 2, so x1 = 1 - (-1)/2 = 1.500000

Step 3: Iteration 2: f(1.5) = 0.875 and f'(1.5) = 5.75, so x2 = 1.5 - 0.875/5.75 = 1.347826

Step 4: Iteration 3: f(1.347826) = 0.100682 and f'(1.347826) = 4.449905, so x3 = 1.347826 - 0.100682/4.449905 = 1.325200

Final answer: After 3 iterations the approximate root is 1.325200

Question: 3. Use Newton Raphson method to find the root of equation x^3 - 2x - 5 = 0 when x0 = 1. (2 iterations)

Answer: Step 1: Let f(x) = x^3 - 2x - 5, so f'(x) = 3x^2 - 2

Step 2: Iteration 1: x0 = 1, f(1) = -6 and f'(1) = 1, so x1 = 1 - (-6)/1 = 7.000000 (the first step jumps far away because f'(1) is small)

Step 3: Iteration 2: f(7) = 324 and f'(7) = 145, so x2 = 7 - 324/145 = 4.765517

Final answer: After 2 iterations the approximate root is 4.765517

Question: 4. Use Newton Raphson method to find the root of equation 2 cos x - x = 0 when x0 = 0.

Answer: Step 1: Let f(x) = 2 cos x - x, so f'(x) = -2 sin x - 1

Step 2: Iteration 1: x0 = 0, f(0) = 2 and f'(0) = -1, so x1 = 0 - 2/(-1) = 2.000000

Step 3: Iteration 2: f(2) = -2.832294 and f'(2) = -2.818595, so x2 = 2 - (-2.832294)/(-2.818595) = 0.995140

Step 4: Iteration 3: f(0.995140) = 0.093631 and f'(0.995140) = -2.677670, so x3 = 0.995140 - 0.093631/(-2.677670) = 1.030107

Step 5: Iteration 4: f(1.030107) = -0.000654 and f'(1.030107) = -2.714708, so x4 = 1.030107 - (-0.000654)/(-2.714708) = 1.029867

Final answer: The approximate root is 1.029867

Question: 5. Use Newton Raphson method to find the root of equation 2^x - x - 1.7 = 0 when x0 = 1.5.

Answer: Step 1: Let f(x) = 2^x - x - 1.7, so f'(x) = 2^x ln 2 - 1

Step 2: Iteration 1: x0 = 1.5, f(1.5) = -0.371573 and f'(1.5) = 0.960516, so x1 = 1.5 - (-0.371573)/0.960516 = 1.886847

Step 3: Iteration 2: f(1.886847) = 0.111414 and f'(1.886847) = 1.563439, so x2 = 1.886847 - 0.111414/1.563439 = 1.815585

Step 4: Iteration 3: f(1.815585) = 0.004438 and f'(1.815585) = 1.439894, so x3 = 1.815585 - 0.004438/1.439894 = 1.812503

Step 5: Iteration 4: f(1.812503) = 0.000008 and f'(1.812503) = 1.434687, so x4 = 1.812503 - 0.000008/1.434687 = 1.812497

Final answer: The approximate root is 1.812497

Question: 6. Use Newton Raphson method to find the root of equation 3x - e^x = 0 when x0 = 0.

Answer: Step 1: Let f(x) = 3x - e^x, so f'(x) = 3 - e^x

Step 2: Iteration 1: x0 = 0, f(0) = -1 and f'(0) = 2, so x1 = 0 - (-1)/2 = 0.500000

Step 3: Iteration 2: f(0.5) = -0.148721 and f'(0.5) = 1.351279, so x2 = 0.5 - (-0.148721)/1.351279 = 0.610060

Step 4: Iteration 3: f(0.610060) = -0.010362 and f'(0.610060) = 1.159459, so x3 = 0.610060 - (-0.010362)/1.159459 = 0.618997

Step 5: Iteration 4: f(0.618997) = -0.000074 and f'(0.618997) = 1.142936, so x4 = 0.618997 - (-0.000074)/1.142936 = 0.619061

Final answer: The approximate root is 0.619061

Question: 7. Use Newton Raphson method to find the root of equation 3x -

sqrt(1 + sin x) = 0 when x0 = 1.

Answer: Step 1: Let f(x) = 3x - sqrt(1 + sin x), so f'(x) = 3 - cos x / (2 sqrt(1 + sin x))

Step 2: Iteration 1: x0 = 1, f(1) = 1.642992 and f'(1) = 2.800921, so x1 = 1 - 1.642992/2.800921 = 0.413410

Step 3: Iteration 2: f(0.413410) = 0.056282 and f'(0.413410) = 2.613262, so x2 = 0.413410 - 0.056282/2.613262 = 0.391873

Step 4: Iteration 3: f(0.391873) = 0.000068 and f'(0.391873) = 2.606910, so x3 = 0.391873 - 0.000068/2.606910 = 0.391847

Step 5: Iteration 4: f(0.391847) is practically 0, so the value is confirmed

Final answer: The approximate root is 0.391847

Exercise 12.4

Question: 1. Evaluate the following integrals by Trapezoidal Rule: (i) integral from 0 to 2 of e^x dx (6 intervals) (ii) integral from 1 to 3 of 1/sqrt(x) dx (5 intervals) (iii) integral from 0 to pi/2 of sin x dx (7 intervals) (iv) integral from 0 to 1.4 of e^(-x^2) dx (5 intervals)

Answer: Step 1: The Trapezoidal Rule is I = (h/2)[y0 + yn + 2(y1 + y2 + ... + y(n-1))], where h = (b - a)/n

Step 2: (i) h = (2 - 0)/6 = 1/3 = 0.3333; the table of values is x: 0, 0.3333, 0.6667, 1, 1.3333, 1.6667, 2 and y = e^x: 1.0000, 1.3956, 1.9477, 2.7183, 3.7937, 5.2945, 7.3891

Step 3: (i) Substitute: I = (0.3333/2)[1.0000 + 7.3891 + 2(1.3956 + 1.9477 + 2.7183 + 3.7937 + 5.2945)] = 6.448105

Step 4: (ii) h = (3 - 1)/5 = 0.4; the table is x: 1, 1.4, 1.8, 2.2, 2.6, 3 and y = 1/sqrt(x): 1.0000, 0.8452, 0.7454, 0.6742, 0.6202, 0.5774

Step 5: (ii) Substitute: I = (0.4/2)[1.0000 + 0.5774 + 2(0.8452 + 0.7454 + 0.6742 + 0.6202)] = 1.469424

Step 6: (iii) h = (pi/2)/7 = 0.2244; the table is x: 0, 0.2244, 0.4488, 0.6732, 0.8976, 1.1220, 1.3464, 1.5708 and y = sin x: 0, 0.2225, 0.4339, 0.6235, 0.7818, 0.9010, 0.9749, 1.0000

Step 7: (iii) Substitute: I = (0.2244/2)[0 + 1.0000 + 2(0.2225 + 0.4339 + 0.6235 + 0.7818 + 0.9010 + 0.9749)] = 0.995800

Step 8: (iv) h = (1.4 - 0)/5 = 0.28; the table is x: 0, 0.28, 0.56, 0.84, 1.12, 1.4 and y = e^(-x^2): 1.0000, 0.9246, 0.7308, 0.4938, 0.2852, 0.1409

Step 9: (iv) Substitute: I = (0.28/2)[1.0000 + 0.1409 + 2(0.9246 + 0.7308 + 0.4938 + 0.2852)] = 0.841370

Final answer: (i) 6.448105 (ii) 1.469424 (iii) 0.995800 (iv) 0.841370

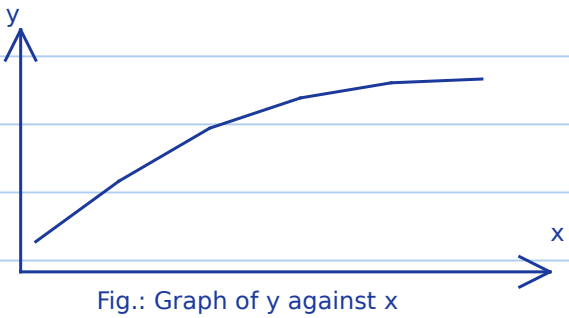


Fig.: Graph of y against x

Question: 2. Evaluate the following integrals by Simpson's 1/3 Rule: (i) integral from 2 to 3 of (4x^2 + 6) dx (4 intervals) (ii) integral from 0 to 2 of e^(-x) dx (6 intervals) (iii) integral from 0 to pi/3 of sqrt(sin x) dx (6 intervals) (iv) integral from 0 to 1 of 2/(1 + x^2) dx (8 intervals)

Answer: Step 1: Simpson's 1/3 Rule is I = (h/3)[y0 + yn + 4(y1 + y3 + ...) + 2(y2 + y4 + ...)], with n even

Step 2: (i) h = (3 - 2)/4 = 0.25; the table is x: 2, 2.25, 2.5, 2.75, 3 and y = 4x^2 + 6: 22, 26.25, 31, 36.25, 42

Step 3: (i) Substitute: I = (0.25/3)[22 + 42 + 4(26.25 + 36.25) + 2(31)] =

(0.25/3)(376) = 31.3333

Step 4: (ii) h = (2 - 0)/6 = 0.3333; the table is x: 0, 0.3333, 0.6667, 1, 1.3333, 1.6667, 2 and y = e^(-x): 1.0000, 0.7165, 0.5134, 0.3679, 0.2636, 0.1889, 0.1353

Step 5: (ii) Substitute: I = (0.3333/3)[1.0000 + 0.1353 + 4(0.7165 + 0.3679 + 0.1889) + 2(0.5134 + 0.2636)] = 0.864723

Step 6: (iii) h = (pi/3)/6 = 0.1745; the table is x: 0, 0.1745, 0.3491, 0.5236, 0.6981, 0.8727, 1.0472 and y = sqrt(sin x): 0, 0.4167, 0.5848, 0.7071, 0.8017, 0.8752, 0.9306

Step 7: (iii) Substitute: I = (0.1745/3)[0 + 0.9306 + 4(0.4167 + 0.7071 + 0.8752) + 2(0.5848 + 0.8017)] = 0.680676

Step 8: (iv) h = (1 - 0)/8 = 0.125; the table is x: 0, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 0.875, 1 and y = 2/(1 + x^2): 2.0000, 1.9692, 1.8824, 1.7534, 1.6000, 1.4382, 1.2800, 1.1327, 1.0000

Step 9: (iv) Substitute: I = (0.125/3)[2.0000 + 1.0000 + 4(1.9692 + 1.7534 + 1.4382 + 1.1327) + 2(1.8824 + 1.6000 + 1.2800)] = 1.570796

Final answer: (i) 31.3333 (ii) 0.864723 (iii) 0.680676 (iv) 1.570796

Question: 3. Evaluate the following integrals by Simpson's 3/8 Rule: (i) integral from 1 to 3 of sqrt(x) dx (6 intervals) (ii) integral from 1 to 6 of (ln x)/x dx (3 subintervals) (iii) integral from 0 to 2 of e^(2x)/(1 + x^2) dx (9 subintervals) (iv) integral from 0 to pi/4 of sin x dx (6 subintervals)

Answer: Step 1: Simpson's 3/8 Rule is I = (3h/8)[y0 + yn + 3(sum of y at indices not divisible by 3) + 2(sum of y at indices divisible by 3)], with n a multiple of 3

Step 2: (i) h = (3 - 1)/6 = 0.3333; the table is x: 1, 1.3333, 1.6667, 2, 2.3333, 2.6667, 3 and y = sqrt(x): 1.0000, 1.1547, 1.2910, 1.4142,

1.5275, 1.6330, 1.7321

Step 3: (i) Substitute: I = (3(0.3333)/8)[1.0000 + 1.7321 + 3(1.1547 + 1.2910 + 1.5275 + 1.6330) + 2(1.4142)] = 2.797390

Step 4: (ii) h = (6 - 1)/3 = 1.6667; the table is x: 1, 2.6667, 4.3333, 6 and y = (ln x)/x: 0, 0.3678, 0.3384, 0.2986

Step 5: (ii) Substitute: I = (3(1.6667)/8)[0 + 0.2986 + 3(0.3678 + 0.3384)] = (0.625)(2.4173) = 1.510760

Step 6: (iii) h = (2 - 0)/9 = 0.2222; the table is x: 0, 0.2222, 0.4444, 0.6667, 0.8889, 1.1111, 1.3333, 1.5556, 1.7778, 2 and y = e^(2x)/(1 + x^2): 1.0000, 1.4862, 2.0312, 2.6264, 3.3052, 4.1296, 5.1811, 6.5636, 8.4142, 10.9196

Step 7: (iii) Substitute: I = (3(0.2222)/8)[1.0000 + 10.9196 + 3(1.4862 + 2.0312 + 3.3052 + 4.1296 + 6.5636 + 8.4142) + 2(2.6264 + 5.1811)] = 8.777054

Step 8: (iv) h = (pi/4)/6 = 0.1309; the table is x: 0, 0.1309, 0.2618, 0.3927, 0.5236, 0.6545, 0.7854 and y = sin x: 0, 0.1305, 0.2588, 0.3827, 0.5000, 0.6088, 0.7071

Step 9: (iv) Substitute: I = (3(0.1309)/8)[0 + 0.7071 + 3(0.1305 + 0.2588 + 0.5000 + 0.6088) + 2(0.3827)] = 0.292894

Final answer: (i) 2.797390 (ii) 1.510760 (iii) 8.777054 (iv) 0.292894

Question: 4. Write MAPLE Command of the following: (i) Trapezoid of (x^2 - 2)/2 ; x = 0..1 ; n = 10 (ii) Trapezoid of sqrt(9 + x^2) ; x = 0..4 ; n = 10 (iii) SIMPSON of 1/(x^2 + 4x + 3) ; x = 0..2 ; n = 10 (iv) SIMPSON of e^(-x^2) ; x = 0..2 ; n = 10

Answer: Step 1: In MAPLE (student package), the trapezoidal approximation command is trapezoid(f, x = a..b, n) and the Simpson

approximation command is simpson(f, x = a..b, n); the command is followed by value(%) to get the decimal value

Step 2: (i) trapezoid((x^2 - 2)/2, x = 0..1, 10); the output is -333/400 = -0.8325

Step 3: (ii) trapezoid(sqrt(9 + x^2), x = 0..4, 10); the output is 14.9544232

Step 4: (iii) simpson(1/(x^2 + 4*x + 3), x = 0..2, 10); the output is 0.2938949

Step 5: (iv) simpson(exp(-x^2), x = 0..2, 10); the output is 0.88208098

Final answer: (i) trapezoid((x^2 - 2)/2, x = 0..1, 10) = -0.8325 (ii) trapezoid(sqrt(9 + x^2), x = 0..4, 10) = 14.9544232 (iii) simpson(1/(x^2 + 4*x + 3), x = 0..2, 10) = 0.2938949 (iv) simpson(exp(-x^2), x = 0..2, 10) = 0.88208098

Review Exercise 12

Question: 1(i). Select the correct option: If real root of an equation f(x) = 0 lies in the interval [a, b] then f(a) f(b) will be (a) positive (b) negative (c) zero (d) All of them

Answer: Step 1: A root lies in [a, b] when the function changes sign across the interval, that is f(a) and f(b) have opposite signs

Step 2: The product of two quantities of opposite signs is negative

Final answer: (b) negative

Question: 1(ii). In bisection method, the approximate root is a/an ---- of end point of an interval in which actual root lies (a) Arithmetic mean (b) Geometric mean (c) Sum (d) Product

Answer: Step 1: In the bisection method each new approximation is the

midpoint of the current interval

Step 2: The midpoint (a + b)/2 is the arithmetic mean of the two endpoints

Final answer: (a) Arithmetic mean

Question: 1(iii). Iterative formula for False Position Method to solve the equation f(x) = 0 at interval [a, b] is (a) (a f(a) - b f(b))/(f(a) - f(b)) (b) (a f(b) - b f(a))/(a - b) (c) (a f(a) - b f(b))/(f(b) - f(a)) (d) (a f(b) - b f(a))/(f(b) - f(a))

Answer: Step 1: The false position formula joins the points (a, f(a)) and (b, f(b)) by a chord and takes its x-intercept

Step 2: That intercept is x = (a f(b) - b f(a))/(f(b) - f(a))

Final answer: (d) (a f(b) - b f(a))/(f(b) - f(a))

Question: 1(iv). The fastest method to solve the nonlinear equation numerically is (a) Bisection Method (b) False Position Method (c) Newton Raphson Method (d) Both a and b

Answer: Step 1: The bisection method converges linearly and slowly, and the false position method is faster but still linear in general

Step 2: The Newton Raphson method converges quadratically near the root, making it the fastest of these methods

Final answer: (c) Newton Raphson Method

Question: 1(v). Newton Raphson Method fails when derivative value of f(x) becomes (a) positive (b) negative (c) zero (d) All of them

Answer: Step 1: The Newton formula divides by f'(x(n)) at every step

Step 2: If f'(x(n)) = 0 the division is impossible and the method fails

Final answer: (c) zero

Question: 1(vi). Iterative formula of Newton Raphson method of solve f(x) = 0 is ---- (a) x(n+1) = x(n) + f'(x(n))/f(x(n)) (b) x(n+1) = x(n) -

f'(x(n))/f(x(n)) (c) x(n+1) = x(n) + f(x(n))/f'(x(n)) (d) x(n+1) = x(n) - f(x(n))/f'(x(n))

Answer: Step 1: Newton's iteration subtracts the correction term f(x(n))/f'(x(n)) from the current approximation

Step 2: Hence x(n+1) = x(n) - f(x(n))/f'(x(n))

Final answer: (d) x(n+1) = x(n) - f(x(n))/f'(x(n))

Question: 1(vii). Numerical integration comprises a broad family of algorithms for calculating the numerical value of a? (a) Definite integral (b) Indefinite integral (c) Simple integral (d) Compound integral

Answer: Step 1: Numerical integration methods such as the Trapezoidal and Simpson rules approximate the area under a curve between two fixed limits

Step 2: An integral with fixed limits is a definite integral

Final answer: (a) Definite integral

Question: 1(viii). In Trapezoidal rule the number of sub interval is the multiple of: (a) 0 (b) 1 (c) 2 (d) 3

Answer: Step 1: The Trapezoidal rule joins consecutive points by straight line segments and works for any number n of subintervals, with no evenness or divisibility restriction

Step 2: Hence n can be any positive integer, that is a multiple of 1

Final answer: (b) 1

Question: 1(ix). In Simpson One Third Method, the number of subinterval is the multiple of (a) 4 (b) 1 (c) 2 (d) 3

Answer: Step 1: Simpson's one-third rule fits parabolas over pairs of subintervals, so the subintervals must be grouped in twos

Step 2: Hence the number of subintervals must be even, that is a multiple

of 2

Final answer: (c) 2

Question: 1(x). The fastest method to solve the definite integral numerically is (a) Trapezoidal Rule (b) Simpson One Third Rule (c) Simpson Three Eight Rule (d) Both a and b

Answer: Step 1: The Trapezoidal rule uses straight line segments and is the least accurate of the three for smooth curves

Step 2: Simpson's one-third rule uses parabolic arcs and gives exact results up to cubic polynomials, making it the fastest (most accurate) of these methods for a definite integral

Final answer: (b) Simpson One Third Rule

Question: 2. Using Bisection method find the root of cos x - x e^x with a = 0 and b = 1, by taking 5 iterations.

Answer: Step 1: Let f(x) = cos x - x e^x; f(0) = 1 and f(1) = -2.1780, so a root lies in [0, 1]

Step 2: Iteration 1: c = 0.5, f(0.5) = 0.053222, positive, so the root lies in [0.5, 1]

Step 3: Iteration 2: c = 0.75, f(0.75) = -0.856061, so the root lies in [0.5, 0.75]

Step 4: Iteration 3: c = 0.625, f(0.625) = -0.356691, so the root lies in [0.5, 0.625]

Step 5: Iteration 4: c = 0.5625, f(0.5625) = -0.141294, so the root lies in [0.5, 0.5625]

Step 6: Iteration 5: c = 0.53125, f(0.53125) = -0.041512, so the root lies in [0.5, 0.53125]

Final answer: After 5 iterations the approximate root is 0.53125 (root lies

in [0.5, 0.53125])

Question: 3. Find a root for the equation 2 e^x sin x = 1 using the false position method and correct it to three decimal places with three iterations, taking [1, 2] as an interval.

Answer: Step 1: Let f(x) = 2 e^x sin x - 1; f(1) = 3.574711 and f(2) = 12.437699

Step 2: Iteration 1: apply x = (a f(b) - b f(a))/(f(b) - f(a)) with a = 1, b = 2: x1 = (1(12.437699) - 2(3.574711))/(12.437699 - 3.574711) = 5.288277/8.862988 = 0.596670, and f(0.596670) = 1.040856

Step 3: Iteration 2: apply the formula with the points x = 0.596670 (f = 1.040856) and x = 1 (f = 3.574711): x2 = (0.596670(3.574711) - 1(1.040856))/(3.574711 - 1.040856) = 0.430990, and f(0.430990) = 0.285714

Step 4: Iteration 3: apply the formula with the points x = 0.430990 (f = 0.285714) and x = 1 (f = 3.574711): x3 = (0.430990(3.574711) - 1(0.285714))/(3.574711 - 0.285714) = 0.381560

Final answer: After three iterations the root, correct to three decimal places, is 0.38156 (0.382)

Question: 4. Find the cube root of 12 using the Newton Raphson method assuming x0 = 2.5.

Answer: Step 1: The cube root of 12 is the root of f(x) = x^3 - 12 = 0, with f'(x) = 3x^2

Step 2: Iteration 1: x0 = 2.5, f(2.5) = 3.625 and f'(2.5) = 18.75, so x1 = 2.5 - 3.625/18.75 = 2.306667

Step 3: Iteration 2: f(2.306667) = 0.273107 and f'(2.306667) = 15.962133, so x2 = 2.306667 - 0.273107/15.962133 = 2.289557

Step 4: Iteration 3: f(2.289557) = 0.002021 and f'(2.289557) = 15.726214, so x3 = 2.289557 - 0.002021/15.726214 = 2.289428

Final answer: The cube root of 12 is approximately 2.289

Question: 5. Solve integral from 0 to 1 of cos(x^2) dx using trapezoidal rule for n = 5.

Answer: Step 1: h = (1 - 0)/5 = 0.2; the table is x: 0, 0.2, 0.4, 0.6, 0.8, 1 and y = cos(x^2): 1.0000, 0.9992, 0.9872, 0.9359, 0.8021, 0.5403

Step 2: Apply the Trapezoidal Rule: I = (h/2)[y0 + y5 + 2(y1 + y2 + y3 + y4)]

Step 3: Substitute: I = (0.2/2)[1.0000 + 0.5403 + 2(0.9992 + 0.9872 + 0.9359 + 0.8021)] = (0.1)(8.9891) = 0.898914

Final answer: The integral is approximately 0.898914

Question: 6. Solve integral from -2 to 4 of e^(x^2) dx using Simpson one Third as well as Simpson Three Eight rule for n = 6.

Answer: Step 1: h = (4 - (-2))/6 = 1; the table is x: -2, -1, 0, 1, 2, 3, 4 and y = e^(x^2): 54.598150, 2.718282, 1.000000, 2.718282, 54.598150, 8103.083928, 8886110.520508

Step 2: Simpson's one-third rule: I = (h/3)[y0 + y6 + 4(y1 + y3 + y5) + 2(y2 + y4)]

Step 3: Substitute: I = (1/3)[54.598150 + 8886110.520508 + 4(2.718282 + 2.718282 + 8103.083928) + 2(1.000000 + 54.598150)] = 2972903.4656

Step 4: Simpson's three-eight rule: I = (3h/8)[y0 + y6 + 3(y1 + y2 + y4 + y5) + 2(y3)]

Step 5: Substitute: I = (3/8)[54.598150 + 8886110.520508 + 3(2.718282 + 1.000000 + 54.598150 + 8103.083928) + 2(2.718282)] =

3341495.5336

Final answer: By Simpson's one-third rule the integral is 2972903.47 and by Simpson's three-eight rule it is 3341495.53